

```

1 classdef FUNMAP_Output < matlab.apps.AppBase      2

3     % Properties that correspond to app components
4     properties (Access = public)
5         UIFigure                                matlab.ui.Figure
6         TabGroup                                matlab.ui.container.TabGroup
7         WaveHeightMapTab                        matlab.ui.container.Tab
8         GeneralLayoutLabel_3                    matlab.ui.control.Label
9         Panel_19                                matlab.ui.container.Panel
10        ButtonGroup_11                           matlab.ui.container.ButtonGroup
11        PlotseparatelyButton_2                   matlab.ui.control.RadioButton
12        PlotalldataainonefigureButton            matlab.ui.control.RadioButton
13        SettoDefaultButton_2                     matlab.ui.control.Button
14        CloseFiguresButton_4                     matlab.ui.control.Button
15        FigureSizeLabel_2                        matlab.ui.control.Label
16        ThicknessLabel_9                         matlab.ui.control.Label
17        Width_2                                  matlab.ui.control.NumericEditField
18        AutoSetCheckBox_2                         matlab.ui.control.CheckBox
19        ThicknessLabel_8                         matlab.ui.control.Label
20        Height_2                                 matlab.ui.control.NumericEditField
21        SouthEditField                           matlab.ui.control.NumericEditField
22        SouthEditFieldLabel                      matlab.ui.control.Label
23        NorthEditField                           matlab.ui.control.NumericEditField
24        NorthEditFieldLabel                      matlab.ui.control.Label
25        WestEditField                            matlab.ui.control.NumericEditField
26        WestEditFieldLabel                       matlab.ui.control.Label
27        EastEditField                            matlab.ui.control.NumericEditField
28        EastEditFieldLabel                       matlab.ui.control.Label
29        PlotindegreesCheckBox_2                  matlab.ui.control.CheckBox
30        BoundaryLimitsLabel                      matlab.ui.control.Label
31        GENERATEButton                           matlab.ui.control.Button
32        SaveMapLabel                             matlab.ui.control.Label
33        Panel_6                                  matlab.ui.container.Panel
34        mp4CheckBox                              matlab.ui.control.CheckBox
35        FramerateEditField_2                     matlab.ui.control.NumericEditField
36        FramerateEditField_2Label                matlab.ui.control.Label
37        OutputDirectoryEditField                 matlab.ui.control.EditField
38        OutputDirectoryLabel                     matlab.ui.control.Label
39        Button_2                                 matlab.ui.control.Button
40        tifCheckBox_2                             matlab.ui.control.CheckBox
41        jpgCheckBox                              matlab.ui.control.CheckBox
42        FileFormatLabel                          matlab.ui.control.Label
43        pngCheckBox                              matlab.ui.control.CheckBox
44        BasemapLabel                             matlab.ui.control.Label
45        Panel_5                                  matlab.ui.container.Panel
46        ColorbarTextSize                         matlab.ui.control.NumericEditField
47        TextSizeLabel_2                          matlab.ui.control.Label
48        CoastlinecolourDropDown_2                matlab.ui.control.DropDown
49        FlipBasemapLabel_2                       matlab.ui.control.Label
50        VerticalCheckBox                          matlab.ui.control.CheckBox
51        HorizontalCheckBox                       matlab.ui.control.CheckBox
52        CoastlinecolourDropDownLabel_9           matlab.ui.control.Label
53        mLabel_2                                 matlab.ui.control.Label
54        mLabel                                    matlab.ui.control.Label
55        toLabel_2                                matlab.ui.control.Label
56        MinEditField_2                           matlab.ui.control.NumericEditField
57        MaxEditField_2                           matlab.ui.control.NumericEditField
58        DivisionLabel                            matlab.ui.control.Label

```

59	ColorinterpolationEditField_2	matlab.ui.control.NumericEditField
60	FlipCheckBox_2	matlab.ui.control.CheckBox
61	CoastlinecolourDropDown_7	matlab.ui.control.DropDown
62	etaLabel	matlab.ui.control.Label
63	toLabel	matlab.ui.control.Label
64	CoastlinecolourDropDownLabel_10	matlab.ui.control.Label
65	toEditField	matlab.ui.control.NumericEditField
66	MinEditField	matlab.ui.control.NumericEditField
67	hmaxLabel	matlab.ui.control.Label
68	CoastlinecolourDropDownLabel_8	matlab.ui.control.Label
69	CoastlinecolourDropDown_8	matlab.ui.control.DropDown
70	FlipCheckBox_5	matlab.ui.control.CheckBox
71	DivisionEditField	matlab.ui.control.NumericEditField
72	DivisionEditFieldLabel	matlab.ui.control.Label
73	OverlayFeaturesLabel	matlab.ui.control.Label
74	Panel_7	matlab.ui.container.Panel
75	GaugesCheckBox	matlab.ui.control.CheckBox
76	ButtonGroup_17	matlab.ui.container.ButtonGroup
77	NoneButton	matlab.ui.control.RadioButton
78	ArrivalTimeButton	matlab.ui.control.RadioButton
79	WaveHeightButton	matlab.ui.control.RadioButton
80	BathymetryButton_2	matlab.ui.control.RadioButton
81	TabGroup2	matlab.ui.container.TabGroup
82	BathymetryTab	matlab.ui.container.Tab
83	IntervalEditField_3	matlab.ui.control.NumericEditField
84	LabelSpacingEditFieldLabel	matlab.ui.control.Label
85	SpacingEditField_4Label_4	matlab.ui.control.Label
86	TextLabelSize	matlab.ui.control.NumericEditField
87	AddLabelsCheckBox_2	matlab.ui.control.CheckBox
88	ColorDropDown	matlab.ui.control.DropDown
89	ColorDropDownLabel	matlab.ui.control.Label
90	StyleDropDown	matlab.ui.control.DropDown
91	StyleDropDownLabel	matlab.ui.control.Label
92	LineLabel	matlab.ui.control.Label
93	WidthEditField	matlab.ui.control.NumericEditField
94	WidthEditFieldLabel	matlab.ui.control.Label
95	LabelSpacingEditField	matlab.ui.control.NumericEditField
96	LabelSpacingEditFieldLabel_2	matlab.ui.control.Label
97	DepthRangeLabel	matlab.ui.control.Label
98	IntervalEditField_5	matlab.ui.control.NumericEditField
99	IntervalEditField_5Label	matlab.ui.control.Label
100	MinimumEditField	matlab.ui.control.NumericEditField
101	MinimumEditFieldLabel	matlab.ui.control.Label
102	MaximumEditField	matlab.ui.control.NumericEditField
103	MaximumEditFieldLabel	matlab.ui.control.Label
104	WaveHeightTab	matlab.ui.container.Tab
105	wh_interval	matlab.ui.control.NumericEditField
106	LabelSpacingEditField_2Label	matlab.ui.control.Label
107	LabelSpacingEditField_2	matlab.ui.control.NumericEditField
108	LabelSpacingEditField_2Label_2	matlab.ui.control.Label
109	SpacingEditField_4Label_5	matlab.ui.control.Label
110	TextLabelSize_2	matlab.ui.control.NumericEditField
111	AddLabelsCheckBox_3	matlab.ui.control.CheckBox
112	StyleDropDown_2	matlab.ui.control.DropDown
113	StyleDropDown_2Label	matlab.ui.control.Label
114	ColorDropDown_2	matlab.ui.control.DropDown
115	ColorDropDown_2Label	matlab.ui.control.Label
116	ThicknessEditField 2	matlab.ui.control.NumericEditField

117	ThicknessEditField_2Label	matlab.ui.control.Label
118	LineLabel_2	matlab.ui.control.Label
119	ContourRangeLabel_2	matlab.ui.control.Label
120	MaximumdepthEditFieldLabel_3	matlab.ui.control.Label
121	MinimumdepthEditFieldLabel_2	matlab.ui.control.Label
122	MaximumdepthEditFieldLabel_2	matlab.ui.control.Label
123	LineintervalEditField_2	matlab.ui.control.NumericEditField
124	MaximumdepthEditField_2	matlab.ui.control.NumericEditField
125	MinimumdepthEditField_2	matlab.ui.control.NumericEditField
126	ArrivalTimeTab	matlab.ui.container.Tab
127	LabelSpacingEditField_2Label_3	matlab.ui.control.Label
128	LabelSpacingEditField_4	matlab.ui.control.NumericEditField
129	StyleDropDown_4	matlab.ui.control.DropDown
130	StyleDropDown_4Label_2	matlab.ui.control.Label
131	ColorDropDown_6	matlab.ui.control.DropDown
132	ColorDropDown_6Label	matlab.ui.control.Label
133	ThicknessEditField_2Label_2	matlab.ui.control.Label
134	ThicknessEditField_3	matlab.ui.control.NumericEditField
135	LineLabel_5	matlab.ui.control.Label
136	AddLabelsCheckBox_5	matlab.ui.control.CheckBox
137	SpacingEditField_4Label_6	matlab.ui.control.Label
138	TextLabelSize_3	matlab.ui.control.NumericEditField
139	IntervalEditField_7	matlab.ui.control.NumericEditField
140	LabelSpacingEditField_2Label_4	matlab.ui.control.Label
141	MinimumdepthEditFieldLabel_5	matlab.ui.control.Label
142	MinimumdepthEditField_3	matlab.ui.control.NumericEditField
143	MaximumdepthEditFieldLabel_10	matlab.ui.control.Label
144	LineintervalEditField_3	matlab.ui.control.NumericEditField
145	MaximumdepthEditFieldLabel_9	matlab.ui.control.Label
146	MaximumdepthEditField_3	matlab.ui.control.NumericEditField
147	ContourRangeLabel	matlab.ui.control.Label
148	FileEditField_3	matlab.ui.control.EditField
149	FileEditField_3Label	matlab.ui.control.Label
150	TimeFileButton	matlab.ui.control.Button
151	GaugesTab	matlab.ui.container.Tab
152	GLspacing	matlab.ui.control.NumericEditField
153	GLspacingLabel	matlab.ui.control.Label
154	SizeEditField_3	matlab.ui.control.NumericEditField
155	SizeEditField_3Label	matlab.ui.control.Label
156	CoastlinecolourDropDown_5	matlab.ui.control.DropDown
157	CoastlinecolourDropDown_4	matlab.ui.control.DropDown
158	AlignmentLabel	matlab.ui.control.Label
159	AddLabelsCheckBox_4	matlab.ui.control.CheckBox
160	SizeEditField	matlab.ui.control.NumericEditField
161	SizeEditFieldLabel	matlab.ui.control.Label
162	CoastlinecolourDropDown_3	matlab.ui.control.DropDown
163	CoastlinecolourDropDownLabel_3	matlab.ui.control.Label
164	ColorDropDown_3	matlab.ui.control.DropDown
165	ColorDropDown_3Label	matlab.ui.control.Label
166	FileEditField	matlab.ui.control.EditField
167	GaugeFileLabel	matlab.ui.control.Label
168	Button_8	matlab.ui.control.Button
169	Tab	matlab.ui.container.Tab
170	InputDataLabel	matlab.ui.control.Label
171	Panel_3	matlab.ui.container.Panel
172	gridY	matlab.ui.control.NumericEditField
173	YEditField_2Label	matlab.ui.control.Label
174	gridX	matlab.ui.control.NumericEditField

175	XEditField_2Label	matlab.ui.control.Label
176	GridSizeLabel_2	matlab.ui.control.Label
177	StartLabel	matlab.ui.control.Label
178	StartTime	matlab.ui.control.NumericEditField
179	IntervalLabel	matlab.ui.control.Label
180	TotalSimuilationTimesecEditField_3	matlab.ui.control.NumericEditField
181	TotalSimuilationTimesecEditFieldLabel_3	matlab.ui.control.Label
182	BathymetryLabel	matlab.ui.control.Label
183	FilesLabel	matlab.ui.control.Label
184	DepthFileEditField	matlab.ui.control.EditField
185	Button_15	matlab.ui.control.Button
186	LatitudeEditField	matlab.ui.control.NumericEditField
187	LatitudeEditFieldLabel	matlab.ui.control.Label
188	LongitudeEditField	matlab.ui.control.NumericEditField
189	LongitudeEditFieldLabel	matlab.ui.control.Label
190	SouthwestCornerLabel	matlab.ui.control.Label
191	Button_14	matlab.ui.control.Button
192	FileTextArea	matlab.ui.control.TextArea
193	GaugeRecordsTab	matlab.ui.container.Tab
194	PlotStyleandLayoutLabel	matlab.ui.control.Label
195	SavePlotLabel	matlab.ui.control.Label
196	Panel_13	matlab.ui.container.Panel
197	epsCheckBox	matlab.ui.control.CheckBox
198	txtCheckBox	matlab.ui.control.CheckBox
199	OutputDirectoryEditField_2	matlab.ui.control.EditField
200	OutputDirectoryLabel_3	matlab.ui.control.Label
201	Button_18	matlab.ui.control.Button
202	pdfCheckBox_2	matlab.ui.control.CheckBox
203	jpgCheckBox_2	matlab.ui.control.CheckBox
204	FileFormatLabel_2	matlab.ui.control.Label
205	pngCheckBox_2	matlab.ui.control.CheckBox
206	InputDataLabel_3	matlab.ui.control.Label
207	Panel_11	matlab.ui.container.Panel
208	XaxisUseDataDropDown	matlab.ui.control.DropDown
209	YaxisUseDataDropDown	matlab.ui.control.DropDown
210	YaxisSetcolumncountLabel	matlab.ui.control.Label
211	Label_11	matlab.ui.control.Label
212	Label_10	matlab.ui.control.Label
213	Label_7	matlab.ui.control.Label
214	Label_6	matlab.ui.control.Label
215	Label_5	matlab.ui.control.Label
216	denominatorX	matlab.ui.control.NumericEditField
217	numeratorX	matlab.ui.control.NumericEditField
218	CheckBox_4	matlab.ui.control.CheckBox
219	CheckBox_3	matlab.ui.control.CheckBox
220	YaxisEditFieldLabel	matlab.ui.control.Label
221	Label_9	matlab.ui.control.Label
222	denominatorY	matlab.ui.control.NumericEditField
223	numeratorY	matlab.ui.control.NumericEditField
224	ConvertLabel	matlab.ui.control.Label
225	XaxisdataLabel	matlab.ui.control.Label
226	SelectColumnLabel	matlab.ui.control.Label
227	XaxisLabel	matlab.ui.control.Label
228	Button_16	matlab.ui.control.Button
229	FileTextArea_2	matlab.ui.control.TextArea
230	FilesLabel_2	matlab.ui.control.Label
231	PLOTButton	matlab.ui.control.Button
232	Panel_10	matlab.ui.container.Panel

233	MarkerDropDown	matlab.ui.control.DropDown
234	MarkerDropDownLabel	matlab.ui.control.Label
235	ThicknessLabel_4	matlab.ui.control.Label
236	LegendSize	matlab.ui.control.NumericEditField
237	LocationDropDown	matlab.ui.control.DropDown
238	LocationDropDownLabel	matlab.ui.control.Label
239	legendfirsttext	matlab.ui.control.EditField
240	legendfirsttextlabel	matlab.ui.control.Label
241	LegendLabel	matlab.ui.control.Label
242	ThicknessLabel_12	matlab.ui.control.Label
243	LineYThickness_2	matlab.ui.control.NumericEditField
244	CloseFiguresButton_2	matlab.ui.control.Button
245	AutoSetCheckBox	matlab.ui.control.CheckBox
246	ThicknessLabel_7	matlab.ui.control.Label
247	Width	matlab.ui.control.NumericEditField
248	FigureSizeLabel	matlab.ui.control.Label
249	ThicknessLabel_6	matlab.ui.control.Label
250	Height	matlab.ui.control.NumericEditField
251	FlipCheckBox_3	matlab.ui.control.CheckBox
252	GaugeLineColor_multiple	matlab.ui.control.DropDown
253	GridLabelSize	matlab.ui.control.NumericEditField
254	ThicknessLabel_5	matlab.ui.control.Label
255	ThicknessLabel_3	matlab.ui.control.Label
256	LineYThickness	matlab.ui.control.NumericEditField
257	Lineaty0DropDown	matlab.ui.control.DropDown
258	Lineaty0DropDownLabel	matlab.ui.control.Label
259	GridStyle	matlab.ui.control.DropDown
260	StyleDropDown_4Label	matlab.ui.control.Label
261	BackgroundGridLabel	matlab.ui.control.Label
262	XAxisEditField	matlab.ui.control.EditField
263	YAxisLabelEditField_2Label	matlab.ui.control.Label
264	XandYLimitsLabel_2	matlab.ui.control.Label
265	YAxisEditField	matlab.ui.control.EditField
266	LabelEditFieldLabel	matlab.ui.control.Label
267	XAutoSetLabel	matlab.ui.control.Label
268	YAutoSetLabel	matlab.ui.control.Label
269	CheckBox	matlab.ui.control.CheckBox
270	yMaxLimit	matlab.ui.control.NumericEditField
271	MinimumdepthEditFieldLabel_3	matlab.ui.control.Label
272	yMinLimit	matlab.ui.control.NumericEditField
273	MaximumdepthEditFieldLabel_8	matlab.ui.control.Label
274	yLimInterval	matlab.ui.control.NumericEditField
275	MaximumdepthEditFieldLabel_4	matlab.ui.control.Label
276	CheckBox_2	matlab.ui.control.CheckBox
277	MaximumdepthEditFieldLabel_6	matlab.ui.control.Label
278	xLimInterval	matlab.ui.control.NumericEditField
279	MaximumdepthEditFieldLabel_5	matlab.ui.control.Label
280	xMaxLimit	matlab.ui.control.NumericEditField
281	MinimumdepthEditFieldLabel_4	matlab.ui.control.Label
282	xMinLimit	matlab.ui.control.NumericEditField
283	TextArea2_4	matlab.ui.control.TextArea
284	AxesLabels	matlab.ui.control.Label
285	WidthLabel	matlab.ui.control.Label
286	DataLineWidth	matlab.ui.control.NumericEditField
287	ButtonGroup_10	matlab.ui.container.ButtonGroup
288	PlotseparatelyButton_4	matlab.ui.control.RadioButton
289	PlotalldataainonegraphButton	matlab.ui.control.RadioButton
290	GaugeLineStyle	matlab.ui.control.DropDown

291	LineStyleLabel	matlab.ui.control.Label
292	GaugeLineColor	matlab.ui.control.DropDown
293	ColorDropDownLabel_2	matlab.ui.control.Label
294	PlotPropertiesLabel	matlab.ui.control.Label
295	VelocityMapTab	matlab.ui.container.Tab
296	GeneralLayoutLabel_2	matlab.ui.control.Label
297	Panel_18	matlab.ui.container.Panel
298	ButtonGroup_12	matlab.ui.container.ButtonGroup
299	PlotseparatelyButton_3	matlab.ui.control.RadioButton
300	PlotalldataainonefigureButton_2	matlab.ui.control.RadioButton
301	SettoDefaultButton	matlab.ui.control.Button
302	CloseFiguresButton_3	matlab.ui.control.Button
303	FigureSizeLabel_3	matlab.ui.control.Label
304	ThicknessLabel_11	matlab.ui.control.Label
305	Width_3	matlab.ui.control.NumericEditField
306	AutoSetCheckBox_3	matlab.ui.control.CheckBox
307	ThicknessLabel_10	matlab.ui.control.Label
308	Height_3	matlab.ui.control.NumericEditField
309	SouthEditField_2	matlab.ui.control.NumericEditField
310	SouthEditField_2Label	matlab.ui.control.Label
311	NorthEditField_2	matlab.ui.control.NumericEditField
312	NorthEditField_2Label	matlab.ui.control.Label
313	WestEditField_2	matlab.ui.control.NumericEditField
314	WestEditField_2Label	matlab.ui.control.Label
315	EastEditField_2	matlab.ui.control.NumericEditField
316	EastEditField_2Label	matlab.ui.control.Label
317	PlotindegreesCheckBox	matlab.ui.control.CheckBox
318	BoundaryLimitsLabel_2	matlab.ui.control.Label
319	BasemapandOverlaysLabel	matlab.ui.control.Label
320	SaveMapLabel_2	matlab.ui.control.Label
321	Panel_17	matlab.ui.container.Panel
322	tifCheckBox	matlab.ui.control.CheckBox
323	txtCheckBox_2	matlab.ui.control.CheckBox
324	OutputDirectoryEditField_3	matlab.ui.control.EditField
325	OutputDirectoryLabel_2	matlab.ui.control.Label
326	mp4CheckBox_2	matlab.ui.control.CheckBox
327	FramerateEditField	matlab.ui.control.NumericEditField
328	FramerateEditFieldLabel	matlab.ui.control.Label
329	Button_26	matlab.ui.control.Button
330	jpgCheckBox_3	matlab.ui.control.CheckBox
331	FileFormatLabel_3	matlab.ui.control.Label
332	pngCheckBox_3	matlab.ui.control.CheckBox
333	Panel_16	matlab.ui.container.Panel
334	TabGroup3	matlab.ui.container.TabGroup
335	BasemapTab	matlab.ui.container.Tab
336	LandColor	matlab.ui.control.DropDown
337	CoastlinecolourDropDownLabel_12	matlab.ui.control.Label
338	FlipBasemapLabel	matlab.ui.control.Label
339	VerticalCheckBox_2	matlab.ui.control.CheckBox
340	HorizontalCheckBox_2	matlab.ui.control.CheckBox
341	FileTextArea_5	matlab.ui.control.TextArea
342	ColorbarLabel	matlab.ui.control.Label
343	ColormapLabel	matlab.ui.control.Label
344	ThirdTabColorMapPanel	matlab.ui.container.Panel
345	TextSizeLabel	matlab.ui.control.Label
346	LimitsLabel	matlab.ui.control.Label
347	MaxBarValue	matlab.ui.control.NumericEditField
348	mLabel_3	matlab.ui.control.Label

349	toLabel_3	matlab.ui.control.Label
350	MinBarValue	matlab.ui.control.NumericEditField
351	InterpolationDivisionEditField_2	matlab.ui.control.NumericEditField
352	InterpolationDivisionEditFieldLabel_2	matlab.ui.control.Label
353	FlipCheckBox_4	matlab.ui.control.CheckBox
354	BackgroundMapColorDropDown	matlab.ui.control.DropDown
355	maplabelsize	matlab.ui.control.NumericEditField
356	ButtonGroup_16	matlab.ui.container.ButtonGroup
357	VorticityButton	matlab.ui.control.RadioButton
358	BathymetryButton	matlab.ui.control.RadioButton
359	hmaxButton	matlab.ui.control.RadioButton
360	etaButton	matlab.ui.control.RadioButton
361	VelocityButton	matlab.ui.control.RadioButton
362	BathymetryContoursTab_2	matlab.ui.container.Tab
363	IntervalEditField_8	matlab.ui.control.NumericEditField
364	SpacingEditField	matlab.ui.control.NumericEditField
365	SpacingEditField_4Label_2	matlab.ui.control.Label
366	SpacingEditField_4Label_3	matlab.ui.control.Label
367	LabelSizeCont	matlab.ui.control.NumericEditField
368	IntervalEditField_8Label	matlab.ui.control.Label
369	WidthEditField_2	matlab.ui.control.NumericEditField
370	WidthEditField_2Label	matlab.ui.control.Label
371	StyleDropDown_3	matlab.ui.control.DropDown
372	StyleDropDown_3Label_2	matlab.ui.control.Label
373	ColorDropDown_4	matlab.ui.control.DropDown
374	ColorDropDown_4Label	matlab.ui.control.Label
375	AddLabelCheckBox_2	matlab.ui.control.CheckBox
376	LineLabel_4	matlab.ui.control.Label
377	IntervalEditField_6	matlab.ui.control.NumericEditField
378	IntervalEditField_6Label	matlab.ui.control.Label
379	MaximumEditField_2	matlab.ui.control.NumericEditField
380	MaximumEditField_2Label	matlab.ui.control.Label
381	MinimumEditField_2	matlab.ui.control.NumericEditField
382	MinimumEditField_2Label	matlab.ui.control.Label
383	PlotBathymetryContoursCheckBox	matlab.ui.control.CheckBox
384	RangeLabel	matlab.ui.control.Label
385	GaugesTab_2	matlab.ui.container.Tab
386	SizeEditField_2	matlab.ui.control.NumericEditField
387	SizeEditField_2Label	matlab.ui.control.Label
388	CoastlinecolourDropDown_11	matlab.ui.control.DropDown
389	CoastlinecolourDropDown_10	matlab.ui.control.DropDown
390	AlignmentLabel_2	matlab.ui.control.Label
391	GLspacingLabel_2	matlab.ui.control.Label
392	GLspacing_2	matlab.ui.control.NumericEditField
393	FontSizeEditField_2	matlab.ui.control.NumericEditField
394	FontSizeEditField_2Label	matlab.ui.control.Label
395	AddLabelCheckBox	matlab.ui.control.CheckBox
396	PlotGaugesCheckBox	matlab.ui.control.CheckBox
397	FileEditField_2	matlab.ui.control.EditField
398	GaugeFileLabel_2	matlab.ui.control.Label
399	Button_27	matlab.ui.control.Button
400	ColorDropDown_7	matlab.ui.control.DropDown
401	ColorDropDown_7Label	matlab.ui.control.Label
402	gaugemarkerVelocityTab	matlab.ui.control.DropDown
403	CoastlinecolourDropDownLabel_19	matlab.ui.control.Label
404	ArrowsLabel	matlab.ui.control.Label
405	InputDataLabel_4	matlab.ui.control.Label
406	Panel 15	matlab.ui.container.Panel

```

407         Button_19                                matlab.ui.control.Button
408         FileTextArea_4                            matlab.ui.control.TextArea
409         gridX_2                                    matlab.ui.control.NumericEditField
410         XEditField_2Label_2                        matlab.ui.control.Label
411         TotalSimuilationTimesecEditField_4        matlab.ui.control.NumericEditField
412         IntervalLabel_2                            matlab.ui.control.Label
413         StartLabel_2                               matlab.ui.control.Label
414         StartTime2                                 matlab.ui.control.NumericEditField
415         TotalSimuilationTimesecEditFieldLabel_4    matlab.ui.control.Label
416         gridY_2                                    matlab.ui.control.NumericEditField
417         YEditField_2Label_2                        matlab.ui.control.Label
418         GridSizeLabel                              matlab.ui.control.Label
419         LatitudeEditField_2                        matlab.ui.control.NumericEditField
420         LatitudeEditField_2Label                    matlab.ui.control.Label
421         LongitudeEditField_2                       matlab.ui.control.NumericEditField
422         LongitudeEditField_2Label                  matlab.ui.control.Label
423         SouthwestCornerLabel_2                      matlab.ui.control.Label
424         Button_22                                  matlab.ui.control.Button
425         BathymetryLabel_2                           matlab.ui.control.Label
426         DepthFileEditField_2                       matlab.ui.control.EditField
427         UvectorsLabel                              matlab.ui.control.Label
428         FileTextArea_3                             matlab.ui.control.TextArea
429         Panel_14                                    matlab.ui.container.Panel
430         ArrowSpacing                                matlab.ui.control.NumericEditField
431         SpacingLabel                                matlab.ui.control.Label
432         PlotvectorsCheckBox                         matlab.ui.control.CheckBox
433         ArrowHeadSize                              matlab.ui.control.NumericEditField
434         ArrowSizeLabel                             matlab.ui.control.Label
435         ColorLabel_2                                matlab.ui.control.Label
436         ArrowThickness                              matlab.ui.control.NumericEditField
437         ArrowThicknessLabel                         matlab.ui.control.Label
438         arrowscale                                  matlab.ui.control.NumericEditField
439         XEditField_2Label_3                         matlab.ui.control.Label
440         QuiverColorDropDown                         matlab.ui.control.DropDown
441         GENERATEButton_2                           matlab.ui.control.Button
442     end
443
444
445     properties (Access = private)
446         ArrivalTimeConverted
447         ArrivalTimename
448         bathyCol
449         bathymetrycheck
450         bathyRow
451         colFirstFile
452         ETAfilenames
453         ETFullfile
454         FigureDirectoryVector
455         FileINPUT
456         FileINPUT1
457         FileINPUT2
458         FileINPUT3
459         FileINPUT4
460         FileName
461         FileName1
462         FileName2
463         FileName3
464         FileName4

```



```
465     FileName5
466     GAUGEFILE
467     GAUGEFILE2
468     GAUGEFILE3
469     GaugeDirectorylist
470     GaugeFname
471     GaugeFname3
472     GaugeNames
473     HMAXfilenames
474     HMAXfullfile
475     INfile
476     LegendGauge
477     LegendLocation
478     MainDirectoryVector
479     STATION_FILE
480     Ufiles
481     Uvectorfullfile
482     Vfile
483     Vvectorfilenames
484     Vvectorfullfile
485     WORKFOLDER0
486     WORKFOLDER
487     WORKFOLDER2
488     WORKFOLDER3
489     WORKFOLDER4
490     WORKFOLDER5
491     XCOLGAUGE
492     YCOLGAUGE
493     bathymetrydata
494     bathymetrydata2
495     bathymetryinputdata
496     bathymetryinputdata2
497     bathymetryname
498     bathymetryname2
499     col1
500     col1vel
501     colorcombi
502     fin
503     inputtxt_directory0
504     inputtxt_directory1
505     LandAreaColor
506     latGauge
507     latGauge2
508     longGauge
509     longGauge2
510     row1
511     row1vel
512     rowFirstFile
513     vectorpath
514     x
515     xcoord
516     y
517     ycoord
518 end
519
520
521 % Callbacks that handle component events
522 methods (Access = private)
```

```

523
524     % Code that executes after component creation
525     function startupFcn(app)
526         %Set the screenshot mode to 'manual' to disable automatic screenshots,
527         %to improve app efficiency
528         currentAppModel.MetadataModel.ScreenshotMode = 'manual';
529
530     end
531
532     % Button pushed function: Button_14
533     function Button_14Pushed(app, event)
534         %Disable all the buttons
535         app.CoastlinecolourDropDown_8.Enable = "off";
536         app.FlipCheckBox_5.Enable = "off";
537         app.toEditField.Enable = "off";
538         app.MinEditField.Enable = "off";
539         app.CoastlinecolourDropDown_7.Enable = "off";
540         app.FlipCheckBox_2.Enable = "off";
541         app.MaxEditField_2.Enable = "off";
542         app.MinEditField_2.Enable = "off";
543         app.DivisionEditField.Enable = "off";
544         app.DivisionEditFieldLabel.Enable = "off";
545         app.DivisionLabel.Enable = "off";
546         app.ColorinterpolationEditField_2.Enable = "off";
547         drawnow
548
549
550         %Find files to import from a directory
551         [files, path] = uigetfile('*..*', 'Select the files', '
'MultiSelect', 'on');
552         if isequal(files, 0) || isequal(path, 0)
553             return;
554         end
555
556         cd(fullfile(path))
557
558         if ischar(files) % Only 1 file is uploaded
559             files = {files};
560         end
561
562         %Sort the files by snapshot timing, and display the list of
563         filenames in the app's textbox
564         app.FileName = sort(string(files));
565         allfiles = fullfile(path, app.FileName);
566         app.FileINPUT = allfiles;
567         app.FileTextArea.Value = app.FileName; %
568         app.FileTextArea.FontColor = 'k';
569
570         %Set variables for tracking the file types
571         containsHmax = false;
572         containsEta = false;
573         containsTime = false;
574
575         %Verify that each file is readable
576         for i = 1:length(allfiles)
577             if exist(allfiles{i}, 'file') ~= 2
578                 errorldq(['Unable to find or open file: ' allfiles{i}],

```

```

'File Error', 'modal');
579         continue; % Skip to the next file
580     end
581
582     %Update T/F statement for file-type tracking
583     str = app.FileName(i);
584     if contains(str, "hmax")
585         containsHmax = true;
586     end
587     if contains(str, "eta")
588         containsEta = true;
589     end
590     if contains(str, "time")
591         containsTime = true;
592     end
593 end
594
595 %Enable or disable colormap options based on file type
596 if containsHmax || containsTime
597     app.CoastlinecolourDropDown_8.Enable = "on";
598     app.FlipCheckBox_5.Enable = "on";
599     app.toEditField.Enable = "on";
600     app.MinEditField.Enable = "on";
601     app.ColorbarTextSize.Enable = "on";
602     app.DivisionEditFieldLabel.Enable = "on";
603     app.DivisionEditField.Enable = "on";
604 end
605 if containsEta
606     app.CoastlinecolourDropDown_7.Enable = "on";
607     app.FlipCheckBox_2.Enable = "on";
608     app.MaxEditField_2.Enable = "on";
609     app.MinEditField_2.Enable = "on";
610     app.ColorinterpolationEditField_2.Enable = "on";
611     app.DivisionLabel.Enable = "on";
612     app.ColorbarTextSize.Enable = "on";
613 else
614     app.CoastlinecolourDropDown_8.Enable = "on";
615     app.FlipCheckBox_5.Enable = "on";
616     app.toEditField.Enable = "on";
617     app.MinEditField.Enable = "on";
618     app.ColorbarTextSize.Enable = "on";
619     app.DivisionEditFieldLabel.Enable = "on";
620     app.DivisionEditField.Enable = "on";
621
622 end
623
624
625 %Read the first file to extract its matrix size, which will be used to set up the map boundary limit
626 try
627     f = readmatrix(app.FileName(1));
628     [row,col] = size(f);
629     app.col1 = col-1;
630     app.row1 = row-1;
631 end
632 end
633
634 % Button pushed function: GENERATEButton

```

```

635         function GENERATEButtonPushed(app, event)
636 %Please cite this paper when you use the applications:
637 %Felix, R., Tan, E. H. Z., Watanabe, M., Verolino, A., Puah, J. Y., & Switzer,
A. D. (2025). Funwave-based open-access mapping applications (FUNMAP) applied to
Tsunami modelling from the Manila Trench to Manila Bay, Philippines. Geoscience
Letters. https://doi.org/10.1186/s40562-025-00422-5
638
639
640
641
642 % NANYANG TECHNOLOGICAL UNIVERSITY - NTUITIVE PTE LTD (NTUITIVE) Dual License
Agreement
643 % Non-Commercial Use Only
644 % This NTUITIVE License Agreement, including all exhibits ("NTUITIVE-LA") is a
legal agreement between you and NTUITIVE (or "we") located at 71 Nanyang Drive, NTU
Innovation Centre, #01-109, Singapore 637722, a wholly owned subsidiary of Nanyang
Technological University ("NTU") for the software or data identified above, which
may include source code, and any associated materials, text or speech files,
associated media and "online" or electronic documentation and any updates we
provide in our discretion (together, the "Software").
645 %
646 % By installing, copying, or otherwise using this Software, found at https://github.com/raquelpfelix, you agree to be bound by the terms of this NTUITIVE-LA.
If you do not agree, do not install copy or use the Software. The Software is
protected by copyright and other intellectual property laws and is licensed, not
sold. If you wish to obtain a commercial royalty bearing license to this software
please contact us at rf761@bath.ac.uk.
647 %
648 % SCOPE OF RIGHTS:
649 % You may use, copy, reproduce, and distribute this Software for any non-
commercial purpose, subject to the restrictions in this NTUITIVE-LA. Some purposes
which can be non-commercial are teaching, academic research, public demonstrations
and personal experimentation. You may also distribute this Software with books or
other teaching materials, or publish the Software on websites, that are intended to
teach the use of the Software for academic or other non-commercial purposes.
650 % You may not use or distribute this Software or any derivative works in any
form for commercial purposes. Examples of commercial purposes would be running
business operations, licensing, leasing, or selling the Software, distributing the
Software for use with commercial products, using the Software in the creation or
use of commercial products or any other activity which purpose is to procure a
commercial gain to you or others.
651 % If the Software includes source code or data, you may create derivative
works of such portions of the Software and distribute the modified Software for
non-commercial purposes, as provided herein.
652 % If you distribute the Software or any derivative works of the Software, you
will distribute them under the same terms and conditions as in this license, and
you will not grant other rights to the Software or derivative works that are
different from those provided by this NTUITIVE-LA.
653 % If you have created derivative works of the Software, and distribute such
derivative works, you will cause the modified files to carry prominent notices so
that recipients know that they are not receiving the original Software. Such
notices must state: (i) that you have changed the Software; and (ii) the date of
any changes.
654 %
655 % You may not distribute this Software or any derivative works.
656 % In return, we simply require that you agree:
657 % 1. That you will not remove any copyright or other notices from the
Software.

```

658 % 2. That if any of the Software is in binary format, you will not attempt to modify such portions of the Software, or to reverse engineer or decompile them, except and only to the extent authorized by applicable law.

659 % 3. That NTUITIVE is granted back, without any restrictions or limitations, a non-exclusive, perpetual, irrevocable, royalty-free, assignable and sub-licensable license, to reproduce, publicly perform or display, install, use, modify, post, distribute, make and have made, sell and transfer your modifications to and/or derivative works of the Software source code or data, for any purpose.

660 % 4. That any feedback about the Software provided by you to us is voluntarily given, and NTUITIVE shall be free to use the feedback as it sees fit without obligation or restriction of any kind, even if the feedback is designated by you as confidential.

661 % 5. THAT THE SOFTWARE COMES "AS IS", WITH NO WARRANTIES. THIS MEANS NO EXPRESS, IMPLIED OR STATUTORY WARRANTY, INCLUDING WITHOUT LIMITATION, WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, ANY WARRANTY AGAINST INTERFERENCE WITH YOUR ENJOYMENT OF THE SOFTWARE OR ANY WARRANTY OF TITLE OR NON-INFRINGEMENT. THERE IS NO WARRANTY THAT THIS SOFTWARE WILL FULFILL ANY OF YOUR PARTICULAR PURPOSES OR NEEDS. ALSO, YOU MUST PASS THIS DISCLAIMER ON WHENEVER YOU DISTRIBUTE THE SOFTWARE OR DERIVATIVE WORKS.

662 % 6. THAT NEITHER NTUITIVE NOR NTU NOR ANY CONTRIBUTOR TO THE SOFTWARE WILL BE LIABLE FOR ANY DAMAGES RELATED TO THE SOFTWARE OR THIS NTUITIVE-LA, INCLUDING DIRECT, INDIRECT, SPECIAL, CONSEQUENTIAL OR INCIDENTAL DAMAGES, TO THE MAXIMUM EXTENT THE LAW PERMITS, NO MATTER WHAT LEGAL THEORY IT IS BASED ON. ALSO, YOU MUST PASS THIS LIMITATION OF LIABILITY ON WHENEVER YOU DISTRIBUTE THE SOFTWARE OR DERIVATIVE WORKS.

663 % 7. That we have no duty of reasonable care or lack of negligence, and we are not obligated to (and will not) provide technical support for the Software.

664 % 8. That if you breach this NTUITIVE-LA or if you sue anyone over patents that you think may apply to or read on the Software or anyone's use of the Software, this NTUITIVE-LA (and your license and rights obtained herein) terminate automatically. Upon any such termination, you shall destroy all of your copies of the Software immediately. Sections 3, 4, 5, 6, 7, 8, 11 and 12 of this NTUITIVE-LA shall survive any termination of this NTUITIVE-LA.

665 % 9. That the patent rights, if any, granted to you in this NTUITIVE-LA only apply to the Software, not to any derivative works you make.

666 % 10. That the Software may be subject to U.S. export jurisdiction at the time it is licensed to you, and it may be subject to additional export or import laws in other places. You agree to comply with all such laws and regulations that may apply to the Software after delivery of the software to you.

667 % 11. That all rights not expressly granted to you in this NTUITIVE-LA are reserved.

668 % 12. That this NTUITIVE-LA shall be construed and controlled by the laws of the Republic of Singapore without regard to conflicts of law. If any provision of this NTUITIVE-LA shall be deemed unenforceable or contrary to law, the rest of this NTUITIVE-LA shall remain in full effect and interpreted in an enforceable manner that most nearly captures the intent of the original language.

669 %

670 %

671 % Do you accept all of the terms of the preceding NTUITIVE-LA license agreement? If you accept the terms, click "I Agree," then "Next." Otherwise click "Cancel."

672 %

673 % Copyright (c) NTUITIVE. All rights reserved.

674

675

676

677 %A copy of FUNWAVE's BDS license agreement:

678 % ! Copyright (c) 2016, FUNWAVE Development Team

```

679 % !
680 % !      (See http://www.udel.edu/kirby/programs/funwave/funwave.html
681 % !      for Development Team membership)
682 % !
683 % !      All rights reserved.
684 % !
685 % !      FUNWAVE_TVD is free software: you can redistribute it and/or modify
686 % !      it under the terms of the Simplified BSD License as released by
687 % !      the Berkeley Software Distribution (BSD).
688 % !
689 % !      Redistribution and use in source and binary forms, with or without
690 % !      modification, are permitted provided that the following conditions are
met:
691 % !
692 % !      1. Redistributions of source code must retain the above copyright
notice, this
693 % !      list of conditions and the following disclaimer.
694 % !      2. Redistributions in binary form must reproduce the above copyright
notice,
695 % !      this list of conditions and the following disclaimer in the
documentation
696 % !      and/or other materials provided with the distribution.
697 % !
698 % !      THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS
IS" AND
699 % !      ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE
IMPLIED
700 % !      WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE
701 % !      DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT OWNER OR CONTRIBUTORS BE
LIABLE FOR
702 % !      ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL
DAMAGES
703 % !      (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR
SERVICES;
704 % !      LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED
AND
705 % !      ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR
TORT
706 % !      (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
OF THIS
707 % !      SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
708 % !
709 % !      The views and conclusions contained in the software and documentation
are those
710 % !      of the authors and should not be interpreted as representing official
policies,
711 % !      either expressed or implied, of the FreeBSD Project.
712
713
714
715 %This is the start of the app's code:
716 close all
717
718      %-----
719      %-----BASEMAP INPUT-----
720      %-----
721      %List of the colormap items in the dropdown list
722      keys = {'autumn', 'bone', 'colorcube', 'cool', 'copper', 'flag',

```



```

'gray', 'hot', 'hsv', 'jet', ...
723         'parula', 'pink', 'prism', 'spring', 'summer', 'turbo', ↵
'winter', 'blue', 'blue - green', ...
724         'blue - purple', 'green - blue', 'greens', 'grays', 'oranges', ↵
'orange - red', 'purple - blue', ...
725         'purple - blue - green', 'purple - red', 'purples', 'red ↵
purple', 'reds', 'yellow - green', ...
726         'yellow - green - blue', 'yellow - orange - brown', 'yellow - ↵
orange - red', 'brown - teal', ...
727         'pink - light green', 'purple - green', 'purple - orange', ↵
'red - blue', 'red - gray', ...
728         'red - yellow - blue', 'red - yellow - green', 'spectral', ↵
'accent', 'dark 2', 'paired', ...
729         'pastel 1', 'pastel 2', 'set 1', 'set 2', 'set 3', '--- MATLAB ↵
default ----', '--- CBREWER 2 ---', ...
730         '< sequential >', '< divergent >', '< qualitative >'};
731
732         %The equivalent colormap code
733         keysColor = {'autumn', 'bone', 'colorcube', 'cool', 'copper', ↵
'flag', 'gray', 'hot', 'hsv', 'jet', ...
734         'parula', 'pink', 'prism', 'spring', 'summer', 'turbo', ↵
'winter', 'Blues', 'BuGn', 'BuPu', 'GnBu', 'Greens', ...
735         'Greys', 'Oranges', 'OrRd', 'PuBu', 'PuBuGn', ...
736 ↵
'PuRd', 'Purples', 'RdPu', 'Reds', 'YlGn', 'YlGnBu', 'YlOrBr', 'YlOrRd', 'BrBG', 'PiYG', 'PRG ↵
n', 'PuOr', 'RdBu', 'RdGy', ...
737         'RdYlBu', 'RdYlGn', 'Spectral', 'Accent', ...
738 ↵
'Dark2', 'Paired', 'Pastel1', 'Pastel2', 'Set1', 'Set2', 'Set3', 'parula', 'parula', 'parula ↵
', 'parula', 'parula'};
739
740         %The colorbrewer
741         keysBrewer = {'none', 'none', 'none', 'none', 'none', 'none', ↵
'none', 'none', 'none', 'none', ...
742         'none', 'none', 'none', 'none', 'none', 'none', 'none', 'seq', ↵
'seq', ...
743         'seq', 'seq', 'seq', 'seq', 'seq', 'seq', ...
744         'seq', 'seq', 'seq', 'seq', 'seq', ...
745         'seq', 'seq', 'seq', 'seq', 'seq', 'div', 'div', 'div', ...
746         'div', 'div', 'div', 'div', 'div', ...
747         'div', 'qual', 'qual', 'qual', ...
748         'qual', 'qual', 'qual', 'qual', 'qual', ...
749         'none', 'none', 'none', 'none', 'none'};
750
751         % Match dropdown list items to their corresponding colors and ↵
ColorBrewer
752         colorMap = containers.Map(keys, keysColor);
753         colorBrewer = containers.Map(keys, keysBrewer);
754
755         %For eta colormap
756         datacolorETA = colorMap(app.CoastlinecolourDropDown_7.Value);
757         datacolorbrewerETA = colorBrewer(app.CoastlinecolourDropDown_7. ↵
Value);
758
759         %For hmax colormap
760         datacolor = colorMap(app.CoastlinecolourDropDown_8.Value);
761         datacolorbrewer = colorBrewer(app.CoastlinecolourDropDown_8. ↵
Value);

```

```

762
763
764             %-----
765             %-----LAND AREA COLOR-----
766             %-----
767
768             %List of commonly used colors and styles
769             colorMap = containers.Map({'Black', 'Dark gray', 'Medium gray', '
'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'}, {'k', '
'[0.4 0.4 0.4]', '[0.3 0.3 0.3]', '[0.8 0.8 0.8]', 'r', 'g', 'b', 'y', 'c', 'm', '
'w'}); % [0.7 0.7 0.7]
770             lineStyleMap = containers.Map({'Solid', 'Dashed', 'Dotted', 'Dash-
dotted'}, {'-', '--', ':', '-.'});
771
772             %Landmass color
773             app.LandAreaColor = colorMap(app.CoastlinecolourDropDown_2.Value);
774
775             %-----
776             %-----GAUGES-----
777             %-----
778             %Marker Style
779             GaugeMarkerOptions = containers.Map({'none', 'o', '+', '*', '.', '
'x', '-', '|', '^', 'v', '>', '<', 'diamond', 'hexagram', 'pentagram', 'square'}, '
...
780             {'none', 'o', '+', '*', '.', 'x', '-', '|', '^', 'v', '>', '
'<', 'diamond', 'hexagram', 'pentagram', 'square'});
781             GaugeMarker = GaugeMarkerOptions(app.CoastlinecolourDropDown_3.
Value);
782
783             %Marker Color
784             GaugeColor = colorMap(app.ColorDropDown_3.Value);
785
786             %Label - horizontal alignment
787             horzLabelMap = containers.Map({'Centre', 'Right', 'Left'}, '
{'center', 'left', 'right'});
788             HorzLabel = horzLabelMap(app.CoastlinecolourDropDown_4.Value);
789
790             %Label - vertical alignment
791             vertLabelMap = containers.Map({'Centre', 'Top', 'Bottom'}, '
{'middle', 'bottom', 'top'});
792             VertLabel = vertLabelMap(app.CoastlinecolourDropDown_5.Value);
793
794
795             %-----
796             %-----BATHYMETRY CONTOURS-----
797             %-----
798             %Line Colours
799             ContourColor = colorMap(app.ColorDropDown.Value);
800             %Line style
801             ContourLineStyle = lineStyleMap(app.StyleDropDown.Value);
802
803
804             %-----
805             %-----WAVE HEIGHT CONTOURS-----
806             %-----
807             %Line Colours
808             WaveContourColor = colorMap(app.ColorDropDown_2.Value);
809             %Line style

```

```

810 WaveContourLineStyle = lineStyleMap(app.StyleDropDown_2.Value);
811
812
813 %-----
814 %-----FIRST WAVE ARRIVAL CONTOURS-----
815 %-----
816 %Default threshold: 0.001 m
817 %Line Colours
818 ArrivalTimeContourColor = colorMap(app.ColorDropDown_6.Value);
819 %Line Style
820 ArrivalTimeLineStyle = lineStyleMap(app.StyleDropDown_4.Value);
821
822 %-----
823 %-----SAVE MAP: Output Directory-----
824 %-----
825 %Check if the output directory has been manually set by the user
826 if app.OutputDirectoryEditField.Value == string(app.WORKFOLDER0)
827     app.inputtxt_directory0 = fullfile(app.WORKFOLDER);
828     app.inputtxt_directory1 = fullfile(app.WORKFOLDER0);
829
830 else % Set the default directory to the Desktop
831     if ismac %macOS
832         defaultDir = fullfile(getenv('HOME'), 'Desktop');
833     elseif ispc %Windows
834         defaultDir = fullfile(getenv('USERPROFILE'), 'Desktop');
835     else %Others
836         defaultDir = pwd;
837     end
838
839     %Create 'OUTPUT_FILES' folder and 'Figures' subfolder
840     Dir0 = fullfile(defaultDir, 'OUTPUT_FILES');
841     Dir1 = fullfile(defaultDir, 'OUTPUT_FILES/Figures');
842     if ~exist(Dir0, 'dir')
843         mkdir(Dir0);
844     end
845     if ~exist(Dir1, 'dir')
846         mkdir(Dir1);
847     end
848
849     app.inputtxt_directory0 = Dir0;
850     app.inputtxt_directory1 = Dir1;
851
852     %Display the path directory of the 'Figures' subfolder in the
853     app.OutputDirectoryEditField.Value = deblank(string(Dir1));
854     app.OutputDirectoryEditField.FontColor = 'k';
855 end
856
857
858
859 %Create animation/ creating and opening the file
860 if app.mp4CheckBox.Value
861     vidObj = VideoWriter(fullfile(char(app.inputtxt_directory1),
'animation_waveheight.mp4'), 'MPEG-4');
862     vidObj.FrameRate = app.FramerateEditField_2.Value;
863     open(vidObj);
864 end
865

```

```

866         %-----
867         %-----LOG REPORT-----
868         %-----
869         %Create 'Log_Files' subfolder in 'OUTPUT_FILES'
870         LogFolder = fullfile(app.inputtxt_directory0, 'Log_Files');
871         if ~exist(LogFolder, 'dir')
872             mkdir(LogFolder);
873         end
874         %Create the text file
875         filePath = fullfile(LogFolder, 'Log_Report_WaveHeightMap.txt');
876         fileId = fopen(filePath, 'w');
877
878         % Template for the default section of the log report (includes
information from the 'Input Data' section)
879         headerTitle = '***** LOG REPORT
*****';
880         timestamp = datestr(now, 'yyyy-mm-dd HH:MM:SS');
881         endHeaderLine =
'*****';
882         footerSeparator =
'
';
883         logContent = [
884             sprintf('%s\n', headerTitle), ...
885             sprintf('%-30s-s\n', 'Map Type:', 'Wave Height Map'), ...
886             sprintf('%-30s-s\n', 'Timestamp:', timestamp), ...
887             sprintf('%s\n\n', endHeaderLine), ...
888             sprintf('%-30s-s\n', 'Southwest Corner:', sprintf('Long: %f
Lat: %f', app.LongitudeEditField.Value, app.LatitudeEditField.Value)), ...
889             sprintf('%-30s-s\n', 'Grid Size:', sprintf('x: %d y: %d',
app.gridX.Value, app.gridY.Value)), ...
890             sprintf('%-30s-s\n', 'Simulation Time Start:', sprintf('%d
sec', app.StartTime.Value)), ...
891             sprintf('%-30s-s\n', 'Simulation Time Interval:', sprintf('%d
sec', app.TotalSimulationTimesecEditField_3.Value)), ...
892             sprintf('%s\n\n', footerSeparator)
893         ];
894         fprintf(fileId, '%s', logContent);
895
896
897         %-----LOG REPORT-----
898         %-----Input Data: Import Files-----
899         %-----
900         %Check the type of output file loaded in the app
901         patterns = {'hmax_', 'eta_', 'time_'};
902         matches1 = false(size(string(app.FileName)));
903         nonMatchingFiles = ~contains(string(app.FileName), patterns);
904         for i = 1:numel(patterns)
905             pattern = patterns{i};
906             matches1 = matches1 | contains(string(app.FileName), pattern);
907         end
908
909         if any(nonMatchingFiles)
910             if contains(app.FileTextArea.Value, "eta_xxxx")
911                 fprintf(fileId, '%-30s-s\n\n', 'IMPORT FILE/S:', 'Empty.
No basemap is plotted'); %Add an error note in the log report
912             else
913                 % if (app.FileTextArea.Value ~= "0") % If non- eta/hmax
file is loaded

```

```

914             fprintf(fileId, '%-30s%-s\n', 'IMPORT FILE/S:', 'Use
filenames with 'eta_', 'hmax_' initials');
915             fprintf(fileId, '%-30s%-s\n', ' ', 'Incorrect files are
loaded:');
916             fprintf(fileId, '%-30s%-s\n\n', ' ', join(app.FileName
(nonMatchingFiles), ', '));% List all filenames with errors in the log report
917
918             end
919         end
920
921         %-----LOG REPORT-----
922         %-----Input Data: Southwest Corner-----
923         %-----
924         %Check if the values have not changed and add an error note to the
log report
925         if app.LongitudeEditField.Value == 0 && app.LatitudeEditField.
Value == 0
926             fprintf(fileId, '%-30s%-s\n\n', 'SOUTHWEST CORNER:', 'Both
latitude and longitude values are zeroes');
927             elseif app.LongitudeEditField.Value == 0 && app.LatitudeEditField.
Value ~= 0
928                 fprintf(fileId, '%-30s%-s\n\n', 'SOUTHWEST CORNER:', 'A default
zero value is used for longitude');
929             elseif app.LongitudeEditField.Value ~= 0 && app.LatitudeEditField.
Value == 0
930                 fprintf(fileId, '%-30s%-s\n\n', 'SOUTHWEST CORNER:', 'A default
zero value is used for latitude');
931             end
932
933         %-----LOG REPORT-----
934         %-----Input Data: Grid Size-----
935         %-----
936         %Check if the values have not changed and add an error note to the
log report
937         if app.gridX.Value == 0 && app.gridY.Value == 0
938             fprintf(fileId, '%-30s%-s\n\n', 'GRID SIZE:', 'Both x and y
values are zeroes');
939             elseif app.gridX.Value == 0 && app.gridY.Value ~= 0
940                 fprintf(fileId, '%-30s%-s\n\n', 'GRID SIZE:', 'X value is
zero');
941             elseif app.gridX.Value ~= 0 && app.gridY.Value == 0
942                 fprintf(fileId, '%-30s%-s\n\n', 'GRID SIZE:', 'Y value is
zero');
943             end
944         end
945
946         %-----LOG REPORT-----
947         %-----Input Data: Load Bathymetry-----
948         %-----
949         %Check if the matrix size of the loaded bathymetry file matches
the imported eta/hmax file
950         if matches(app.FileTextArea.Value, string(app.FileName)) % Check if
the imported eta/hmax files match the names in the textbox
951             if matches(app.DepthFileEditField.Value, string(app.
bathymetryname)) %Check if the imported bathymetry matches the name in the textbox
952                 if app.rowFirstFile ~= app.bathyRow
953                     fprintf(fileId, '%-30s%-s\n', 'MATRIX ROWS:', 'Rows in
eta/hmax/time/ file and bathymetry do not match');

```

```

954         end
955         if app.colFirstFile ~= app.bathyCol
956             fprintf(fileId, '%-30s%-s\n', 'MATRIX↵
COLUMNS:', 'Columns in eta/hmax/time file and bathymetry do not match');
957         end
958     end
959 end
960     if matches(app.DepthFileEditField.Value, "0") || ~matches(app.↵
DepthFileEditField.Value, string(app.bathymetryname)) % Add a log note that no↵
bathymetry is loaded, so the features to edit land area and bathymetry contours are ↵
unavailable.
961         fprintf(fileId, '%-30s%-s\n\n', 'BATHYMETRY:', 'Empty. Land↵
color and bathymetry contours are not applicable.');
```

```

962         app.DepthFileEditField.FontColor = 'w';
963         app.DepthFileEditField.BackgroundColor = 'r';
964         app.DepthFileEditField.Value = 'NO FILE';
965     end
966
967
968     %-----LOG REPORT-----
969     %-----Basemap: Wave heights-----
970     %-----
971     if all(contains(string(app.FileName), patterns)) %If all files↵
are in eta_ /hmax_ format
972         %Verify that the minimum, interval, and maximum input values↵
are correct; update values if conditions are not met
973         %hmax option
974         if app.MinEditField.Value > app.toEditField.Value
975             temp = app.MinEditField.Value;
976             app.MinEditField.Value = app.toEditField.Value;
977             app.toEditField.Value = temp;
978             drawnow
979             fprintf(fileId, '%-30s%-s\n', 'BASEMAP HMAX↵
COLORBAR:', 'Max value must exceed min value');
```

```

980             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Min value↵
adjusted to load the map');
```

```

981         end
982         %eta option
983         if app.MinEditField_2.Value > app.MaxEditField_2.Value
984             temp = app.MinEditField_2.Value;
985             app.MinEditField_2.Value = app.MaxEditField_2.Value;
986             app.MaxEditField_2.Value = temp;
987             drawnow
988             fprintf(fileId, '%-30s%-s\n', 'BASEMAP ETA COLORBAR:', 'Max↵
value must exceed min value');
```

```

989             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Min value↵
adjusted to load the map');
```

```

990         end
991     end
992
993     %-----LOG REPORT-----
994     %-----Overlay Features: Bathymetry contours-----
995     %-----
996     if matches(app.DepthFileEditField.Value, string(app.↵
bathymetryname))
997         if app.MaximumEditField.Value < app.MinimumEditField.Value %↵
Update the max value if it is less than the user-defined minimum value.
998             app.MaximumEditField.Value = 10000;
```



```

999             fprintf(fileId, '%-30s%-s\n', 'BATHYMETRY CONTOURS:', 'Max value must exceed min value');
1000             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Max value set to 10,000');
1001             end
1002             if app.IntervalEditField_5.Value > (app.MaximumEditField.Value - app.MinimumEditField.Value) %Update the contour interval value when the condition is not met
1003                 app.IntervalEditField_5.Value = (app.MaximumEditField.Value - app.MinimumEditField.Value)/2;
1004                 fprintf(fileId, '%-30s%-s\n', 'BATHYMETRY CONTOUR INTERVAL:', 'Value must be less than the max-min difference');
1005                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value adjusted to half the max-min difference');
1006             end
1007
1008             if app.AddLabelsCheckBox_2.Value %Add labels
1009                 %Check if the label interval is divisible by the value provided in the 'Range' section
1010                 if mod(app.IntervalEditField_3.Value, app.IntervalEditField_5.Value) ~= 0
1011                     app.IntervalEditField_3.Value = app.IntervalEditField_5.Value;
1012                     fprintf(fileId, '%-30s%-s\n', 'BATHYMETRY CONTOUR LABEL:', 'Value must be divisible by the Interval value in the Range section');
1013                     fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value adjusted to be the same as the Interval Value in the Range section');
1014                 end
1015
1016                 %Check if the label's interval value is less than the maximum value in 'Contour Range'
1017                 if app.IntervalEditField_3.Value > app.MaximumEditField.Value
1018                     app.IntervalEditField_3.Value = app.IntervalEditField_5.Value;
1019                     fprintf(fileId, '%-30s%-s\n', 'BATHYMETRY CONTOUR LABEL:', 'Value must be less than the maximum value of the contour line assigned in the Range section');
1020                     fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value adjusted to be the same as the Interval Value in the Range section');
1021                 end
1022             end
1023         end
1024
1025         if app.BathymetryButton_2.Value && (strcmp(app.DepthFileEditField.Value, "0") || strcmp(app.DepthFileEditField.Value, "00"))
1026             %No bathymetry file found, show "NO FILE" warning in the bathymetry input window
1027             app.DepthFileEditField.Value = "NO FILE";
1028             app.DepthFileEditField.FontColor = 'w';
1029             app.DepthFileEditField.BackgroundColor = 'r';
1030         end
1031
1032         %-----LOG REPORT-----
1033         %-----Overlay Features: Wave height contours-----
1034         %-----
1035         if app.MaximumdepthEditField_2.Value < app.MinimumdepthEditField_2.Value

```

```

1036         app.MaximumdepthEditField_2.Value = 1;
1037         app.MinimumdepthEditField_2.Value = 0;
1038         fprintf(fileId, '%-30s%-s\n', 'WAVE HEIGHT CONTOURS:', 'Max
value must exceed min value');
1039         fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Max and min
values reset to default [0,1]');
1040         end
1041
1042         if app.LineintervalEditField_2.Value > app.
MaximumdepthEditField_2.Value - app.MinimumdepthEditField_2.Value
1043             app.LineintervalEditField_2.Value = (app.
MaximumdepthEditField_2.Value - app.MinimumdepthEditField_2.Value)/2;
1044             fprintf(fileId, '%-30s%-s\n', 'WAVE HEIGHT CONTOUR
INTERVAL:', 'Value must be less than the max-min difference');
1045             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value adjusted
to half the max-min difference');
1046         end
1047
1048         if app.AddLabelsCheckBox_3.Value %Add labels
1049             if mod(app.wh_interval.Value, app.LineintervalEditField_2.
Value) ~= 0
1050                 app.wh_interval.Value = app.LineintervalEditField_2.Value;
1051                 fprintf(fileId, '%-30s%-s\n', 'WAVE HEIGHT CONTOUR
LABEL:', 'Value must be divisible by the Interval value in the Range section');
1052                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value
adjusted to be the same as the Interval Value in the Range section');
1053             end
1054
1055             %Check if the label's interval value is less than the maximum
value in 'Contour Range'
1056             if app.wh_interval.Value > app.MaximumdepthEditField_2.Value
1057                 app.wh_interval.Value = app.LineintervalEditField_2.Value;
1058                 fprintf(fileId, '%-30s%-s\n', 'WAVE HEIGHT CONTOUR
LABEL:', 'Value must be less than the maximum value of the contour line assigned in
the Range section');
1059                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value
adjusted to be the same as the Interval Value in the Range section');
1060             end
1061         end
1062
1063         %-----LOG REPORT-----
1064         %-----Overlay Features: Arrival Time-----
1065         %-----
1066         if app.ArrivalTimeButton.Value
1067             if ~matches(app.FileEditField_3.Value, string(app.
ArrivalTimeName)) || matches(app.FileEditField_3.Value, "0")%Check if there is an
uploaded file
1068                 fprintf(fileId, '%-30s%-s\n\n', 'ARRIVAL TIME:', 'Empty.
No Arrival Time Contour plotted');
1069                 app.FileEditField_3.FontColor = 'w';
1070                 app.FileEditField_3.BackgroundColor = 'r';
1071                 app.FileEditField_3.Value = 'NO FILE';
1072             end
1073             if app.MinimumdepthEditField_3.Value > app.
MaximumdepthEditField_3.Value
1074                 % Swap the values
1075                 tempArrivalTime = app.MinimumdepthEditField_3.Value;
1076                 app.MinimumdepthEditField_3.Value = app.

```

```

MaximumdepthEditField_3.Value;
1077         app.MaximumdepthEditField_3.Value = tempArrivalTime;
1078         fprintf(fileId, '%-30s%-s\n', 'ARRIVAL TIME:', 'Max value
must exceed min value');
1079         fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Values are
swapped');
1080         end
1081         if app.LineintervalEditField_3.Value > app.
MaximumdepthEditField_3.Value - app.MinimumdepthEditField_3.Value
1082             app.LineintervalEditField_3.Value = (app.
MaximumdepthEditField_3.Value - app.MinimumdepthEditField_3.Value)/2;
1083             fprintf(fileId, '%-30s%-s\n', 'ARRIVAL TIME
INTERVAL:', 'Value must be less than the max-min difference');
1084             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value
adjusted to half the max-min difference');
1085         end
1086
1087         if app.AddLabelsCheckBox_5.Value %Add Labels
1088             if mod(app.IntervalEditField_7.Value, app.
LineintervalEditField_3.Value) ~= 0
1089                 app.IntervalEditField_7.Value = app.
LineintervalEditField_3.Value;
1090                 fprintf(fileId, '%-30s%-s\n', 'ARRIVAL TIME
LABEL:', 'Value must be divisible by the Interval value in the Range section');
1091                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value
adjusted to be the same as the Interval Value in the Range section');
1092             end
1093
1094             %Check if the label's interval value is less than the
maximum value in 'Contour Range'
1095             if app.IntervalEditField_7.Value > app.
MaximumdepthEditField_3.Value
1096                 app.IntervalEditField_7.Value = app.
LineintervalEditField_3.Value;
1097                 fprintf(fileId, '%-30s%-s\n', 'ARRIVAL TIME
LABEL:', 'Value must be less than the maximum value of the contour line assigned in
the Range section');
1098                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value
adjusted to be the same as the Interval Value in the Range section');
1099             end
1100         end
1101     end
1102
1103
1104     %-----LOG REPORT-----
1105     %-----Overlay Features: Gauges-----
1106     %-----
1107     if app.GaugesCheckBox.Value && ~matches(app.FileEditField.Value ,
string(app.GAUGEFILE))
1108         fprintf(fileId, '%-30s%-s\n\n', 'GAUGES:', 'Empty. No virtual
gauges plotted');
1109         app.FileEditField.Value = "NO FILE";
1110         app.FileEditField.FontColor = 'w';
1111         app.FileEditField.BackgroundColor = 'r';
1112     end
1113
1114     %-----LOG REPORT-----
1115     %-----General Layout: Boundary Limits-----

```

```

1116         %-----
1117         if app.WestEditField.Value > app.EastEditField.Value
1118             fprintf(fileId, '%-30s%-s\n', 'BOUNDARY LIMIT:', 'East boundary
must be greater than West boundary');
1119             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Values are
swapped');
1120         elseif app.EastEditField.Value == app.WestEditField.Value
1121             fprintf(fileId, '%-30s%-s\n', 'BOUNDARY LIMIT:', 'East and
West values are the same');
1122             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved:', 'The default
values are used');
1123         end
1124
1125         if app.SouthEditField.Value > app.NorthEditField.Value
1126             fprintf(fileId, '%-30s%-s\n', 'BOUNDARY LIMIT:', 'North
boundary must be greater than South boundary');
1127             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Values are
swapped');
1128         elseif app.NorthEditField.Value == app.SouthEditField.Value
1129             fprintf(fileId, '%-30s%-s\n', 'BOUNDARY LIMIT:', 'North and
South values are the same');
1130             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved:', 'The default
values are used');
1131         end
1132
1133         %-----
1134         %-----SAVE MAP: .tif-----
1135         %-----
1136         if app.tifCheckBox_2.Value && (app.LatitudeEditField.Value > 90 ||
app.LatitudeEditField.Value < -90)
1137             fprintf(fileId, '%-30s%-s\n\n', 'BOUNDARY LIMIT:', 'The
Latitude values should be within -90 to 90. ');
1138         end
1139
1140         fclose(fileId); %Close the log report file
1141
1142         %-----
1143         %-----PRE-SET MAP OPTIONS-----
1144         %-----
1145         clear depth fl
1146         FileLength = length(app.FileINPUT); %Read all the imported
eta/hmax files
1147
1148         %-----Plot all data in one figure-----
1149         if app.PlotalldatainonefigureButton.Value
1150             %Create the figure
1151             figure(1);
1152
1153             %Formatting the subplots
1154             if FileLength == 1
1155                 plotCol = 1;
1156                 plotRow = 1;
1157             elseif FileLength > 1 && FileLength <= 10
1158                 plotCol = 2;
1159                 plotRow = ceil(FileLength/2);
1160             elseif FileLength > 10
1161                 plotCol = 3;
1162                 plotRow = ceil(FileLength/3);

```

```

1163         end
1164
1165     end
1166
1167     %-----Bathymetry data-----
1168     % bathymetryFileExists = exist(app.bathymetryinputdata, 'file') == 0
1169     2;
1170
1171     % if bathymetryFileExists
1172     depth = app.bathymetrydata;
1173
1174     % Flip the data horizontally
1175     if app.HorizontalCheckBox.Value
1176         depth = fliplr(depth);
1177     end
1178
1179     % Flip the data vertically
1180     if app.VerticalCheckBox.Value
1181         depth = flipud(depth);
1182     end
1183     % end
1184
1185     %-----Basemap Colormap-----
1186     colorbarHandles = [];
1187     if strcmp(datacolorbrewerETA, 'none') %Using default matlab
1188     colormap options
1189         datacolor2ETA = evalin('base',datacolorETA);
1190     else %Using cbrewer2
1191         datacolor2ETA = cbrewer2(datacolorbrewerETA,datacolorETA,app.
1192     ColorinterpolationEditField_2.Value, 'pchip');
1193     end
1194
1195     if strcmp(datacolorbrewer, 'none') %Using default matlab colormap
1196     options
1197         datacolor2 = evalin('base',datacolor);
1198     else %Using cbrewer2
1199         datacolor2 = cbrewer2(datacolorbrewer,datacolor,app.
1200     DivisionEditField.Value, 'pchip');
1201     end
1202
1203     colormapMinHMAX = app.MinEditField.Value;
1204     colormapMaxHMAX = app.toEditField.Value;
1205     colorbartitleHMAX = 'Maximum \eta (m)';
1206
1207     colormapMinETA = app.MinEditField_2.Value;
1208     colormapMaxETA = app.MaxEditField_2.Value;
1209     colorbartitleETA = '\eta (m)';
1210
1211     colorbarFontSize = app.ColorbarTextSize.Value;
1212     flipCheckBoxETA = app.FlipCheckBox_2.Value;
1213     flipCheckBoxHMAX = app.FlipCheckBox_5.Value;
1214
1215     %-----Bathymetry Contours-----
1216     if app.BathymetryButton_2.Value && matches(app.DepthFileEditField.
1217     Value,string(app.bathymetryname))
1218         kernel = ones(6) / 36; %Make the contour smoother
1219         wavebathymetry= filter2(kernel, depth);
1220         contourLevels = app.MinimumEditField.Value:app.

```

```

IntervalEditField_5.Value:app.MaximumEditField.Value;
1215         manualLabels = app.MinimumEditField.Value:app.
IntervalEditField_3.Value:app.MaximumEditField.Value;
1216     end
1217
1218
1219         isBathymetryChecked = app.BathymetryButton_2.Value;
1220         isMatchingBathymetry = matches(app.DepthFileEditField.Value,
string(app.bathymetryname));
1221         isAddingLabels = app.AddLabelsCheckBox_2.Value;
1222         fontSize = app.TextLabelSize.Value;
1223         labelSpacing = app.LabelSpacingEditField.Value;
1224         lineWidth = app.WidthEditField.Value;
1225         contourCol = ContourColor; % Capture contour color if it's same
for all
1226         contourStyle = ContourLineStyle; % Capture contour style if it's
same for all
1227
1228
1229         %-----Wave Height Contours-----
1230         if app.WaveHeightButton.Value
1231             wavecontour = app.fin;
1232             if exist('app.bathymetrydata', 'var')
1233                 wavecontour(depth<=0) = NaN; %Remove values on land
1234             end
1235             contourLevelsWH = app.MinimumdepthEditField_2.Value:app.
LineintervalEditField_2.Value:app.MaximumdepthEditField_2.Value;
1236             manualLabels2 = app.MinimumdepthEditField_2.Value:app.
wh_interval.Value:app.MaximumdepthEditField_2.Value;
1237         end
1238
1239         waveContourColor = WaveContourColor;
1240         waveContourStyle = WaveContourLineStyle;
1241         waveLineWidth = app.ThicknessEditField_2.Value;
1242         fontSize2 = app.TextLabelSize_2.Value;
1243         labelSpacing2 = app.LabelSpacingEditField_2.Value;
1244
1245         %-----Arrival Time Contours-----
1246         if app.ArrivalTimeButton.Value && matches(app.FileEditField_3.
Value, string(app.ArrivalTimename))
1247             atc = app.ArrivalTimeConverted;
1248
1249             %Flip the data vertically
1250             if app.VerticalCheckBox.Value
1251                 atc = flipud(app.ArrivalTimeConverted);
1252             end
1253
1254             %Flip the data horizontally
1255             if app.HorizontalCheckBox.Value
1256                 atc = fliplr(app.ArrivalTimeConverted);
1257             end
1258
1259             contourLevelsATC = app.MinimumdepthEditField_3.Value:app.
LineintervalEditField_3.Value:app.MaximumdepthEditField_3.Value;
1260             manualLabels3 = app.MinimumdepthEditField_3.Value:app.
IntervalEditField_7.Value:app.MaximumdepthEditField_3.Value;
1261         end
1262

```

```

1263         atcContourColor = ArrivalTimeContourColor;
1264         atcContourStyle = ArrivalTimeLineStyle;
1265         atcLineWidth = app.ThicknessEditField_3.Value;
1266         fontSize3 = app.TextLabelSize_3.Value;
1267         labelSpacing3 = app.LabelSpacingEditField_4.Value;
1268
1269         %-----Gauges-----
1270         markerSize = app.SizeEditField.Value;
1271         labelFontSize = app.SizeEditField_3.Value;
1272
1273         %-----Boundary limit-----
1274         eastValue = app.EastEditField.Value;
1275         westValue = app.WestEditField.Value;
1276
1277         if eastValue < westValue
1278             % Swap West and East input values if the West value is greater
1279             % than the East
1280             xMin_WH = eastValue;
1281             xMax_WH = westValue;
1282             app.EastEditField.Value = xMax_WH;
1283             app.WestEditField.Value = xMin_WH;
1284         elseif eastValue == westValue
1285             % Set the limits to the default values
1286             app.EastEditField.Value = max(app.x);
1287             app.WestEditField.Value = min(app.x);
1288             xMin_WH = app.WestEditField.Value;
1289             xMax_WH = app.EastEditField.Value;
1290         else
1291             xMin_WH = westValue;
1292             xMax_WH = eastValue;
1293         end
1294
1295         northValue = app.NorthEditField.Value;
1296         southValue = app.SouthEditField.Value;
1297
1298         if northValue < southValue
1299             % Swap North and South input values if the South value is
1300             % greater than the North
1301             yMin_WH = northValue;
1302             yMax_WH = southValue;
1303             app.NorthEditField.Value = yMax_WH;
1304             app.SouthEditField.Value = yMin_WH;
1305         elseif northValue == southValue
1306             % Set the limits to the default values
1307             app.NorthEditField.Value = max(app.y);
1308             app.SouthEditField.Value = min(app.y);
1309         else
1310             yMin_WH = southValue;
1311             yMax_WH = northValue;
1312         end
1313
1314         %-----Map Title-----
1315         totalSimulationTime = app.TotalSimuilationTimesecEditField_3.
1316         Value;
1317         startTime = app.StartTime.Value;
1318         colorbarTextSize = app.ColorbarTextSize.Value + 1;
1319         pats = digitsPattern;
1320

```



```

1318
1319
1320 %-----
1321 %-----Start of the loop for plotting-----
1322 %-----
1323 for i = 1:FileLength
1324     if all(contains(string(app.FileName), patterns)) % Check if
each file is either an eta or hmax file
1325         f1 = readmatrix(app.FileINPUT(i));
1326
1327         % Flip the data horizontally
1328         if app.HorizontalCheckBox.Value
1329             f1 = fliplr(f1);
1330         end
1331
1332         % Flip the data vertically
1333         if app.VerticalCheckBox.Value
1334             f1 = flipud(f1);
1335         end
1336
1337         if matches(app.DepthFileEditField.Value, string(app.
bathymetryname))
1338             %If the bathymetry file is loaded, delete negative
land values
1339             f1(depth <= 0) = NaN;
1340         end
1341
1342         % if matches(app.DepthFileEditField.Value, string(app.
bathymetryname))
1343         %     % Ensure depth is defined and non-empty
1344         %     if ~isempty(depth)
1345         %         % Preallocate f1 to match the size of depth
1346         %         f1 = NaN(size(depth)); % Initialize with NaNs
1347         %         % Optionally assign valid values where depth > 0
1348         %         % f1(depth > 0) = someComputation(depth(depth >
0));
1349         %     % else
1350         %         %     warning('Depth data is empty. Skipping
bathymetry filtering.');
```

```

1351         %     end
1352         % end
1353         app.fin = f1;
1354
1355         %-----
1356         %-----PLOT ALL DATA IN ONE FIGURE-----
1357         %-----
1358         if app.PlotalldatainonefigureButton.Value
1359             gca(i) = subplot(plotRow, plotCol, i);
1360             %-----
1361             %-----PLOT SEPARATELY-----
1362             %-----
1363         elseif app.PlotseparatelyButton_2.Value
1364             figure();
1365             gca(i) = subplot(1, 1, 1);
1366         end
1367
1368         %-----
1369         %-----MAIN PLOT-----

```

```

1370                                     %-----
1371
1372                                     f = pcolor(app.x, app.y, (app.fin)); %(X coordinates, X
coordinates, eta/hmax values)
1373                                     shading interp;
1374                                     set(f, 'EdgeColor', 'none', 'FaceColor', 'interp');
1375                                     set(gca, 'Color', app.LandAreaColor) %Set the background
color for the land
1376                                     hold on
1377
1378
1379                                     %-----
1380                                     %-----BASEMAP: Colormap-----
1381                                     %-----
1382                                     cb = colorbar;
1383                                     colorbarHandles = [colorbarHandles, cb];
1384
1385                                     % Set the properties once
1386                                     cb.Label.FontSize = colorbarFontSize; % Colorbar label
1387                                     cb.FontSize = colorbarFontSize; % Size of the tick
number% Size of the tick number
1388
1389                                     if contains(string(app.FileName(i)), "eta") % For ETA/sea
surface displacement at time t
1390                                     % Flip the colorbar if needed
1391                                     if flipCheckBoxETA
1392                                         colormap(gca(i), flipud(datacolor2ETA));
1393                                     else
1394                                         colormap(gca(i), datacolor2ETA);
1395                                     end
1396                                     caxis(gca(i), [colormapMinETA, colormapMaxETA]); %
Colorbar min and max values
1397                                     cb.Label.String = colorbartitleETA;
1398                                     else % For hmax/maximum wave height
1399                                     % Flip the colorbar if needed
1400                                     if flipCheckBoxHMAX
1401                                         colormap(gca(i), flipud(datacolor2));
1402                                     else
1403                                         colormap(gca(i), datacolor2);
1404                                     end
1405                                     caxis(gca(i), [colormapMinHMAX, colormapMaxHMAX]); %
Colorbar min and max values
1406                                     cb.Label.String = colorbartitleHMAX;
1407                                     end
1408
1409                                     %-----
1410                                     %-----OVERLAY FEATURES: Bathymetry Contours-----
1411                                     %-----
1412                                     if isBathymetryChecked && isMatchingBathymetry
1413                                         % hold on
1414                                         % Plot the contours
1415                                         [C, h] = contour(app.x, app.y, wavebathymetry,
contourLevels, 'EdgeColor', contourCol, 'LineStyle', contourStyle, 'LineWidth',
lineWidth);
1416                                         % Add contour labels if needed
1417                                         if isAddingLabels
1418                                             clabel(C, h, manualLabels, 'FontSize', fontSize,
'Color', 'k');

```

```

1419             set(findobj(h, '-property', 'LabelSpacing'), ↵
'LabelSpacing', labelSpacing); % Adjust the label spacing
1420         end
1421     end
1422
1423     %-----
1424     %-----OVERLAY FEATURES: Wave Height Contours-----
1425     %-----
1426     if app.WaveHeightButton.Value
1427         % Plot the contours
1428         [D, g] = contour(app.x, app.y, wavecontour, ↵
contourLevelsWH, 'EdgeColor', waveContourColor, 'LineStyle', waveContourStyle, ↵
'LineWidth', waveLineWidth);
1429
1430         % Add contour labels
1431         if app.AddLabelsCheckBox_3.Value
1432             clabel(D, g, manualLabels2, 'FontSize', fontSize2, ↵
'Color', 'k');
1433             set(findobj(g, '-property', 'LabelSpacing'), ↵
'LabelSpacing', labelSpacing2); % Adjust the label spacing
1434         end
1435     end
1436
1437     %-----
1438     %-----OVERLAY FEATURES: Arrival Time Contours-----
1439     %-----
1440     if app.ArrivalTimeButton.Value && matches(app. ↵
FileEditField_3.Value, string(app.ArrivalTimename))
1441         % Plot the contours
1442         [E, ggg] = contour(app.x, app.y, atc, ↵
contourLevelsATC, 'EdgeColor', atcContourColor, 'LineStyle', atcContourStyle, ↵
'LineWidth', atcLineWidth);
1443
1444         if app.AddLabelsCheckBox_5.Value
1445             clabel(E, ggg, manualLabels3, 'FontSize', ↵
fontSize3, 'Color', 'k');
1446             set(findobj(ggg, '-property', 'LabelSpacing'), ↵
'LabelSpacing', labelSpacing3); % Adjust the label spacing
1447         end
1448     end
1449
1450     %-----
1451     %-----OVERLAY FEATURES: Gauges-----
1452     %-----
1453     if app.GaugesCheckBox.Value && matches(app.FileEditField. ↵
Value, string(app.GAUGEFILE))
1454         % Plot the points
1455         gaugePoint = plot(app.longGauge, app.latGauge, ↵
GaugeMarker);
1456         set(gaugePoint, 'MarkerSize', markerSize, ↵
'MarkerFaceColor', GaugeColor, 'MarkerEdgeColor', GaugeColor); % Size and color of ↵
the points
1457
1458         % Add labels
1459         if app.AddLabelsCheckBox_4.Value
1460             horzLabel = HorzLabel; % Ensure HorzLabel is ↵
precomputed
1461             vertLabel = VertLabel; % Ensure VertLabel is ↵

```

```

precomputed
1462         gSpacing = app.GLspacing.Value;
1463
1464         for k = 1:length(app.latGauge)
1465             gtxt = text(app.longGauge(k) + gSpacing, app.
latGauge(k), num2str(k), 'FontSize', labelFontSize, 'HorizontalAlignment',
horzLabel, 'VerticalAlignment', vertLabel);
1466             set(gtxt, 'Clipping', 'on');
1467         end
1468     end
1469 end
1470 hold on
1471
1472     %-----
1473     %-----GENERAL LAYOUT: Boundary limit-----
1474     %-----
1475     xlim([xMin_WH,xMax_WH])
1476     ylim([yMin_WH,yMax_WH])
1477     hold on
1478
1479     %-----
1480     %-----GENERAL LAYOUT: Ticks-----
1481     %-----
1482     %Reduce the number of ticks to 2 on the X axis and 2
on the Y axis for simplicity
1483     % Get the current ticks from the first plot
1484     if i == 1
1485         xAllTicks = xticks(gca);
1486         yAllTicks = yticks(gca);
1487
1488         % Ensure the ticks are numeric arrays
1489         if iscell(xAllTicks)
1490             xAllTicks = cellfun(@str2double, xAllTicks,
'UniformOutput', true);
1491         end
1492         if iscell(yAllTicks)
1493             yAllTicks = cellfun(@str2double, yAllTicks,
'UniformOutput', true);
1494         end
1495
1496         % Remove any NaN values that might result from
str2double conversion
1497         xAllTicks = xAllTicks(~isnan(xAllTicks));
1498         yAllTicks = yAllTicks(~isnan(yAllTicks));
1499
1500         % Select only the 2nd and second-to-last ticks
1501         % For X axis
1502         if length(xAllTicks) >= 4 % Check first if there
are enough ticks to select from
1503             xticksSelected = sort([xAllTicks(2), xAllTicks
(end-1)]);
1504         elseif length(xAllTicks) >= 2
1505             xticksSelected = sort([xAllTicks(1), xAllTicks
(end)]); % Use the first and last if less than 4
1506         else
1507             xticksSelected = xAllTicks; % Use whatever
ticks are available
1508         end

```

```

1509
1510         % For Y axis
1511         if length(yAllTicks) >= 4
1512             yticksSelected = sort([yAllTicks(2), yAllTicks
(end-1)]);
1513         elseif length(yAllTicks) >= 2
1514             yticksSelected = sort([yAllTicks(1), yAllTicks
(end)]); % Use the first and last if less than 4
1515         else
1516             yticksSelected = yAllTicks; % Use whatever
ticks are available
1517         end
1518
1519     end
1520
1521     % Update the number of ticks on the axes
1522     xticks(gca, xticksSelected);
1523     yticks(gca, yticksSelected);
1524
1525     % Determine the maximum decimal places for x and y
ticks
1526     xmaxDecimals = max(cellfun(@(x) length(regexp(x, '(?
<=\.)\d+', 'match', 'once'))), ...
1527         cellstr(num2str(xticksSelected'))));
1528     ymaxDecimals = max(cellfun(@(y) length(regexp(y, '(?
<=\.)\d+', 'match', 'once'))), ...
1529         cellstr(num2str(yticksSelected'))));
1530
1531     % Convert tick labels to their equivalent in degrees
1532     if app.PlotIndegreesCheckBox_2.Value
1533         % Convert selected ticks to formatted strings and
set them for degrees
1534         xticklabels(gca, arrayfun(@(x) sprintf('%.1f°',
mod(x + 360, 360)), xticksSelected, 'UniformOutput', false));
1535         yticklabels(gca, arrayfun(@(y) sprintf('%.1f°',
y), yticksSelected, 'UniformOutput', false));
1536     else
1537         % Correct the format string and apply calculated
maximum decimals
1538         xticklabels(gca, arrayfun(@(x) sprintf(['%. '
num2str(xmaxDecimals) 'f'], x), xticksSelected, 'UniformOutput', false));
1539         yticklabels(gca, arrayfun(@(y) sprintf(['%. '
num2str(ymaxDecimals) 'f'], y), yticksSelected, 'UniformOutput', false));
1540     end
1541
1542     % Set the y-axis tick angle
1543     ytickangle(gca, -270);
1544
1545     %Fontsize of the tick labels
1546     set(gca, 'Layer', 'top', 'LineWidth', 1, 'FontSize', app.
ColorbarTextSize.Value-1)
1547
1548     %Adjust tick lengths based on the number of subplots
if 'Plot all data in one figure' is selected
1549     if FileLength <= 2 && app.
PlotalldatainonefigureButton.Value
1550         set(gca, 'TickLength', [0.015 0.015])
1551     else

```

```

1552         set(gca, 'TickLength', [0.01 0.01])
1553     end
1554
1555     grid off
1556     box on
1557     daspect([1 1 1]) %check if it distorts the countries
1558     hold off
1559
1560     %-----
1561     %-----GENERAL LAYOUT: Map Title-----
1562     %-----
1563     str = string(app.FileName(i));
1564     dbl1 = extract(str, pats); % Extract numbers from the
eta/hmax filename to calculate the time frame
1565
1566     if startTime == 0 && str2double(dbl1) == 0 %Set the
time to zero
1567         timeT = startTime + (str2double(dbl1)-1);
1568     else
1569         originalTimeT = totalSimulationTime * (str2double
(dbl1)-1); % Convert the filename to time
1570         timeT = startTime + originalTimeT; % Adjust the
start of the time count
1571     end
1572
1573     % Time conversion
1574     if timeT < 60
1575         timeStr = sprintf("%d sec", timeT);
1576     elseif timeT < 3600
1577         minutes = round(timeT / 60);
1578         if minutes == 1
1579             timeStr = "1 min";
1580         else
1581             timeStr = sprintf("%d mins", minutes);
1582         end
1583     else
1584         raw_hours = timeT/3600;
1585         hours = fix(timeT/3600); %Extract the whole
number%Extract the whole number
1586         hours_decimal = raw_hours - floor(raw_hours); %
Extract the decimal places
1587         minutes = round(hours_decimal * 60);
1588
1589         if hours > 0
1590             if minutes > 0
1591                 timeStr = sprintf("%d hr, %d mins", hours,
minutes);
1592             else
1593                 timeStr = sprintf("%d hr", hours);
1594             end
1595         end
1596     end
1597
1598     title(timeStr, 'FontSize', colorbarTextSize); % Add
the title to the top of the map; Adjust the font size of the title
1599
1600     %-----
1601     %-----GENERAL LAYOUT: Figure Size-----

```

```

1602                                     %-----
1603                                     if ~app.AutoSetCheckBox_2.Value
1604                                         %Manually set the figure size
1605                                         figureHandle = ancestor(f, 'figure'); %
1606                                         set(figureHandle, 'Units', 'Inches', 'Position', [
1607 [0, 0, app.Width_2.Value, app.Height_2.Value], 'PaperUnits', 'Inches', 'PaperSize', [
1608 [app.Width_2.Value, app.Height_2.Value]);
1609                                         set(gcf, 'Units', 'Inches', 'Position', [0, 0,
1610 app.Width_2.Value, app.Height_2.Value], 'PaperUnits', 'Inches', 'PaperSize', [app.
1611 Width_2.Value, app.Height_2.Value]);
1612                                     end
1613                                     %-----
1614                                     %-----SAVE MAP: Plot Separately-----
1615                                     %-----
1616                                     if app.PlotseparatelyButton_2.Value
1617                                         %Save the maps with filenames matching the loaded
1618 file
1619                                         OF = fullfile(app.inputtxt_directory1,app.
1620 FileNAME(i));
1621
1622                                         %Options for saving format
1623                                         %PNG
1624                                         if app.pngCheckBox.Value
1625                                             outputfile = OF + ".png";
1626                                             exportgraphics(gcf,outputfile, 'Resolution',
1627 300)
1628                                         end
1629
1630                                         %JPG
1631                                         if app.jpgCheckBox.Value
1632                                             outputfile = OF + ".jpg";
1633                                             exportgraphics(gcf,outputfile, 'Resolution',
1634 300)
1635                                         end
1636
1637                                         %TIF
1638                                         if app.tifCheckBox_2.Value
1639                                             XXX = app.x;
1640                                             YYY = app.y;
1641
1642                                             west_limit = app.WestEditField.Value;
1643                                             east_limit = app.EastEditField.Value;
1644                                             north_limit = app.NorthEditField.Value;
1645                                             south_limit = app.SouthEditField.Value;
1646
1647                                             %Make the longitude values within -180 to 180
1648 range
1649                                             if west_limit ~= 180
1650                                                 west_limit = mod(west_limit + 180, 360) -
1651 180;
1652                                             end
1653
1654                                             if east_limit ~= 180
1655                                                 east_limit = mod(east_limit + 180, 360) -
1656 180;
1657                                             end
1658                                         end

```



```

1649
1650 % Convert x coordinates to correct degree
range
1651 XXX(XXX > 180) = XXX(XXX > 180) - 360;
1652
1653 % Create meshgrid
1654 [X, Y] = meshgrid(XXX, YYY);
1655 Z = f1;
1656 %Check if the latitude is within the
Geographic limits
1657 if app.LatitudeEditField.Value > 90 || app.
LatitudeEditField.Value < -90
1658     app.LatitudeEditField.Value = 0;
1659 else
1660     %Clip the matrix based on boundary limits
1661     if west_limit >= 0 && east_limit >=0 ||
west_limit < 0 && east_limit < 0 % if east and west values are either both (+) or
both (-) values
1662         x_indices_positive = XXX >=
west_limit & XXX <= east_limit;
1663         y_indices = YYY >= south_limit & YYY
<= north_limit;
1664         Z_positive = Z(y_indices,
x_indices_positive);
1665         X_positive = X(y_indices,
x_indices_positive);
1666         Y = Y(y_indices, x_indices_positive);
% Keep the same for both file
1667         if ~isempty(Z_positive)
1668             % Define spatial referencing
information
1669             R_positive = georasterref
('RasterSize', size(Z_positive), 'LatitudeLimits', [min(Y(:)), max(Y(:))],
'LongitudeLimits', [min(X_positive(:)), max(X_positive(:))]);
1670             R_positive.ColumnsStartFrom =
'south'; % Set the column orientation to start from the north
1671             R_positive.RowsStartFrom = 'west';
1672
1673             % Save as geotiff
1674             filename_positive = OF + ".tif";
1675             geotiffwrite(filename_positive,
Z_positive, R_positive);
1676         end
1677
1678 elseif west_limit >= 0 && east_limit < 0
%if longitude values are a combination of (+) west limit and (-) east limit
1679     x_indices_positive = XXX >= 0 & XXX >=
west_limit;
1680     x_indices_negative = XXX < 0 & XXX <=
east_limit;
1681     y_indices = YYY >= south_limit & YYY
<= north_limit;
1682     Z_positive = Z(y_indices,
x_indices_positive);
1683     Z_negative = Z(y_indices,
x_indices_negative);
1684     X_positive = X(y_indices,
x indices positive);

```

```

1685 X_negative = X(y_indices, ↵
x_indices_negative);
1686 Y = Y(y_indices, x_indices_positive); ↵
% Keep the same for both file
1687
1688 if ~isempty(Z_positive)
1689     % Define spatial referencing ↵
information
1690     R_positive = georasterref ↵
('RasterSize', size(Z_positive), 'LatitudeLimits', [min(Y(:)), max(Y(:))], ↵
'LongitudeLimits', [min(X_positive(:)), max(X_positive(:))]);
1691     R_positive.ColumnsStartFrom = ↵
'south'; % Set the column orientation to start from the north
1692     R_positive.RowsStartFrom = 'west';
1693
1694     % Save as geotiff
1695     filename_positive = OF + "_1.tif";
1696     geotiffwrite(filename_positive, ↵
Z_positive, R_positive);
1697 end
1698
1699 if ~isempty(Z_negative)
1700     % Define spatial referencing ↵
information
1701     R_negative = georasterref ↵
('RasterSize', size(Z_negative), 'LatitudeLimits', [min(Y(:)), max(Y(:))], ↵
'LongitudeLimits', [min(X_negative(:)), max(X_negative(:))]);
1702     R_negative.ColumnsStartFrom = ↵
'south'; % Set the column orientation to start from the north
1703     R_negative.RowsStartFrom = 'west';
1704
1705     % Save as geotiff
1706     filename_negative = OF + "_2.tif";
1707     geotiffwrite(filename_negative, ↵
Z_negative, R_negative);
1708 end
1709
1710 elseif west_limit < 0 && east_limit >= 0 ↵
%if longitude values are a combination of (-) west limit and (+) east limit
1711     x_indices_positive = XXX >= 0 & XXX <= ↵
east_limit;
1712     x_indices_negative = XXX < 0 & XXX >= ↵
west_limit;
1713     y_indices = YYY >= south_limit & YYY ↵
<= north_limit;
1714     Z_positive = Z(y_indices, ↵
x_indices_positive);
1715     Z_negative = Z(y_indices, ↵
x_indices_negative);
1716     X_positive = X(y_indices, ↵
x_indices_positive);
1717     X_negative = X(y_indices, ↵
x_indices_negative);
1718     Y = Y(y_indices, x_indices_positive); ↵
% Keep the same for both file
1719
1720 if ~isempty(Z_positive)
1721     % Define spatial referencing ↵

```

```
information
1722                                     R_positive = georasterref ↵
('RasterSize', size(Z_positive), 'LatitudeLimits', [min(Y(:)), max(Y(:))], ↵
'LongitudeLimits', [min(X_positive(:)), max(X_positive(:))]));
1723                                     R_positive.ColumnsStartFrom = ↵
'south'; % Set the column orientation to start from the north
1724                                     R_positive.RowsStartFrom = 'west';
1725
1726                                     % Save as geotiff
1727                                     filename_positive = OF + "_1.tif";
1728                                     geotiffwrite(filename_positive, ↵
Z_positive, R_positive);
1729                                 end
1730                                 if ~isempty(Z_negative)
1731                                     % Define spatial referencing ↵
information
1732                                     R_negative = georasterref ↵
('RasterSize', size(Z_negative), 'LatitudeLimits', [min(Y(:)), max(Y(:))], ↵
'LongitudeLimits', [min(X_negative(:)), max(X_negative(:))]));
1733                                     R_negative.ColumnsStartFrom = ↵
'south'; % Set the column orientation to start from the north
1734                                     R_negative.RowsStartFrom = 'west';
1735
1736                                     % Save as geotiff
1737                                     filename_negative = OF + "_2.tif";
1738                                     geotiffwrite(filename_negative, ↵
Z_negative, R_negative);
1739                                 end
1740                             end
1741                         end
1742                     end
1743
1744                     %Create animation / closing the file
1745                     if app.mp4CheckBox.Value
1746                         set(gcf,'Renderer','zbuffer');
1747                         f = getframe(gcf);
1748                         writeVideo(vidObj,f);
1749                     end
1750                 end
1751             end
1752         end
1753     end
1754
1755     %Close the animation
1756     if app.mp4CheckBox.Value
1757         close(vidObj);
1758     end
1759
1760     %-----SAVE MAP: Plot all data in one figure-----
1761     %-----SAVE MAP: Plot all data in one figure-----
1762     if app.PlotalldatainonefigureButton.Value
1763         %Set up the filename
1764         if FileLength == 1 %Use raw hmax/eta filename
1765             OF = fullfile(app.inputtxt_directory1,string(app.↵
FileNAME));
1766         else %Use a generic name
```

```

1769             OF = fullfile(app.inputtxt_directory1, ↵
"WaveHeight_Output");
1770         end
1771
1772         %Save the figure
1773         if app.pngCheckBox.Value
1774             outputfile = OF + ".png";
1775             exportgraphics(gcf,outputfile,'Resolution',300)
1776         end
1777
1778         if app.jpgCheckBox.Value
1779             outputfile = OF + ".jpg";
1780             exportgraphics(gcf,outputfile,'Resolution',300)
1781         end
1782     end
1783
1784
1785
1786 end
1787
1788 % Button pushed function: Button_15
1789 function Button_15Pushed(app, event)
1790
1791
1792     %Load the bathymetry file and display its filename in the textbox
1793     [bathymetryfile,path2] = uigetfile('*', 'Select the files',↵
'MultiSelect', 'on');
1794     if ~(ismcc || isdeployed)
1795         addpath(genpath(fullfile(string(path2))));
1796     end
1797     app.DepthFileEditField.Value = string(bathymetryfile);
1798     app.bathymetryinputdata = string(path2) + string(bathymetryfile);
1799     app.bathymetryname = string(bathymetryfile);
1800     app.DepthFileEditField.BackgroundColor = 'w';
1801     app.DepthFileEditField.FontColor = 'k';
1802
1803     %Extract information from the bathymetry
1804     if app.DepthFileEditField.Value == string(bathymetryfile)
1805         if ~strcmp(app.DepthFileEditField.Value, "0") || ~strcmp(app.↵
DepthFileEditField.Value, "00")
1806             % If the file is .tif, read the matrix, extract the ↵
geographic values and grid sizes, and display them in the app textboxes
1807             if contains(app.bathymetryinputdata, '.tif')
1808                 [A1,R2] = readgeoraster(app.bathymetryinputdata);
1809                 app.bathymetrydata = flipud(A1);
1810
1811                 %Extract lat and long of the southwest corner
1812                 app.LongitudeEditField.Value = min(R2.↵
LongitudeLimits);
1813                 app.LatitudeEditField.Value = min(R2.LatitudeLimits);
1814                 app.LongitudeEditField.FontColor = 'k';
1815                 app.LatitudeEditField.FontColor = 'k';
1816
1817                 %Extract grid size
1818                 app.gridX.Value = R2.CellExtentInLatitude;
1819                 app.gridY.Value = R2.CellExtentInLongitude;
1820                 app.gridX.FontColor = 'k';
1821                 app.gridY.FontColor = 'k';

```

```

1822         drawnow
1823
1824         %Update the X-axis limits in the 'General Layout'
section
1825         if isprop(app, 'col1')
1826             app.x = app.LongitudeEditField.Value + [0:app.col1] * app.gridX.Value;
1827             app.EastEditField.Value = max(app.x);
1828             app.WestEditField.Value = min(app.x);
1829             app.EastEditField.FontColor = 'k';
1830             app.WestEditField.FontColor = 'k';
1831         end
1832
1833         %Update the Y-axis limits in the 'General Layout'
section
1834         if isprop(app, 'row1')
1835             app.y = app.LatitudeEditField.Value + [0:app.row1] * app.gridY.Value;
1836             app.NorthEditField.Value = max(app.y);
1837             app.SouthEditField.Value = min(app.y);
1838             app.NorthEditField.FontColor = 'k';
1839             app.SouthEditField.FontColor = 'k';
1840         end
1841         drawnow
1842
1843         else %Only read the matrix for use as a basemap and for plotting bathymetry contours
1844             app.bathymetrydata = readmatrix(app.bathymetryinputdata);
1845             app.MaximumEditField.Value = max(max(readmatrix(app.bathymetryinputdata)));
1846         end
1847     end
1848     %Enable the 'Land Color' dropdown list in 'Basemap' section
1849     app.CoastlinecolourDropDown_2.Enable = "on";
1850 end
1851
1852     if isprop(app, 'bathymetrydata')
1853         [app.bathyRow, app.bathyCol] = size(app.bathymetrydata);
1854     end
1855
1856 end
1857
1858 % Callback function
1859 function ImportinfoButtonPushed(app, event)
1860
1861 end
1862
1863 % Value changed function: FileTextArea
1864 function FileTextAreaValueChanged(app, event)
1865     if contains(app.FileTextArea.Value, "hmax") || contains(app.FileTextArea.Value, "eta")
1866         app.WaveHeightContoursButton.Enable = "on";
1867     end
1868
1869 end
1870
1871 % Callback function

```

```

1872     function ButtonGroup_2SelectionChanged(app, event)
1873
1874     end
1875
1876     % Callback function
1877     function MinimumEditFieldValueChanged(app, event)
1878         app.MinimumEditField.FontColor = 'k';
1879
1880     end
1881
1882     % Callback function
1883     function MaximumEditFieldValueChanged(app, event)
1884         app.MaximumEditField.FontColor = 'k';
1885     end
1886
1887     % Callback function
1888     function IntervalEditField_5ValueChanged(app, event)
1889         app.IntervalEditField_5.FontColor = 'k';
1890         app.IntervalEditField_3.Value = app.IntervalEditField_5.Value; %
Copy the values to the contour label interval textbox
1891         drawnow
1892
1893     end
1894
1895     % Value changed function: CoastlinecolourDropDown_8
1896     function CoastlinecolourDropDown_6ValueChanged(app, event)
1897         value = app.CoastlinecolourDropDown_8.Value;
1898         %List of values based on the cbrewer2 colormap
1899         validValues = {'blue', 'blue - green', 'blue - purple', 'green -
blue', ...
1900             'greens', 'grays', 'oranges', 'orange - red', ...
1901             'purple - blue', 'purple - blue - green', 'purple - red', ...
1902             'purples', 'red - purple', 'reds', 'yellow - green', ...
1903             'yellow - green - blue', 'yellow - orange - brown', ...
1904             'yellow - orange - red', 'brown - teal', 'pink - light green',
...
1905             'purple - green', 'purple - orange', 'red - blue', ...
1906             'red - gray', 'red - yellow - blue', 'red - yellow - green',
...
1907             'spectral', 'accent', 'dark 2', 'paired', 'pastel 1', ...
1908             'pastel 2', 'set 1', 'set 2', 'set 3'};
1909
1910         % If the selected value in the dropdown list is part of the
cbrewer2 colormap, enable the color interpolation textbox
1911         if ismember(value, validValues)
1912             app.DivisionEditField.Enable = "on";
1913             app.DivisionEditFieldLabel.Enable = "on";
1914         else
1915             app.DivisionEditField.Enable= "off";
1916             app.DivisionEditFieldLabel.Enable = "off";
1917         end
1918
1919     end
1920
1921     % Callback function
1922     function Button_8Pushed(app, event)
1923         try
1924             [filename3,path3] = uigetfile('*.txt;*.shp');

```

```

1925         fullname = fullfile(path3,filename3);
1926
1927         [~, name, ext] = fileparts(filename3);
1928         if ~(ismcc || isdeployed)
1929             addpath(genpath(string(path3)));
1930         end
1931
1932         app.GaugeFname = name;
1933         app.GAUGEFILE = filename3;
1934         app.FileEditField.Value = filename3;
1935         app.FileEditField.FontColor = 'k';
1936         app.FileEditField.BackgroundColor = 'w';
1937
1938
1939         if strcmp(ext, '.shp')
1940             S = shaperead(fullname);
1941             latitudes= [S.Y]';
1942             longitudes= [S.X]';
1943             list = [round(latitudes, 4), round(longitudes, 4)];
1944             list = unique(list, 'rows');
1945             app.latGauge = list(:, 1);
1946             app.longGauge = list(:, 2);
1947         elseif strcmp(ext, '.txt')
1948             fileID = fopen(fullname, 'r');
1949             dataArray = textscan(fileID, '%f %f', 'Delimiter', '\t',
'whitespace'); %Read text file and ensure that it is tab delimited
1950             fclose(fileID);
1951             app.STATION_FILE = string(filename3);
1952             app.latGauge = dataArray{1};
1953             app.longGauge = dataArray{2};
1954         end
1955     end
1956
1957
1958
1959 end
1960
1961 % Value changed function: CoastlinecolourDropDown_2
1962 function CoastlinecolourDropDown_2ValueChanged(app, event)
1963
1964
1965 end
1966
1967 % Callback function
1968 function ButtonGroup_6SelectionChanged(app, event)
1969
1970 end
1971
1972 % Button pushed function: Button_2
1973 function Button_2Pushed(app, event)
1974     % Set up the directory path for saving the files
1975     workingfolder = uigetdir;
1976     Dir1 = fullfile(string(workingfolder), 'OUTPUT_FILES');
1977     app.WORKFOLDER = Dir1;
1978
1979     %Create the 'OUTPUT_FILES/Figures' folder in the working directory
1980     FigureFolder = fullfile(Dir1, 'Figures');
1981     app.WORKFOLDER0 = FigureFolder;

```

```

1982         app.OutputDirectoryEditField.Value = fullfile(deblank(
(FigureFolder)));
1983         app.OutputDirectoryEditField.FontColor = 'k';
1984
1985         %Create the folders
1986         if ~exist(Dir1, 'dir')
1987             mkdir(Dir1);
1988         end
1989         if ~exist(FigureFolder, 'dir')
1990             mkdir(FigureFolder);
1991         end
1992     end
1993
1994     % Callback function
1995     function VirtualGaugesButtonValueChanged(app, event)
1996         %Display the gauges tab
1997         app.TabGroup2.SelectedTab = app.GaugesTab;
1998
1999
2000     end
2001
2002     % Callback function
2003     function BathymetryContoursButtonValueChanged(app, event)
2004
2005     end
2006
2007     % Callback function
2008     function WaveHeightContoursButtonValueChanged(app, event)
2009
2010
2011     end
2012
2013     % Value changed function: DepthFileEditField
2014     function DepthFileEditFieldValueChanged(app, event)
2015         value = app.DepthFileEditField.Value;
2016         if value == app.bathymetryname
2017             app.BathymetryContoursButton.Enable = "on";
2018         end
2019     end
2020
2021     % Callback function
2022     function LineintervalEditField_2ValueChanged(app, event)
2023         app.wh_interval.Value = app.LineintervalEditField_2.Value;
2024         drawnow
2025         app.LineintervalEditField_2.FontColor = 'k';
2026     end
2027
2028     % Callback function
2029     function MaximumdepthEditField_2ValueChanged(app, event)
2030
2031
2032         app.MaximumdepthEditField_2.FontColor = 'k';
2033     end
2034
2035     % Button pushed function: PLOTButton
2036     function PLOTButtonPushed(app, event)
2037         close all
2038         %-----

```



```

2039         %-----Marker Style-----
2040         %-----
2041         GaugeMarkerOptions = containers.Map({'none', 'o', '+', '*', '.', 'x',
'x', '-', '|', '^', 'v', '>', '<', 'diamond', 'hexagram', 'pentagram', 'square'},
...
2042         {'none', 'o', '+', '*', '.', 'x', '-', '|', '^', 'v', '>',
'<', 'diamond', 'hexagram', 'pentagram', 'square'});
2043         GaugeMarker2 = GaugeMarkerOptions(app.MarkerDropDown.Value);
2044
2045
2046         %-----
2047         %-----Line Style at y=0-----
2048         %-----
2049         YLineStyleOptions = containers.Map({'None', 'Solid', 'Dashed',
'Dotted', 'Dash-dotted'}, {'off', '-', '--', ':', '-.'});
2050         YLineStyle = YLineStyleOptions(app.LineatY0DropDown.Value);
2051
2052         %-----
2053         %-----Line Style-----
2054         %-----
2055         GLineStyleOptions = containers.Map({'Solid', 'Dashed', 'Dotted',
'Dash-dotted'}, {'-', '--', ':', '-.'});
2056         GLineStyle = GLineStyleOptions(app.GaugeLineStyle.Value);
2057
2058         %-----
2059         %-----Line Color for one data per plot-----
2060         %-----
2061         colors = {'Black', 'Dark gray', 'Medium gray', 'Light gray',
'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
2062         colorCodes = {'k', '[0.4 0.4 0.4]', '[0.7 0.7 0.7]', '[0.8 0.8
0.8]', 'r', 'g', 'b', 'y', 'c', 'm', 'w'};
2063         index = find(strcmp(colors, app.GaugeLineColor.Value));
2064         GLineColor0 = colorCodes{index};
2065
2066         %-----
2067         %-----Line Color for multiple data in one graph-----
2068         %-----
2069         LineColorOptions = containers.Map({'autumn', 'bone', 'colorcube',
'cool', 'copper', 'flag', ...
2070         'gray', 'hot', 'hsv', 'jet', 'parula', 'pink', 'prism',
'spring', 'summer', ...
2071         'turbo', 'winter', 'lines'}, ...
2072         {'autumn', 'bone', 'colorcube', 'cool', 'copper', 'flag',
'gray', 'hot', ...
2073         'hsv', 'jet', 'parula', 'pink', 'prism', 'spring', 'summer',
'turbo', 'winter', 'lines'});
2074
2075         GLineColor = LineColorOptions(app.GaugeLineColor_multiple.Value);
2076
2077         %-----
2078         %-----Legend Location-----
2079         %-----
2080         LegendLocationOptions = containers.Map({'North', 'South', 'East',
'West', 'Northeast', ...
2081         'Northwest', 'Southeast', 'Southwest', 'Northoutside',
'Southoutside', 'Eastoutside', ...
2082         'Westoutside', 'Northeastoutside', 'Northwestoutside',
'Southeastoutside', 'Southwestoutside', ...

```

```

2083         'Best', 'Bestoutside', 'None'}, ...
2084         {'north', 'south', 'east', 'west', 'northeast', 'northwest', '↵
'southeast', 'southwest', ...
2085         'northoutside', 'southoutside', 'eastoutside', 'westoutside', '↵
'northeastoutside', ...
2086         'northwestoutside', 'southeastoutside', 'southwestoutside', '↵
'best', 'bestoutside', 'none'}});
2087     legloc = LegendLocationOptions(app.LocationDropDown.Value);
2088     app.LegendLocation = legloc;
2089
2090     %-----
2091     %-----SAVE MAP: Output Directory-----
2092     %-----
2093     %Check if the output directory has been manually set by the user
2094     if app.OutputDirectoryEditField_2.Value == string(app.↵
WORKFOLDER5)
2095         inputtxt_directory3 = fullfile(app.WORKFOLDER2);
2096         Dir1 = fullfile(app.WORKFOLDER5);
2097         app.OutputDirectoryEditField_2.FontColor = 'k';
2098         if ~exist(inputtxt_directory3, 'dir')
2099             mkdir(inputtxt_directory3);
2100         end
2101         if ~exist(Dir1, 'dir')
2102             mkdir(Dir1);
2103         end
2104     else
2105         % Set the default directory to the Desktop
2106         if ismac
2107             %macOS
2108             defaultDir = fullfile(getenv('HOME'), 'Desktop');
2109         elseif ispc
2110             %Windows
2111             defaultDir = fullfile(getenv('USERPROFILE'), 'Desktop');
2112         else
2113             %Others
2114             defaultDir = pwd;
2115         end
2116
2117         %Create 'OUTPUT_FILES' folder and 'Figures' subfolder
2118         inputtxt_directory3 = fullfile(defaultDir, 'OUTPUT_FILES');
2119         % Dir1 = fullfile(defaultDir, 'OUTPUT_FILES/Figures');
2120         Dir1 = fullfile(inputtxt_directory3, 'Figures');
2121         if ~exist(inputtxt_directory3, 'dir')
2122             mkdir(inputtxt_directory3);
2123         end
2124         if ~exist(Dir1, 'dir')
2125             mkdir(Dir1);
2126         end
2127
2128         %Display the path directory of the 'Figures' subfolder in the ↵
textbox
2129         app.OutputDirectoryEditField_2.Value = deblank(string(Dir1));
2130         app.OutputDirectoryEditField_2.FontColor = 'k';
2131     end
2132
2133
2134     %-----
2135     %-----LOG REPORT-----

```

```

2136         %-----
2137         %Create 'Log_Files' subfolder in 'OUTPUT_FILES'
2138         LogFolder = fullfile(inputtxt_directory3, 'Log_Files');
2139         if ~exist(LogFolder, 'dir')
2140             mkdir(LogFolder);
2141         end
2142
2143         %Create the text file
2144         fileId = fopen(fullfile(LogFolder, 'Log_Report_GaugeRecords.
2145 txt'),'w'); %if same file exists, overwrite the file
2146         fileList = app.FileTextArea_2.Value;
2147
2148         %Template for the default section of the log report (includes
2149 information from the 'Input Data' section)
2150         fprintf(fileId, '%s\n', '***** LOG REPORT
2151 ***** ');
2152         fprintf(fileId, '%-30s-s\n', 'Type:', 'Gauge Records');
2153         timestamp = datestr(now, 'yyyy-mm-dd HH:MM:SS'); % Get the current
2154 timestamp
2155         fprintf(fileId, '%-30s-s\n', 'Timestamp:', timestamp); %Add
2156 timestamp in the header
2157         fprintf(fileId, '%
2158 s\n', '*****');
2159         fprintf(fileId, '%-s\n\n', '');
2160         fprintf(fileId, '%-30s-s\n', 'X axis:', app.XAxisEditField.Value);
2161         fprintf(fileId, '%-30s-s\n', 'Y axis:', app.YAxisEditField.Value);
2162
2163         %-----LOG REPORT-----
2164         %-----Input Data: Import Files-----
2165         %-----
2166         %If no file is uploaded
2167         if any(app.FileTextArea_2.Value == "0" | app.FileTextArea_2.Value
2168 == "00") || isempty(fileList)
2169             fprintf(fileId, '%-30s-s\n', 'IMPORT FILE/S:', 'Empty. No
2170 data plotted');
2171         end
2172
2173         %Check if the name of each uploaded file starts with 'sta_'
2174         for i = 1:numel(fileList)
2175             if any(contains(fileList, 'sta_xxxx')) || ~any(contains
2176 (fileList(i), 'sta_'))
2177                 fprintf(fileId, '%-30s-s\n', 'IMPORT FILE/S:', 'Invalid
2178 filename format. Use filenames with "sta_"');
2179                 fprintf(fileId, '%-30s-s\n', '', string(fileList(i)));
2180             end
2181         end
2182
2183         %List all the files loaded
2184         fprintf(fileId, '%-30s-s\n', 'Loaded Files:', [num2str(length
2185 (app.GaugeDirectorylist)), ' files']);
2186         fprintf(fileId, '%
2187 s\n\n', '_____');
2188
2189         %-----LOG REPORT-----
2190         %-----General Layout: Boundary Limits-----
2191         %-----
2192         %Check the X limits and interval of the plot if 'Auto Set' is not

```

```

selected for the limits
2182         if ~app.CheckBox_2.Value
2183             if app.xMinLimit.Value > app.xMaxLimit.Value
2184                 fprintf(fileId, '%-30s%-s\n', 'X Limit:', 'Max value must
exceed min value');
2185                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Max and min
values are swapped');
2186             end
2187             if app.xLimInterval.Value > app.xMaxLimit.Value - app.
xMinLimit.Value
2188                 fprintf(fileId, '%-30s%-s\n', 'X Limit Interval:', 'Value
must be less than the max-min difference');
2189                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value
adjusted to half the max-min difference');
2190             end
2191         end
2192
2193         %Check the Y limits and interval of the plot if 'Auto Set' is not
selected for the limits
2194         if ~app.CheckBox.Value
2195             if app.yMinLimit.Value > app.yMaxLimit.Value
2196                 fprintf(fileId, '%-30s%-s\n', 'Y Limit:', 'Max value must
exceed min value');
2197                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Max and min
values are swapped');
2198             end
2199             if app.yLimInterval.Value > app.yMinLimit.Value - app.
yMaxLimit.Value
2200                 fprintf(fileId, '%-30s%-s\n', 'Y Limit Interval:', 'Value
must be less than the max-min difference');
2201                 fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Value
adjusted to half the max-min difference');
2202             end
2203         end
2204
2205         %% PROCESS THE INPUT FILES AND VALUES, THEN GENERATE THE FIGURES
2206         %-----
2207         %-----READ THE INPUT FILES-----
2208         %-----
2209         clear xData yData yData2
2210         %Check if any file is loaded
2211         if app.FileTextArea_2.Value ~= string(app.GaugeNames)
2212             app.FileTextArea_2.Value = "NO FILE";
2213         else
2214             %Read the data
2215             FileLength = length(app.GaugeDirectorylist);
2216             plotHandles = [];
2217             legendEntries = {};
2218
2219             for i = 1:FileLength
2220                 g = readmatrix(app.GaugeDirectorylist(i)); %Read all
matrices
2221
2222                 %Calculate the magnitude of the Z-component of velocity
using the U and V velocity values in columns 3 and 4.
2223                 Zvelocity = sqrt(g(:,3).^2 + g(:,4).^2);
2224                 %Include the Z velocity as the fifth column
2225                 a = [a, Zvelocity];

```

```

2226 %Extracting the station numbers from the filename
2227 GN = erase(app.GaugeNames(i), "sta_"); %remove 'sta_'
from the name
2228 GN2 = strip(GN, "left", '0'); %remove the trailing
zeroes on the left side
2229
2230 if contains(GN2, '_')
2231     GaugeNumber{i} = regexprep(GN2, '_', ' '); %Replace
"_" to space
2232 else
2233     GaugeNumber{i} = GN2;
2234 end
2235
2236 %Choose the column from which to extract data points
for the X plot values
2237 app.XCOLGAUGE = str2double(app.XaxisUseDataDropDown.
Value);
2238 app.YCOLGAUGE = str2double(app.YaxisUseDataDropDown.
Value);
2239
2240 %Convert data loaded for X axis
2241 if app.CheckBox_3.Value
2242     g(:, app.XCOLGAUGE) = g(:, app.XCOLGAUGE) * (app.
numeratorX.Value/app.denominatorX.Value);
2243 end
2244
2245 %If '3,4,5' or '5' option is selected, calculate Z
velocity vector
2246 if strcmp(app.YaxisUseDataDropDown.Value, '3,4,5') ||
strcmp(app.YaxisUseDataDropDown.Value, '5')
2247     %Calculate the magnitude of the Z-component of
velocity using the U and V velocity values in columns 3 and 4.
2248     Zvelocity = sqrt(g(:,3).^2 + g(:,4).^2);
2249     %Include the Z velocity as the fifth column
2250     g = [g, Zvelocity];
2251 end
2252
2253 %Convert data loaded for Y axis
2254 if app.CheckBox_4.Value %Convert data loaded for Y
axis
2255     if ~strcmp(app.YaxisUseDataDropDown.Value,
'3,4,5') %If only one column of data will be used
2256         g(:, app.YCOLGAUGE) = g(:, app.YCOLGAUGE) *
(app.numeratorY.Value / app.denominatorY.Value);
2257     else
2258         %Convert the values
2259         g(:, 3) = g(:, 3) * (app.numeratorY.Value /
app.denominatorY.Value);
2260         g(:, 4) = g(:, 4) * (app.numeratorY.Value /
app.denominatorY.Value);
2261         g(:, 5) = g(:, 5) * (app.numeratorY.Value /
app.denominatorY.Value);
2262     end
2263 end
2264
2265 %Extract the required Y data based on dropdown
selection
2266 if ~strcmp(app.YaxisUseDataDropDown.Value, '3,4,5')

```

```

2267         yData = g(:,str2double(app.YaxisUseDataDropDown.Value));
2268     else
2269         yData = g(:, 3);
2270         y2Data = g(:, 4);
2271         y3Data = g(:, 5);
2272     end
2273
2274     %Sort data list to avoid connecting lines between the
first and last points
2275     xData = double(g(:, app.XCOLGAUGE));
2276     [xDataArranged, sortIndex] = sort(xData);
2277     yDataArranged= yData(sortIndex);
2278     if strcmp(app.YaxisUseDataDropDown.Value, '3,4,5') %If
columns 3, 4, 5 are to be plotted together
2279         y2DataArranged= y2Data(sortIndex);
2280         y3DataArranged= y3Data(sortIndex);
2281     end
2282
2283     %-----
2284     %-----MAIN PLOT-----
2285     %-----
2286     %Select the line color of the plots
2287     if app.PlotalldatainonegraphButton.Value
2288         a = str2func(GLineColor);
2289         rndm = a(FileLength); %divide the colormap based
on the number of uploaded dataset
2290
2291         if app.FlipCheckBox_3.Value % Flip colors
2292             app.colorcombi = rndm(FileLength-i+1,:);
2293         else
2294             app.colorcombi = rndm(i,:);
2295         end
2296     elseif app.PlotseparatelyButton_4.Value
2297         app.colorcombi = GLineColor0; %For single line
plot
2298     end
2299
2300     %-----
2301     %-----PLOT ALL DATA IN ONE GRAPH-----
2302     %-----
2303     if app.PlotalldatainonegraphButton.Value %Plot all
sta_ data into one figure
2304         figure(1);
2305         set(gcf, 'Color', 'w');
2306         hold on;
2307
2308         if strcmp(app.YaxisUseDataDropDown.Value,
'3,4,5')
2309             %Main plotting loop
2310             %U vector
2311             ph1 = plot(xDataArranged, yDataArranged,
'LineWidth', app.DataLineWidth.Value, 'LineStyle', '--', 'Marker', GaugeMarker2, ...
2312                 'MarkerEdgeColor', app.colorcombi,
'MarkerFaceColor', app.colorcombi, 'Color', [0.596, 0.404, 0.553]);
2313             hold on
2314             %V vector
2315             ph2 = plot(xDataArranged, y2DataArranged,

```

```

'LineWidth', app.DataLineWidth.Value, 'LineStyle', '--', 'Marker', ↵
GaugeMarker2, 'MarkerEdgeColor', app.colorcombi, 'MarkerFaceColor', app.↵
colorcombi, 'Color', [0.4, 0.596, 0.447]);
2316             hold on
2317             %Z vector
2318             ph3 = plot(xDataArranged, y3DataArranged, ↵
'LineWidth', app.DataLineWidth.Value, 'LineStyle', GLineStyle, 'Marker', ↵
GaugeMarker2, 'MarkerEdgeColor', app.colorcombi, 'MarkerFaceColor', app.↵
colorcombi, 'Color', 'k');
2319             hold off
2320
2321
2322             plotHandles(end+1) = ph1;
2323             plotHandles(end+1) = ph2;
2324             plotHandles(end+1) = ph3;
2325
2326             legendEntries{end+1} = sprintf('%s %s U↵
vector', string(app.legendfirsttext.Value), string(GaugeNumber(min(i, length↵
(GaugeNumber)))));
2327             legendEntries{end+1} = sprintf('%s %s V↵
vector', string(app.legendfirsttext.Value), string(GaugeNumber(min(i, length↵
(GaugeNumber)))));
2328             legendEntries{end+1} = sprintf('%s %s Z↵
vector', string(app.legendfirsttext.Value), string(GaugeNumber(min(i, length↵
(GaugeNumber)))));
2329
2330             else %Plot data from only one column
2331             ph1 = plot(xDataArranged, yDataArranged, ↵
'LineWidth', app.DataLineWidth.Value, 'LineStyle', GLineStyle, 'Marker', ↵
GaugeMarker2, ...
2332             'MarkerEdgeColor', app.colorcombi, ↵
'MarkerFaceColor', app.colorcombi, 'Color', app.colorcombi);
2333             plotHandles(end+1) = ph1; % Store first↵
plot handle
2334             legendEntries{end+1} = sprintf('%s %s', ↵
string(app.legendfirsttext.Value), string(GaugeNumber(min(i, length↵
(GaugeNumber)))));
2335             end
2336
2337             % Update legend outside the loop with all↵
entries
2338             legend(plotHandles, legendEntries, 'Location', ↵
legloc, 'FontSize', app.LegendSize.Value);
2339             legend box off;
2340             hold off;
2341
2342             %-----
2343             %-----GENERAL LAYOUT: Figure Size-----
2344             %-----
2345             if ~app.AutoSetCheckBox.Value
2346             %Manually set the figure size
2347             figureHandle = ancestor(ph1, 'figure');
2348             set(figureHandle, 'Units', 'Inches', ↵
'Position', [0, 0, app.Width.Value, app.Height.Value], 'PaperUnits', 'Inches', ↵
'PaperSize', [app.Width.Value, app.Height.Value]);
2349             set(gcf, 'Units', 'Inches', 'Position', ↵
[0, 0, app.Width.Value, app.Height.Value], 'PaperUnits', 'Inches', 'PaperSize', ↵
[app.Width.Value, app.Height.Value]);

```

```

2350         end
2351
2352     end
2353
2354     %-----
2355     %-----PLOT SEPARATELY-----
2356     %-----
2357     if app.PlotseparatelyButton_4.Value %Plot each
sta_ data in separate figures
2358
2359         % Open a new figure for each file
2360         figure(i);
2361         clf %clean figure so legend won't repeat
2362         set(gcf, 'Color', 'w'); % Set figure
background to white
2363
2364         % Reset legend values for each figure
2365         plotHandles = [];
2366         legendEntries = {};
2367
2368         hold on
2369         if strcmp(app.YaxisUseDataDropDown.Value,
'3,4,5')
2370             %Main plotting loop
2371             %U vector
2372             ph1 = plot(xDataArranged, yDataArranged,
'LineWidth', app.DataLineWidth.Value, 'LineStyle', '--', 'Marker', GaugeMarker2, ...
2373             'MarkerEdgeColor', app.colorcombi,
'MarkerFaceColor', app.colorcombi, 'Color', [0.596, 0.404, 0.553]);
2374             hold on
2375             %V vector
2376             ph2 = plot(xDataArranged, y2DataArranged,
'LineWidth', app.DataLineWidth.Value, 'LineStyle', '--', 'Marker',
GaugeMarker2, 'MarkerEdgeColor', app.colorcombi, 'MarkerFaceColor', app.
colorcombi, 'Color', [0.4, 0.596, 0.447]);
2377             disp(ph2);
2378             hold on
2379             %Z vector
2380             ph3 = plot(xDataArranged, y3DataArranged,
'LineWidth', app.DataLineWidth.Value, 'LineStyle', GLineStyle, 'Marker',
GaugeMarker2, 'MarkerEdgeColor', app.colorcombi, 'MarkerFaceColor', app.
colorcombi, 'Color', 'k');
2381             hold off
2382
2383             plotHandles(end+1) = ph1;
2384             plotHandles(end+1) = ph2;
2385             plotHandles(end+1) = ph3;
2386
2387             %Add each legend entry
2388             legendEntries{end+1} = 'U vector';
2389             legendEntries{end+1} = 'V vector';
2390             legendEntries{end+1} = 'Z vector';
2391
2392             legend(plotHandles, legendEntries,
'Location', legloc, 'FontSize', app.LegendSize.Value);
2393             legend box off;
2394
2395         else %Plot data from only one column

```



```

2396                                     ph1 = plot(xDataArranged, yDataArranged, ↵
'LineWidth', app.DataLineWidth.Value, 'LineStyle', GLineStyle, 'Marker', ↵
GaugeMarker2, ...
2397                                     'MarkerEdgeColor', app.colorcombi, ↵
'MarkerFaceColor', app.colorcombi, 'Color', app.colorcombi);
2398                                     end
2399
2400                                     titleText = sprintf('%s %s', string(app.↵
legendfirstttext.Value), string(GaugeNumber(min(i, length(GaugeNumber)))));
2401                                     title(titleText);
2402
2403                                     %Figure Size
2404                                     if ~app.AutoSetCheckBox.Value
2405                                         figureHandle = ancestor(ph1, 'figure'); %↵
Get the figure handle containing the axes
2406                                         set(figureHandle, 'Units', 'Inches', ↵
'Position', [0, 0, app.Width.Value, app.Height.Value], 'PaperUnits', 'Inches', ↵
'PaperSize', [app.Width.Value, app.Height.Value]);
2407                                         set(gcf, 'Units', 'Inches', 'Position', ↵
[0, 0, app.Width.Value, app.Height.Value], 'PaperUnits', 'Inches', 'PaperSize', ↵
[app.Width.Value, app.Height.Value]);
2408                                     end
2409                                     end
2410
2411
2412                                     %-----
2413                                     %-----GENERAL LAYOUT: Boundary Limit-----
2414                                     %-----
2415                                     %Set X axis limit
2416                                     if app.CheckBox_2.Value
2417                                         xlim("auto")
2418                                         xticks('auto')
2419                                     else %Using the input values
2420                                         %Swap the minimum and maximum input values of ↵
X-axis limits when needed
2421                                         if app.xMinLimit.Value > app.xMaxLimit.Value
2422                                             temp3 = app.xMinLimit.Value;
2423                                             app.xMinLimit.Value = app.xMaxLimit.Value;
2424                                             app.xMaxLimit.Value = temp3;
2425                                         end
2426
2427                                     %Check the interval value if it is within the ↵
limits
2428                                     if app.xLimInterval.Value > (app.xMaxLimit.↵
Value-app.xMinLimit.Value)
2429                                         app.xLimInterval.Value = (app.xMaxLimit.↵
Value-app.xMinLimit.Value)/2;
2430                                     end
2431
2432                                     xlim([app.xMinLimit.Value app.xMaxLimit.↵
Value])
2433                                     xticksLOC = app.xMinLimit.Value:app.↵
xLimInterval.Value:app.xMaxLimit.Value;
2434                                     xticks(xticksLOC)
2435                                     tickXname = string(xticksLOC);
2436                                     xticklabels(tickXname)
2437                                     end
2438

```

```

2439         %Set Y axis limit
2440         if app.CheckBox.Value
2441             ylim("auto")
2442             yticks('auto')
2443         else %Using the input values
2444             % Swap the minimum and maximum input values of
Y-axis limits when needed
2445             if app.yMinLimit.Value > app.yMaxLimit.Value
2446                 temp4 = app.yMinLimit.Value;
2447                 app.yMinLimit.Value = app.yMaxLimit.Value;
2448                 app.yMaxLimit.Value = temp4;
2449             end
2450
2451             %Check the interval value if it is within the
limits
2452             if app.yLimInterval.Value > (app.yMaxLimit.
Value-app.yMinLimit.Value) %Check the interval value if it is within the limits
2453                 app.yLimInterval.Value = (app.yMaxLimit.
Value-app.yMinLimit.Value)/2;
2454             end
2455
2456             ylim([app.yMinLimit.Value app.yMaxLimit.
Value]);
2457             yticksLOC = [app.yMinLimit.Value:app.
yLimInterval.Value:app.yMaxLimit.Value];
2458             yticks(yticksLOC)
2459             tickYname = string(yticksLOC);
2460             yticklabels(tickYname)
2461             ytickformat('%1f')
2462         end
2463
2464
2465         %-----
2466         %-----GENERAL LAYOUT: Background Grid-----
2467         %-----
2468         ax = gca;
2469         ax.FontSize = app.GridLabelSize.Value;
2470         ax.LineWidth = app.LineYThickness_2.Value;
2471         if strcmp(app.GridStyle.Value, 'none')
2472             grid(ax, 'off');
2473         elseif strcmp(app.GridStyle.Value, 'X axis only')
2474             ax.XGrid = 'on';
2475             ax.YGrid = 'off';
2476             ax.XMinorGrid = 'off';
2477             ax.YMinorGrid = 'off';
2478         elseif strcmp(app.GridStyle.Value, 'Y axis only')
2479             ax.XGrid = 'off';
2480             ax.YGrid = 'on';
2481             ax.XMinorGrid = 'off';
2482             ax.YMinorGrid = 'off';
2483         elseif strcmp(app.GridStyle.Value, 'Both - major
lines')
2484             grid(ax, 'on');
2485             ax.XMinorGrid = 'off';
2486             ax.YMinorGrid = 'off';
2487         elseif strcmp(app.GridStyle.Value, 'Both - with minor
lines')
2488             grid(ax, 'on');

```

```

2489         ax.XMinorGrid = 'on';
2490         ax.YMinorGrid = 'on';
2491     end
2492
2493     %Grid line at y = 0 for all data in one graph
2494     if ~strcmp(app.LineatY0DropDown.Value, "None")
2495         yhorzline = yline(0,YLineStyle,'LineWidth',app.
LineYThickness.Value);
2496         yhorzline.Annotation.LegendInformation.
IconDisplayStyle = 'off';
2497     end
2498
2499
2500     %-----
2501     %-----GENERAL LAYOUT: Ticks-----
2502     %-----
2503     %Labels of Axes and Ticks
2504     xlabel(app.XAxisEditField.Value,'FontSize',app.
GridLabelSize.Value);
2505     ylabel(app.YAxisEditField.Value,'FontSize',app.
GridLabelSize.Value);
2506
2507
2508     %-----
2509     %-----SAVE MAP: Plot Separately-----
2510     %-----
2511     if app.PlotseparatelyButton_4.Value
2512         %Save the maps with filenames matching the loaded
file
2513         PlotsFolder = fullfile(Dir1);
2514         OF = fullfile(PlotsFolder, app.GaugeNames(i));
2515
2516         %Options for saving format
2517         %PNG
2518         if app.pngCheckBox_2.Value
2519             outputfile = OF + ".png";
2520             exportgraphics(ax,outputfile,'Resolution',300)
2521         end
2522
2523         %JPG
2524         if app.jpgCheckBox_2.Value
2525             outputfile = OF + ".jpg";
2526             exportgraphics(ax,outputfile,'Resolution',300)
2527         end
2528
2529         %PDF
2530         if app.pdfCheckBox_2.Value
2531             outputfile = OF + ".pdf";
2532             exportgraphics(ax,
outputfile,'ContentType','vector','Resolution',300)
2533         end
2534
2535         %EPS
2536         if app.epsCheckBox.Value %save as .eps
2537             outputfile = OF + ".eps";
2538             saveas(ax,outputfile,'epsc')
2539         end
2540

```

```

2541         %TXT for saving the raw 'sta_' files
2542         if app.txtCheckBox.Value
2543             outputfile = OF + ".txt";
2544             writematrix(g, outputfile, 'Delimiter', '\t');
2545             dlmwrite(outputfile,g, 'delimiter', '\t');
2546         end
2547     end
2548 end
2549
2550 %Add legend in the plot
2551 if app.PlotseparatelyButton_4.Value
2552     if strcmp(app.YaxisUseDataDropDown.Value, '3,4,5')
2553         legend(plotHandles, legendEntries, 'Location', legloc, 'FontSize', app.LegendSize.Value);
2554         legend box off;
2555     end
2556 end
2557
2558 %-----
2559 %-----SAVE MAP: Plot all data in one graph-----
2560 %-----
2561 if app.PlotalldataainonegraphButton.Value
2562     if length(app.GaugeNames) == 1
2563         OF = fullfile(Dir1, app.GaugeNames);
2564     else
2565         OF = fullfile(Dir1, 'GaugeRecord');
2566     end
2567
2568     if app.jpgCheckBox_2.Value
2569         outputfile = OF + ".jpg";
2570         exportgraphics(gcf,outputfile, 'Resolution',300)
2571     end
2572
2573     if app.pngCheckBox_2.Value
2574         outputfile = OF + ".png";
2575         exportgraphics(gcf,outputfile, 'Resolution',300)
2576     end
2577     if app.pdfCheckBox_2.Value %save as .pdf
2578         outputfile = OF + ".pdf";
2579         exportgraphics(ax, outputfile, 'ContentType', 'vector', 'Resolution',300)
2580     end
2581     if app.epsCheckBox.Value %save as .eps
2582         outputfile = OF + ".eps";
2583         saveas(gcf,outputfile, 'epsc')
2584         pause(2)
2585         clf;
2586     end
2587     if app.txtCheckBox.Value %save as .txt
2588         outputfile = OF + ".txt";
2589         writematrix(g,outputfile);
2590         dlmwrite(outputfile,g, 'delimiter', '\t');
2591     end
2592 end
2593 end
2594 end
2595
2596 % Callback function

```

```

2597     function BathymetryCheckBoxValueChanged(app, event)
2598
2599     end
2600
2601     % Callback function
2602     function WaveHeightCheckBoxValueChanged(app, event)
2603
2604
2605     end
2606
2607     % Value changed function: GaugesCheckBox
2608     function GaugesCheckBoxValueChanged(app, event)
2609         if app.GaugesCheckBox.Value
2610             %Display and enable the parameter options for the gauges tab
2611             app.TabGroup2.SelectedTab = app.GaugesTab;
2612             for i = 1:length(app.GaugesTab.Children)
2613                 app.GaugesTab.Children(i).Enable = 'on';
2614             end
2615             %If 'Add Labels' is checked, enable the items below it
2616             components = {app.GLspacing, app.GLspacingLabel, app.↵
SizeEditField_3, app.SizeEditField_3Label, app.CoastlinecolourDropDown_5, app.↵
CoastlinecolourDropDown_4, app.AlignmentLabel};
2617
2618             for i = 1:length(components)
2619                 if app.AddLabelsCheckBox_4.Value
2620                     components{i}.Enable = 'on';
2621                 else
2622                     components{i}.Enable = 'off';
2623                 end
2624             end
2625         else
2626             for i = 1:length(app.GaugesTab.Children)
2627                 if isprop(app.GaugesTab.Children(i), 'Enable')
2628                     app.GaugesTab.Children(i).Enable = 'off';
2629                 end
2630             end
2631         end
2632     end
2633
2634     % Button pushed function: Button_16
2635     function Button_16Pushed(app, event)
2636         %Find files to import from a directory
2637         [files, path] = uigetfile('*..*', 'Select the files', 'MultiSelect', '↵
'on');
2638
2639         allfiles = string(path) + string(files);
2640         if ~(ismcc || isdeployed)
2641             addpath(genpath(fullfile(string(path))));
2642         end
2643
2644         cd(fullfile(path))
2645
2646         %Sort the files by snapshot timing, and display the list of ↵
filenames in the app's textbox
2647         app.GaugeDirectorylist = sort(allfiles);
2648         app.GaugeNames = sort(string(files));
2649         app.FileTextArea_2.Value = string(app.GaugeNames);
2650         app.FileTextArea_2.FontColor = 'k';
2651     end

```

```

2651
2652 % Value changed function: FileTextArea_2
2653 function FileTextArea_2ValueChanged(app, event)
2654
2655
2656 end
2657
2658 % Value changed function: CheckBox
2659 function CheckBoxValueChanged(app, event)
2660     if app.CheckBox.Value
2661         app.yMaxLimit.Enable = "off";
2662         app.yMinLimit.Enable = "off";
2663         app.yLimInterval.Enable = "off";
2664     else
2665         app.yMaxLimit.Enable = "on";
2666         app.yMinLimit.Enable = "on";
2667         app.yLimInterval.Enable = "on";
2668     end
2669 end
2670
2671 % Value changed function: CheckBox_2
2672 function CheckBox_2ValueChanged(app, event)
2673     if app.CheckBox_2.Value
2674         app.xMaxLimit.Enable = "off";
2675         app.xMinLimit.Enable = "off";
2676         app.xLimInterval.Enable = "off";
2677     else
2678         app.xMaxLimit.Enable = "on";
2679         app.xMinLimit.Enable = "on";
2680         app.xLimInterval.Enable = "on";
2681     end
2682 end
2683
2684 % Value changed function: yLimInterval
2685 function yLimIntervalValueChanged(app, event)
2686     app.yLimInterval.FontColor = 'k';
2687
2688
2689 end
2690
2691 % Value changed function: xLimInterval
2692 function xLimIntervalValueChanged(app, event)
2693
2694
2695 end
2696
2697 % Button pushed function: Button_18
2698 function Button_18Pushed(app, event)
2699     % Set up the directory path for saving the files
2700     workingfolder = uigetdir;
2701     Dir1 = fullfile(string(workingfolder), 'OUTPUT_FILES');
2702     app.WORKFOLDER2 = Dir1;
2703
2704     %Create the 'Figures' folder in the 'OUTPUT_FILES' working
2705     directory
2706     FigureFolder = fullfile(Dir1, 'Figures');
2707     app.WORKFOLDER5 = FigureFolder;
2708     app.OutputDirectoryEditField 2.Value = FigureFolder;

```

```

2708         app.OutputDirectoryEditField_2.FontColor = 'k';
2709
2710         %Create the folders
2711         if ~exist(Dir1, 'dir')
2712             mkdir(Dir1);
2713         end
2714         if ~exist(FigureFolder, 'dir')
2715             mkdir(FigureFolder);
2716         end
2717
2718     end
2719
2720     % Value changed function: yMinLimit
2721     function yMinLimitValueChanged(app, event)
2722         app.yMinLimit.FontColor = 'k';
2723     end
2724
2725     % Value changed function: yMaxLimit
2726     function yMaxLimitValueChanged(app, event)
2727         app.yMaxLimit.FontColor = 'k';
2728     end
2729
2730     % Value changed function: xMinLimit
2731     function xMinLimitValueChanged(app, event)
2732         app.xMinLimit.FontColor = 'k';
2733
2734
2735     end
2736
2737     % Value changed function: xMaxLimit
2738     function xMaxLimitValueChanged(app, event)
2739
2740     end
2741
2742     % Value changed function: GridStyle
2743     function GridStyleValueChanged(app, event)
2744         value = app.GridStyle.Value;
2745         if strcmp(value, "None")
2746             app.LineYThickness_2.Enable = "off";
2747             app.LineYThickness_2.Editable = "off";
2748         else
2749             app.LineYThickness_2.Enable = "on";
2750             app.LineYThickness_2.Editable = "on";
2751         end
2752
2753     end
2754
2755     % Drop down opening function: GridStyle
2756     function GridStyleDropDownOpening(app, event)
2757
2758     end
2759
2760     % Value changed function: Lineaty0DropDown
2761     function Lineaty0DropDownValueChanged(app, event)
2762         value = app.Lineaty0DropDown.Value;
2763         if strcmp(value, "None")
2764             app.LineYThickness.Enable = "off";
2765         else

```

```

2766         app.LineYThickness.Enable = "on";
2767     end
2768 end
2769
2770 % Value changed function: LocationDropDown
2771 function LocationDropDownValueChanged(app, event)
2772     value = app.LocationDropDown.Value;
2773     if strcmp(value, "None")
2774         app.LegendSize.Enable = "off";
2775     else
2776         app.LegendSize.Enable = "on";
2777     end
2778
2779 end
2780
2781 % Callback function
2782 function XCOLValueChanged(app, event)
2783
2784
2785 end
2786
2787 % Callback function
2788 function YCOLValueChanged(app, event)
2789
2790 end
2791
2792 % Value changed function: DataLineWidth
2793 function DataLineWidthValueChanged(app, event)
2794     app.DataLineWidth.FontColor = 'k';
2795
2796 end
2797
2798 % Value changed function: XAxisEditField
2799 function XAxisEditFieldValueChanged(app, event)
2800     app.XAxisEditField.FontColor = 'k';
2801
2802 end
2803
2804 % Value changed function: YAxisEditField
2805 function YAxisEditFieldValueChanged(app, event)
2806     app.YAxisEditField.FontColor = 'k';
2807
2808 end
2809
2810 % Value changed function: GridLabelSize
2811 function GridLabelSizeValueChanged(app, event)
2812     app.GridLabelSize.FontColor = 'k';
2813
2814 end
2815
2816 % Value changed function: LineYThickness
2817 function LineYThicknessValueChanged(app, event)
2818     app.LineYThickness.FontColor = 'k';
2819
2820 end
2821
2822 % Callback function
2823 function GridThicknessValueChanged(app, event)

```



```

2824         app.GridThickness.FontColor = 'k';
2825
2826     end
2827
2828     % Value changed function: LegendSize
2829     function LegendSizeValueChanged(app, event)
2830         app.LegendSize.FontColor = 'k';
2831
2832     end
2833
2834     % Value changed function: NorthEditField
2835     function NorthEditFieldValueChanged(app, event)
2836         app.NorthEditField.FontColor = 'k';
2837
2838     end
2839
2840     % Value changed function: SouthEditField
2841     function SouthEditFieldValueChanged(app, event)
2842         app.SouthEditField.FontColor = 'k';
2843
2844     end
2845
2846     % Value changed function: EastEditField
2847     function EastEditFieldValueChanged(app, event)
2848         app.EastEditField.FontColor = 'k';
2849
2850     end
2851
2852     % Value changed function: WestEditField
2853     function WestEditFieldValueChanged(app, event)
2854         app.WestEditField.FontColor = 'k';
2855
2856     end
2857
2858     % Value changed function: toEditField
2859     function toEditFieldValueChanged(app, event)
2860         app.toEditField.FontColor = 'k';
2861
2862     end
2863
2864     % Value changed function: MinEditField
2865     function MinEditFieldValueChanged(app, event)
2866         app.MinEditField.FontColor = 'k';
2867
2868     end
2869
2870     % Value changed function: MaxEditField_2
2871     function MaxEditField_2ValueChanged(app, event)
2872         app.MaxEditField_2.FontColor = 'k';
2873
2874     end
2875
2876     % Value changed function: MinEditField_2
2877     function MinEditField_2ValueChanged(app, event)
2878         app.MinEditField_2.FontColor = 'k';
2879
2880     end
2881

```

```

2882
2883 % Callback function
2884 function LabelSpacingEditFieldValueChanged(app, event)
2885     app.LabelSpacingEditField.FontColor = 'k';
2886
2887 end
2888
2889 % Callback function
2890 function IntervalEditField_3ValueChanged(app, event)
2891     app.IntervalEditField_3.FontColor = 'k';
2892
2893 end
2894
2895 % Callback function
2896 function WidthEditFieldValueChanged(app, event)
2897     app.WidthEditField.FontColor = 'k';
2898
2899 end
2900
2901 % Value changed function: ColorinterpolationEditField_2
2902 function ColorinterpolationEditField_2ValueChanged(app, event)
2903     app.ColorinterpolationEditField_2.FontColor = 'k';
2904
2905 end
2906
2907 % Value changed function: DivisionEditField
2908 function DivisionEditFieldValueChanged(app, event)
2909     app.DivisionEditField.FontColor = 'k';
2910
2911 end
2912
2913 % Callback function
2914 function MinimumdepthEditField_2ValueChanged(app, event)
2915     app.MinimumdepthEditField_2.FontColor = 'k';
2916
2917 end
2918
2919 % Callback function
2920 function ThicknessEditField_2ValueChanged(app, event)
2921     app.ThicknessEditField_2.FontColor = 'k';
2922
2923 end
2924
2925 % Callback function
2926 function wh_intervalValueChanged(app, event)
2927     app.wh_interval.FontColor = 'k';
2928
2929 end
2930
2931 % Callback function
2932 function LabelSpacingEditField_2ValueChanged(app, event)
2933     app.LabelSpacingEditField_2.FontColor = 'k';
2934
2935 end
2936
2937 % Callback function
2938 function SizeEditFieldValueChanged(app, event)
2939     app.SizeEditField.FontColor = 'k';

```

```

2940         end
2941
2942         % Callback function
2943         function SizeEditField_3ValueChanged(app, event)
2944             app.SizeEditField_3.FontColor = 'k';
2945
2946         end
2947
2948         % Value changed function: CheckBox_3
2949         function CheckBox_3ValueChanged(app, event)
2950             %If checked, activate the textboxes for X-axis data in the
2951             'Convert Values' section of the 'Input Data'
2952             if app.CheckBox_3.Value
2953                 app.denominatorX.Enable = "on";
2954                 app.numeratorX.Enable = "on";
2955             else
2956                 app.denominatorX.Enable = "off";
2957                 app.numeratorX.Enable = "off";
2958             end
2959         end
2960
2961         % Value changed function: CheckBox_4
2962         function CheckBox_4ValueChanged(app, event)
2963             %If checked, activate the textboxes for Y-axis data in the
2964             'Convert Values' section of the 'Input Data'
2965             if app.CheckBox_4.Value
2966                 app.denominatorY.Enable = "on";
2967                 app.numeratorY.Enable = "on";
2968             else
2969                 app.denominatorY.Enable = "off";
2970                 app.numeratorY.Enable = "off";
2971             end
2972         end
2973
2974         % Callback function
2975         function XaxisEditFieldValueChanging(app, event)
2976
2977             if event.Value
2978                 app.egconvertcolumn1datafromsecondstominutesLabel.Visible =
2979                 "off";
2980             end
2981         end
2982
2983         % Callback function
2984         function YaxisEditFieldValueChanging(app, event)
2985
2986             if event.Value
2987                 app.egconvertcolumn2datafrommeterstocentimetersLabel.Visible =
2988                 "off";
2989             end
2990         end
2991
2992         % Value changed function: denominatorY
2993         function denominatorYValueChanged(app, event)
2994
2995         end
2996
2997         % Selection changed function: ButtonGroup_10
2998         function ButtonGroup_10SelectionChanged(app, event)

```

```

2994         %Activate/deactivate the dropdown list for col if str2double(app.
YaxisUseDataDropDown.Value) == 1 or line depending on the number of files uploaded
and the selected plotting style.
2995         value = str2double(app.YaxisUseDataDropDown.Value);
2996         isSingleStation = length(app.GaugeDirectorylist) == 1;
2997         isMultipleStations = length(app.GaugeDirectorylist) > 1;
2998         plotAllData = app.PlotalldatainonegraphButton.Value;
2999         plotSeparately = app.PlotseparatelyButton_4.Value;
3000
3001         if ~ismember(value, [3, 4, 5]) % Will only plot data from one
column
3002             if isSingleStation
3003                 setGaugeLineColorVisibility(true, false, false);
3004             elseif isMultipleStations
3005                 if plotAllData
3006                     setGaugeLineColorVisibility(false, true, true);
3007                 elseif plotSeparately
3008                     setGaugeLineColorVisibility(true, false, false,
false);
3009                 end
3010             end
3011         else % Use columns 3, 4, 5 to plot in Y-axis
3012             if isSingleStation
3013                 setGaugeLineColorVisibility(true, false, false);
3014             elseif isMultipleStations
3015                 if plotAllData
3016                     setGaugeLineColorVisibility(false, true, true);
3017                 elseif plotSeparately
3018                     setGaugeLineColorVisibility(true, false, false,
false);
3019                 end
3020             end
3021         end
3022
3023         function setGaugeLineColorVisibility(lineColorVisible,
lineColorMultipleVisible, flipCheckBoxVisible, flipCheckBoxEnable)
3024             if nargin < 4
3025                 flipCheckBoxEnable = true; % Default to true if not
specified
3026             end
3027             app.GaugeLineColor.Visible = lineColorVisible;
3028             app.GaugeLineColor_multiple.Visible =
lineColorMultipleVisible;
3029             app.FlipCheckBox_3.Visible = flipCheckBoxVisible;
3030             app.FlipCheckBox_3.Enable = flipCheckBoxEnable;
3031         end
3032 %         if ~ismember(str2double(app.YaxisUseDataDropDown.Value), [3, 4,
5])%Will only plot data from one column
3033 %             if length(app.GaugeDirectorylist) == 1 %one station, use
dropdown for selecting one color
3034 %                 app.GaugeLineColor.Visible = "on";
3035 %                 app.GaugeLineColor_multiple.Visible = "off";
3036 %                 app.FlipCheckBox_3.Visible = "off";
3037 %             elseif length(app.GaugeDirectorylist) > 1 && app.
PlotalldatainonegraphButton.Value %multiple stations, use dropdown for selecting a
set of color
3038 %                 app.GaugeLineColor.Visible = "off";
3039 %                 app.GaugeLineColor_multiple.Visible = "on";

```

```

3040 %             app.FlipCheckBox_3.Visible = "on";
3041 %             elseif length(app.GaugeDirectorylist) > 1  && app.
PlotseparatelyButton_4.Value
3042 %                 app.FlipCheckBox_3.Enable = "off";
3043 %                 app.GaugeLineColor.Visible = "on";
3044 %                 app.GaugeLineColor_multiple.Visible = "off";
3045 %             end
3046 %         else %Use columns 3,4,5 to plot in Y-axis
3047 %             if length(app.GaugeDirectorylist) == 1 %one station, use
dropdown for selecting one color
3048 %                 app.GaugeLineColor.Visible = "on";
3049 %                 app.GaugeLineColor_multiple.Visible = "off";
3050 %                 app.FlipCheckBox_3.Visible = "off";
3051 %             elseif length(app.GaugeDirectorylist) > 1  && app.
PlotalldatainonegraphButton.Value %multiple stations, use dropdown for selecting a
set of color
3052 %                 app.GaugeLineColor.Visible = "off";
3053 %                 app.GaugeLineColor_multiple.Visible = "on";
3054 %                 app.FlipCheckBox_3.Visible = "on";
3055 %             elseif length(app.GaugeDirectorylist) > 1  && app.
PlotseparatelyButton_4.Value
3056 %                 app.FlipCheckBox_3.Enable = "off";
3057 %                 app.GaugeLineColor.Visible = "on";
3058 %                 app.GaugeLineColor_multiple.Visible = "off";
3059 %             end
3060 %         end
3061
3062 % Activate/deactivate the options for the plot legend and the
option to save the sta_ file as a .txt file.
3063 if app.PlotseparatelyButton_4.Value && length(app.
GaugeDirectorylist) == 1
3064     app.legendfirsttext.Enable = "off";
3065     app.LegendSize.Enable = "off";
3066     app.LocationDropDown.Enable = "off";
3067     app.txtCheckBox.Enable = "on";
3068 elseif app.PlotseparatelyButton_4.Value && length(app.
GaugeDirectorylist) > 1
3069     app.legendfirsttext.Enable = "on";
3070     app.LegendSize.Enable = "on";
3071     app.LocationDropDown.Enable = "on";
3072     app.txtCheckBox.Enable = "on";
3073 elseif app.PlotalldatainonegraphButton.Value
3074     app.legendfirsttext.Enable = "on";
3075     app.LegendSize.Enable = "on";
3076     app.LocationDropDown.Enable = "on";
3077     app.txtCheckBox.Enable = "off";
3078 end
3079
3080 end
3081
3082 % Value changed function: GaugeLineColor_multiple
3083 function GaugeLineColor_multipleValueChanged(app, event)
3084
3085 end
3086
3087 % Value changed function: AutoSetCheckBox
3088 function AutoSetCheckBoxValueChanged(app, event)
3089     if app.AutoSetCheckBox.Value

```

```

3090         app.Width.Enable = "off";
3091         app.Height.Enable = "off";
3092     else
3093         app.Width.Enable = "on";
3094         app.Height.Enable = "on";
3095     end
3096 end
3097
3098 % Callback function
3099 function CloseFiguresButtonPushed(app, event)
3100     close all %figures
3101 end
3102
3103 % Value changed function: mp4CheckBox
3104 function mp4CheckBoxValueChanged(app, event)
3105     if app.mp4CheckBox.Value
3106         app.FramerateEditField_2.Visible = "on";
3107         app.FramerateEditField_2Label.Visible="on";
3108     else
3109         app.FramerateEditField_2.Visible = "off";
3110         app.FramerateEditField_2Label.Visible="off";
3111     end
3112 end
3113
3114 % Callback function
3115 function CloseFiguresButton_2Pushed(app, event)
3116     close all
3117 end
3118
3119 % Selection changed function: ButtonGroup_11
3120 function ButtonGroup_11SelectionChanged(app, event)
3121     % Enable the tif and animation options in the 'Save Map' section ✓
3122     when 'Plot all data in one figure' is selected
3123         if app.PlotseparatelyButton_2.Value
3124             app.mp4CheckBox.Enable = "on";
3125             app.tifCheckBox_2.Enable = "on";
3126             app.FramerateEditField_2.Enable = "on";
3127             app.FramerateEditField_2Label.Enable = "on";
3128         elseif app.PlotalldatainonefigureButton.Value
3129             app.mp4CheckBox.Enable = "off";
3130             app.tifCheckBox_2.Enable = "off";
3131             app.FramerateEditField_2.Enable = "off";
3132             app.FramerateEditField_2Label.Enable = "off";
3133         end
3134     end
3135
3136 % Value changed function: AutoSetCheckBox_2
3137 function AutoSetCheckBox_2ValueChanged(app, event)
3138     %Disable the Width and Height textboxes
3139     if app.AutoSetCheckBox_2.Value
3140         app.Width_2.Enable = "off";
3141         app.Height_2.Enable = "off";
3142     else
3143         app.Width_2.Enable = "on";
3144         app.Height_2.Enable = "on";
3145     end
3146 end

```

```

3147
3148     % Button pushed function: CloseFiguresButton_4
3149     function CloseFiguresButton_4Pushed2(app, event)
3150         close all
3151     end
3152
3153     % Button pushed function: Button_19
3154     function Button_19Pushed(app, event)
3155
3156         % Find U-vector files to import from a directory
3157         [files, path] = uigetfile('*..*', 'Select the files',
3158 'MultiSelect', 'on');
3159
3160         % Convert path and files to strings once
3161         pathStr = string(path);
3162         filesStr = string(files);
3163
3164         if exist(pathStr, 'dir')
3165             cd(fullfile(pathStr));
3166
3167             % Collect all the files
3168             allfiles = fullfile(pathStr + filesStr); % fullfile is used to
3169 handle folder names with spaces
3170             if ~(ismcc || isdeployed)
3171                 addpath(genpath(fullfile(pathStr)));
3172             end
3173
3174             % Sort the files by snapshot timing, and display the list of
3175 filenames in the app's textbox
3176             sortedFiles = sort(filesStr);
3177             app.Uvectorfullfile = sort(allfiles);
3178             app.vectorpath = pathStr;
3179             app.Ufiles = sortedFiles;
3180             app.FileName5 = sortedFiles;
3181             app.FileTextArea_3.Value = sortedFiles;
3182             app.FileTextArea_3.FontColor = 'k';
3183             app.FileTextArea_3.BackgroundColor = 'w';
3184         end
3185
3186         app.MinBarValue.Enable = "on";
3187         app.MaxBarValue.Enable = "on";
3188
3189         ArrowSection = app.Panel_14.Children; % Get all the items in the
3190 Arrow section
3191         if contains(app.FileName5, "umax") % if the file is for plotting
3192 the maximum velocity, do not find the corresponding V-vector file
3193             % Disable the Arrow Section
3194             set(ArrowSection, 'Enable', 'off');
3195             app.PlotvectorsCheckBox.Value = false; % Uncheck the box
3196
3197             % Change default value of the colormap and the bar limits
3198             app.BackgroundMapColorDropDown.Value = 'blue - purple';
3199             app.MaxBarValue.Value = 6;
3200             app.InterpolationDivisionEditField_2.Value = 4;
3201             app.InterpolationDivisionEditField_2.Enable = "on";
3202             app.InterpolationDivisionEditFieldLabel_2.Enable = "on";
3203             app.BackgroundMapColorDropDown.Enable = "on";
3204             app.FlipCheckBox_4.Enable = "on";

```

```

3200         else
3201             % Enable the Arrow Section
3202             set(ArrowSection, 'Enable', 'on');
3203
3204             % Find the equivalent V Vector files, display the list of ↵
filenames in the app's textbox
3205             vfilenames = regexprep(sortedFiles, "u", "v", 'once'); % ↵
replaces only the first letter "u" in each of the filenames
3206             app.Vvectorfullfile = fullfile(pathStr + vfilenames);
3207             app.Vvectorfilenames = vfilenames;
3208             app.FileTextArea_4.Value = vfilenames;
3209             app.FileTextArea_4.FontColor = 'k';
3210             app.FileTextArea_4.BackgroundColor = 'w';
3211             app.BackgroundMapColorDropDown.Enable = "on";
3212             app.InterpolationDivisionEditField_2.Enable = "on";
3213             app.InterpolationDivisionEditFieldLabel_2.Enable = "on";
3214         end
3215
3216         % Find corresponding 'eta' files when 'eta' is selected as the ↵
basemap option
3217         ETAfiles = regexprep(sortedFiles, "u", "eta", 'once'); % change ↵
'u' to 'eta' in the filename
3218         app.ETAfullfile = fullfile(pathStr + ETAfiles);
3219         app.ETAfilenames = ETAfiles;
3220
3221         % Find corresponding 'hmax' files when 'hmax' is selected as the ↵
basemap option
3222         HMAXfiles = regexprep(sortedFiles, "u", "hmax", 'once'); % change ↵
'u' to 'hmax' in the filename
3223         app.HMAXfullfile = fullfile(pathStr + HMAXfiles);
3224         app.HMAXfilenames = HMAXfiles;
3225
3226         % Read the first file to extract its matrix size, which will be ↵
used to set up the map boundary limit
3227         if exist(app.Uvectorfullfile{1}, 'file') == 2
3228             % Read the matrix from the file
3229             f = readmatrix(app.Uvectorfullfile{1});
3230             % Setup of the coordinates
3231             [row, col] = size(f);
3232             app.col1vel = col - 1;
3233             app.row1vel = row - 1;
3234         end
3235
3236     end
3237
3238     % Callback function
3239     function Button_20Pushed(app, event)
3240
3241     end
3242
3243     % Button pushed function: GENERATEButton_2
3244     function GENERATEButton_2Pushed(app, event)
3245         %Defining Parameters
3246         close all
3247         clear depthVM Uvector Vvector VMdata
3248
3249         %-----
3250         %-----ARROW COLOR-----

```



```

3251         %-----
3252         quivercolorOptions = containers.Map({'Black', 'Dark gray', 'Medium
gray', 'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'},
{'k', '[0.4 0.4 0.4]', '[0.7 0.7 0.7]', '[0.8 0.8 0.8]', 'r', 'g', 'b', 'y', 'c',
'm', 'w'});
3253         quivercolor = quivercolorOptions(app.QuiverColorDropDown.Value);
3254
3255
3256         %-----
3257         %-----LAND AREA COLOR-----
3258         %-----
3259         LandAreaColorOptions = containers.Map({'Black', 'Dark gray',
'Medium gray', 'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta',
'White'}, {'k', '[0.4 0.4 0.4]', '[0.7 0.7 0.7]', '[0.8 0.8 0.8]', 'r', 'g', 'b',
'y', 'c', 'm', 'w'});
3260         LandAreaColor2 = LandAreaColorOptions(app.LandColor.Value);
3261
3262
3263         %-----
3264         %-----BASEMAP COLORMAP-----
3265         %-----
3266         %List of the colormap items in the dropdown list
3267         keys = {'autumn', 'bone', 'colorcube', 'cool', 'copper', 'flag',
'gray', 'hot', 'hsv', 'jet', ...
3268             'parula', 'pink', 'prism', 'spring', 'summer', 'turbo',
'winter', 'blue', 'blue - green', ...
3269             'blue - purple', 'green - blue', 'greens', 'gray', 'oranges',
'orange - red', 'purple - blue', ...
3270             'purple - blue - green', 'purple - red', 'purples', 'red -
purple', 'reds', 'yellow - green', ...
3271             'yellow - green - blue', 'yellow - orange - brown', 'yellow -
orange - red', 'brown - teal', ...
3272             'pink - light green', 'purple - green', 'purple - orange',
'red - blue', 'red - gray', ...
3273             'red - yellow - blue', 'red - yellow - green', 'spectral',
'accent', 'dark 2', 'paired', ...
3274             'pastel 1', 'pastel 2', 'set 1', 'set 2', 'set 3', '--- MATLAB
default ----', '--- CBREWER 2 ---', ...
3275             '< sequential >', '< divergent >', '< qualitative >'};
3276
3277         %The equivalent colormap code
3278         keysColor = {'autumn', 'bone', 'colorcube', 'cool', 'copper',
'flag', 'gray', 'hot', 'hsv', 'jet', ...
3279             'parula', 'pink', 'prism', 'spring', 'summer', 'turbo',
'winter', 'Blues', 'BuGn', 'BuPu', 'GnBu', 'Greens', ...
3280             'Greys', 'Oranges', 'OrRd', 'PuBu', 'PuBuGn', ...
3281             'PuRd', 'Purples', 'RdPu', 'Reds', 'YlGn', 'YlGnBu', 'YlOrBr', 'YlOrRd', 'BrBG', 'PiYG', 'PRG
n', 'PuOr', 'RdBu', 'RdGy', ...
3282             'RdYlBu', 'RdYlGn', 'Spectral', 'Accent', ...
3283             'Dark2', 'Paired', 'Pastel1', 'Pastel2', 'Set1', 'Set2', 'Set3', 'parula', 'parula', 'parula
', 'parula', 'parula'};
3284
3285         %The colorbrewer
3286         keysBrewer = {'none', 'none', 'none', 'none', 'none', 'none',
'none', 'none', 'none', 'none', ...
3287             'none', 'none', 'none', 'none', 'none', 'none', 'none', 'seq',

```

```

'seq', ...
3288         'seq', 'seq', 'seq', 'seq', 'seq', 'seq', ...
3289         'seq', 'seq', 'seq', 'seq', 'seq', ...
3290         'seq', 'seq', 'seq', 'seq', 'seq', 'div', 'div', 'div', ...
3291         'div', 'div', 'div', 'div', 'div', ...
3292         'div', 'qual', 'qual', 'qual', ...
3293         'qual', 'qual', 'qual', 'qual', 'qual', ...
3294         'none', 'none', 'none', 'none', 'none'};
3295
3296     % Match dropdown list items to their corresponding colors and
ColorBrewer
3297     colorMap = containers.Map(keys, keysColor);
3298     colorBrewer = containers.Map(keys, keysBrewer);
3299
3300     %Set up the colormap
3301     datacolor2 = colorMap(app.BackgroundMapColorDropDown.Value);
3302     datacolorbrewer2 = colorBrewer(app.BackgroundMapColorDropDown.
Value);
3303
3304
3305     %-----
3306     %-----BATHYMETRY CONTOURS-----
3307     %-----
3308     %Line Colours
3309     ContourColor2Options = containers.Map({'Black', 'Dark gray',
'Medium gray', 'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta',
'White'}, {'k', '[0.4 0.4 0.4]', '[0.7 0.7 0.7]', '[0.8 0.8 0.8]', 'r', 'g', 'b',
'y', 'c', 'm', 'w'});
3310     ContourColor2 = ContourColor2Options(app.ColorDropDown_4.Value);
3311
3312     %Line style
3313     ContourLineStyle2Options = containers.Map({'Solid', 'Dashed',
'Dotted', 'Dash-dotted'}, {'-', '--', ':', '-.'});
3314     ContourLineStyle2 = ContourLineStyle2Options(app.StyleDropDown_3.
Value);
3315
3316
3317     %-----
3318     %-----GAUGES-----
3319     %-----
3320     %Marker Style
3321     GaugeMarkerOptions = containers.Map({'none', 'o', '+', '*', '.',
'x', '-', '|', '^', 'v', '>', '<', 'diamond', 'hexagram', 'pentagram', 'square'},
...
3322         {'none', 'o', '+', '*', '.', 'x', '-', '|', '^', 'v', '>',
'<', 'diamond', 'hexagram', 'pentagram', 'square'});
3323     GaugeMarker2 = GaugeMarkerOptions(app.gaugemarkerVelocityTab.
Value);
3324
3325     %Marker Color
3326     GaugeColor2Options = containers.Map({'Black', 'Dark gray', 'Medium
gray', 'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'},
{'k', '[0.4 0.4 0.4]', '[0.7 0.7 0.7]', '[0.8 0.8 0.8]', 'r', 'g', 'b', 'y', 'c',
'm', 'w'});
3327     GaugeColor2 = GaugeColor2Options(app.ColorDropDown_7.Value);
3328
3329     %Label - horizontal alignment
3330     HorzLabel2Options = containers.Map({'Centre', 'Right', 'Left'},

```

```

{'center', 'left', 'right'});
3331         HorzLabel2 = HorzLabel2Options(app.CoastlinecolourDropDown_10.↵
Value);
3332
3333         %Label - vertical alignment
3334         VertLabel2Options = containers.Map({'Centre', 'Top', 'Bottom'},↵
{'middle', 'bottom', 'top'});
3335         VertLabel2 = VertLabel2Options(app.CoastlinecolourDropDown_11.↵
Value);
3336
3337
3338         %-----
3339         %-----SAVE MAP: Output Directory-----
3340         %-----
3341         %Check if the output directory has been manually set by the user
3342         if app.OutputDirectoryEditField_3.Value == string(app.↵
WORKFOLDER4)
3343             app.MainDirectoryVector= fullfile(app.WORKFOLDER3);
3344             app.FigureDirectoryVector = fullfile(app.WORKFOLDER4);
3345         else % Set the default directory to the Desktop
3346             if ismac %macOS
3347                 defaultDir = fullfile(getenv('HOME'), 'Desktop');
3348             elseif ispc %Windows
3349                 defaultDir = fullfile(getenv('USERPROFILE'), 'Desktop');
3350             else %Others
3351                 defaultDir = pwd;
3352             end
3353
3354             %Create 'OUTPUT_FILES' folder and 'Figures' subfolder
3355             Dir0 = fullfile(defaultDir, 'OUTPUT_FILES');
3356             Dir1 = fullfile(defaultDir, 'OUTPUT_FILES/Figures');
3357             app.MainDirectoryVector= Dir0;
3358             app.FigureDirectoryVector = Dir1;
3359             if ~exist(Dir0, 'dir')
3360                 mkdir(Dir0);
3361             end
3362             if ~exist(Dir1, 'dir')
3363                 mkdir(Dir1);
3364             end
3365
3366             %Display the path directory of the 'Figures' subfolder in the↵
textbox
3367             app.OutputDirectoryEditField_3.Value = deblank(string(Dir1));
3368             app.OutputDirectoryEditField_3.FontColor = 'k';
3369         end
3370
3371
3372
3373         %Create animation/ creating and opening the file
3374         if app.mp4CheckBox_2.Value
3375             vidObj = VideoWriter(fullfile(char(app.FigureDirectoryVector),↵
'animation_vector.mp4'), 'MPEG-4');
3376             vidObj.FrameRate = app.FramerateEditField_2.Value;
3377             open(vidObj);
3378         end
3379
3380         %-----
3381         %-----LOG REPORT-----

```

```

3382         %-----
3383         %Create 'Log_Files' subfolder in 'OUTPUT_FILES'
3384         LogFolder = fullfile(app.MainDirectoryVector, 'Log_Files');
3385         if ~exist(LogFolder, 'dir')
3386             mkdir(LogFolder);
3387         end
3388         %Create the text file
3389         filePath = fullfile(LogFolder, 'Log_Report_VelocityMap.txt');
3390         fileId = fopen(filePath, 'w');
3391
3392         % Template for the default section of the log report (includes
information from the 'Input Data' section)
3393         headerTitle = '***** LOG REPORT
*****';
3394         timestamp = datestr(now, 'yyyy-mm-dd HH:MM:SS');
3395         endHeaderLine =
'*****';
3396         footerSeparator =
'_____';
3397         logContent = [
3398             sprintf('%s\n', headerTitle), ...
3399             sprintf('%-30s-s\n', 'Map Type:', 'Velocity Map'), ...
3400             sprintf('%-30s-s\n', 'Timestamp:', timestamp), ...
3401             sprintf('%s\n\n', endHeaderLine), ...
3402             sprintf('%-30s-s\n', 'Southwest Corner:', sprintf('Long: %f
Lat: %f', app.LongitudeEditField_2.Value, app.LatitudeEditField_2.Value)), ...
3403             sprintf('%-30s-s\n', 'Grid Size:', sprintf('x: %d y: %d',
app.gridX_2.Value, app.gridY_2.Value)), ...
3404             sprintf('%-30s-s\n', 'Simulation Time Start:', sprintf('%d
sec', app.StartTime2.Value)), ...
3405             sprintf('%-30s-s\n', 'Simulation Time Interval:', sprintf('%d
sec', app.TotalSimulationTimesecEditField_4.Value)), ...
3406             sprintf('%s\n\n', footerSeparator)
3407         ];
3408         fprintf(fileId, '%s', logContent);
3409
3410
3411         %-----LOG REPORT-----
3412         %-----Input Data: Import Files-----
3413         %-----
3414         %Check loaded files in the U vector input form
3415         patterns = {'u_', 'umax_', 'umean_'}; %Check each loaded files if
they contain "u"
3416         matches1 = false(size(app.FileName5));
3417         nonMatchingFiles = ~contains(string(app.FileName5), patterns);
3418         for i = 1:numel(patterns)
3419             pattern = patterns{i};
3420             matches1 = matches1 | contains(string(app.FileName5),
pattern);
3421         end
3422
3423         if any(nonMatchingFiles)
3424             fprintf(fileId, '%-30s-s\n', 'U VECTOR:', 'Use filenames with
'u_', 'umax_', or 'umean_' initials');
3425             fprintf(fileId, '%-30s-s\n', ' ', 'Incorrect files are
loaded:');
3426             fprintf(fileId, '%-30s-s\n', ' ', join(app.FileName5
(nonMatchingFiles), ', ')); % List all files that do not start with 'u' in the

```

```

filename
3427             fprintf(fileId, '%-30s%-s\n\n', ' ', 'No vectors are
plotted');
3428             elseif all(contains(string(app.FileName5), patterns)) %If all
files are in U vector format
3429                 %Check if V vector files exists in the same directory as the U
vectors
3430                 missingFiles = {};
3431                 for i = 1:length(app.Vvectorfullfile) % Check each filename in
the list
3432                     filePath = fullfile(app.Vvectorfullfile{i});
3433                     if exist(filePath, 'file') ~= 2
3434                         missingFiles = [missingFiles, app.Vvectorfilenames
{i}]; %Make a list all corresponding V vector files that are not found
3435                     end
3436                 end
3437                 if ~isempty(missingFiles)
3438                     fprintf(fileId, '%-30s%-s\n', 'V VECTOR :','Missing Files.
Ensure they are in the same folder as the U vector files:');
3439                     fprintf(fileId, '%-30s%-s\n\n', ' ', strjoin(missingFiles,
', '));
3440                 end
3441             end
3442
3443             %-----LOG REPORT-----
3444             %-----Input Data: Southwest Corner-----
3445             %-----
3446             %Check if the values have not changed and add an error note to
the log report
3447             if app.LongitudeEditField_2.Value == 0 && app.
LatitudeEditField_2.Value == 0
3448                 fprintf(fileId, '%-30s%-s\n\n', 'SOUTHWEST CORNER:', 'Both
latitude and longitude values are zeroes');
3449             elseif app.LongitudeEditField_2.Value == 0 && app.
LatitudeEditField_2.Value ~= 0
3450                 fprintf(fileId, '%-30s%-s\n\n', 'SOUTHWEST
CORNER:', 'Longitude value is zero');
3451             elseif app.LatitudeEditField_2.Value ~= 0 && app.
LatitudeEditField_2.Value == 0
3452                 fprintf(fileId, '%-30s%-s\n\n', 'SOUTHWEST
CORNER:', 'Latitude value is zero');
3453             end
3454
3455             %-----LOG REPORT-----
3456             %-----Input Data: Grid Size-----
3457             %-----
3458             %Check if the values have not changed and add an error note to
the log report
3459             if app.gridX_2.Value == 0 && app.gridY_2.Value == 0
3460                 fprintf(fileId, '%-30s%-s\n\n', 'GRID SIZE:', 'Both x and y
values are zeroes');
3461             elseif app.gridX.Value == 0 && app.gridY.Value ~= 0
3462                 fprintf(fileId, '%-30s%-s\n\n', 'GRID SIZE:', 'X value is
zero');
3463             elseif app.gridX.Value ~= 0 && app.gridY.Value == 0
3464                 fprintf(fileId, '%-30s%-s\n\n', 'GRID SIZE:', 'Y value is
zero');
3465             end

```

```

3466
3467         %-----LOG REPORT-----
3468         %-----Input Data: Load Bathymetry-----
3469         %-----
3470         %Check if the matrix size of the loaded bathymetry file
matches the imported eta/hmax file
3471         if matches(app.FileTextArea_3.Value,string(app.FileName5)) %
Check if the imported V files match the names in the textbox
3472         if matches(app.DepthFileEditField_2.Value,string(app.
bathymetryname2)) %Check if the imported bathymetry matches the name in the textbox
3473             [rowbath,colbath] = size(app.bathymetryname2);
3474             if app.rowlvel ~= rowbath
3475                 fprintf(fileId, '%-30s%-s\n\n', 'MATRIX
ROWS:', 'Rows in U-vector file and bathymetry do not match');
3476             end
3477             if app.collvel ~= colbath
3478                 fprintf(fileId, '%-30s%-s\n\n', 'MATRIX
COLUMNS:', 'Columns in U-vector file and bathymetry do not match');
3479             end
3480         end
3481     end
3482
3483     if app.DepthFileEditField_2.Value == "0" || ~matches(app.
DepthFileEditField_2.Value,string(app.bathymetryname2)) % Add a log note that no
bathymetry is loaded, so the features to edit land area and bathymetry contours are
unavailable.
3484         fprintf(fileId, '%-30s%-s\n\n', 'BATHYMETRY:', 'Empty.
Land color and bathymetry contours are not applicable. ');
3485         app.DepthFileEditField_2.Value = "NO FILE";
3486         app.DepthFileEditField_2.FontColor = 'w';
3487         app.DepthFileEditField_2.BackgroundColor = 'r';
3488     end
3489
3490     %Check if the bathymetry is has the correct file format
3491     if app.bathymetrycheck == 1
3492         fprintf(fileId, '%-30s%-s\n\n', 'BATHYMETRY:', 'File must
be .txt, .tif or mask_ ');
3493     end
3494
3495     %-----LOG REPORT-----
3496     %-----Basemap and Overlays: Basemap-----
3497     %-----
3498     if app.MinBarValue.Value > app.MaxBarValue.Value
3499         fprintf(fileId, '%-30s%-s\n', 'BASEMAP COLORBAR:', 'Max
value must exceed min value ');
3500         fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Max and min
values are swapped ');
3501     end
3502
3503     %-----LOG REPORT-----
3504     %-----Basemap and Overlays: Bathymetry Contours-----
3505     %-----
3506     if app.PlotBathymetryContoursCheckBox.Value
3507         if matches(app.DepthFileEditField_2.Value ,string(app.
bathymetryname2))
3508             % Check if the minimum and maximum values are in the
correct order
3509             if app.MinimumEditField 2.Value > app.

```

```

MaximumEditField_2.Value
3510                                     temp = app.MaximumEditField_2.Value;
3511                                     app.MaximumEditField_2.Value = app.
MinimumEditField_2.Value;
3512                                     app.MinimumEditField_2.Value = temp;
3513                                     fprintf(fileId, '%-30s%-s\n', 'BATHYMETRY
CONTOURS:', 'Max value must exceed min value');
3514                                     fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved:
Values are swapped');
3515                                     end
3516
3517                                     %Check the interval value is within the limit
3518                                     maxMinDifference = app.MaximumEditField_2.Value - app.
MinimumEditField_2.Value;
3519                                     if app.IntervalEditField_6.Value > maxMinDifference
3520                                         app.IntervalEditField_6.Value = maxMinDifference /
2;
3521                                     fprintf(fileId, '%-30s%-s\n', 'BATHYMETRY CONTOUR
INTERVAL:', 'Input must be less than max-min difference');
3522                                     fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved:
Value adjusted to half the max-min difference');
3523                                     end
3524
3525                                     if app.AddLabelCheckBox_2.Value %Add labels
3526                                         %Check if the label interval is divisible by the
value provided in the 'Range' section
3527                                         if mod(app.IntervalEditField_8.Value, app.
IntervalEditField_6.Value) ~= 0
3528                                             app.IntervalEditField_8.Value = app.
IntervalEditField_6.Value;
3529                                             fprintf(fileId, '%-30s%-s\n', 'BATHYMETRY
CONTOUR LABEL:', 'Value must be divisible by the Interval value in the Range
section');
3530                                             fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved:
Value adjusted to be the same as the Interval Value in the Range section');
3531                                             end
3532                                         end
3533                                     end
3534                                     end
3535
3536                                     %-----LOG REPORT-----
3537                                     %-----Basemap and Overlays: Gauges-----
3538                                     %-----
3539                                     if app.PlotGaugesCheckBox.Value && ~matches(app.
FileEditField_2.Value ,string(app.GAUGEFILE3))
3540                                         fprintf(fileId, '%-30s%-s\n\n', 'GAUGES:', 'Empty. No
virtual gauges plotted');
3541                                         app.FileEditField_2.Value = "NO FILE";
3542                                         app.FileEditField_2.FontColor = 'w';
3543                                         app.FileEditField_2.BackgroundColor = 'r';
3544                                     end
3545
3546                                     %-----LOG REPORT-----
3547                                     %-----General Layout: Boundary Limits-----
3548                                     %-----
3549                                     if app.WestEditField_2.Value > app.EastEditField_2.Value
3550                                         fprintf(fileId, '%-30s%-s\n', 'BOUNDARY LIMIT:', 'East
boundary must be greater than West boundary');

```

```

3551         fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Values are↵
swapped');
3552     elseif app.EastEditField_2.Value == app.WestEditField_2.Value
3553         fprintf(fileId, '%-30s%-s\n', 'BOUNDARY LIMIT:', 'East and↵
West values are the same');
3554         fprintf(fileId, '%-30s%-s\n\n', 'Resolved:', 'The default↵
values are used');
3555     end
3556
3557     if app.SouthEditField_2.Value > app.NorthEditField_2.Value
3558         fprintf(fileId, '%-30s%-s\n', 'BOUNDARY LIMIT:', 'North↵
boundary must be greater than South boundary');
3559         fprintf(fileId, '%-30s%-s\n\n', ' ', 'Resolved: Values are↵
swapped');
3560     elseif app.NorthEditField_2.Value == app.SouthEditField_2.↵
Value
3561         fprintf(fileId, '%-30s%-s\n', 'BOUNDARY LIMIT:', 'North↵
and South values are the same');
3562         fprintf(fileId, '%-30s%-s\n\n', 'Resolved:', 'The default↵
values are used');
3563     end
3564
3565     fclose(fileId); %Close the log report file
3566
3567     %% PROCESS THE INPUT FILES AND VALUES, THEN GENERATE THE ↵
FIGURES
3568     %-----
3569     %-----READ THE INPUT FILES-----
3570     %-----
3571     % Set up the bathymetry data
3572     bathymetryFileExists = exist(app.bathymetryinputdata2, 'file')↵
== 2;
3573     if bathymetryFileExists
3574         depthVM = app.bathymetrydata2;
3575         if app.HorizontalCheckBox_2.Value
3576             depthVM = fliplr(depthVM);
3577         end
3578         if app.VerticalCheckBox_2.Value
3579             depthVM = flipud(depthVM);
3580         end
3581     end
3582
3583     FileLength1 = length(app.Uvectorfullfile);
3584     patterns = {'u_', 'umax_', 'umean_'};
3585
3586     %-----
3587     %-----PLOT ALL DATA IN ONE FIGURE-----
3588     %-----
3589     if app.PlotalldataainonefigureButton_2.Value
3590
3591         %Create the figure
3592         f = figure(1);
3593         f.Visible = 'off';
3594
3595         %Formatting the subplots
3596         if FileLength1 == 1
3597             plotCol = 1;
3598             plotRow = 1;

```



```

3599         elseif FileLength1 > 1 && FileLength1 <= 10
3600             plotCol = 2;
3601             plotRow = ceil(FileLength1/2);
3602         elseif FileLength1 > 10
3603             plotCol = 3;
3604             plotRow = ceil(FileLength1/3);
3605         end
3606
3607         %-----
3608         %-----PLOT SEPARATELY-----
3609         %-----
3610     elseif app.PlotseparatelyButton_3.Value
3611         f = figure();
3612         f.Visible = 'off';
3613     end
3614
3615     %-----
3616     %-----PRE-SET MAP OPTIONS-----
3617     %-----
3618     %Allocate memory for the vectors
3619     Uvector = NaN(app.rowlvel, app.col1vel);
3620     Vvector = NaN(app.rowlvel, app.col1vel);
3621     colorbarHandles = [];
3622
3623     %-----Basemap Colormap-----
3624     if strcmp(datacolorbrewer2,'none') %Using default matlab
3625         datacolor3 = evalin('base',datacolor2);
3626     else %Using cbrewer2
3627         datacolor3 = cbrewer2(datacolorbrewer2,datacolor2,app.
InterpolationDivisionEditField_2.Value,'pchip');
3628     end
3629
3630     %-----Basemap Colorbar-----
3631     colormapMin = app.MinBarValue.Value;
3632     colormapMax = app.MaxBarValue.Value;
3633     %Options for label
3634     if app.BathymetryButton.Value
3635         colorbartitle = 'Bathymetry(m)';
3636     elseif app.etaButton.Value
3637         colorbartitle = '\eta (m)';
3638     elseif app.hmaxButton.Value
3639         colorbartitle = 'Maximum \eta (m)';
3640     elseif app.VelocityButton.Value
3641         if contains(string(app.Ufiles(i)),'umax')
3642             colorbartitle = 'Maximum Velocity (m/s)';
3643         else
3644             colorbartitle = 'Velocity (m/s)';
3645         end
3646     elseif app.VorticityButton.Value
3647         colorbartitle = 'Vorticity (s^{-1})';
3648     end
3649
3650     %Check if the minimum and maximum values are in the correct
order; if not, swap them
3651     if app.BathymetryButton.Value
3652         if -app.MaxBarValue.Value > -app.MinBarValue.Value %Min
and Max values of the colorbar limit are swapped

```

```

3653             %Swap the values
3654             temp = app.MaxBarValue.Value;
3655             app.MaxBarValue.Value = app.MinBarValue.Value;
3656             app.MinBarValue.Value = temp;
3657         end
3658     else
3659         if app.MaxBarValue.Value < app.MinBarValue.Value %Min and
Max values of the colorbar limit are swapped
3660             %Swap the values
3661             temp = app.MaxBarValue.Value;
3662             app.MaxBarValue.Value = app.MinBarValue.Value;
3663             app.MinBarValue.Value = temp;
3664         end
3665     end
3666
3667     %-----Boundary limit-----
3668     eastValue = app.EastEditField_2.Value;
3669     westValue = app.WestEditField_2.Value;
3670
3671     if eastValue < westValue
3672         % Swap West and East input values if the West value is
greater than the East
3673         xMin_VM = eastValue;
3674         xMax_VM = westValue;
3675         app.EastEditField_2.Value = xMax_VM;
3676         app.WestEditField_2.Value = xMin_VM;
3677     elseif eastValue == westValue
3678         % Set the limits to the default values
3679         app.EastEditField_2.Value = max(app.x);
3680         app.WestEditField_2.Value = min(app.x);
3681         xMin_VM = app.WestEditField_2.Value;
3682         xMax_VM = app.EastEditField_2.Value;
3683     else
3684         xMin_VM = westValue;
3685         xMax_VM = eastValue;
3686     end
3687
3688     northValue = app.NorthEditField_2.Value;
3689     southValue = app.SouthEditField_2.Value;
3690
3691     if northValue < southValue
3692         % Swap North and South input values if the South value is
greater than the North
3693         yMin_VM = northValue;
3694         yMax_VM = southValue;
3695         app.NorthEditField_2.Value = yMax_VM;
3696         app.SouthEditField_2.Value = yMin_VM;
3697     elseif northValue == southValue
3698         % Set the limits to the default values
3699         app.NorthEditField_2.Value = max(app.y);
3700         app.SouthEditField_2.Value = min(app.y);
3701     else
3702         yMin_VM = southValue;
3703         yMax_VM = northValue;
3704     end
3705
3706     %-----Map Title-----
3707     totalSimulationTime = app.TotalSimuilationTimesecEditField 4.

```

```

Value;
3708         startTime = app.StartTime2.Value;
3709         colorbarTextSize = app.maplabelsSize.Value + 1;
3710         pat = digitsPattern;
3711
3712
3713         %-----
3714         %-----Start of the loop for plotting-----
3715         %-----
3716         for i = 1:FileLength1
3717             if all(contains(string(app.FileTextArea_3.Value), ↵
patterns)) % Check if each file is a U vector
3718                 % Load the U vector files
3719                 if exist(app.Uvectorfullfile{i}, 'file') == 2
3720                     Uvector = readmatrix(app.Uvectorfullfile{i});
3721
3722                 %Remove values on land (negative values)
3723                 if bathymetryFileExists
3724                     Uvector(depthVM <= 0) = NaN; % Set land ↵
values to NaN or another appropriate value
3725                 end
3726             end
3727
3728
3729             % Load the V vector files
3730             if ~contains(char(app.Ufiles{i}), 'umax') && exist ↵
(app.Vvectorfullfile{i}, 'file') == 2
3731                 Vvector = readmatrix(app.Vvectorfullfile{i});
3732
3733             %-----Flip the Matrix-----
3734             if app.HorizontalCheckBox_2.Value
3735                 % Flip horizontally
3736                 Vvector = fliplr(Vvector);
3737             end
3738             if app.VerticalCheckBox_2.Value
3739                 Vvector = flipud(Vvector);
3740             end
3741
3742             % Remove values on land if the bathymetry file ↵
exists
3743             if bathymetryFileExists
3744                 Vvector(depthVM <= 0) = NaN; % Set land ↵
values to NaN or another appropriate value
3745             end
3746
3747         end
3748
3749         %-----
3750         %-----PLOT ALL DATA IN ONE FIGURE-----
3751         %-----
3752         if app.PlotalldatainonefigureButton_2.Value
3753             % Create a subplot in the specified position
3754             gca(i) = subplot(plotRow, plotCol, i);
3755             f.Visible = 'on';
3756
3757         %-----
3758         %-----PLOT SEPARATELY-----
3759         %-----

```

```

3760 elseif app.PlotseparatelyButton_3.Value
3761     f = figure();
3762     f.Visible = 'off';
3763     gca(i) = subplot(1,1,1);
3764 end
3765
3766 %-----
3767 %-----Basemap and Overlays: Basemap-----
3768 %-----
3769 %Check which option is selected for the basemap
3770 clear VMdata
3771 %Option 1: W Velocity
3772 if app.VelocityButton.Value
3773     if ~contains(char(app.Ufiles(i)), 'umax') && exist
3774 (app.Vvectorfullfile(i), 'file')
3775         VMdata = sqrt(Uvector.^2 + Vvector.^2); %
3776 Calculate the Z magnitude based on the magnitudes of the U and V vectors
3777     elseif contains(char(app.Ufiles(i)), 'umax')
3778         VMdata = Uvector;
3779     end
3780
3781 %Option 2: eta / sea surface displacement at time
3782 t
3783 elseif app.etaButton.Value
3784     VMdata = readmatrix(app.ETAfullfile(i));
3785
3786 %Option 3: hmax / maximum wave height
3787 elseif app.hmaxButton.Value
3788     if exist(app.HMAXfullfile(i), 'file')
3789         VMdata = readmatrix(app.HMAXfullfile(i));
3790     end
3791
3792 %Option 4: Bathymetry
3793 elseif app.BathymetryButton.Value
3794     if matches(app.DepthFileEditField_2.Value, string
3795 (app.bathymetryname2))
3796         VMdata = -depthVM;
3797     end
3798
3799 %Option 5: Vorticity
3800 elseif app.VorticityButton.Value
3801     [VMdata, ~]=curl(app.xcoord, app.ycoord, Uvector,
3802 Vvector); %from simple_cases/rip_2d
3803 end
3804
3805 %-----Flip the Matrix-----
3806 % Flip horizontally
3807 if app.HorizontalCheckBox_2.Value && ~app.
3808 BathymetryButton.Value
3809     VMdata = fliplr(VMdata);
3810 end
3811
3812 % Flip vertically
3813 if app.VerticalCheckBox_2.Value && ~app.
3814 BathymetryButton.Value
3815     VMdata = flipud(VMdata);
3816 end

```

```

3811
3812 %Remove land values from the selected basemap
3813 try
3814     if isprop(app, 'bathymetryinputdata2')
3815         VMdata(depthVM<=0) = NaN;
3816     end
3817 end
3818
3819 %-----
3820 %-----MAIN PLOT-----
3821 %-----
3822 p = pcolor(app.xcoord,app.ycoord,VMdata);
3823 shading interp
3824 set(p, 'EdgeColor', 'none','FaceColor', 'interp');
3825 set(gca,'Color', LandAreaColor2) %Set the background
color for the land, %Adjust tick size label
3826 hold on
3827
3828
3829 %-----
3830 %-----Basemap and Overlays: Basemap Colormap-----
3831 %-----
3832 %Flip the colorbar
3833 if app.FlipCheckBox_4.Value
3834     colormap(gca(i), flipud(datacolor3));
3835 else
3836     colormap(gca(i), datacolor3);
3837 end
3838
3839 %-----
3840 %-----Basemap and Overlays: Basemap Colorbar-----
3841 %-----
3842 hold on
3843 %Check if the minimum and maximum values are in the
correct order; if not, swap them
3844 if app.BathymetryButton.Value
3845     caxis(gca(i), [-colormapMax,-colormapMin]); %
Colorbar min and max values
3846 else
3847     caxis(gca(i), [colormapMin,colormapMax]); %
Colorbar min and max values
3848 end
3849
3850 cb = colorbar;
3851 colorbarHandles = [colorbarHandles, cb];
3852
3853 if cb.FontSize ~= colorbarTextSize
3854     cb.FontSize = colorbarTextSize; % size of the tick
numbers
3855 end
3856 cb.Label.String = colorbartitle; % Set the colorbar
title
3857
3858 %-----
3859 %-----Basemap and Overlays: Bathymetry Contours-----
3860 %-----
3861 if app.PlotBathymetryContoursCheckBox.Value && matches
(app.DepthFileEditField 2.Value,string(app.bathymetryname2))

```

```

3862         hold on
3863         %Plot the contours
3864         depth0 = depthVM;
3865         kernel = ones(6) / 36; % Use averaging kernel to
make the contour smoother
3866         depthfiltered = filter2(kernel, depth0);
3867         [C1,h1] = contour(app.xcoord,app.ycoord,
depthfiltered,app.MinimumEditField_2.Value:app.IntervalEditField_6.Value:app.
MaximumEditField_2.Value, 'EdgeColor',ContourColor2, 'LineStyle',
ContourLineStyle2, 'LineWidth', app.WidthEditField_2.Value);
3868         %Add contour labels
3869         if app.AddLabelCheckBox_2.Value
3870             hold on
3871             manualLabels3 = app.MinimumEditField_2.Value:
app.IntervalEditField_6.Value:app.MaximumEditField_2.Value;
3872             clabel(C1, h1, manualLabels3, 'FontSize', app.
LabelSizeCont.Value, 'Color', 'k');
3873             set(handle(h1), 'LabelSpacing', app.
SpacingEditField.Value); % Adjust the label spacing
3874         end
3875     end
3876
3877
3878     %-----
3879     %-----Basemap and Overlays: Gauges-----
3880     %-----
3881     hold on
3882     if app.PlotGaugesCheckBox.Value && matches(app.
FileEditField_2.Value,string(app.GAUGEFILE3))
3883         %Plot the points
3884         gaugepoint = plot(app.longGauge2,app.latGauge2,
GaugeMarker2);
3885         set(gaugepoint, 'MarkerSize', app.SizeEditField_2.
Value, 'MarkerFaceColor', GaugeColor2, 'MarkerEdgeColor', GaugeColor2); %Size and
color of the points
3886         %Add labels
3887         if app.AddLabelCheckBox.Value
3888             hold on
3889             for k = 1:length(app.latGauge2)
3890                 gtxt = text(app.longGauge2(k)+ app.
GLspacing_2.Value,app.latGauge2(k,1),num2str(k), 'FontSize', app.FontSizeEditField_2.
Value, 'HorizontalAlignment',HorzLabel2,...
3891                     'VerticalAlignment',VertLabel2); %
GLspacing_2.Value is added to longitude so there's space between the point and the
label
3892                 set(gtxt, 'Clipping', 'on');
3893             end
3894         end
3895     end
3896
3897     %-----
3898     %-----Arrows: Plot Vector-----
3899     %-----
3900     if app.PlotvectorsCheckBox.Value
3901         if any(~contains(app.FileNAME5, "umax") | app.
VorticityButton.Value == false)
3902             %-----Flip the Matrix-----
3903             % Flip horizontally

```

```

3904         if app.HorizontalCheckBox_2.Value
3905             UvectorArrow = flipplr(Uvector);
3906             VvectorArrow = flipplr(Vvector);
3907         else
3908             UvectorArrow = Uvector;
3909             VvectorArrow = Vvector;
3910         end
3911
3912         % Flip vertically
3913         if app.VerticalCheckBox_2.Value
3914             UvectorArrow = flipud(Uvector);
3915             VvectorArrow = flipud(Vvector);
3916         else
3917             UvectorArrow = Uvector;
3918             VvectorArrow = Vvector;
3919         end
3920
3921         q1 = quiver(gca(i),app.xcoord(1:app.
ArrowSpacing.Value:end,1:app.ArrowSpacing.Value:end),app.ycoord(1:app.ArrowSpacing.
Value:end,1:app.ArrowSpacing.Value:end),UvectorArrow(1:app.ArrowSpacing.Value:end,
1:app.ArrowSpacing.Value:end),VvectorArrow(1:app.ArrowSpacing.Value:end,1:app.
ArrowSpacing.Value:end),app.arrowscale.Value);
3922         q1.Color = quivercolor;
3923         q1.LineWidth = app.ArrowThickness.Value;
3924         q1.MaxHeadSize = app.ArrowHeadSize.Value;
3925         axis equal %use equal unit lengths
3926     end
3927 end
3928
3929 %-----
3930 %-----GENERAL LAYOUT: Boundary limit-----
3931 %-----
3932 xlim([xMin_VM,xMax_VM])
3933 ylim([yMin_VM,yMax_VM])
3934 hold on
3935 f.Visible = 'on';
3936
3937 %-----
3938 %-----GENERAL LAYOUT: Ticks-----
3939 %-----
3940 %Reduce the number of ticks to 2 on the X axis and 2
on the Y axis for simplicity
3941 % Get the current ticks from the first plot
3942 if i == 1
3943     xAllTicks = xticks(gca);
3944     yAllTicks = yticks(gca);
3945
3946     % Ensure the ticks are numeric arrays
3947     if iscell(xAllTicks)
3948         xAllTicks = cellfun(@str2double, xAllTicks,
'UniformOutput', true);
3949     end
3950     if iscell(yAllTicks)
3951         yAllTicks = cellfun(@str2double, yAllTicks,
'UniformOutput', true);
3952     end
3953
3954     % Remove any NaN values that might result from

```

```

str2double conversion
3955         xAllTicks = xAllTicks(~isnan(xAllTicks));
3956         yAllTicks = yAllTicks(~isnan(yAllTicks));
3957
3958         % Select only the 2nd and second-to-last ticks
3959         % For X axis
3960         if length(xAllTicks) >= 4 % Check first if there
are enough ticks to select from
3961             xticksSelected = sort([xAllTicks(2), xAllTicks
(end-1)]);
3962         elseif length(xAllTicks) >= 2
3963             xticksSelected = sort([xAllTicks(1), xAllTicks
(end)]); % Use the first and last if less than 4
3964         else
3965             xticksSelected = xAllTicks; % Use whatever
ticks are available
3966         end
3967
3968         % For Y axis
3969         if length(yAllTicks) >= 4
3970             yticksSelected = sort([yAllTicks(2), yAllTicks
(end-1)]);
3971         elseif length(yAllTicks) >= 2
3972             yticksSelected = sort([yAllTicks(1), yAllTicks
(end)]); % Use the first and last if less than 4
3973         else
3974             yticksSelected = yAllTicks; % Use whatever
ticks are available
3975         end
3976
3977     end
3978
3979     % Update the number of ticks on the axes
3980     xticks(gca, xticksSelected);
3981     yticks(gca, yticksSelected);
3982
3983     % Determine the maximum decimal places for x and y
ticks
3984     xmaxDecimals = max(cellfun(@(x) length(regexpi(x, '(?
<=\.)\d+', 'match', 'once'))), ...
3985         cellstr(num2str(xticksSelected'))));
3986     ymaxDecimals = max(cellfun(@(y) length(regexpi(y, '(?
<=\.)\d+', 'match', 'once'))), ...
3987         cellstr(num2str(yticksSelected'))));
3988
3989     % Convert tick labels to their equivalent in degrees
3990     if app.PlotIndegreesCheckBox.Value
3991         % Convert selected ticks to formatted strings and
set them for degrees
3992         xticklabels(gca, arrayfun(@(x) sprintf('%.1f°',
mod(x + 360, 360)), xticksSelected, 'UniformOutput', false));
3993         yticklabels(gca, arrayfun(@(y) sprintf('%.1f°',
y), yticksSelected, 'UniformOutput', false));
3994     else
3995         % Correct the format string and apply calculated
maximum decimals
3996         xticklabels(gca, arrayfun(@(x) sprintf(['%. '
num2str(xmaxDecimals) 'f'], x), xticksSelected, 'UniformOutput', false));

```



```

3997             yticklabels(gca, arrayfun(@(y) sprintf(['%.', app.
num2str(ymaxDecimals) 'f'], y), yticksSelected, 'UniformOutput', false));
3998         end
3999
4000         % Set the y-axis tick angle
4001         ytickangle(gca, -270);
4002
4003         %Fontsize of the tick labels
4004         set(gca, 'Layer', 'top', 'LineWidth', 1, 'FontSize', app.
ColorbarTextSize.Value-1)
4005
4006         %Adjust tick lengths based on the number of subplots
if 'Plot all data in one figure' is selected
4007             if FileLength1 <= 2 && app.
PlotalldatainonefigureButton_2.Value
4008                 set(gca, 'TickLength', [0.015 0.015])
4009             else
4010                 set(gca, 'TickLength', [0.01 0.01])
4011             end
4012
4013             grid off
4014             box on
4015             daspect([1 1 1])
4016
4017             %-----
4018             %-----GENERAL LAYOUT: Map Title-----
4019             %-----
4020             str = string(app.FileName5(i));
4021
4022             dbl1 = extract(str, pat); % Extract numbers from the
filename to calculate the time frame
4023
4024             if startTime == 0 && str2double(dbl1) == 0 %Set the
time to zero
4025                 timeT = startTime + (str2double(dbl1)-1);
4026             else
4027                 OriginaltimeT2 = totalSimulationTime * (str2double
(dbl1)-1); % Convert the filename to time
4028                 timeT = startTime + OriginaltimeT2; % Adjust the
start of the time count
4029             end
4030
4031
4032             % Time conversion
4033             if timeT < 60
4034                 timeStr = sprintf("%d sec", timeT);
4035             elseif timeT < 3600
4036                 minutes = round(timeT / 60);
4037                 if minutes == 1
4038                     timeStr = "1 min";
4039                 else
4040                     timeStr = sprintf("%d mins", minutes);
4041                 end
4042             else
4043                 raw_hours = timeT/3600;
4044                 hours = fix(timeT/3600); %Extract the whole
number%Extract the whole number
4045                 hours decimal = raw hours - floor(raw hours); %

```

Extract the decimal places

```
4046         minutes = round(hours_decimal * 60);
4047
4048         if hours > 0
4049             if minutes > 0
4050                 timeStr = sprintf("%d hr, %d mins", hours,
minutes);
4051             else
4052                 timeStr = sprintf("%d hr", hours);
4053             end
4054         end
4055     end
4056
4057     title(timeStr, 'FontSize',colorbarTextSize); %Add the
title to the top of the map; Adjust the font size of the title
4058     hold on
4059
4060
4061     %-----
4062     %-----GENERAL LAYOUT: Figure Size-----
4063     %-----
4064     %Figure Size
4065     if ~app.AutoSetCheckBox_3.Value
4066         figureHandle = ancestor(p, 'figure'); % Get the
figure handle containing the axes
4067         set(figureHandle, 'Units', 'Inches', 'Position',
[0, 0, app.Width_3.Value, app.Height_3.Value], 'PaperUnits', 'Inches', 'PaperSize',
[app.Width_3.Value, app.Height_3.Value]);
4068         %         set(gcf, 'Units', 'Inches', 'Position', [0, 0,
app.Width_3.Value, app.Height_3.Value], 'PaperUnits', 'Inches', 'PaperSize', [app.
Width_3.Value, app.Height_3.Value]);
4069     end
4070
4071     %Create animation / closing the file
4072     if app.mp4CheckBox_2.Value
4073         set(gcf, 'Renderer', 'zbuffer');
4074         f = getframe(gcf);
4075         writeVideo(vidObj,f);
4076     end
4077
4078     %-----
4079     %-----SAVE MAP: Plot Separately-----
4080     %-----
4081     if app.PlotseparatelyButton_3.Value
4082         %Save the maps with filenames matching the loaded
file
4083         OF = fullfile(app.FigureDirectoryVector,app.
Ufiles(i));
4084
4085
4086         % Set the figure size for exporting
4087         if ~app.AutoSetCheckBox_3.Value
4088             set(figureHandle, 'Units', 'Inches',
'Position', [0, 0, app.Width_3.Value, app.Height_3.Value], ...
4089                 'PaperUnits', 'Inches', 'PaperSize', [app.
Width_3.Value, app.Height_3.Value]);
4090             set(gcf, 'Units', 'Inches', 'Position', [0, 0,
app.Width_3.Value, app.Height_3.Value], ...
```

```

4091         'PaperUnits', 'Inches', 'PaperSize', [app.Width_3.Value, app.Height_3.Value]);
4092     end
4093
4094     %Options for saving format
4095     %PNG
4096     if app.pngCheckBox_3.Value
4097         outputfile = OF + ".png";
4098         exportgraphics(gcf,outputfile,'Resolution',300)
4099     end
4100
4101     %JPG
4102     if app.jpgCheckBox_3.Value
4103         outputfile = OF + ".jpg";
4104         exportgraphics(gcf,outputfile,'Resolution',300)
4105     end
4106
4107     %PDF
4108     if app.pdfCheckBox_3.Value
4109         outputfile = OF + ".pdf";
4110         exportgraphics(gcf,outputfile,'ContentType','vector','Resolution',300)
4111     end
4112
4113     %TXT - to save the raw vector files
4114     if app.txtCheckBox_2.Value
4115         outputfile1 = OF + "_U.txt";
4116         outputfile2 = OF + "_V.txt";
4117         writematrix(Uvector,outputfile1);
4118         dlmwrite(outputfile1,Uvector,'delimiter','\t');
4119         writematrix(Vvector,outputfile2);
4120         dlmwrite(outputfile2,Vvector,'delimiter','\t');
4121     end
4122
4123     %TIF
4124     if app.tifCheckBox.Value
4125         XXX = app.xcoord;
4126         YYY = app.ycoord;
4127
4128         west_limit = app.WestEditField_2.Value;
4129         east_limit = app.EastEditField_2.Value;
4130         north_limit = app.NorthEditField_2.Value;
4131         south_limit = app.SouthEditField_2.Value;
4132
4133         %Make the longitude values within -180 to 180
4134         if west_limit ~= 180
4135             west_limit = mod(west_limit + 180, 360) - 180;
4136         end
4137
4138         if east_limit ~= 180
4139             east_limit = mod(east_limit + 180, 360) - 180;

```

```

4140                                     end
4141
4142                                     % Convert x coordinates to correct degree
range
4143                                     XXX(XXX > 180) = XXX(XXX > 180) - 360;
4144
4145                                     % Create meshgrid
4146                                     [X, Y] = meshgrid(XXX, YYY);
4147                                     Z = VMdata;
4148
4149                                     %Check if the latitude is within the
Geographic limits
4150                                     if app.LatitudeEditField_2.Value > 90 || app.
LatitudeEditField_2.Value < -90
4151                                         app.LatitudeEditField_2.Value = 0;
4152                                     else
4153                                         %Clip the matrix based on boundary limits
4154                                         if west_limit >= 0 && east_limit >=0 ||
west_limit < 0 && east_limit < 0 % if east and west values are either both (+) or
both (-) values
4155                                             x_indices_positive = XXX >=
west_limit & XXX <= east_limit;
4156                                             y_indices = YYY >= south_limit & YYY
<= north_limit;
4157                                             Z_positive = Z(y_indices,
x_indices_positive);
4158                                             X_positive = X(y_indices,
x_indices_positive);
4159                                             Y = Y(y_indices, x_indices_positive);
% Keep the same for both file
4160                                             if ~isempty(Z_positive)
4161                                                 % Define spatial referencing
information
4162                                                 R_positive = georasterref
('RasterSize', size(Z_positive), 'LatitudeLimits', [min(Y(:)), max(Y(:))],
'LongitudeLimits', [min(X_positive(:)), max(X_positive(:))]);
4163                                                 R_positive.ColumnsStartFrom =
'south'; % Set the column orientation to start from the north
4164                                                 R_positive.RowsStartFrom = 'west';
4165
4166                                                 % Save as geotiff
4167                                                 filename_positive = OF + ".tif";
4168                                                 geotiffwrite(filename_positive,
Z_positive, R_positive);
4169                                     end
4170
4171                                     elseif west_limit >= 0 && east_limit < 0
%if longitude values are a combination of (+) west limit and (-) east limit
4172                                         x_indices_positive = XXX >= 0 & XXX >=
west_limit;
4173                                         x_indices_negative = XXX < 0 & XXX <=
east_limit;
4174                                         y_indices = YYY >= south_limit & YYY
<= north_limit;
4175                                         Z_positive = Z(y_indices,
x_indices_positive);
4176                                         Z_negative = Z(y_indices,
x_indices_negative);

```

```

4177         X_positive = X(y_indices, ↵
x_indices_positive);
4178         X_negative = X(y_indices, ↵
x_indices_negative);
4179         Y = Y(y_indices, x_indices_positive); ↵
% Keep the same for both file
4180
4181         if ~isempty(Z_positive)
4182             % Define spatial referencing ↵
information
4183             R_positive = georasterref ↵
('RasterSize', size(Z_positive), 'LatitudeLimits', [min(Y(:)), max(Y(:))], ↵
'LongitudeLimits', [min(X_positive(:)), max(X_positive(:))]);
4184             R_positive.ColumnsStartFrom = ↵
'south'; % Set the column orientation to start from the north
4185             R_positive.RowsStartFrom = 'west';
4186
4187             % Save as geotiff
4188             filename_positive = OF + "_1.tif";
4189             geotiffwrite(filename_positive, ↵
Z_positive, R_positive);
4190         end
4191
4192         if ~isempty(Z_negative)
4193             % Define spatial referencing ↵
information
4194             R_negative = georasterref ↵
('RasterSize', size(Z_negative), 'LatitudeLimits', [min(Y(:)), max(Y(:))], ↵
'LongitudeLimits', [min(X_negative(:)), max(X_negative(:))]);
4195             R_negative.ColumnsStartFrom = ↵
'south'; % Set the column orientation to start from the north
4196             R_negative.RowsStartFrom = 'west';
4197
4198             % Save as geotiff
4199             filename_negative = OF + "_2.tif";
4200             geotiffwrite(filename_negative, ↵
Z_negative, R_negative)
4201         end
4202
4203         elseif west_limit < 0 && east_limit >= 0 ↵
%if longitude values are a combination of (-) west limit and (+) east limit
4204             x_indices_positive = XXX >= 0 & XXX <= ↵
east_limit;
4205             x_indices_negative = XXX < 0 & XXX >= ↵
west_limit;
4206             y_indices = YYY >= south_limit & YYY ↵
<= north_limit;
4207             Z_positive = Z(y_indices, ↵
x_indices_positive);
4208             Z_negative = Z(y_indices, ↵
x_indices_negative);
4209             X_positive = X(y_indices, ↵
x_indices_positive);
4210             X_negative = X(y_indices, ↵
x_indices_negative);
4211             Y = Y(y_indices, x_indices_positive); ↵
% Keep the same for both file
4212

```

```

4213         if ~isempty(Z_positive)
4214             % Define spatial referencing
information
4215             R_positive = georasterref
('RasterSize', size(Z_positive), 'LatitudeLimits', [min(Y(:)), max(Y(:))],
'LongitudeLimits', [min(X_positive(:)), max(X_positive(:))]);
4216             R_positive.ColumnsStartFrom =
'south'; % Set the column orientation to start from the north
4217             R_positive.RowsStartFrom = 'west';
4218
4219             % Save as geotiff
4220             filename_positive = OF + "_1.tif";
4221             geotiffwrite(filename_positive,
Z_positive, R_positive);
4222         end
4223         if ~isempty(Z_negative)
4224             % Define spatial referencing
information
4225             R_negative = georasterref
('RasterSize', size(Z_negative), 'LatitudeLimits', [min(Y(:)), max(Y(:))],
'LongitudeLimits', [min(X_negative(:)), max(X_negative(:))]);
4226             R_negative.ColumnsStartFrom =
'south'; % Set the column orientation to start from the north
4227             R_negative.RowsStartFrom = 'west';
4228
4229             % Save as geotiff
4230             filename_negative = OF + "_2.tif";
4231             geotiffwrite(filename_negative,
Z_negative, R_negative);
4232         end
4233     end
4234 end
4235 end
4236 end
4237 end
4238 end
4239
4240 %Change colorbar size
4241     if cb.Label.FontSize ~= colorbarTextSize
4242         cb.Label.FontSize =colorbarTextSize;
4243     end
4244
4245 %-----
4246 %-----SAVE MAP: Plot all data in one figure-----
4247 %-----
4248 if app.PlotalldatainonefigureButton_2.Value
4249
4250     % Set the figure size for exporting
4251     if ~app.AutoSetCheckBox_3.Value
4252         set(gcf, 'Units', 'Inches', 'Position', [0, 0, app.Width_3.Value, app.Height_3.Value], ...
4253             'PaperUnits', 'Inches', 'PaperSize', [app.Width_3.Value, app.Height_3.Value]);
4254     end
4255
4256     if length(app.Ufiles) == 1
4257         OF = fullfile(app.FigureDirectoryVector,app.Ufiles);
4258     else

```

```

4259             OF = fullfile(app.FigureDirectoryVector, ↵
"Vector_Output");
4260         end
4261
4262         %PNG
4263         if app.pngCheckBox_3.Value
4264             outputfile = OF + ".png";
4265             exportgraphics(gcf,outputfile,'Resolution',300)
4266         end
4267
4268         %JPG
4269         if app.jpgCheckBox_3.Value
4270             outputfile = OF + ".jpg";
4271             exportgraphics(gcf,outputfile,'Resolution',300)
4272         end
4273
4274         %PDF
4275 %         if app.pdfCheckBox_3.Value
4276 %             outputfile = OF + ".pdf";
4277 %             exportgraphics(gcf, ↵
outputfile,'ContentType','vector','Resolution',300)
4278 %         end
4279     end
4280
4281     end
4282
4283     % Value changed function: QuiverColorDropDown
4284     function QuiverColorDropDownValueChanged(app, event)
4285
4286     end
4287
4288     % Button pushed function: CloseFiguresButton_3
4289     function CloseFiguresButton_3Pushed(app, event)
4290         close all
4291     end
4292
4293     % Button pushed function: Button_22
4294     function Button_22Pushed(app, event)
4295         clear app.bathymetryinputdata2 app.bathymetrydata2
4296
4297         %Load the bathymetry file and display its filename in the textbox
4298         [bathymetryfile,path2] = uigetfile('*'); %*.txt;*.out;*.tif'
4299         if ~(ismcc || isdeployed)
4300             addpath(genpath(fullfile(string(path2))));
4301         end
4302         app.DepthFileEditField_2.Value = string(bathymetryfile);
4303         app.bathymetryinputdata2 = fullfile(string(path2), string ↵
(bathymetryfile));
4304         app.bathymetryname2 = string(bathymetryfile);
4305         app.DepthFileEditField_2.BackgroundColor = 'w';
4306         app.DepthFileEditField_2.FontColor = 'k';
4307
4308
4309         if app.DepthFileEditField_2.Value == string(bathymetryfile)
4310             if ~strcmp(app.DepthFileEditField_2.Value, "0") && ~strcmp ↵
(app.DepthFileEditField_2.Value, "00")
4311                 try
4312                     if contains(app.bathymetryinputdata2, '.tif')

```

```

4313         [A1,R2] = readgeoraster(app.bathymetryinputdata2);
4314         app.bathymetrydata2 = flipud(A1);
4315
4316         %Extract lat and long of the southwest corner
4317         app.LongitudeEditField_2.Value = min(R2.␣
LongitudeLimits);
4318         app.LatitudeEditField_2.Value = min(R2.␣
LatitudeLimits);
4319         app.LongitudeEditField_2.FontColor = 'k';
4320         app.LatitudeEditField_2.FontColor = 'k';
4321
4322         %Extract grid size
4323         app.gridX_2.Value = R2.CellExtentInLatitude;
4324         app.gridY_2.Value = R2.CellExtentInLongitude;
4325         app.gridX_2.FontColor = 'k';
4326         app.gridY_2.FontColor = 'k';
4327
4328         %Update the X-axis limits in the 'General Layout'␣
section
4329         app.xcoord = [0:app.col1vel] * app.gridX_2.Value+␣
app.LongitudeEditField_2.Value;
4330         app.EastEditField_2.Value = max(app.xcoord);
4331         app.WestEditField_2.Value = min(app.xcoord);
4332
4333         %Update the Y-axis limits in the 'General Layout'␣
section
4334         app.ycoord = [0:app.row1vel] * app.gridY_2.Value+␣
app.LatitudeEditField_2.Value;
4335         app.NorthEditField_2.Value = max(app.ycoord);
4336         app.SouthEditField_2.Value = min(app.ycoord);
4337
4338         app.EastEditField_2.FontColor = 'k';
4339         app.WestEditField_2.FontColor = 'k';
4340         app.NorthEditField_2.FontColor = 'k';
4341         app.SouthEditField_2.FontColor = 'k';
4342         drawnow
4343
4344         if app.BathymetryButton.Value %If bathymetry is␣
selected as the basemap
4345             app.FileTextArea_5.Value = app.␣
DepthFileEditField_2.Value; %show the bathymetry filename in the textbox in␣
'Basemap and Overlays' section
4346         else
4347             app.FileTextArea_5.Value = '';
4348         end
4349         drawnow
4350
4351         else %Only read the matrix for use as a basemap and␣
for plotting bathymetry contours
4352             app.bathymetrydata2 = readmatrix(app.␣
bathymetryinputdata2);
4353         end
4354
4355     end
4356     %Enable the 'Land Color' dropdown list & the Plot␣
Bathymetry Contours option in 'Basemap and Overlays' section
4357     app.LandColor.Enable = "on";
4358     app.PlotBathymetryContoursCheckBox.Enable = "on";

```



```

4359         end
4360     end
4361
4362     %Check if the uploaded file has the required file format to log
4363     error
4364     if ~contains(bathymetryfile, '.txt') && ~contains(bathymetryfile,
4365     '.tif') && ~contains(bathymetryfile, 'mask_')
4366         app.bathymetrycheck = 1;
4367     end
4368
4369     end
4370
4371     % Callback function
4372     function ButtonGroup_13SelectionChanged(app, event)
4373
4374     end
4375
4376     % Value changed function: PlotindegreesCheckBox_2
4377     function PlotindegreesCheckBox_2ValueChanged(app, event)
4378
4379     end
4380
4381     % Callback function
4382     function Button_23Pushed(app, event)
4383
4384     end
4385
4386     % Value changed function: FileTextArea_5
4387     function FileTextArea_5ValueChanged(app, event)
4388
4389     end
4390
4391     % Selection changed function: ButtonGroup_12
4392     function ButtonGroup_12SelectionChanged(app, event)
4393         if app.PlotalldatainonefigureButton_2.Value
4394             app.mp4CheckBox_2.Enable = "off";
4395             app.FramerateEditFieldLabel.Visible = "off";
4396             app.FramerateEditField.Visible = "off";
4397             app.txtCheckBox_2.Enable = "off";
4398             app.tifCheckBox.Enable = "off";
4399         elseif app.PlotseparatelyButton_3.Value
4400             app.mp4CheckBox_2.Enable = "on";
4401             app.txtCheckBox_2.Enable = "on";
4402             app.tifCheckBox.Enable = "on";
4403             if app.mp4CheckBox_2.Value %If it is left checked, show the
4404             frame rate option
4405                 app.FramerateEditFieldLabel.Visible = "on";
4406                 app.FramerateEditField.Visible = "on";
4407             else
4408                 app.FramerateEditFieldLabel.Visible = "off";
4409                 app.FramerateEditField.Visible = "off";
4410             end
4411         end
4412     end
4413

```

```

4414         end
4415     end
4416
4417     % Callback function
4418     function CurrentDirectionsCheckBoxValueChanged(app, event)
4419
4420     end
4421
4422     % Callback function
4423     function Button_24Pushed(app, event)
4424         %LOAD THE U VECTOR FILES
4425         [files, path] = uigetfile('*..*','Select the files','Code
4426 files','MultiSelect', 'on');
4427
4428         %Collect all the files
4429         allfiles = string(path) + string(files);
4430         app.Uvectorfullfile = sort(allfiles);
4431         FileName = sort(string(files));
4432         app.FileTextArea_6.Value = string(FileName); %display the filename
4433 list
4434         app.FileTextArea_6.FontColor = 'k';
4435         app.FileTextArea_6.BackgroundColor = 'w';
4436
4437         %Find the equivalent V Vector files
4438         vfiles = string(files);
4439         vfilenames = regexprep(vfiles,"u","v",'once'); %replaces only the
4440 first letter "u" in each of the filenames
4441         app.Vvectorfullfile = string(path) + vfilenames;
4442         app.FileTextArea_7.Value = vfilenames;
4443         app.FileTextArea_7.FontColor = 'k';
4444         app.FileTextArea_7.BackgroundColor = 'w';
4445
4446         %Check if any file is loaded
4447         if matches(app.FileTextArea_3.Value,"0") || matches(app.
4448 FileTextArea_3.Value,"00")
4449             %do nothing
4450         else
4451             %READ THE FIRST U FILE TO SETUP MATRIX SIZE AND MAP BOUNDARY
4452 LIMIT
4453             f = readmatrix(app.Ufile(1));
4454             %SETUP OF THE COORDINATES
4455             [row,col] = size(f);
4456             col1 = col-1;
4457             row1 = row-1;
4458             app.xcoord = (0:col1) * app.gridX_2.Value+ app.
4459 LongitudeEditField_2.Value;
4460             app.ycoord = [0:row1] * app.gridY_2.Value+ app.
4461 LatitudeEditField_2.Value;
4462
4463             %SET UP THE MAP BOUNDARY
4464             app.EastEditField_2.Value = max(app.xcoord);
4465             app.WestEditField_2.Value = min(app.xcoord);
4466             app.NorthEditField_2.Value = max(app.ycoord);
4467             app.SouthEditField_2.Value = min(app.ycoord);
4468         end
4469     end
4470 end

```

```

4465
4466 % Value changed function: FileTextArea_3
4467 function FileTextArea_3ValueChanged(app, event)
4468
4469
4470 end
4471
4472 % Size changed function: VelocityMapTab
4473 function VelocityMapTabSizeChanged(app, event)
4474
4475
4476 end
4477
4478 % Callback function
4479 function PlotColormapCheckBoxValueChanged(app, event)
4480     if app.PlotColormapCheckBox.Value
4481         app.BackgroundMapColorDropDown.Enable = "on";
4482         app.FlipCheckBox_4.Enable = "on";
4483         app.InterpolationDivisionEditField_2.Enable = "on";
4484         app.MaxBarValue.Enable = "on";
4485         app.MinBarValue.Enable = "on";
4486     else
4487         app.BackgroundMapColorDropDown.Enable = "off";
4488         app.FlipCheckBox_4.Enable = "off";
4489         app.InterpolationDivisionEditField_2.Enable = "off";
4490         app.MaxBarValue.Enable = "off";
4491         app.MinBarValue.Enable = "off";
4492     end
4493
4494     if app.PlotColormapCheckBox.Value && app.etaButton.Value
4495         app.Button_23.Enable = "on";
4496     else
4497         app.Button_23.Enable = "off";
4498     end
4499
4500     if app.BathymetryButton.Value
4501         app.Button_23.Enable = "off";
4502     end
4503 end
4504
4505 % Value changed function: PlotBathymetryContoursCheckBox
4506 function PlotBathymetryContoursCheckBoxValueChanged(app, event)
4507     if app.PlotBathymetryContoursCheckBox.Value
4508         app.WidthEditField_2.Enable = "on";
4509         app.StyleDropDown_3.Enable = "on";
4510         app.IntervalEditField_6.Enable = "on";
4511         app.MaximumEditField_2.Enable = "on";
4512         app.MinimumEditField_2.Enable = "on";
4513         app.ColorDropDown_4.Enable = "on";
4514         app.AddLabelCheckBox_2.Enable = "on";
4515         if app.AddLabelCheckBox_2.Value
4516             app.IntervalEditField_8.Enable = "on";
4517             app.LabelSizeCont.Enable = "on";
4518             app.SpacingEditField.Enable = "on";
4519         else
4520             app.IntervalEditField_8.Enable = "off";
4521             app.LabelSizeCont.Enable = "off";
4522             app.SpacingEditField.Enable = "off";

```

```

4523         end
4524     else
4525         app.IntervalEditField_8.Enable = "off";
4526         app.WidthEditField_2.Enable = "off";
4527         app.StyleDropDown_3.Enable = "off";
4528         app.IntervalEditField_6.Enable = "off";
4529         app.MaximumEditField_2.Enable = "off";
4530         app.MinimumEditField_2.Enable = "off";
4531         app.ColorDropDown_4.Enable = "off";
4532         app.AddLabelCheckBox_2.Enable = "off";
4533         app.LabelSizeCont.Enable = "off";
4534         app.SpacingEditField.Enable = "off";
4535     end
4536 end
4537
4538 % Size changed function: Panel_16
4539 function Panel_16SizeChanged(app, event)
4540
4541
4542 end
4543
4544 % Size changed function: BasemapTab
4545 function BasemapTabSizeChanged(app, event)
4546
4547 end
4548
4549 % Value changed function: DepthFileEditField_2
4550 function DepthFileEditField_2ValueChanged(app, event)
4551
4552 end
4553
4554 % Value changed function: AutoSetCheckBox_3
4555 function AutoSetCheckBox_3ValueChanged(app, event)
4556     if app.AutoSetCheckBox_3.Value
4557         app.Height_3.Enable = "off";
4558         app.Width_3.Enable = "off";
4559     else
4560         app.Height_3.Enable = "on";
4561         app.Width_3.Enable = "on";
4562     end
4563 end
4564
4565 % Size changed function: ButtonGroup_12
4566 function ButtonGroup_12SizeChanged(app, event)
4567
4568 end
4569
4570 % Button pushed function: Button_26
4571 function Button_26Pushed(app, event)
4572     workingfolder = uigetdir;
4573     Dir1 = fullfile(string(workingfolder), 'OUTPUT_FILES');
4574     app.WORKFOLDER3 = Dir1;
4575     FigureFolder = fullfile(Dir1, 'Figures');
4576     app.OutputDirectoryEditField_3.Value = FigureFolder;
4577     app.WORKFOLDER4 = fullfile(deblank(FigureFolder));
4578     app.OutputDirectoryEditField_3.FontColor = 'k';
4579
4580     %Create the folders

```

```

4581         if ~exist(Dir1, 'dir')
4582             mkdir(Dir1);
4583         end
4584         if ~exist(FigureFolder, 'dir')
4585             mkdir(FigureFolder);
4586         end
4587     end
4588
4589     % Value changed function: CoastlinecolourDropDown_7
4590     function CoastlinecolourDropDown_7ValueChanged(app, event)
4591         value = app.CoastlinecolourDropDown_7.Value;
4592         %List of values based on the cbrewer2 colormap
4593         validValues = {'blue', 'blue - green', 'blue - purple', 'green -
blue', ...
4594             'greens', 'grays', 'oranges', 'orange - red', ...
4595             'purple - blue', 'purple - blue - green', 'purple - red', ...
4596             'purples', 'red - purple', 'reds', 'yellow - green', ...
4597             'yellow - green - blue', 'yellow - orange - brown', ...
4598             'yellow - orange - red', 'brown - teal', 'pink - light green',
...
4599             'purple - green', 'purple - orange', 'red - blue', ...
4600             'red - gray', 'red - yellow - blue', 'red - yellow - green',
...
4601             'spectral', 'accent', 'dark 2', 'paired', 'pastel 1', ...
4602             'pastel 2', 'set 1', 'set 2', 'set 3'};
4603
4604         % If the selected value in the dropdown list is part of the
cbrewer2 colormap, enable the color interpolation textbox
4605         if ismember(value, validValues)
4606             app.ColorinterpolationEditField_2.Enable = "on";
4607             app.DivisionLabel.Enable = "on";
4608         else
4609             app.ColorinterpolationEditField_2.Enable = "off";
4610             app.DivisionLabel.Enable = "off";
4611         end
4612     end
4613
4614     % Value changed function: BackgroundMapColorDropDown
4615     function BackgroundMapColorDropDownValueChanged(app, event)
4616         cbrewercolors = {'blue', 'blue - green', 'blue - purple', 'green -
blue', 'greens', ...
4617             'grays', 'oranges', 'orange - red', 'purple - blue', 'purple -
blue - green', ...
4618             'purple - red', 'purples', 'red - purple', 'reds', 'yellow -
green', ...
4619             'yellow - green - blue', 'yellow - orange - brown', 'yellow -
orange - red', ...
4620             'brown - teal', 'pink - light green', 'purple - green',
'purple - orange', ...
4621             'red - blue', 'red - gray', 'red - yellow - blue', 'red -
yellow - green', ...
4622             'spectral', 'accent', 'dark 2', 'paired', 'pastel 1', 'pastel
2', ...
4623             'set 1', 'set 2', 'set 3'};
4624
4625         %Activate the interpolation division when the selected colormap is
of cbrewer type
4626         if ismember(app.BackgroundMapColorDropDown.Value, cbrewercolors)

```

```

4627         app.InterpolationDivisionEditField_2.Enable = "on";
4628         app.InterpolationDivisionEditFieldLabel_2.Enable = "on";
4629     else
4630         app.InterpolationDivisionEditField_2.Enable = "off";
4631         app.InterpolationDivisionEditFieldLabel_2.Enable = "off";
4632     end
4633
4634
4635
4636     end
4637
4638     % Drop down opening function: BackgroundMapColorDropDown
4639     function BackgroundMapColorDropDownOpening(app, event)
4640         app.IntervalEditField_6.FontColor = 'k';
4641     end
4642
4643     % Value changed function: IntervalEditField_6
4644     function IntervalEditField_6ValueChanged(app, event)
4645         app.IntervalEditField_6.FontColor = 'k';
4646         app.IntervalEditField_8.Value = app.IntervalEditField_6.Value;
4647     end
4648
4649     % Button pushed function: SettoDefaultButton
4650     function SettoDefaultButtonPushed(app, event)
4651         %Restore the boundary extent using input from the 'Input Data' ↵
section
4652
4653         %Extract row and column dimensions from the first imported file
4654         f = readmatrix(app.Uvectorfullfile(1));
4655
4656         %Determine the coordinates
4657         [row,col] = size(f);
4658         col1 = col-1;
4659         row1 = row-1;
4660         app.xcoord = [0:col1] * app.gridX_2.Value+ app.↵
LongitudeEditField_2.Value;
4661         app.ycoord = [0:row1] * app.gridY_2.Value+ app.↵
LatitudeEditField_2.Value;
4662
4663         %Show the map boundary in the textboxes in 'General Layout'
4664         app.EastEditField_2.Value = max(app.xcoord);
4665         app.WestEditField_2.Value = min(app.xcoord);
4666         app.NorthEditField_2.Value = max(app.ycoord);
4667         app.SouthEditField_2.Value = min(app.ycoord);
4668
4669         app.EastEditField_2.FontColor = 'k';
4670         app.WestEditField_2.FontColor = 'k';
4671         app.NorthEditField_2.FontColor = 'k';
4672         app.SouthEditField_2.FontColor = 'k';
4673         drawnow
4674
4675     end
4676
4677     % Button pushed function: SettoDefaultButton_2
4678     function SettoDefaultButton_2Pushed(app, event)
4679         %Restore the boundary extent using input from the 'Input Data' ↵
section
4680

```

```

4681         %Extract row and column dimensions from the first imported file
4682         f = readmatrix(app.FileINPUT(1));
4683         [row,col] = size(f);
4684
4685         %Determine the coordinates
4686         col1 = col-1;
4687         row1 = row-1;
4688         app.x = app.LongitudeEditField.Value + [0:col1] * app.gridX.Value;
4689         app.y = app.LatitudeEditField.Value + [0:row1] * app.gridY.Value;
4690
4691
4692         %Show the map boundary in the textboxes in 'General Layout'
4693         app.EastEditField.Value = max(app.x);
4694         app.WestEditField.Value = min(app.x);
4695         app.NorthEditField.Value = max(app.y);
4696         app.SouthEditField.Value = min(app.y);
4697         drawnow
4698
4699         app.EastEditField.FontColor = 'k';
4700         app.WestEditField.FontColor = 'k';
4701         app.NorthEditField.FontColor = 'k';
4702         app.SouthEditField.FontColor = 'k';
4703
4704     end
4705
4706     % Size changed function: Panel_15
4707     function Panel_15SizeChanged(app, event)
4708
4709
4710     end
4711
4712     % Value changed function: PlotGaugesCheckBox
4713     function PlotGaugesCheckBoxValueChanged(app, event)
4714         if app.PlotGaugesCheckBox.Value
4715             app.FileEditField_2.Enable = "on";
4716             app.Button_27.Enable = "on";
4717             app.ColorDropDown_7.Enable = "on";
4718             app.SizeEditField_2.Enable = "on";
4719             app.gaugemarkerVelocityTab.Enable = "on";
4720             app.AddLabelCheckBox.Enable = "on";
4721             if app.AddLabelCheckBox.Value
4722                 app.GLspacing_2.Enable = "on";
4723                 app.GLspacing_2.Editable = "on";
4724             else
4725                 app.GLspacing_2.Editable = "off";
4726                 app.GLspacingLabel_2.Enable = "off";
4727             end
4728
4729         else
4730             app.FileEditField_2.Enable = "off";
4731             app.Button_27.Enable = "off";
4732             app.ColorDropDown_7.Enable = "off";
4733             app.SizeEditField_2.Enable = "off";
4734             app.gaugemarkerVelocityTab.Enable = "off";
4735             app.AddLabelCheckBox.Enable = "off";
4736         end
4737
4738

```

```

4739     end
4740
4741     % Value changed function: AddLabelCheckBox
4742     function AddLabelCheckBoxValueChanged(app, event)
4743         if app.AddLabelCheckBox.Value
4744             app.FontSizeEditField_2.Enable = "on";
4745             app.FontSizeEditField_2.Editable = "on";
4746             app.CoastlinecolourDropDown_11.Enable = "on";
4747             app.CoastlinecolourDropDown_10.Enable = "on";
4748             app.GLspacing_2.Enable = "on";
4749             app.GLspacing_2.Editable = "on";
4750             app.GLspacingLabel_2.Enable = "on";
4751             app.FontSizeEditField_2Label.Enable = "on";
4752             app.AlignmentLabel_2.Enable = "on";
4753         else
4754             app.FontSizeEditField_2.Enable = "off";
4755             app.FontSizeEditField_2.Editable = "off";
4756             app.CoastlinecolourDropDown_11.Enable = "off";
4757             app.CoastlinecolourDropDown_10.Enable = "off";
4758             app.GLspacing_2.Enable = "off";
4759             app.GLspacingLabel_2.Enable = "off";
4760             app.FontSizeEditField_2Label.Enable = "off";
4761             app.AlignmentLabel_2.Enable = "off";
4762         end
4763     end
4764 end
4765
4766 % Callback function
4767 function AddLabelsCheckBox_2ValueChanged(app, event)
4768     if app.AddLabelsCheckBox_2.Value
4769         app.IntervalEditField_3.Enable = "on";
4770         app.TextLabelSize.Enable = "on";
4771         app.LabelSpacingEditField.Enable = "on";
4772         app.SpacingEditField_4Label_4.Enable = "on";
4773         app.LabelSpacingEditFieldLabel.Enable = "on";
4774         app.LabelSpacingEditFieldLabel_2.Enable = "on";
4775     else
4776         app.IntervalEditField_3.Enable = "off";
4777         app.TextLabelSize.Enable = "off";
4778         app.LabelSpacingEditField.Enable = "off";
4779         app.SpacingEditField_4Label_4.Enable = "off";
4780         app.LabelSpacingEditFieldLabel.Enable = "off";
4781         app.LabelSpacingEditFieldLabel_2.Enable = "off";
4782     end
4783
4784 end
4785
4786 % Callback function
4787 function AddLabelsCheckBox_3ValueChanged(app, event)
4788     if app.AddLabelsCheckBox_3.Value
4789         app.TextLabelSize_2.Enable = "on";
4790         app.wh_interval.Enable = "on";
4791         app.LabelSpacingEditField_2.Enable = "on";
4792         app.SpacingEditField_4Label_5.Enable = "on";
4793         app.LabelSpacingEditField_2Label_2.Enable = "on";
4794         app.LabelSpacingEditField_2Label.Enable = "on";
4795     else
4796         app.TextLabelSize_2.Enable = "off";

```



```

4797         app.wh_interval.Enable = "off";
4798         app.LabelSpacingEditField_2.Enable = "off";
4799         app.SpacingEditField_4Label_5.Enable = "off";
4800         app.LabelSpacingEditField_2Label_2.Enable = "off";
4801         app.LabelSpacingEditField_2Label.Enable = "off";
4802     end
4803 end
4804
4805 % Callback function
4806 function AddLabelsCheckBox_4ValueChanged(app, event)
4807     if app.AddLabelsCheckBox_4.Value
4808         app.CoastlinecolourDropDown_5.Enable = "on";
4809         app.CoastlinecolourDropDown_4.Enable = "on";
4810         app.SizeEditField_3Label.Enable = "on";
4811         app.AlignmentLabel.Enable = "on";
4812         app.SizeEditField_3.Enable = "on";
4813         app.SizeEditField_3.Editable = "on";
4814         app.GLspacing.Enable = "on";
4815         app.GLspacing.Editable = "on";
4816         app.GLspacingLabel.Enable = "on";
4817         app.FontSizeEditField_2Label.Enable = "on";
4818         app.AlignmentLabel_2.Enable = "on";
4819     else
4820         app.CoastlinecolourDropDown_5.Enable = "off";
4821         app.CoastlinecolourDropDown_4.Enable = "off";
4822         app.SizeEditField_3Label.Enable = "off";
4823         app.AlignmentLabel.Enable = "off";
4824         app.SizeEditField_3.Enable = "off";
4825         app.SizeEditField_3.Editable = "off";
4826         app.GLspacing.Enable = "off";
4827         app.GLspacing.Editable = "off";
4828         app.GLspacingLabel.Enable = "off";
4829         app.FontSizeEditField_2Label.Enable = "off";
4830         app.AlignmentLabel_2.Enable = "off";
4831     end
4832 end
4833
4834 % Size changed function: GaugesTab_2
4835 function GaugesTab_2SizeChanged(app, event)
4836
4837
4838 end
4839
4840 % Button pushed function: Button_27
4841 function Button_27Pushed(app, event)
4842     [filename3,path3] = uigetfile('*.txt;*.shp');
4843     fullname = fullfile(path3,filename3);
4844
4845     [~, name, ext] = fileparts(filename3);
4846     if ~(ismcc || isdeployed)
4847         addpath(genpath(string(path3)));
4848     end
4849
4850     app.GaugeFname3 = name;
4851     app.FileEditField_2.Value = filename3;
4852     app.GAUGEFILE3= filename3;
4853     app.FileEditField_2.FontColor = 'k';
4854     app.FileEditField_2.BackgroundColor = 'w';

```

```

4855
4856         if strcmp(ext, '.shp')
4857             S = shaperead(fullname);
4858             latitudes= [S.Y]';
4859             longitudes= [S.X]';
4860             list = [round(latitudes, 4), round(longitudes, 4)];
4861             list = unique(list, 'rows');
4862             app.latGauge2 = list(:, 1);
4863             app.longGauge2 = list(:, 2);
4864
4865         elseif strcmp(ext, '.txt')
4866             fileID = fopen(fullname, 'r');
4867             dataArray = textscan(fileID, '%f %f', 'Delimiter', '\t',
4868 'whitespace'); %Read text file and ensure that it is tab delimited
4869             fclose(fileID);
4870             app.latGauge2 = dataArray{1};
4871             app.longGauge2 = dataArray{2};
4872         end
4873
4874         app.FileEditField_2.FontColor = 'k';
4875         app.FileEditField_2.BackgroundColor = 'w';
4876
4877     end
4878
4879     % Value changed function: AddLabelCheckBox_2
4880     function AddLabelCheckBox_2ValueChanged(app, event)
4881         if app.AddLabelCheckBox_2.Value
4882             app.LabelSizeCont.Enable = "on";
4883             app.SpacingEditField.Enable = "on";
4884             app.IntervalEditField_8.Enable = "on";
4885         else
4886             app.LabelSizeCont.Enable = "off";
4887             app.SpacingEditField.Enable = "off";
4888             app.IntervalEditField_8.Enable = "off";
4889         end
4890     end
4891
4892     % Value changed function: LongitudeEditField_2
4893     function LongitudeEditField_2ValueChanged(app, event)
4894         app.LongitudeEditField_2.FontColor = 'k';
4895
4896         %Update the X-axis limits in the 'General Layout' section
4897         app.xcoord = [0:app.col1vel] * app.gridX_2.Value+ app.
4898 LongitudeEditField_2.Value;
4899         app.EastEditField_2.Value = max(app.xcoord);
4900         app.WestEditField_2.Value = min(app.xcoord);
4901         drawnow
4902     end
4903
4904     % Value changed function: LatitudeEditField_2
4905     function LatitudeEditField_2ValueChanged(app, event)
4906         app.LatitudeEditField_2.FontColor = 'k';
4907
4908         %Update the Y-axis limits in the 'General Layout' section
4909         app.ycoord = [0:app.row1vel] * app.gridY_2.Value+ app.
4910 LatitudeEditField_2.Value;
4911         app.NorthEditField_2.Value = max(app.ycoord);

```

```

4910         app.SouthEditField_2.Value = min(app.ycoord);
4911         drawnow
4912     end
4913
4914     % Value changed function: gridX_2
4915     function gridX_2ValueChanged(app, event)
4916         app.gridX_2.FontColor = 'k';
4917
4918         %Update the X-axis limits in the 'General Layout' section
4919         app.xcoord = [0:app.col1vel] * app.gridX_2.Value+ app.↵
LongitudeEditField_2.Value;
4920         app.EastEditField_2.Value = max(app.xcoord);
4921         app.WestEditField_2.Value = min(app.xcoord);
4922         drawnow
4923     end
4924
4925     % Value changed function: gridY_2
4926     function gridY_2ValueChanged(app, event)
4927         app.gridY_2.FontColor = 'k';
4928
4929         %Update the Y-axis limits in the 'General Layout' section
4930         app.ycoord = [0:app.row1vel] * app.gridY_2.Value+ app.↵
LatitudeEditField_2.Value;
4931         app.NorthEditField_2.Value = max(app.ycoord);
4932         app.SouthEditField_2.Value = min(app.ycoord);
4933         drawnow
4934     end
4935
4936     % Value changed function: TotalSimuilationTimesecEditField_4
4937     function TotalSimuilationTimesecEditField_4ValueChanged(app, event)
4938         app.TotalSimuilationTimesecEditField_4.FontColor = 'k';
4939
4940     end
4941
4942     % Value changed function: arrowscale
4943     function arrowscaleValueChanged(app, event)
4944         app.arrowscale.FontColor = 'k';
4945
4946     end
4947
4948     % Value changed function: ArrowHeadSize
4949     function ArrowHeadSizeValueChanged(app, event)
4950         app.ArrowHeadSize.FontColor = 'k';
4951
4952     end
4953
4954     % Value changed function: ArrowThickness
4955     function ArrowThicknessValueChanged(app, event)
4956         app.ArrowThickness.FontColor = 'k';
4957
4958     end
4959
4960     % Callback function
4961     function ArrowSizeLegendValueChanged(app, event)
4962         app.ArrowSizeLegend.FontColor = 'k';
4963
4964     end
4965

```

```

4966 % Callback function
4967 function ColorbarTextSize_2ValueChanged(app, event)
4968     app.ColorbarTextSize_2.FontColor = 'k';
4969
4970 end
4971
4972 % Callback function
4973 function ArrowXLabelLocValueChanged(app, event)
4974     app.ArrowXLabelLoc.FontColor = 'k';
4975
4976 end
4977
4978 % Callback function
4979 function ArrowYLabelLocValueChanged(app, event)
4980     app.ArrowYLabelLoc.FontColor = 'k';
4981
4982 end
4983
4984 % Value changed function: MinimumEditField_2
4985 function MinimumEditField_2ValueChanged(app, event)
4986     app.MinimumEditField_2.FontColor = 'k';
4987
4988 end
4989
4990 % Value changed function: MaximumEditField_2
4991 function MaximumEditField_2ValueChanged(app, event)
4992     app.MaximumEditField_2.FontColor = 'k';
4993
4994 end
4995
4996 % Value changed function: WidthEditField_2
4997 function WidthEditField_2ValueChanged(app, event)
4998     app.WidthEditField_2.FontColor = 'k';
4999
5000 end
5001
5002 % Value changed function: SpacingEditField
5003 function SpacingEditFieldValueChanged(app, event)
5004     app.SpacingEditField.FontColor = 'k';
5005
5006 end
5007
5008 % Value changed function: LabelSizeCont
5009 function LabelSizeContValueChanged(app, event)
5010     app.LabelSizeCont.FontColor = 'k';
5011
5012 end
5013
5014 % Value changed function: IntervalEditField_8
5015 function IntervalEditField_8ValueChanged(app, event)
5016     app.IntervalEditField_8.FontColor = 'k';
5017
5018 end
5019
5020 % Value changed function: NorthEditField_2
5021 function NorthEditField_2ValueChanged(app, event)
5022     app.NorthEditField_2.FontColor = 'k';
5023

```

```

5024     end
5025
5026     % Value changed function: WestEditField_2
5027     function WestEditField_2ValueChanged(app, event)
5028         app.WestEditField_2.FontColor = 'k';
5029
5030     end
5031
5032     % Value changed function: EastEditField_2
5033     function EastEditField_2ValueChanged(app, event)
5034         app.EastEditField_2.FontColor = 'k';
5035
5036     end
5037
5038     % Value changed function: SouthEditField_2
5039     function SouthEditField_2ValueChanged(app, event)
5040         app.SouthEditField_2.FontColor = 'k';
5041
5042     end
5043
5044     % Value changed function: Width_3
5045     function Width_3ValueChanged(app, event)
5046         app.Width_3.FontColor = 'k';
5047
5048     end
5049
5050     % Value changed function: Height_3
5051     function Height_3ValueChanged(app, event)
5052         app.Height_3.FontColor = 'k';
5053
5054     end
5055
5056     % Value changed function: OutputDirectoryEditField_3
5057     function OutputDirectoryEditField_3ValueChanged(app, event)
5058         app.OutputDirectoryEditField_3.FontColor = 'k';
5059
5060     end
5061
5062     % Value changed function: FramerateEditField
5063     function FramerateEditFieldValueChanged(app, event)
5064         app.FramerateEditField.FontColor = 'k';
5065
5066     end
5067
5068     % Value changed function: maplabelsize
5069     function maplabelsizeValueChanged(app, event)
5070         app.maplabelsize.FontColor = 'k';
5071
5072     end
5073
5074     % Value changed function: MaxBarValue
5075     function MaxBarValueValueChanged(app, event)
5076         app.MaxBarValue.FontColor = 'k';
5077
5078     end
5079
5080     % Value changed function: MinBarValue
5081     function MinBarValueValueChanged(app, event)

```

```

5082         app.MinBarValue.FontColor = 'k';
5083
5084     end
5085
5086     % Value changed function: InterpolationDivisionEditField_2
5087     function InterpolationDivisionEditField_2ValueChanged(app, event)
5088         app.InterpolationDivisionEditField_2.FontColor = 'k';
5089
5090     end
5091
5092     % Value changed function: FontSizeEditField_2
5093     function FontSizeEditField_2ValueChanged(app, event)
5094         app.FontSizeEditField_2.FontColor = 'k';
5095
5096     end
5097
5098     % Value changed function: SizeEditField_2
5099     function SizeEditField_2ValueChanged(app, event)
5100         app.SizeEditField_2.FontColor = 'k';
5101
5102     end
5103
5104     % Value changed function: FileEditField_2
5105     function FileEditField_2ValueChanged(app, event)
5106
5107     end
5108
5109     % Size changed function: WaveHeightMapTab
5110     function WaveHeightMapTabSizeChanged(app, event)
5111
5112     end
5113
5114     % Button pushed function: CloseFiguresButton_2
5115     function CloseFiguresButton_2Pushed2(app, event)
5116         close all
5117     end
5118
5119     % Value changed function: PlotindegreesCheckBox
5120     function PlotindegreesCheckBoxValueChanged(app, event)
5121
5122     end
5123
5124     % Value changed function: txtCheckBox
5125     function txtCheckBoxValueChanged(app, event)
5126
5127
5128     end
5129
5130     % Value changed function: FlipCheckBox_2
5131     function FlipCheckBox_2ValueChanged(app, event)
5132
5133
5134     end
5135
5136     % Value changed function: YaxisUseDataDropDown
5137     function YaxisUseDataDropDownValueChanged(app, event)
5138         % Determine the Y-axis label based on the dropdown value
5139         switch app.YaxisUseDataDropDown.Value

```

```

5140         case '1'
5141             if ~app.CheckBox_4.Value
5142                 app.YAxisEditField.Value = "Time (sec)";
5143             else %If 'Convert the Values' checkbox is ticked, remove
the unit
5144                 app.YAxisEditField.Value = "Time";
5145             end
5146         case '2'
5147             if ~app.CheckBox_4.Value
5148                 app.YAxisEditField.Value = "\eta (m)";
5149             else
5150                 app.XAxisEditField.Value = "\eta";
5151             end
5152         case {'3', '4', '3,4,5'}
5153             if ~app.CheckBox_4.Value
5154                 app.YAxisEditField.Value = "Velocity (m/s)";
5155             else
5156                 app.YAxisEditField.Value = "Velocity";
5157             end
5158         end
5159
5160         %Change the font color of 'Axes Labels: Y axis' in 'Plot Style and
Layout'
5161         if ~app.CheckBox_4.Value
5162             app.YAxisEditField.FontColor = "k";
5163         end
5164
5165         %Update which dropdown list to show in 'Plot Properties: Color' in
'Plot Style and Layout'
5166         if length(app.GaugeDirectorylist) == 1 && ~strcmp(app.
YaxisUseDataDropDown.Value, '3,4,5')
5167             app.GaugeLineColor.Visible = "on";
5168             app.GaugeLineColor.Enable = "on";
5169             app.GaugeLineColor_multiple.Visible = "off";
5170             app.GaugeLineColor_multiple.Enable = "off";
5171             app.FlipCheckBox_3.Visible = "off";
5172             app.FlipCheckBox_3.Enable = "off";
5173         else
5174             app.GaugeLineColor.Visible = "off";
5175             app.GaugeLineColor.Enable = "off";
5176             app.GaugeLineColor_multiple.Visible = "on";
5177             app.GaugeLineColor_multiple.Enable = "on";
5178             app.FlipCheckBox_3.Visible = "on";
5179             app.FlipCheckBox_3.Enable = "on";
5180         end
5181         drawnow
5182     end
5183
5184     % Callback function
5185     function YCol2ValueChanged(app, event)
5186         app.YCol2.FontColor = 'k';
5187     end
5188
5189     % Callback function
5190     function YCol3ValueChanged(app, event)
5191         app.YCol3.FontColor = 'k';
5192     end
5193

```

```

5194         end
5195
5196         % Callback function
5197         function YCol4ValueChanged(app, event)
5198             app.YCol4.FontColor = 'k';
5199
5200         end
5201
5202         % Size changed function: GaugeRecordsTab
5203         function GaugeRecordsTabSizeChanged(app, event)
5204
5205
5206         end
5207
5208         % Size changed function: Panel_11
5209         function Panel_11SizeChanged(app, event)
5210
5211         end
5212
5213         % Selection changed function: ButtonGroup_16
5214         function ButtonGroup_16SelectionChanged(app, event)
5215             % Enable parameters for the selected basemap option
5216             if ~isempty(app.FileTextArea_3.Value) %Check if there are u_ files ✓
loaded in the app
5217                 if app.etaButton.Value
5218                     %Find the corresponding 'eta' file in the directory of the ✓
u_ file
5219                     matchingFiles = [];
5220                     for i = 1:length(app.ETAfilenames)
5221                         filename = app.ETAfilenames{i};
5222                         fullPath = fullfile(app.vectorpath, filename);
5223                         if exist(fullPath, 'file') == 2 %Append to the list
5224                             matchingFiles = [matchingFiles; filename];
5225                         end
5226                     end
5227
5228                     if isempty(matchingFiles)
5229                         %No matching files found
5230                         app.FileTextArea_5.Value = 'No matching files found';
5231
5232                         %Disable colormap and colorbar options
5233                         children = app.ThirdTabColorMapPanel.Children;
5234                         for i = 1:length(children)
5235                             children(i).Enable = 'off';
5236                         end
5237
5238                         else %At least one matching file was found
5239                             app.FileTextArea_5.Value = strjoin(string ✓
(matchingFiles), newline);
5240
5241                             %Enable colormap and colorbar options
5242                             children = app.ThirdTabColorMapPanel.Children;
5243                             for i = 1:length(children)
5244                                 children(i).Enable = 'on';
5245                             end
5246                             app.BackgroundMapColorDropDown.Value = 'red - blue';
5247                             app.FlipCheckBox_4.Value = true;
5248                         end

```



```

5249
5250         elseif app.hmaxButton.Value
5251             %Find the corresponding 'hmax' file in the directory of
the U_ file
5252             matchingFiles = [];
5253             for i = 1:length(app.HMAXfilenames)
5254                 filename = app.HMAXfilenames{i};
5255                 fullPath = fullfile(app.vectorpath, filename);
5256                 if exist(fullPath, 'file') == 2
5257                     matchingFiles = [matchingFiles; filename];
5258                 end
5259             end
5260
5261             if isempty(matchingFiles)
5262                 % No matching files found
5263                 app.FileTextArea_5.Value = 'No matching files found';
5264
5265                 %Disable colormap and colorbar options
5266                 children = app.ThirdTabColorMapPanel.Children;
5267                 for i = 1:length(children)
5268                     children(i).Enable = 'off';
5269                 end
5270             else
5271                 % At least one matching file was found
5272                 app.FileTextArea_5.Value = strjoin(string
(matchingFiles), newline);
5273
5274                 %Enable colormap and colorbar options
5275                 children = app.ThirdTabColorMapPanel.Children;
5276                 for i = 1:length(children)
5277                     children(i).Enable = 'on';
5278                 end
5279                 app.BackgroundMapColorDropDown.Value = 'parula';
5280                 app.FlipCheckBox_4.Value = false;
5281                 app.InterpolationDivisionEditField_2.Enable = 'off';
5282             end
5283
5284         elseif app.BathymetryButton.Value
5285             %Use bathymetry data as the basemap
5286             if isempty(app.DepthFileEditField_2.Value) %If a bathymetry
file is not loaded in the 'Input Data' section
5287                 % No matching files found
5288                 app.FileTextArea_5.Value = 'No bathymetry file found';
drawnow
5289
5290                 %Disable colormap and colorbar options
5291                 children = app.ThirdTabColorMapPanel.Children;
5292                 for i = 1:length(children)
5293                     children(i).Enable = 'off';
5294                 end
5295                 app.LandColor.Enable = 'off';
5296             else
5297                 %Enable colormap and colorbar options
5298                 children = app.ThirdTabColorMapPanel.Children;
5299                 for i = 1:length(children)
5300                     children(i).Enable = 'on';
5301                 end
5302                 app.FileTextArea_5.Value = ' ';

```

```

5303         app.BackgroundColorDropDown.Value = 'parula';
5304         app.FlipCheckBox_4.Value = false;
5305         app.InterpolationDivisionEditField_2.Enable = 'off';
5306     end
5307
5308     elseif app.VelocityButton.Value
5309         children = app.ThirdTabColorMapPanel.Children;
5310         if ~isempty(app.FileName5)
5311             % Use the Z magnitude, calculated from the U and V
5312             %Enable colormap and colorbar options
5313             for i = 1:length(children)
5314                 children(i).Enable = 'on';
5315             end
5316
5317             %Change the default colormap and bar limits to match
5318             %the damage index ranges in Lynett et al. (2022)
5319             app.BackgroundColorDropDown.Value = 'blue -
purple';
5320             app.InterpolationDivisionEditField_2.Value = 4;
5321             app.InterpolationDivisionEditField_2.Enable = "on";
5322         else
5323             for i = 1:length(children)
5324                 children(i).Enable = 'off';
5325             end
5326         end
5327
5328     elseif app.VorticityButton.Value
5329         children = app.ThirdTabColorMapPanel.Children;
5330         if ~isempty(app.FileName5)
5331             % Use the Z magnitude, calculated from the U and V
5332             %Enable colormap and colorbar options
5333             for i = 1:length(children)
5334                 children(i).Enable = 'on';
5335             end
5336             app.BackgroundColorDropDown.Value = 'jet';
5337             app.FlipCheckBox_4.Value = false;
5338             app.InterpolationDivisionEditField_2.Enable = 'off';
5339         else
5340             for i = 1:length(children)
5341                 children(i).Enable = 'off';
5342             end
5343         end
5344     end
5345
5346     ArrowSection = app.Panel_14.Children;
5347     if ~app.VorticityButton.Value
5348         %Enable the Arrow Section
5349         for i = 1:length(ArrowSection)
5350             ArrowSection(i).Enable = 'on';
5351         end
5352     else
5353         %Disable the Arrow Section
5354         for i = 1:length(ArrowSection)
5355             ArrowSection(i).Enable = 'off';
5356         end

```

```

5357         app.PlotvectorsCheckBox.Value = false; %Uncheck the box
5358     end
5359
5360     if contains(string(app.FileName5), 'umax')
5361         for i = 1:length(ArrowSection)
5362             ArrowSection(i).Enable = 'off';
5363         end
5364     end
5365 end
5366
5367 end
5368
5369 % Value changed function: LongitudeEditField
5370 function LongitudeEditFieldValueChanged(app, event)
5371     app.LongitudeEditField.FontColor = 'k';
5372
5373     %Update the X-axis limits in the 'General Layout' section
5374     app.x = app.LongitudeEditField.Value + [0:app.col1] * app.gridX.
Value;
5375     app.EastEditField.Value = max(app.x);
5376     app.WestEditField.Value = min(app.x);
5377     drawnow
5378 end
5379
5380 % Value changed function: LatitudeEditField
5381 function LatitudeEditFieldValueChanged(app, event)
5382     app.LatitudeEditField.FontColor = 'k';
5383
5384     %Update the Y-axis limits in the 'General Layout' section
5385     app.y = [0:app.row1] * app.gridY.Value+ app.LatitudeEditField.
Value;
5386     app.NorthEditField.Value = max(app.y);
5387     app.SouthEditField.Value = min(app.y);
5388     drawnow
5389 end
5390
5391 % Value changed function: gridX
5392 function gridXValueChanged(app, event)
5393     app.gridX.FontColor = 'k';
5394
5395     %Update the X-axis limits in the 'General Layout' section
5396     app.x = app.LongitudeEditField.Value + [0:app.col1] * app.gridX.
Value;
5397     app.EastEditField.Value = max(app.x);
5398     app.WestEditField.Value = min(app.x);
5399     drawnow
5400
5401 end
5402
5403 % Value changed function: gridY
5404 function gridYValueChanged(app, event)
5405     app.gridY.FontColor = 'k';
5406
5407     %Update the Y-axis limits in the 'General Layout' section
5408     app.y = [0:app.row1] * app.gridY.Value+ app.LatitudeEditField.
Value;
5409     app.NorthEditField.Value = max(app.y);
5410     app.SouthEditField.Value = min(app.y);

```

```

5411         drawnow
5412     end
5413
5414     % Value changed function: TotalSimuilationTimesecEditField_3
5415     function TotalSimuilationTimesecEditField_3ValueChanged(app, event)
5416         app.TotalSimuilationTimesecEditField_3.FontColor = 'k';
5417
5418     end
5419
5420     % Value changed function: tifCheckBox_2
5421     function tifCheckBox_2ValueChanged(app, event)
5422
5423
5424     end
5425
5426     % Callback function
5427     function ArrivalTimeCheckBoxValueChanged(app, event)
5428
5429
5430     end
5431
5432     % Callback function
5433     function TimeFileButtonPushed(app, event)
5434         [timefile,path2] = uigetfile('*..*', 'Select the files',↵
5435         'MultiSelect', 'on'); %uigetfile('*.txt;*.out'); %;*.tiff;*.tif;*.asdf;*.asc;*.grd;↵
5436         *.flt;*.grb;*.grid;*.grib2;*.dt0;*.dt1;*.dt2;*.ddf;*.dem;*.hgt;*.grd;*.grc;*.ers;*.↵
5437         dat;*.img;*.jpg2'); uigetfile('*.txt;*.shp')
5438         app.FileEditField_3.Value = string(timefile);
5439         app.FileEditField_3.FontColor = 'k';
5440
5441         if ~(ismcc || isdeployed)
5442             addpath(genpath(fullfile(string(path2))));
5443         end
5444
5445         conditions = [
5446             ~strcmp(app.FileEditField_3.Value, "0"),
5447             ~strcmp(app.FileEditField_3.Value, "00")
5448         ];
5449
5450         if app.FileEditField_3.Value == string(timefile)
5451             if any(conditions)
5452                 try
5453                     ArrivalTimeinputdata = string(path2) + string↵
5454                     (timefile);
5455                     app.ArrivalTimename = string(timefile);
5456                     app.FileEditField_3.BackgroundColor = 'w';
5457                     app.FileEditField_3.FontColor = 'k';
5458                     ArrivalTimeDATA = readmatrix(ArrivalTimeinputdata);
5459                     app.ArrivalTimeConverted = ArrivalTimeDATA ./ 60; %↵
5460                     convert the values to minutes. Default threshold to read first wave arrival time in↵
5461                     FUNWAVE is 0.001 m
5462                     app.MaximumdepthEditField_3.Value = max(max(app.↵
5463                     ArrivalTimeConverted));
5464                     app.MaximumdepthEditField_3.FontColor = 'k';
5465                 end
5466             end
5467         end
5468     end
5469 end
5470
5471
5472
5473
5474
5475
5476
5477
5478
5479
5480
5481
5482
5483
5484
5485
5486
5487
5488
5489
5490
5491
5492
5493
5494
5495
5496
5497
5498
5499
5500
5501
5502
5503
5504
5505
5506
5507
5508
5509
5510
5511
5512
5513
5514
5515
5516
5517
5518
5519
5520
5521
5522
5523
5524
5525
5526
5527
5528
5529
5530
5531
5532
5533
5534
5535
5536
5537
5538
5539
5540
5541
5542
5543
5544
5545
5546
5547
5548
5549
5550
5551
5552
5553
5554
5555
5556
5557
5558
5559
5560
5561
5562
5563
5564
5565
5566
5567
5568
5569
5570
5571
5572
5573
5574
5575
5576
5577
5578
5579
5580
5581
5582
5583
5584
5585
5586
5587
5588
5589
5590
5591
5592
5593
5594
5595
5596
5597
5598
5599
5600
5601
5602
5603
5604
5605
5606
5607
5608
5609
5610
5611
5612
5613
5614
5615
5616
5617
5618
5619
5620
5621
5622
5623
5624
5625
5626
5627
5628
5629
5630
5631
5632
5633
5634
5635
5636
5637
5638
5639
5640
5641
5642
5643
5644
5645
5646
5647
5648
5649
5650
5651
5652
5653
5654
5655
5656
5657
5658
5659
5660
5661
5662
5663
5664
5665
5666
5667
5668
5669
5670
5671
5672
5673
5674
5675
5676
5677
5678
5679
5680
5681
5682
5683
5684
5685
5686
5687
5688
5689
5690
5691
5692
5693
5694
5695
5696
5697
5698
5699
5700
5701
5702
5703
5704
5705
5706
5707
5708
5709
5710
5711
5712
5713
5714
5715
5716
5717
5718
5719
5720
5721
5722
5723
5724
5725
5726
5727
5728
5729
5730
5731
5732
5733
5734
5735
5736
5737
5738
5739
5740
5741
5742
5743
5744
5745
5746
5747
5748
5749
5750
5751
5752
5753
5754
5755
5756
5757
5758
5759
5760
5761
5762
5763
5764
5765
5766
5767
5768
5769
5770
5771
5772
5773
5774
5775
5776
5777
5778
5779
5780
5781
5782
5783
5784
5785
5786
5787
5788
5789
5790
5791
5792
5793
5794
5795
5796
5797
5798
5799
5800
5801
5802
5803
5804
5805
5806
5807
5808
5809
5810
5811
5812
5813
5814
5815
5816
5817
5818
5819
5820
5821
5822
5823
5824
5825
5826
5827
5828
5829
5830
5831
5832
5833
5834
5835
5836
5837
5838
5839
5840
5841
5842
5843
5844
5845
5846
5847
5848
5849
5850
5851
5852
5853
5854
5855
5856
5857
5858
5859
5860
5861
5862
5863
5864
5865
5866
5867
5868
5869
5870
5871
5872
5873
5874
5875
5876
5877
5878
5879
5880
5881
5882
5883
5884
5885
5886
5887
5888
5889
5890
5891
5892
5893
5894
5895
5896
5897
5898
5899
5900
5901
5902
5903
5904
5905
5906
5907
5908
5909
5910
5911
5912
5913
5914
5915
5916
5917
5918
5919
5920
5921
5922
5923
5924
5925
5926
5927
5928
5929
5930
5931
5932
5933
5934
5935
5936
5937
5938
5939
5940
5941
5942
5943
5944
5945
5946
5947
5948
5949
5950
5951
5952
5953
5954
5955
5956
5957
5958
5959
5960
5961
5962
5963
5964
5965
5966
5967
5968
5969
5970
5971
5972
5973
5974
5975
5976
5977
5978
5979
5980
5981
5982
5983
5984
5985
5986
5987
5988
5989
5990
5991
5992
5993
5994
5995
5996
5997
5998
5999
6000

```

```

5462     end
5463
5464     % Callback function
5465     function AddLabelsCheckBox_5ValueChanged(app, event)
5466         if app.AddLabelsCheckBox_5.Value
5467             app.SpacingEditField_4Label_6.Enable = "on";
5468             app.TextLabelSize_3.Enable = "on";
5469             app.LabelSpacingEditField_2Label_3.Enable = "on";
5470             app.LabelSpacingEditField_4.Enable = "on";
5471             app.LabelSpacingEditField_2Label_4.Enable = "on";
5472             app.IntervalEditField_7.Enable = "on";
5473         else
5474             app.SpacingEditField_4Label_6.Enable = "off";
5475             app.TextLabelSize_3.Enable = "off";
5476             app.LabelSpacingEditField_2Label_3.Enable = "off";
5477             app.LabelSpacingEditField_4.Enable = "off";
5478             app.LabelSpacingEditField_2Label_4.Enable = "off";
5479             app.IntervalEditField_7.Enable = "off";
5480         end
5481     end
5482 end
5483
5484 % Value changed function: XaxisUseDataDropDown
5485 function XaxisUseDataDropDownValueChanged(app, event)
5486     % Determine the X-axis label based on the dropdown value
5487     switch app.XaxisUseDataDropDown.Value
5488     case '1'
5489         if ~app.CheckBox_3.Value
5490             app.XAxisEditField.Value = "Time (sec)";
5491         else %If 'Convert the Values' checkbox is ticked, remove
the unit
5492             app.XAxisEditField.Value = "Time";
5493         end
5494     case '2'
5495         if ~app.CheckBox_3.Value
5496             app.XAxisEditField.Value = "\eta (m)";
5497         else
5498             app.XAxisEditField.Value = "\eta";
5499         end
5500     case {'3', '4'} % Handles both '3' and '4'
5501         if ~app.CheckBox_3.Value
5502             app.XAxisEditField.Value = "Velocity (m/s)";
5503         else
5504             app.XAxisEditField.Value = "Velocity";
5505         end
5506     end
5507
5508     %Change the font color of 'Axes Labels: X axis' in 'Plot Style and
Layout'
5509     if ~app.CheckBox_3.Value
5510         app.XAxisEditField.FontColor = "k";
5511     end
5512
5513
5514 end
5515
5516 % Selection changed function: ButtonGroup_17
5517 function ButtonGroup_17SelectionChanged(app, event)

```

```

5518         %Display parameter options for each tab when selected
5519         if app.BathymetryButton_2.Value
5520             app.TabGroup2.SelectedTab = app.BathymetryTab;
5521         elseif app.WaveHeightButton.Value
5522             app.TabGroup2.SelectedTab = app.WaveHeightTab;
5523         elseif app.ArrivalTimeButton.Value
5524             app.TabGroup2.SelectedTab = app.ArrivalTimeTab;
5525         end
5526
5527         tabs = {app.BathymetryTab, app.WaveHeightTab, app.ArrivalTimeTab};
5528         components1 = {app.LabelSpacingEditField, app.IntervalEditField_3, app.
5529 app.TextLabelSize, app.LabelSpacingEditFieldLabel_2, app.
5530 LabelSpacingEditFieldLabel, app.SpacingEditField_4Label_4};
5531         components2 = {app.LabelSpacingEditField_2, app.wh_interval, app.
5532 TextLabelSize_2, app.LabelSpacingEditField_2Label_2, app.
5533 LabelSpacingEditField_2Label, app.SpacingEditField_4Label_5};
5534         components3 = {app.LabelSpacingEditField_2Label_3, app.
5535 LabelSpacingEditField_4, app.SpacingEditField_4Label_6, app.TextLabelSize_3, app.
5536 IntervalEditField_7, app.LabelSpacingEditField_2Label_4};
5537
5538         for k = 1:length(tabs)
5539             children = tabs{k}.Children;
5540             for i = 1:length(children)
5541                 if app.NoneButton.Value %Disable all the items when
5542 'None' is selected
5543                     children(i).Enable = 'off';
5544                 else %Enable all the items
5545                     children(i).Enable = 'on';
5546             end
5547
5548             %For 'Add Labels' suboptions, enable only when the box
5549 is ticked
5550
5551             %Bathymetry Tab
5552             for i = 1:length(components1)
5553                 if app.AddLabelsCheckBox_2.Value
5554                     components1{i}.Enable = 'on';
5555                 else
5556                     components1{i}.Enable = 'off';
5557                 end
5558             end
5559
5560             %Wave Height Tab
5561             for i = 1:length(components2)
5562                 if app.AddLabelsCheckBox_3.Value
5563                     components2{i}.Enable = 'on';
5564                 else
5565                     components2{i}.Enable = 'off';
5566                 end
5567             end
5568
5569             %Arrival Time Tab
5570             for i = 1:length(components3)
5571                 if app.AddLabelsCheckBox_5.Value
5572                     components3{i}.Enable = 'on';
5573                 else
5574                     components3{i}.Enable = 'off';
5575                 end
5576             end
5577         end

```

```

5568         end
5569     end
5570 end
5571
5572
5573 % Size changed function: ButtonGroup_17
5574 function ButtonGroup_17SizeChanged(app, event)
5575
5576 end
5577
5578 % Callback function
5579 function LineintervalEditField_3ValueChanged(app, event)
5580     app.IntervalEditField_7.Value =    app.LineintervalEditField_3.↵
Value;
5581     app.LineintervalEditField_3.FontColor = 'k';
5582
5583 end
5584
5585 % Callback function
5586 function LabelSpacingEditField_4ValueChanged(app, event)
5587     app.LabelSpacingEditField_4.FontColor = 'k';
5588
5589 end
5590
5591 % Callback function
5592 function TextLabelSize_3ValueChanged(app, event)
5593     app.TextLabelSize_3.FontColor = 'k';
5594
5595 end
5596
5597 % Callback function
5598 function ThicknessEditField_3ValueChanged(app, event)
5599     app.ThicknessEditField_3.FontColor = 'k';
5600
5601 end
5602
5603 % Callback function
5604 function MinimumdepthEditField_3ValueChanged(app, event)
5605     app.MinimumdepthEditField_3.FontColor = 'k';
5606
5607 end
5608
5609 % Callback function
5610 function MaximumdepthEditField_3ValueChanged(app, event)
5611     app.MaximumdepthEditField_3.FontColor = 'k';
5612
5613 end
5614
5615 % Drop down opening function: YaxisUseDataDropDown
5616 function YaxisUseDataDropDownOpening(app, event)
5617
5618 end
5619
5620 % Value changed function: ArrowSpacing
5621 function ArrowSpacingValueChanged(app, event)
5622     app.ArrowSpacing.FontColor = 'k';
5623
5624 end

```

```
5625
5626 % Value changed function: StartTime
5627 function StartTimeValueChanged(app, event)
5628     app.StartTime.FontColor = 'k';
5629 end
5630
5631 % Value changed function: ColorbarTextSize
5632 function ColorbarTextSizeValueChanged(app, event)
5633     app.ColorbarTextSize.FontColor = 'k';
5634
5635 end
5636
5637 % Value changed function: FlipCheckBox_5
5638 function FlipCheckBox_5ValueChanged(app, event)
5639
5640 end
5641
5642 % Value changed function: StartTime2
5643 function StartTime2ValueChanged(app, event)
5644
5645
5646 end
5647
5648 % Value changed function: PlotvectorsCheckBox
5649 function PlotvectorsCheckBoxValueChanged(app, event)
5650
5651
5652 end
5653
5654 % Value changed function: GLspacing_2
5655 function GLspacing_2ValueChanged(app, event)
5656     app.GLspacing_2.FontColor = 'k';
5657
5658 end
5659
5660 % Value changing function: XAxisEditField
5661 function XAxisEditFieldValueChanging(app, event)
5662
5663 end
5664
5665 % Value changed function: legendfirsttext
5666 function legendfirsttextValueChanged(app, event)
5667     app.legendfirsttext.FontColor = 'k';
5668
5669 end
5670
5671 % Value changed function: LineYThickness_2
5672 function LineYThickness_2ValueChanged(app, event)
5673     app.LineYThickness_2.FontColor = 'k';
5674
5675 end
5676
5677 % Value changed function: Width
5678 function WidthValueChanged(app, event)
5679     app.Width.FontColor = 'k';
5680
5681 end
5682
```



```

5683     % Value changed function: Height
5684     function HeightValueChanged(app, event)
5685         app.Height.FontColor = 'k';
5686
5687     end
5688
5689     % Value changed function: Width_2
5690     function Width_2ValueChanged(app, event)
5691         app.Width_2.FontColor = 'k';
5692
5693     end
5694
5695     % Value changed function: Height_2
5696     function Height_2ValueChanged(app, event)
5697         app.Height_2.FontColor = 'k';
5698
5699     end
5700
5701     % Value changed function: OutputDirectoryEditField_2
5702     function OutputDirectoryEditField_2ValueChanged(app, event)
5703
5704
5705     end
5706
5707     % Value changed function: mp4CheckBox_2
5708     function mp4CheckBox_2ValueChanged(app, event)
5709         if app.mp4CheckBox_2.Value
5710             app.FramerateEditFieldLabel.Visible = "on";
5711             app.FramerateEditField.Visible = "on";
5712         else
5713             app.FramerateEditFieldLabel.Visible = "off";
5714             app.FramerateEditField.Visible = "off";
5715         end
5716
5717     end
5718
5719     % Callback function
5720     function IntervalEditField_7ValueChanged(app, event)
5721         app.IntervalEditField_7.FontColor = 'k';
5722     end
5723
5724     % Button pushed function: TimeFileButton
5725     function TimeFileButtonPushed2(app, event)
5726         [timefile,path2] = uigetfile('*..*', 'Select the files',  

5727 'MultiSelect', 'on'); %uigetfile('*.txt;*.out'); %;*.tiff;*.tif;*.asdf;*.asc;*.grd;  

5728 '*.flt;*.grb;*.grid;*.grib2;*.dt0;*.dt1;*.dt2;*.ddf;*.dem;*.hgt;*.grd;*.grc;*.ers;*.  

5729 dat;*.img;*.jp2'); uigetfile('*.txt;*.shp')
5730         app.FileEditField_3.Value = string(timefile);
5731         app.FileEditField_3.FontColor = 'k';
5732
5733         if ~(ismcc || isdeployed)
5734             addpath(genpath(fullfile(string(path2))));
5735         end
5736
5737         conditions = [
5738             ~strcmp(app.FileEditField_3.Value, "0"),
5739             ~strcmp(app.FileEditField_3.Value, "00")
5740         ];

```

```

5738
5739         if app.FileEditField_3.Value == string(timefile)
5740             if any(conditions)
5741                 try
5742                     ArrivalTimeinputdata = string(path2) + string(
5743 (timefile);
5744                     app.ArrivalTimename = string(timefile);
5745                     app.FileEditField_3.BackgroundColor = 'w';
5746                     app.FileEditField_3.FontColor = 'k';
5747                     ArrivalTimeDATA = readmatrix(ArrivalTimeinputdata);
5748                     app.ArrivalTimeConverted = round(ArrivalTimeDATA ./
60, 2); %convert the values to minutes. Default threshold to read first wave
arrival time in FUNWAVE is 0.001 m
5748                     app.MaximumdepthEditField_3.Value = max(max(app.
ArrivalTimeConverted));
5749                     app.MaximumdepthEditField_3.FontColor = 'k';
5750                 end
5751             end
5752         end
5753     end
5754 end
5755
5756 % Button pushed function: Button_8
5757 function Button_8Pushed2(app, event)
5758
5759     [filename3,path3] = uigetfile('*.txt;*.shp');
5760     fullname = fullfile(path3,filename3);
5761
5762     [~, name, ext] = fileparts(filename3);
5763     if ~(ismcc || isdeployed)
5764         addpath(genpath(string(path3)));
5765     end
5766
5767     app.GaugeFname = name;
5768     app.GAUGEFILE = filename3;
5769     app.FileEditField.Value = filename3;
5770     app.FileEditField.FontColor = 'k';
5771     app.FileEditField.BackgroundColor = 'w';
5772
5773
5774     if strcmp(ext, '.shp')
5775         S = shaperead(fullname);
5776         latitudes= [S.Y]';
5777         longitudes= [S.X]';
5778         list = [round(latitudes, 4), round(longitudes, 4)];
5779         list = unique(list, 'rows');
5780         app.latGauge = list(:, 1);
5781         app.longGauge = list(:, 2);
5782     elseif strcmp(ext, '.txt')
5783         fileID = fopen(fullname, 'r');
5784         dataArray = textscan(fileID, '%f %f', 'Delimiter',
'whitespace'); %Read text file and ensure that it is tab delimited
5785         fclose(fileID);
5786         app.STATION_FILE = string(filename3);
5787         app.latGauge = dataArray{1};
5788         app.longGauge = dataArray{2};
5789     end
5790

```

```

5791     end
5792
5793 % Value changed function: AddLabelsCheckBox_4
5794 function AddLabelsCheckBox_4ValueChanged2(app, event)
5795
5796     if app.AddLabelsCheckBox_4.Value
5797         app.CoastlinecolourDropDown_5.Enable = "on";
5798         app.CoastlinecolourDropDown_4.Enable = "on";
5799         app.SizeEditField_3Label.Enable = "on";
5800         app.AlignmentLabel.Enable = "on";
5801         app.SizeEditField_3.Enable = "on";
5802         app.SizeEditField_3.Editable = "on";
5803         app.GLspacing.Enable = "on";
5804         app.GLspacing.Editable = "on";
5805         app.GLspacingLabel.Enable = "on";
5806         app.FontSizeEditField_2Label.Enable = "on";
5807         app.AlignmentLabel_2.Enable = "on";
5808     else
5809         app.CoastlinecolourDropDown_5.Enable = "off";
5810         app.CoastlinecolourDropDown_4.Enable = "off";
5811         app.SizeEditField_3Label.Enable = "off";
5812         app.AlignmentLabel.Enable = "off";
5813         app.SizeEditField_3.Enable = "off";
5814         app.SizeEditField_3.Editable = "off";
5815         app.GLspacing.Enable = "off";
5816         app.GLspacing.Editable = "off";
5817         app.GLspacingLabel.Enable = "off";
5818         app.FontSizeEditField_2Label.Enable = "off";
5819         app.AlignmentLabel_2.Enable = "off";
5820     end
5821 end
5822
5823 % Value changed function: AddLabelsCheckBox_5
5824 function AddLabelsCheckBox_5ValueChanged2(app, event)
5825     if app.AddLabelsCheckBox_5.Value
5826         app.SpacingEditField_4Label_6.Enable = "on";
5827         app.TextLabelSize_3.Enable = "on";
5828         app.LabelSpacingEditField_2Label_3.Enable = "on";
5829         app.LabelSpacingEditField_4.Enable = "on";
5830         app.LabelSpacingEditField_2Label_4.Enable = "on";
5831         app.IntervalEditField_7.Enable = "on";
5832     else
5833         app.SpacingEditField_4Label_6.Enable = "off";
5834         app.TextLabelSize_3.Enable = "off";
5835         app.LabelSpacingEditField_2Label_3.Enable = "off";
5836         app.LabelSpacingEditField_4.Enable = "off";
5837         app.LabelSpacingEditField_2Label_4.Enable = "off";
5838         app.IntervalEditField_7.Enable = "off";
5839     end
5840 end
5841
5842 % Value changed function: AddLabelsCheckBox_3
5843 function AddLabelsCheckBox_3ValueChanged2(app, event)
5844     if app.AddLabelsCheckBox_3.Value
5845         app.TextLabelSize_2.Enable = "on";
5846         app.wh_interval.Enable = "on";
5847         app.LabelSpacingEditField_2.Enable = "on";
5848         app.SpacingEditField_4Label_5.Enable = "on";

```

```

5849         app.LabelSpacingEditField_2Label_2.Enable = "on";
5850         app.LabelSpacingEditField_2Label.Enable = "on";
5851     else
5852         app.TextLabelSize_2.Enable = "off";
5853         app.wh_interval.Enable = "off";
5854         app.LabelSpacingEditField_2.Enable = "off";
5855         app.SpacingEditField_4Label_5.Enable = "off";
5856         app.LabelSpacingEditField_2Label_2.Enable = "off";
5857         app.LabelSpacingEditField_2Label.Enable = "off";
5858     end
5859 end
5860
5861 % Value changed function: AddLabelsCheckBox_2
5862 function AddLabelsCheckBox_2ValueChanged2(app, event)
5863     if app.AddLabelsCheckBox_2.Value
5864         app.IntervalEditField_3.Enable = "on";
5865         app.TextLabelSize.Enable = "on";
5866         app.LabelSpacingEditField.Enable = "on";
5867         app.SpacingEditField_4Label_4.Enable = "on";
5868         app.LabelSpacingEditFieldLabel.Enable = "on";
5869         app.LabelSpacingEditFieldLabel_2.Enable = "on";
5870     else
5871         app.IntervalEditField_3.Enable = "off";
5872         app.TextLabelSize.Enable = "off";
5873         app.LabelSpacingEditField.Enable = "off";
5874         app.SpacingEditField_4Label_4.Enable = "off";
5875         app.LabelSpacingEditFieldLabel.Enable = "off";
5876         app.LabelSpacingEditFieldLabel_2.Enable = "off";
5877     end
5878 end
5879 end
5880
5881 % Component initialization
5882 methods (Access = private)
5883
5884     % Create UIFigure and components
5885     function createComponents(app)
5886
5887         % Create UIFigure and hide until all components are created
5888         app.UIFigure = uifigure('Visible', 'off');
5889         app.UIFigure.AutoResizeChildren = 'off';
5890         app.UIFigure.Position = [1 1 542 861];
5891         app.UIFigure.Name = 'MATLAB App';
5892         app.UIFigure.Scrollable = 'on';
5893
5894         % Create TabGroup
5895         app.TabGroup = uitabgroup(app.UIFigure);
5896         app.TabGroup.AutoResizeChildren = 'off';
5897         app.TabGroup.Position = [1 1 542 861];
5898
5899         % Create WaveHeightMapTab
5900         app.WaveHeightMapTab = uitab(app.TabGroup);
5901         app.WaveHeightMapTab.AutoResizeChildren = 'off';
5902         app.WaveHeightMapTab.SizeChangedFcn = createCallbackFcn(app, ↵
@WaveHeightMapTabSizeChanged, true);
5903         app.WaveHeightMapTab.Title = 'Wave Height Map';
5904         app.WaveHeightMapTab.Scrollable = 'on';
5905

```

```

5906 % Create Panel_3
5907 app.Panel_3 = uipanel(app.WaveHeightMapTab);
5908 app.Panel_3.AutoResizeChildren = 'off';
5909 app.Panel_3.ForegroundColor = [0.149 0.149 0.149];
5910 app.Panel_3.Position = [11 662 520 160];
5911
5912 % Create FileTextArea
5913 app.FileTextArea = uitextarea(app.Panel_3);
5914 app.FileTextArea.ValueChangedFcn = createCallbackFcn(app, ↵
@FileTextAreaValueChanged, true);
5915 app.FileTextArea.Editable = 'off';
5916 app.FileTextArea.FontColor = [0.149 0.149 0.149];
5917 app.FileTextArea.Tooltip = {' '};
5918 app.FileTextArea.Placeholder = 'eta_xxxx; hmax_xxxx;';
5919 app.FileTextArea.Position = [43 65 95 55];
5920
5921 % Create Button_14
5922 app.Button_14 = uibutton(app.Panel_3, 'push');
5923 app.Button_14.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_14Pushed, true);
5924 app.Button_14.FontAngle = 'italic';
5925 app.Button_14.FontColor = [0.651 0.651 0.651];
5926 app.Button_14.Tooltip = {'Load the files'; ''; 'Files must start ↵
with 'eta' or 'hmax' ' '};
5927 app.Button_14.Position = [146 101 19 19];
5928 app.Button_14.Text = '...';
5929
5930 % Create SouthwestCornerLabel
5931 app.SouthwestCornerLabel = uilabel(app.Panel_3);
5932 app.SouthwestCornerLabel.FontWeight = 'bold';
5933 app.SouthwestCornerLabel.Tooltip = {'Coordinates of the southwest ↵
corner of the loaded files.'};
5934 app.SouthwestCornerLabel.Position = [232 121 110 22];
5935 app.SouthwestCornerLabel.Text = 'Southwest Corner';
5936
5937 % Create LongitudeEditFieldLabel
5938 app.LongitudeEditFieldLabel = uilabel(app.Panel_3);
5939 app.LongitudeEditFieldLabel.HorizontalAlignment = 'right';
5940 app.LongitudeEditFieldLabel.WordWrap = 'on';
5941 app.LongitudeEditFieldLabel.Tooltip = {'West boundary'; ''; 'Unit: ↵
degrees / meters'};
5942 app.LongitudeEditFieldLabel.Position = [228 98 60 22];
5943 app.LongitudeEditFieldLabel.Text = 'Longitude';
5944
5945 % Create LongitudeEditField
5946 app.LongitudeEditField = uieditfield(app.Panel_3, 'numeric');
5947 app.LongitudeEditField.ValueDisplayFormat = '%8.4f';
5948 app.LongitudeEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@LongitudeEditFieldValueChanged, true);
5949 app.LongitudeEditField.FontColor = [0.651 0.651 0.651];
5950 app.LongitudeEditField.Tooltip = {' '};
5951 app.LongitudeEditField.Position = [296 104 60 16];
5952
5953 % Create LatitudeEditFieldLabel
5954 app.LatitudeEditFieldLabel = uilabel(app.Panel_3);
5955 app.LatitudeEditFieldLabel.HorizontalAlignment = 'right';
5956 app.LatitudeEditFieldLabel.WordWrap = 'on';
5957 app.LatitudeEditFieldLabel.Tooltip = {'South boundary'; ''; 'Unit: ↵

```

```

degrees / meters'});
5958         app.LatitudeEditFieldLabel.Position = [234 81 44 17];
5959         app.LatitudeEditFieldLabel.Text = 'Latitude';
5960
5961         % Create LatitudeEditField
5962         app.LatitudeEditField = uieditfield(app.Panel_3, 'numeric');
5963         app.LatitudeEditField.ValueDisplayFormat = '%8.4f';
5964         app.LatitudeEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@LatitudeEditFieldValueChanged, true);
5965         app.LatitudeEditField.FontColor = [0.651 0.651 0.651];
5966         app.LatitudeEditField.Tooltip = {''};
5967         app.LatitudeEditField.Position = [296 82 60 16];
5968
5969         % Create Button_15
5970         app.Button_15 = uibutton(app.Panel_3, 'push');
5971         app.Button_15.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_15Pushed, true);
5972         app.Button_15.FontAngle = 'italic';
5973         app.Button_15.FontColor = [0.651 0.651 0.651];
5974         app.Button_15.Tooltip = {'Load the file'; ''; 'Select the ↵
bathymetry used in the simulation, or the output 'mask_xxxx' file.'; ''; 'Format ↵
accepted: .txt, .tif, mask_'};
5975         app.Button_15.Position = [146 11 19 19];
5976         app.Button_15.Text = '...';
5977
5978         % Create DepthFileEditField
5979         app.DepthFileEditField = uieditfield(app.Panel_3, 'text');
5980         app.DepthFileEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@DepthFileEditFieldValueChanged, true);
5981         app.DepthFileEditField.Editable = 'off';
5982         app.DepthFileEditField.FontColor = [0.149 0.149 0.149];
5983         app.DepthFileEditField.Tooltip = {''};
5984         app.DepthFileEditField.Placeholder = '.txt, .tif., mask_';
5985         app.DepthFileEditField.Position = [44 13 96 16];
5986
5987         % Create FilesLabel
5988         app.FilesLabel = uilabel(app.Panel_3);
5989         app.FilesLabel.FontWeight = 'bold';
5990         app.FilesLabel.Tooltip = {'Load the files'; ''; 'Files must start ↵
with ''eta'' or ''hmax'''};
5991         app.FilesLabel.Position = [44 121 32 22];
5992         app.FilesLabel.Text = 'Files';
5993
5994         % Create BathymetryLabel
5995         app.BathymetryLabel = uilabel(app.Panel_3);
5996         app.BathymetryLabel.FontWeight = 'bold';
5997         app.BathymetryLabel.Tooltip = {'Load the file'; ''; 'Select the ↵
bathymetry used in the simulation, or the output 'mask_xxxx' file.'; ''; 'Format ↵
accepted: .txt, .tif, mask_'};
5998         app.BathymetryLabel.Position = [44 31 71 22];
5999         app.BathymetryLabel.Text = 'Bathymetry';
6000
6001         % Create TotalSimuilationTimesecEditFieldLabel_3
6002         app.TotalSimuilationTimesecEditFieldLabel_3 = uilabel(app. ↵
Panel_3);
6003         app.TotalSimuilationTimesecEditFieldLabel_3.FontWeight = 'bold';
6004         app.TotalSimuilationTimesecEditFieldLabel_3.Tooltip = {'For values ↵
>60 seconds, the unit will be automatically converted to minutes or hours. The user ↵

```

```

can verify whether the conversion is correct by double-checking the LOG.TXT file
generated after the simulation.'; ''; 'For example, in the line:'; '"PRINTING FILE
NO. 1 TIME/TOTAL: 3600.000 / 7200.000"'; 'the hmax or eta 00001 entry will be
labelled with a title corresponding to 1 hour.'};
6005         app.TotalSimulationTimesecEditFieldLabel_3.Position = [232 31 98
22];
6006         app.TotalSimulationTimesecEditFieldLabel_3.Text = 'Simulation
Time';
6007
6008         % Create TotalSimulationTimesecEditField_3
6009         app.TotalSimulationTimesecEditField_3 = uicontrol(app.Panel_3,
'numeric');
6010         app.TotalSimulationTimesecEditField_3.Limits = [1 Inf];
6011         app.TotalSimulationTimesecEditField_3.ValueDisplayFormat = '%8.1
f';
6012         app.TotalSimulationTimesecEditField_3.ValueChangedFcn =
createCallbackFcn(app, @TotalSimulationTimesecEditField_3ValueChanged, true);
6013         app.TotalSimulationTimesecEditField_3.FontColor = [0.651 0.651
0.651];
6014         app.TotalSimulationTimesecEditField_3.Tooltip = {'';
6015         app.TotalSimulationTimesecEditField_3.Position = [409 16 60 16];
6016         app.TotalSimulationTimesecEditField_3.Value = 1;
6017
6018         % Create IntervalLabel
6019         app.IntervalLabel = uicontrol(app.Panel_3);
6020         app.IntervalLabel.HorizontalAlignment = 'right';
6021         app.IntervalLabel.Tooltip = {'Snapshot interval.'; ''; 'Same as
PLOT_INTV in input.txt'; ''; 'Unit: seconds'};
6022         app.IntervalLabel.Position = [358 13 45 22];
6023         app.IntervalLabel.Text = 'Interval';
6024
6025         % Create StartTime
6026         app.StartTime = uicontrol(app.Panel_3, 'numeric');
6027         app.StartTime.Limits = [0 Inf];
6028         app.StartTime.ValueDisplayFormat = '%8.1f';
6029         app.StartTime.ValueChangedFcn = createCallbackFcn(app,
@StartTimeValueChanged, true);
6030         app.StartTime.FontColor = [0.651 0.651 0.651];
6031         app.StartTime.Tooltip = {'';
6032         app.StartTime.Position = [296 16 47 16];
6033
6034         % Create StartLabel
6035         app.StartLabel = uicontrol(app.Panel_3);
6036         app.StartLabel.HorizontalAlignment = 'right';
6037         app.StartLabel.Tooltip = {'For imported initial tsunami files with
a non-zero start time, input the value here. Example: for a 30-second snapshot,
input ''30''.'; ''; 'Unit: seconds'};
6038         app.StartLabel.Position = [255 13 31 22];
6039         app.StartLabel.Text = 'Start';
6040
6041         % Create GridSizeLabel_2
6042         app.GridSizeLabel_2 = uicontrol(app.Panel_3);
6043         app.GridSizeLabel_2.FontWeight = 'bold';
6044         app.GridSizeLabel_2.Tooltip = {'Resolution'; 'Unit: degrees /
meters'};
6045         app.GridSizeLabel_2.Position = [386 121 57 22];
6046         app.GridSizeLabel_2.Text = 'Grid Size';
6047

```



```

6048 % Create XEditField_2Label
6049 app.XEditField_2Label = uilabel(app.Panel_3);
6050 app.XEditField_2Label.HorizontalAlignment = 'center';
6051 app.XEditField_2Label.Tooltip = {'X direction'; ''; 'Unit: degrees↵
/ meters'};
6052 app.XEditField_2Label.Position = [388 98 13 22];
6053 app.XEditField_2Label.Text = 'X ';
6054
6055 % Create gridX
6056 app.gridX = uieditfield(app.Panel_3, 'numeric');
6057 app.gridX.Limits = [0 Inf];
6058 app.gridX.ValueDisplayFormat = '%8.5f';
6059 app.gridX.ValueChangedFcn = createCallbackFcn(app, ↵
@gridXValueChanged, true);
6060 app.gridX.FontColor = [0.651 0.651 0.651];
6061 app.gridX.Tooltip = {' '};
6062 app.gridX.Position = [409 104 60 16];
6063
6064 % Create YEditField_2Label
6065 app.YEditField_2Label = uilabel(app.Panel_3);
6066 app.YEditField_2Label.HorizontalAlignment = 'center';
6067 app.YEditField_2Label.Tooltip = {'Y direction'; ''; 'Unit: degrees↵
/ meters'};
6068 app.YEditField_2Label.Position = [386 79 13 21];
6069 app.YEditField_2Label.Text = 'Y';
6070
6071 % Create gridY
6072 app.gridY = uieditfield(app.Panel_3, 'numeric');
6073 app.gridY.Limits = [0 Inf];
6074 app.gridY.ValueDisplayFormat = '%8.5f';
6075 app.gridY.ValueChangedFcn = createCallbackFcn(app, ↵
@gridYValueChanged, true);
6076 app.gridY.FontColor = [0.651 0.651 0.651];
6077 app.gridY.Tooltip = {' '};
6078 app.gridY.Position = [409 82 60 16];
6079
6080 % Create InputDataLabel
6081 app.InputDataLabel = uilabel(app.WaveHeightMapTab);
6082 app.InputDataLabel.BackgroundColor = [0.9412 0.9412 0.9412];
6083 app.InputDataLabel.FontSize = 15;
6084 app.InputDataLabel.FontWeight = 'bold';
6085 app.InputDataLabel.FontColor = [0.0314 0.3686 0.6];
6086 app.InputDataLabel.Tooltip = {' '};
6087 app.InputDataLabel.Position = [38 813 97 22];
6088 app.InputDataLabel.Text = ' Input Data ';
6089
6090 % Create Panel_7
6091 app.Panel_7 = uipanel(app.WaveHeightMapTab);
6092 app.Panel_7.AutoResizeChildren = 'off';
6093 app.Panel_7.TitlePosition = 'righttop';
6094 app.Panel_7.FontSize = 10;
6095 app.Panel_7.Position = [11 315 520 168];
6096
6097 % Create TabGroup2
6098 app.TabGroup2 = uitabgroup(app.Panel_7);
6099 app.TabGroup2.AutoResizeChildren = 'off';
6100 app.TabGroup2.Position = [0 0 520 153];
6101

```



```

6102 % Create BathymetryTab
6103 app.BathymetryTab = uitab(app.TabGroup2);
6104 app.BathymetryTab.AutoResizeChildren = 'off';
6105 app.BathymetryTab.Title = 'Bathymetry';
6106
6107 % Create MaximumEditFieldLabel
6108 app.MaximumEditFieldLabel = uilabel(app.BathymetryTab);
6109 app.MaximumEditFieldLabel.Enable = 'off';
6110 app.MaximumEditFieldLabel.Tooltip = {'Higher value indicate deeper
waters'};
6111
6112 app.MaximumEditFieldLabel.Position = [48 25 58 22];
6113 app.MaximumEditFieldLabel.Text = 'Maximum';
6114
6115 % Create MaximumEditField
6116 app.MaximumEditField = uieditfield(app.BathymetryTab, 'numeric');
6117 app.MaximumEditField.Limits = [1 Inf];
6118 app.MaximumEditField.ValueDisplayFormat = '%8.1f';
6119 app.MaximumEditField.Enable = 'off';
6120 app.MaximumEditField.Tooltip = {' '};
6121 app.MaximumEditField.Position = [107 28 50 16];
6122 app.MaximumEditField.Value = 1000;
6123
6124 % Create MinimumEditFieldLabel
6125 app.MinimumEditFieldLabel = uilabel(app.BathymetryTab);
6126 app.MinimumEditFieldLabel.Enable = 'off';
6127 app.MinimumEditFieldLabel.Tooltip = {'0 value = coastline'; ''};
6128 'Lower values indicate shallower waters.'};
6129 app.MinimumEditFieldLabel.Position = [48 71 55 22];
6130 app.MinimumEditFieldLabel.Text = 'Minimum';
6131
6132 % Create MinimumEditField
6133 app.MinimumEditField = uieditfield(app.BathymetryTab, 'numeric');
6134 app.MinimumEditField.Limits = [0 Inf];
6135 app.MinimumEditField.ValueDisplayFormat = '%8.1f';
6136 app.MinimumEditField.Enable = 'off';
6137 app.MinimumEditField.Tooltip = {' '};
6138 app.MinimumEditField.Position = [107 74 50 16];
6139
6140 % Create IntervalEditField_5Label
6141 app.IntervalEditField_5Label = uilabel(app.BathymetryTab);
6142 app.IntervalEditField_5Label.Enable = 'off';
6143 app.IntervalEditField_5Label.Tooltip = {'Interval between adjacent
contours'};
6144
6145 app.IntervalEditField_5Label.Position = [48 48 45 22];
6146 app.IntervalEditField_5Label.Text = 'Interval';
6147
6148 % Create IntervalEditField_5
6149 app.IntervalEditField_5 = uieditfield(app.BathymetryTab,
'numeric');
6150 app.IntervalEditField_5.Limits = [0 Inf];
6151 app.IntervalEditField_5.ValueDisplayFormat = '%8.1f';
6152 app.IntervalEditField_5.Enable = 'off';
6153 app.IntervalEditField_5.Tooltip = {' '};
6154 app.IntervalEditField_5.Position = [107 51 40 17];
6155 app.IntervalEditField_5.Value = 500;
6156
6157 % Create DepthRangeLabel
6158 app.DepthRangeLabel = uilabel(app.BathymetryTab);

```

```

6156     app.DepthRangeLabel.FontWeight = 'bold';
6157     app.DepthRangeLabel.Enable = 'off';
6158     app.DepthRangeLabel.Position = [48 95 80 22];
6159     app.DepthRangeLabel.Text = 'Depth Range';
6160
6161     % Create LabelSpacingEditFieldLabel_2
6162     app.LabelSpacingEditFieldLabel_2 = uilabel(app.BathymetryTab);
6163     app.LabelSpacingEditFieldLabel_2.WordWrap = 'on';
6164     app.LabelSpacingEditFieldLabel_2.Enable = 'off';
6165     app.LabelSpacingEditFieldLabel_2.Tooltip = {'Adjust the distance↵
between the labels'; ''; 'Higher values reduce label crowding.'; ''; 'Unit: 1 =↵
1/72 inch'};
6166     app.LabelSpacingEditFieldLabel_2.Position = [377 25 58 22];
6167     app.LabelSpacingEditFieldLabel_2.Text = 'Spacing';
6168
6169     % Create LabelSpacingEditField
6170     app.LabelSpacingEditField = uieditfield(app.BathymetryTab, ↵
'numeric');
6171     app.LabelSpacingEditField.Limits = [0 Inf];
6172     app.LabelSpacingEditField.ValueDisplayFormat = '%8.1f';
6173     app.LabelSpacingEditField.Enable = 'off';
6174     app.LabelSpacingEditField.Tooltip = {'';
6175     app.LabelSpacingEditField.Position = [426 28 47 16];
6176     app.LabelSpacingEditField.Value = 1000;
6177
6178     % Create WidthEditFieldLabel
6179     app.WidthEditFieldLabel = uilabel(app.BathymetryTab);
6180     app.WidthEditFieldLabel.Enable = 'off';
6181     app.WidthEditFieldLabel.Tooltip = {'Line thickness'};
6182     app.WidthEditFieldLabel.Position = [196 71 37 22];
6183     app.WidthEditFieldLabel.Text = 'Width';
6184
6185     % Create WidthEditField
6186     app.WidthEditField = uieditfield(app.BathymetryTab, 'numeric');
6187     app.WidthEditField.Limits = [0 Inf];
6188     app.WidthEditField.ValueDisplayFormat = '%3.1f';
6189     app.WidthEditField.Enable = 'off';
6190     app.WidthEditField.Tooltip = {'';
6191     app.WidthEditField.Position = [236 74 35 16];
6192     app.WidthEditField.Value = 0.1;
6193
6194     % Create LineLabel
6195     app.LineLabel = uilabel(app.BathymetryTab);
6196     app.LineLabel.FontWeight = 'bold';
6197     app.LineLabel.Enable = 'off';
6198     app.LineLabel.Position = [196 95 30 22];
6199     app.LineLabel.Text = 'Line';
6200
6201     % Create StyleDropDownLabel
6202     app.StyleDropDownLabel = uilabel(app.BathymetryTab);
6203     app.StyleDropDownLabel.Enable = 'off';
6204     app.StyleDropDownLabel.Position = [196 49 32 22];
6205     app.StyleDropDownLabel.Text = 'Style';
6206
6207     % Create StyleDropDown
6208     app.StyleDropDown = uidropdown(app.BathymetryTab);
6209     app.StyleDropDown.Items = {'Solid', 'Dashed', 'Dotted', 'Dash-↵
dotted'};

```

```

6210     app.StyleDropDown.Enable = 'off';
6211     app.StyleDropDown.Position = [236 52 104 16];
6212     app.StyleDropDown.Value = 'Solid';
6213
6214     % Create ColorDropDownLabel
6215     app.ColorDropDownLabel = uilabel(app.BathymetryTab);
6216     app.ColorDropDownLabel.Enable = 'off';
6217     app.ColorDropDownLabel.Position = [196 25 35 22];
6218     app.ColorDropDownLabel.Text = 'Color';
6219
6220     % Create ColorDropDown
6221     app.ColorDropDown = uidropdown(app.BathymetryTab);
6222     app.ColorDropDown.Items = {'Black', 'Dark gray', 'Medium gray', '
6223     'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
6224     app.ColorDropDown.Enable = 'off';
6225     app.ColorDropDown.Position = [236 28 104 16];
6226     app.ColorDropDown.Value = 'Dark gray';
6227
6228     % Create AddLabelsCheckBox_2
6229     app.AddLabelsCheckBox_2 = uicheckbox(app.BathymetryTab);
6230     app.AddLabelsCheckBox_2.ValueChangedFcn = createCallbackFcn(app, '
6231     @AddLabelsCheckBox_2ValueChanged2', true);
6232     app.AddLabelsCheckBox_2.Enable = 'off';
6233     app.AddLabelsCheckBox_2.Tooltip = {'Unit: meters'};
6234     app.AddLabelsCheckBox_2.Text = ' Add Labels';
6235     app.AddLabelsCheckBox_2.WordWrap = 'on';
6236     app.AddLabelsCheckBox_2.FontWeight = 'bold';
6237     app.AddLabelsCheckBox_2.Position = [377 95 93 22];
6238
6239     % Create TextLabelSize
6240     app.TextLabelSize = uieditfield(app.BathymetryTab, 'numeric');
6241     app.TextLabelSize.Limits = [0 30];
6242     app.TextLabelSize.ValueDisplayFormat = '%8.0f';
6243     app.TextLabelSize.Enable = 'off';
6244     app.TextLabelSize.Tooltip = {''};
6245     app.TextLabelSize.Position = [426 74 30 16];
6246     app.TextLabelSize.Value = 8;
6247
6248     % Create SpacingEditField_4Label_4
6249     app.SpacingEditField_4Label_4 = uilabel(app.BathymetryTab);
6250     app.SpacingEditField_4Label_4.WordWrap = 'on';
6251     app.SpacingEditField_4Label_4.Enable = 'off';
6252     app.SpacingEditField_4Label_4.Position = [377 71 40 22];
6253     app.SpacingEditField_4Label_4.Text = 'Size';
6254
6255     % Create LabelSpacingEditFieldLabel
6256     app.LabelSpacingEditFieldLabel = uilabel(app.BathymetryTab);
6257     app.LabelSpacingEditFieldLabel.Enable = 'off';
6258     app.LabelSpacingEditFieldLabel.Tooltip = {'Add labels at specified
6259     intervals.'; ''; 'Suggestion: Set at intervals divisible by the depth range
6260     interval'};
6261     app.LabelSpacingEditFieldLabel.Position = [377 49 45 22];
6262     app.LabelSpacingEditFieldLabel.Text = 'Interval';
6263
6264     % Create IntervalEditField_3
6265     app.IntervalEditField_3 = uieditfield(app.BathymetryTab, '
6266     numeric');
6267     app.IntervalEditField_3.Limits = [0.01 Inf];

```

```

6263 app.IntervalEditField_3.ValueDisplayFormat = '%8.1f';
6264 app.IntervalEditField_3.Enable = 'off';
6265 app.IntervalEditField_3.Tooltip = {' '};
6266 app.IntervalEditField_3.Position = [426 52 47 16];
6267 app.IntervalEditField_3.Value = 500;
6268
6269 % Create WaveHeightTab
6270 app.WaveHeightTab = uitab(app.TabGroup2);
6271 app.WaveHeightTab.AutoResizeChildren = 'off';
6272 app.WaveHeightTab.Title = 'Wave Height';
6273
6274 % Create MinimumdepthEditField_2
6275 app.MinimumdepthEditField_2 = uieditfield(app.WaveHeightTab, '
'numeric');
6276 app.MinimumdepthEditField_2.ValueDisplayFormat = '%8.1f';
6277 app.MinimumdepthEditField_2.Tooltip = {'Minimum value of the wave
height'};
6278 app.MinimumdepthEditField_2.Position = [107 74 45 16];
6279
6280 % Create MaximumdepthEditField_2
6281 app.MaximumdepthEditField_2 = uieditfield(app.WaveHeightTab, '
'numeric');
6282 app.MaximumdepthEditField_2.Limits = [1 Inf];
6283 app.MaximumdepthEditField_2.ValueDisplayFormat = '%8.1f';
6284 app.MaximumdepthEditField_2.Tooltip = {'Maximum value of the wave
height'};
6285 app.MaximumdepthEditField_2.Position = [107 29 45 16];
6286 app.MaximumdepthEditField_2.Value = 1;
6287
6288 % Create LineintervalEditField_2
6289 app.LineintervalEditField_2 = uieditfield(app.WaveHeightTab, '
'numeric');
6290 app.LineintervalEditField_2.Limits = [0 Inf];
6291 app.LineintervalEditField_2.ValueDisplayFormat = '%8.2f';
6292 app.LineintervalEditField_2.HorizontalAlignment = 'center';
6293 app.LineintervalEditField_2.Tooltip = {'Interval between adjacent
contours'};
6294 app.LineintervalEditField_2.Position = [107 51 34 16];
6295 app.LineintervalEditField_2.Value = 0.5;
6296
6297 % Create MaximumdepthEditFieldLabel_2
6298 app.MaximumdepthEditFieldLabel_2 = uilabel(app.WaveHeightTab);
6299 app.MaximumdepthEditFieldLabel_2.Position = [48 25 53 22];
6300 app.MaximumdepthEditFieldLabel_2.Text = 'Maximum';
6301
6302 % Create MinimumdepthEditFieldLabel_2
6303 app.MinimumdepthEditFieldLabel_2 = uilabel(app.WaveHeightTab);
6304 app.MinimumdepthEditFieldLabel_2.Position = [48 71 51 22];
6305 app.MinimumdepthEditFieldLabel_2.Text = 'Minimum';
6306
6307 % Create MaximumdepthEditFieldLabel_3
6308 app.MaximumdepthEditFieldLabel_3 = uilabel(app.WaveHeightTab);
6309 app.MaximumdepthEditFieldLabel_3.Position = [48 48 48 22];
6310 app.MaximumdepthEditFieldLabel_3.Text = 'Interval';
6311
6312 % Create ContourRangeLabel_2
6313 app.ContourRangeLabel_2 = uilabel(app.WaveHeightTab);
6314 app.ContourRangeLabel_2.FontWeight = 'bold';

```

```

6315     app.ContourRangeLabel_2.Position = [48 95 92 22];
6316     app.ContourRangeLabel_2.Text = 'Contour Range';
6317
6318     % Create LineLabel_2
6319     app.LineLabel_2 = uilabel(app.WaveHeightTab);
6320     app.LineLabel_2.FontWeight = 'bold';
6321     app.LineLabel_2.Position = [196 95 30 22];
6322     app.LineLabel_2.Text = 'Line';
6323
6324     % Create ThicknessEditField_2Label
6325     app.ThicknessEditField_2Label = uilabel(app.WaveHeightTab);
6326     app.ThicknessEditField_2Label.Tooltip = {'Line thickness'};
6327     app.ThicknessEditField_2Label.Position = [196 71 37 22];
6328     app.ThicknessEditField_2Label.Text = 'Width';
6329
6330     % Create ThicknessEditField_2
6331     app.ThicknessEditField_2 = uieditfield(app.WaveHeightTab, '
'numeric');
6332
6333     app.ThicknessEditField_2.Limits = [0 20];
6334     app.ThicknessEditField_2.ValueDisplayFormat = '%3.1f';
6335     app.ThicknessEditField_2.Position = [236 74 35 16];
6336     app.ThicknessEditField_2.Value = 0.1;
6337
6338     % Create ColorDropDown_2Label
6339     app.ColorDropDown_2Label = uilabel(app.WaveHeightTab);
6340     app.ColorDropDown_2Label.Position = [196 25 35 22];
6341     app.ColorDropDown_2Label.Text = 'Color';
6342
6343     % Create ColorDropDown_2
6344     app.ColorDropDown_2 = uidropdown(app.WaveHeightTab);
6345     app.ColorDropDown_2.Items = {'Black', 'Dark gray', 'Medium gray', '
'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
6346     app.ColorDropDown_2.Position = [236 28 104 16];
6347     app.ColorDropDown_2.Value = 'Dark gray';
6348
6349     % Create StyleDropDown_2Label
6350     app.StyleDropDown_2Label = uilabel(app.WaveHeightTab);
6351     app.StyleDropDown_2Label.Position = [196 49 32 22];
6352     app.StyleDropDown_2Label.Text = 'Style';
6353
6354     % Create StyleDropDown_2
6355     app.StyleDropDown_2 = uidropdown(app.WaveHeightTab);
6356     app.StyleDropDown_2.Items = {'Solid', 'Dashed', 'Dotted', 'Dash-
'dotted'};
6357     app.StyleDropDown_2.Position = [236 52 104 16];
6358     app.StyleDropDown_2.Value = 'Solid';
6359
6360     % Create AddLabelsCheckBox_3
6361     app.AddLabelsCheckBox_3 = uicheckbox(app.WaveHeightTab);
6362     app.AddLabelsCheckBox_3.ValueChangedFcn = createCallbackFcn(app, '
'@AddLabelsCheckBox_3ValueChanged2', true);
6363     app.AddLabelsCheckBox_3.Tooltip = {'Unit: meters'};
6364     app.AddLabelsCheckBox_3.Text = ' Add Labels';
6365     app.AddLabelsCheckBox_3.FontWeight = 'bold';
6366     app.AddLabelsCheckBox_3.Position = [377 95 90 22];
6367
6368     % Create TextLabelSize_2
        app.TextLabelSize_2 = uieditfield(app.WaveHeightTab, 'numeric');

```

```

6369     app.TextLabelSize_2.Limits = [0 Inf];
6370     app.TextLabelSize_2.ValueDisplayFormat = '%8.0f';
6371     app.TextLabelSize_2.Enable = 'off';
6372     app.TextLabelSize_2.Position = [426 74 30 16];
6373     app.TextLabelSize_2.Value = 8;
6374
6375     % Create SpacingEditField_4Label_5
6376     app.SpacingEditField_4Label_5 = uilabel(app.WaveHeightTab);
6377     app.SpacingEditField_4Label_5.WordWrap = 'on';
6378     app.SpacingEditField_4Label_5.Enable = 'off';
6379     app.SpacingEditField_4Label_5.Position = [377 71 40 22];
6380     app.SpacingEditField_4Label_5.Text = 'Size';
6381
6382     % Create LabelSpacingEditField_2Label_2
6383     app.LabelSpacingEditField_2Label_2 = uilabel(app.WaveHeightTab);
6384     app.LabelSpacingEditField_2Label_2.WordWrap = 'on';
6385     app.LabelSpacingEditField_2Label_2.Enable = 'off';
6386     app.LabelSpacingEditField_2Label_2.Tooltip = {'Adjust the distance
between the labels'; ''; 'Higher values reduce label crowding.'; ''; 'Unit: 1 =
1/72 inch'};
6387     app.LabelSpacingEditField_2Label_2.Position = [377 27 46 18];
6388     app.LabelSpacingEditField_2Label_2.Text = 'Spacing';
6389
6390     % Create LabelSpacingEditField_2
6391     app.LabelSpacingEditField_2 = uieditfield(app.WaveHeightTab, '
numeric');
6392     app.LabelSpacingEditField_2.Limits = [0 Inf];
6393     app.LabelSpacingEditField_2.ValueDisplayFormat = '%8.1f';
6394     app.LabelSpacingEditField_2.Enable = 'off';
6395     app.LabelSpacingEditField_2.Tooltip = {''};
6396     app.LabelSpacingEditField_2.Position = [426 28 47 16];
6397     app.LabelSpacingEditField_2.Value = 1000;
6398
6399     % Create LabelSpacingEditField_2Label
6400     app.LabelSpacingEditField_2Label = uilabel(app.WaveHeightTab);
6401     app.LabelSpacingEditField_2Label.Enable = 'off';
6402     app.LabelSpacingEditField_2Label.Tooltip = {'Add labels at
specified intervals'; ''; 'Suggestion: Set at intervals divisible by the contour
range interval'};
6403     app.LabelSpacingEditField_2Label.Position = [377 49 45 22];
6404     app.LabelSpacingEditField_2Label.Text = 'Interval';
6405
6406     % Create wh_interval
6407     app.wh_interval = uieditfield(app.WaveHeightTab, 'numeric');
6408     app.wh_interval.Limits = [0.01 Inf];
6409     app.wh_interval.ValueDisplayFormat = '%8.1f';
6410     app.wh_interval.Enable = 'off';
6411     app.wh_interval.Tooltip = {''};
6412     app.wh_interval.Position = [426 52 47 16];
6413     app.wh_interval.Value = 1;
6414
6415     % Create ArrivalTimeTab
6416     app.ArrivalTimeTab = uitab(app.TabGroup2);
6417     app.ArrivalTimeTab.AutoResizeChildren = 'off';
6418     app.ArrivalTimeTab.Title = 'Arrival Time';
6419     app.ArrivalTimeTab.ForegroundColor = [0.9412 0.9412 0.9412];
6420
6421     % Create TimeFileButton

```

```

6422         app.TimeFileButton = uibutton(app.ArrivalTimeTab, 'push');
6423         app.TimeFileButton.ButtonPushedFcn = createCallbackFcn(app, ↵
@TimeFileButtonPushed2, true);
6424         app.TimeFileButton.Tooltip = {'Load the file.'; ''; 'File↵
extension must be .shp or .txt'; ''; 'Unit: automatically converted to minutes. '};↵
''; ''; 'The default threshold for detecting the first wave arrival time in FUNWAVE↵
is 0.001 meters.'};
6425         app.TimeFileButton.Position = [211 105 19 19];
6426         app.TimeFileButton.Text = '...';
6427
6428         % Create FileEditField_3Label
6429         app.FileEditField_3Label = uilabel(app.ArrivalTimeTab);
6430         app.FileEditField_3Label.FontWeight = 'bold';
6431         app.FileEditField_3Label.Position = [48 103 29 22];
6432         app.FileEditField_3Label.Text = 'File:';
6433
6434         % Create FileEditField_3
6435         app.FileEditField_3 = uieditfield(app.ArrivalTimeTab, 'text');
6436         app.FileEditField_3.Editable = 'off';
6437         app.FileEditField_3.Tooltip = {'Unit: automatically converted to↵
meters '; ''; 'Default threshold for the first wave arrival time in FUNWAVE is↵
0.001 m.'};
6438         app.FileEditField_3.Placeholder = 'time_xxxxx';
6439         app.FileEditField_3.Position = [80 106 125 16];
6440
6441         % Create ContourRangeLabel
6442         app.ContourRangeLabel = uilabel(app.ArrivalTimeTab);
6443         app.ContourRangeLabel.FontWeight = 'bold';
6444         app.ContourRangeLabel.Tooltip = {'Unit: minutes'};
6445         app.ContourRangeLabel.Position = [49 78 92 22];
6446         app.ContourRangeLabel.Text = 'Contour Range';
6447
6448         % Create MaximumdepthEditField_3
6449         app.MaximumdepthEditField_3 = uieditfield(app.ArrivalTimeTab, ↵
'numeric');
6450         app.MaximumdepthEditField_3.Limits = [1 Inf];
6451         app.MaximumdepthEditField_3.ValueDisplayFormat = '%8.1f';
6452         app.MaximumdepthEditField_3.Tooltip = {'Unit: minutes'};
6453         app.MaximumdepthEditField_3.Position = [106 18 55 16];
6454         app.MaximumdepthEditField_3.Value = 1;
6455
6456         % Create MaximumdepthEditFieldLabel_9
6457         app.MaximumdepthEditFieldLabel_9 = uilabel(app.ArrivalTimeTab);
6458         app.MaximumdepthEditFieldLabel_9.Tooltip = {'Unit: minutes'};
6459         app.MaximumdepthEditFieldLabel_9.Position = [49 15 53 22];
6460         app.MaximumdepthEditFieldLabel_9.Text = 'Maximum';
6461
6462         % Create LineintervalEditField_3
6463         app.LineintervalEditField_3 = uieditfield(app.ArrivalTimeTab, ↵
'numeric');
6464         app.LineintervalEditField_3.Limits = [0 Inf];
6465         app.LineintervalEditField_3.ValueDisplayFormat = '%8.1f';
6466         app.LineintervalEditField_3.Tooltip = {'Interval between adjacent↵
contours.'; ''; 'Unit: minutes'};
6467         app.LineintervalEditField_3.Position = [106 38 55 16];
6468         app.LineintervalEditField_3.Value = 100;
6469
6470         % Create MaximumdepthEditFieldLabel 10

```



```

6471 app.MaximumdepthEditFieldLabel_10 = uilabel(app.ArrivalTimeTab);
6472 app.MaximumdepthEditFieldLabel_10.Tooltip = {'Unit: minutes'};
6473 app.MaximumdepthEditFieldLabel_10.Position = [49 35 48 22];
6474 app.MaximumdepthEditFieldLabel_10.Text = 'Interval';
6475
6476 % Create MinimumdepthEditField_3
6477 app.MinimumdepthEditField_3 = uieditfield(app.ArrivalTimeTab, '
numeric');
6478 app.MinimumdepthEditField_3.Limits = [0 Inf];
6479 app.MinimumdepthEditField_3.ValueDisplayFormat = '%8.1f';
6480 app.MinimumdepthEditField_3.Tooltip = {'Unit: minutes'};
6481 app.MinimumdepthEditField_3.Position = [106 59 55 16];
6482 app.MinimumdepthEditField_3.Value = 10;
6483
6484 % Create MinimumdepthEditFieldLabel_5
6485 app.MinimumdepthEditFieldLabel_5 = uilabel(app.ArrivalTimeTab);
6486 app.MinimumdepthEditFieldLabel_5.Tooltip = {'Unit: minutes'};
6487 app.MinimumdepthEditFieldLabel_5.Position = [49 56 51 22];
6488 app.MinimumdepthEditFieldLabel_5.Text = 'Minimum';
6489
6490 % Create LabelSpacingEditField_2Label_4
6491 app.LabelSpacingEditField_2Label_4 = uilabel(app.ArrivalTimeTab);
6492 app.LabelSpacingEditField_2Label_4.Enable = 'off';
6493 app.LabelSpacingEditField_2Label_4.Tooltip = {'Add labels at
specified intervals.'; ''; 'Suggestion: Set at intervals divisible by the contour
range interval'; ''; 'Unit: meters'};
6494 app.LabelSpacingEditField_2Label_4.Position = [392 35 45 22];
6495 app.LabelSpacingEditField_2Label_4.Text = 'Interval';
6496
6497 % Create IntervalEditField_7
6498 app.IntervalEditField_7 = uieditfield(app.ArrivalTimeTab, '
numeric');
6499 app.IntervalEditField_7.Limits = [0.01 Inf];
6500 app.IntervalEditField_7.ValueDisplayFormat = '%8.1f';
6501 app.IntervalEditField_7.Enable = 'off';
6502 app.IntervalEditField_7.Tooltip = {''};
6503 app.IntervalEditField_7.Position = [441 38 41 16];
6504 app.IntervalEditField_7.Value = 1;
6505
6506 % Create TextLabelSize_3
6507 app.TextLabelSize_3 = uieditfield(app.ArrivalTimeTab, 'numeric');
6508 app.TextLabelSize_3.Limits = [0 Inf];
6509 app.TextLabelSize_3.ValueDisplayFormat = '%8.0f';
6510 app.TextLabelSize_3.Enable = 'off';
6511 app.TextLabelSize_3.Tooltip = {''};
6512 app.TextLabelSize_3.Position = [441 59 30 16];
6513 app.TextLabelSize_3.Value = 8;
6514
6515 % Create SpacingEditField_4Label_6
6516 app.SpacingEditField_4Label_6 = uilabel(app.ArrivalTimeTab);
6517 app.SpacingEditField_4Label_6.WordWrap = 'on';
6518 app.SpacingEditField_4Label_6.Enable = 'off';
6519 app.SpacingEditField_4Label_6.Tooltip = {'Unit: meters'};
6520 app.SpacingEditField_4Label_6.Position = [392 59 36 17];
6521 app.SpacingEditField_4Label_6.Text = 'Size';
6522
6523 % Create AddLabelsCheckBox_5
6524 app.AddLabelsCheckBox_5 = uicheckbox(app.ArrivalTimeTab);

```



```

6525         app.AddLabelsCheckBox_5.ValueChangedFcn = createCallbackFcn(app, ↵
@AddLabelsCheckBox_5ValueChanged2, true);
6526         app.AddLabelsCheckBox_5.Tooltip = {'Unit: meters'};
6527         app.AddLabelsCheckBox_5.Text = 'Add Labels';
6528         app.AddLabelsCheckBox_5.FontWeight = 'bold';
6529         app.AddLabelsCheckBox_5.Position = [392 78 86 22];
6530
6531         % Create LineLabel_5
6532         app.LineLabel_5 = uilabel(app.ArrivalTimeTab);
6533         app.LineLabel_5.FontWeight = 'bold';
6534         app.LineLabel_5.Position = [201 78 30 22];
6535         app.LineLabel_5.Text = 'Line';
6536
6537         % Create ThicknessEditField_3
6538         app.ThicknessEditField_3 = uieditfield(app.ArrivalTimeTab, ↵
'numeric');
6539         app.ThicknessEditField_3.Limits = [0 20];
6540         app.ThicknessEditField_3.ValueDisplayFormat = '%3.1f';
6541         app.ThicknessEditField_3.Position = [239 59 35 16];
6542         app.ThicknessEditField_3.Value = 0.1;
6543
6544         % Create ThicknessEditField_2Label_2
6545         app.ThicknessEditField_2Label_2 = uilabel(app.ArrivalTimeTab);
6546         app.ThicknessEditField_2Label_2.Position = [201 56 39 22];
6547         app.ThicknessEditField_2Label_2.Text = 'Width';
6548
6549         % Create ColorDropDown_6Label
6550         app.ColorDropDown_6Label = uilabel(app.ArrivalTimeTab);
6551         app.ColorDropDown_6Label.Position = [200 15 35 22];
6552         app.ColorDropDown_6Label.Text = 'Color';
6553
6554         % Create ColorDropDown_6
6555         app.ColorDropDown_6 = uidropdown(app.ArrivalTimeTab);
6556         app.ColorDropDown_6.Items = {'Black', 'Dark gray', 'Medium gray', ↵
'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
6557         app.ColorDropDown_6.Position = [239 18 104 16];
6558         app.ColorDropDown_6.Value = 'Dark gray';
6559
6560         % Create StyleDropDown_4Label_2
6561         app.StyleDropDown_4Label_2 = uilabel(app.ArrivalTimeTab);
6562         app.StyleDropDown_4Label_2.Position = [201 35 32 22];
6563         app.StyleDropDown_4Label_2.Text = 'Style';
6564
6565         % Create StyleDropDown_4
6566         app.StyleDropDown_4 = uidropdown(app.ArrivalTimeTab);
6567         app.StyleDropDown_4.Items = {'Solid', 'Dashed', 'Dotted', 'Dash-↵
dotted'};
6568         app.StyleDropDown_4.Position = [239 38 104 16];
6569         app.StyleDropDown_4.Value = 'Solid';
6570
6571         % Create LabelSpacingEditField_4
6572         app.LabelSpacingEditField_4 = uieditfield(app.ArrivalTimeTab, ↵
'numeric');
6573         app.LabelSpacingEditField_4.Limits = [0 Inf];
6574         app.LabelSpacingEditField_4.ValueDisplayFormat = '%8.0f';
6575         app.LabelSpacingEditField_4.Enable = 'off';
6576         app.LabelSpacingEditField_4.Tooltip = {' '};
6577         app.LabelSpacingEditField_4.Position = [441 18 41 16];

```

```

6578         app.LabelSpacingEditField_4.Value = 1000;
6579
6580         % Create LabelSpacingEditField_2Label_3
6581         app.LabelSpacingEditField_2Label_3 = uilabel(app.ArrivalTimeTab);
6582         app.LabelSpacingEditField_2Label_3.WordWrap = 'on';
6583         app.LabelSpacingEditField_2Label_3.Enable = 'off';
6584         app.LabelSpacingEditField_2Label_3.Tooltip = {'Adjust the distance↵
between the labels'; ''; 'Higher values reduce label crowding.'; ''; 'Unit: 1 =↵
1/72 inch'};
6585         app.LabelSpacingEditField_2Label_3.Position = [392 12 45 29];
6586         app.LabelSpacingEditField_2Label_3.Text = 'Spacing';
6587
6588         % Create GaugesTab
6589         app.GaugesTab = uitab(app.TabGroup2);
6590         app.GaugesTab.AutoSizeChildren = 'off';
6591         app.GaugesTab.Title = 'Gauges';
6592
6593         % Create Button_8
6594         app.Button_8 = uibutton(app.GaugesTab, 'push');
6595         app.Button_8.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_8Pushed2, true);
6596         app.Button_8.Tooltip = {'Load the file'; ''; 'File extension must↵
be .shp or .txt (tab delimited)'};
6597         app.Button_8.Position = [209 96 19 19];
6598         app.Button_8.Text = '...';
6599
6600         % Create GaugeFileLabel
6601         app.GaugeFileLabel = uilabel(app.GaugesTab);
6602         app.GaugeFileLabel.FontWeight = 'bold';
6603         app.GaugeFileLabel.Tooltip = {'File extension must be .shp or .↵
txt'};
6604         app.GaugeFileLabel.Position = [38 94 26 22];
6605         app.GaugeFileLabel.Text = 'File';
6606
6607         % Create FileEditField
6608         app.FileEditField = uieditfield(app.GaugesTab, 'text');
6609         app.FileEditField.Editable = 'off';
6610         app.FileEditField.Tooltip = {''};
6611         app.FileEditField.Placeholder = '.shp or .txt';
6612         app.FileEditField.Position = [77 97 125 16];
6613
6614         % Create ColorDropDown_3Label
6615         app.ColorDropDown_3Label = uilabel(app.GaugesTab);
6616         app.ColorDropDown_3Label.Tooltip = {'Marker color'};
6617         app.ColorDropDown_3Label.Position = [38 24 35 22];
6618         app.ColorDropDown_3Label.Text = 'Color';
6619
6620         % Create ColorDropDown_3
6621         app.ColorDropDown_3 = uidropdown(app.GaugesTab);
6622         app.ColorDropDown_3.Items = {'Black', 'Dark gray', 'Medium gray',↵
'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
6623         app.ColorDropDown_3.Position = [77 27 103 16];
6624         app.ColorDropDown_3.Value = 'Black';
6625
6626         % Create CoastlinecolourDropDownLabel_3
6627         app.CoastlinecolourDropDownLabel_3 = uilabel(app.GaugesTab);
6628         app.CoastlinecolourDropDownLabel_3.Tooltip = {'Marker style'};
6629         app.CoastlinecolourDropDownLabel_3.Position = [38 46 32 22];

```

```

6630         app.CoastlinecolourDropDownLabel_3.Text = 'Style';
6631
6632         % Create CoastlinecolourDropDown_3
6633         app.CoastlinecolourDropDown_3 = uidropdown(app.GaugesTab);
6634         app.CoastlinecolourDropDown_3.Items = {'o', '+', '*', '.', 'x', '◀',
        '-|', '|', '^', 'v', '>', '<', 'diamond', 'hexagram', 'pentagram', 'square'};
6635         app.CoastlinecolourDropDown_3.Tooltip = {''};
6636         app.CoastlinecolourDropDown_3.Position = [79 49 92 16];
6637         app.CoastlinecolourDropDown_3.Value = 'o';
6638
6639         % Create SizeEditFieldLabel
6640         app.SizeEditFieldLabel = uilabel(app.GaugesTab);
6641         app.SizeEditFieldLabel.Tooltip = {'Marker size'};
6642         app.SizeEditFieldLabel.Position = [38 69 28 22];
6643         app.SizeEditFieldLabel.Text = 'Size';
6644
6645         % Create SizeEditField
6646         app.SizeEditField = uieditfield(app.GaugesTab, 'numeric');
6647         app.SizeEditField.Limits = [0 50];
6648         app.SizeEditField.ValueDisplayFormat = '%3.0f';
6649         app.SizeEditField.HorizontalAlignment = 'left';
6650         app.SizeEditField.Tooltip = {''};
6651         app.SizeEditField.Position = [79 72 40 16];
6652         app.SizeEditField.Value = 5;
6653
6654         % Create AddLabelsCheckBox_4
6655         app.AddLabelsCheckBox_4 = uicheckbox(app.GaugesTab);
6656         app.AddLabelsCheckBox_4.ValueChangedFcn = createCallbackFcn(app, @AddLabelsCheckBox_4ValueChanged2, true);
6657         app.AddLabelsCheckBox_4.Tooltip = {'The label follows the sequence
        of latitude-longitude combinations from the loaded file.'; ''; 'The labels
        correspond to the numeric values in the sta_files.'};
6658         app.AddLabelsCheckBox_4.Text = ' Add Labels';
6659         app.AddLabelsCheckBox_4.WordWrap = 'on';
6660         app.AddLabelsCheckBox_4.FontWeight = 'bold';
6661         app.AddLabelsCheckBox_4.Position = [267 91 96 28];
6662
6663         % Create AlignmentLabel
6664         app.AlignmentLabel = uilabel(app.GaugesTab);
6665         app.AlignmentLabel.Enable = 'off';
6666         app.AlignmentLabel.Tooltip = {'Horizontal and vertical alignments
        relative to the marker location'};
6667         app.AlignmentLabel.Position = [267 24 59 22];
6668         app.AlignmentLabel.Text = 'Alignment';
6669
6670         % Create CoastlinecolourDropDown_4
6671         app.CoastlinecolourDropDown_4 = uidropdown(app.GaugesTab);
6672         app.CoastlinecolourDropDown_4.Items = {'Centre', 'Right', 'Left'};
6673         app.CoastlinecolourDropDown_4.Enable = 'off';
6674         app.CoastlinecolourDropDown_4.Tooltip = {''};
6675         app.CoastlinecolourDropDown_4.Position = [326 27 74 16];
6676         app.CoastlinecolourDropDown_4.Value = 'Right';
6677
6678         % Create CoastlinecolourDropDown_5
6679         app.CoastlinecolourDropDown_5 = uidropdown(app.GaugesTab);
6680         app.CoastlinecolourDropDown_5.Items = {'Centre', 'Top', 'Bottom'};
6681         app.CoastlinecolourDropDown_5.Enable = 'off';
6682         app.CoastlinecolourDropDown_5.Tooltip = {''};

```

```

6683     app.CoastlinecolourDropDown_5.Position = [406 27 74 16];
6684     app.CoastlinecolourDropDown_5.Value = 'Centre';
6685
6686     % Create SizeEditField_3Label
6687     app.SizeEditField_3Label = uilabel(app.GaugesTab);
6688     app.SizeEditField_3Label.Enable = 'off';
6689     app.SizeEditField_3Label.Position = [267 69 30 22];
6690     app.SizeEditField_3Label.Text = 'Size';
6691
6692     % Create SizeEditField_3
6693     app.SizeEditField_3 = uieditfield(app.GaugesTab, 'numeric');
6694     app.SizeEditField_3.Limits = [0 100];
6695     app.SizeEditField_3.ValueDisplayFormat = '%8.0f';
6696     app.SizeEditField_3.Enable = 'off';
6697     app.SizeEditField_3.Position = [326 72 40 16];
6698     app.SizeEditField_3.Value = 12;
6699
6700     % Create GLspacingLabel
6701     app.GLspacingLabel = uilabel(app.GaugesTab);
6702     app.GLspacingLabel.Enable = 'off';
6703     app.GLspacingLabel.Tooltip = {'Horizontal distance between the
point and the label.'; ''; 'A value of 0.5 means that a label for a point at 120°E
will be placed at 120.5°E.'; ' '; 'Unit: degrees / meters'};
6704     app.GLspacingLabel.Position = [267 46 50 22];
6705     app.GLspacingLabel.Text = 'Spacing';
6706
6707     % Create GLspacing
6708     app.GLspacing = uieditfield(app.GaugesTab, 'numeric');
6709     app.GLspacing.ValueDisplayFormat = '%8.3f';
6710     app.GLspacing.Enable = 'off';
6711     app.GLspacing.Tooltip = {' '};
6712     app.GLspacing.Position = [326 49 40 16];
6713     app.GLspacing.Value = 1;
6714
6715     % Create Tab
6716     app.Tab = uitab(app.TabGroup2);
6717     app.Tab.AutoSizeChildren = 'off';
6718     app.Tab.Title = 'Tab';
6719
6720     % Create ButtonGroup_17
6721     app.ButtonGroup_17 = uibuttongroup(app.Panel_7);
6722     app.ButtonGroup_17.AutoSizeChildren = 'off';
6723     app.ButtonGroup_17.SelectionChangedFcn = createCallbackFcn(app,
@ButtonGroup_17SelectionChanged, true);
6724     app.ButtonGroup_17.BorderType = 'none';
6725     app.ButtonGroup_17.SizeChangedFcn = createCallbackFcn(app,
@ButtonGroup_17SizeChanged, true);
6726     app.ButtonGroup_17.Position = [1 126 518 37];
6727
6728     % Create BathymetryButton_2
6729     app.BathymetryButton_2 = uiradiobutton(app.ButtonGroup_17);
6730     app.BathymetryButton_2.Tooltip = {' '};
6731     app.BathymetryButton_2.Text = 'Bathymetry';
6732     app.BathymetryButton_2.Position = [44 3 84 22];
6733
6734     % Create WaveHeightButton
6735     app.WaveHeightButton = uiradiobutton(app.ButtonGroup_17);
6736     app.WaveHeightButton.Tooltip = {'This is based on the uploaded

```

```

hmax/eta files'};
6737         app.WaveHeightButton.Text = 'Wave Height';
6738         app.WaveHeightButton.Position = [139 3 90 22];
6739
6740         % Create ArrivalTimeButton
6741         app.ArrivalTimeButton = uiradiobutton(app.ButtonGroup_17);
6742         app.ArrivalTimeButton.Tooltip = {'Arrival time threshold set by'
FUNWAVE: 0.001 m'};
6743         app.ArrivalTimeButton.Text = 'Arrival Time';
6744         app.ArrivalTimeButton.Position = [239 3 85 22];
6745
6746         % Create NoneButton
6747         app.NoneButton = uiradiobutton(app.ButtonGroup_17);
6748         app.NoneButton.Text = 'None';
6749         app.NoneButton.Position = [338 3 51 22];
6750         app.NoneButton.Value = true;
6751
6752         % Create GaugesCheckBox
6753         app.GaugesCheckBox = uicheckbox(app.Panel_7);
6754         app.GaugesCheckBox.ValueChangedFcn = createCallbackFcn(app,
@GaugesCheckBoxValueChanged, true);
6755         app.GaugesCheckBox.Tooltip = {' '};
6756         app.GaugesCheckBox.Text = 'Gauges';
6757         app.GaugesCheckBox.Position = [419 129 64 22];
6758
6759         % Create OverlayFeaturesLabel
6760         app.OverlayFeaturesLabel = uilabel(app.WaveHeightMapTab);
6761         app.OverlayFeaturesLabel.BackgroundColor = [0.9412 0.9412 0.9412];
6762         app.OverlayFeaturesLabel.FontSize = 15;
6763         app.OverlayFeaturesLabel.FontWeight = 'bold';
6764         app.OverlayFeaturesLabel.FontColor = [0.0314 0.3686 0.6];
6765         app.OverlayFeaturesLabel.Position = [38 471 143 22];
6766         app.OverlayFeaturesLabel.Text = ' Overlay Features ';
6767
6768         % Create Panel_5
6769         app.Panel_5 = uipanel(app.WaveHeightMapTab);
6770         app.Panel_5.AutoResizeChildren = 'off';
6771         app.Panel_5.Position = [11 494 520 157];
6772
6773         % Create DivisionEditFieldLabel
6774         app.DivisionEditFieldLabel = uilabel(app.Panel_5);
6775         app.DivisionEditFieldLabel.FontSize = 11;
6776         app.DivisionEditFieldLabel.Enable = 'off';
6777         app.DivisionEditFieldLabel.Tooltip = {'Defines the color'
distribution:; ''; 'Low value:; 'Distinct color boundaries; ''; 'Higher value:;
'Smoother color transition'};
6778         app.DivisionEditFieldLabel.Position = [179 85 44 22];
6779         app.DivisionEditFieldLabel.Text = 'Division';
6780
6781         % Create DivisionEditField
6782         app.DivisionEditField = uieditfield(app.Panel_5, 'numeric');
6783         app.DivisionEditField.Limits = [0 200];
6784         app.DivisionEditField.ValueDisplayFormat = '%3.0f';
6785         app.DivisionEditField.ValueChangedFcn = createCallbackFcn(app,
@DivisionEditFieldValueChanged, true);
6786         app.DivisionEditField.FontColor = [0.651 0.651 0.651];
6787         app.DivisionEditField.Enable = 'off';
6788         app.DivisionEditField.Tooltip = {' '};

```

```

6789         app.DivisionEditField.Position = [223 88 30 16];
6790         app.DivisionEditField.Value = 100;
6791
6792         % Create FlipCheckBox_5
6793         app.FlipCheckBox_5 = uicheckbox(app.Panel_5);
6794         app.FlipCheckBox_5.ValueChangedFcn = createCallbackFcn(app, ↵
@FlipCheckBox_5ValueChanged, true);
6795         app.FlipCheckBox_5.Enable = 'off';
6796         app.FlipCheckBox_5.Tooltip = {'Reverse the color sequence'};
6797         app.FlipCheckBox_5.Text = 'Flip';
6798         app.FlipCheckBox_5.FontSize = 11;
6799         app.FlipCheckBox_5.Position = [112 85 40 22];
6800
6801         % Create CoastlinecolourDropDown_8
6802         app.CoastlinecolourDropDown_8 = uidropdown(app.Panel_5);
6803         app.CoastlinecolourDropDown_8.Items = {'--- MATLAB default ----', ↵
'autumn', 'bone', 'colorcube', 'cool', 'copper', 'flag', 'gray', 'hot', 'hsv', ↵
'jet', 'parula', 'pink', 'prism', 'spring', 'summer', 'turbo', 'winter', '--- ↵
CBREWER 2 ---', '< sequential >', 'blue', 'blue - green', 'blue - purple', 'green - ↵
blue', 'greens', 'grays', 'oranges', 'orange - red', 'purple - blue', 'purple - ↵
blue - green', 'purple - red', 'purples', 'red - purple', 'reds', 'yellow - green', ↵
'yellow - green - blue', 'yellow - orange - brown', 'yellow - orange - red', '< ↵
divergent >', 'brown - teal', 'pink - light green', 'purple - green', 'purple - ↵
orange', 'red - blue', 'red - gray', 'red - yellow - blue', 'red - yellow - green', ↵
'spectral', '< qualitative >', 'accent', 'dark 2', 'paired', 'pastel 1', 'pastel ↵
2', 'set 1', 'set 2', 'set 3'};
6804         app.CoastlinecolourDropDown_8.ValueChangedFcn = createCallbackFcn ↵
(app, @CoastlinecolourDropDown_6ValueChanged, true);
6805         app.CoastlinecolourDropDown_8.Enable = 'off';
6806         app.CoastlinecolourDropDown_8.Tooltip = {' '};
6807         app.CoastlinecolourDropDown_8.Position = [112 106 141 17];
6808         app.CoastlinecolourDropDown_8.Value = 'parula';
6809
6810         % Create CoastlinecolourDropDownLabel_8
6811         app.CoastlinecolourDropDownLabel_8 = uilabel(app.Panel_5);
6812         app.CoastlinecolourDropDownLabel_8.FontWeight = 'bold';
6813         app.CoastlinecolourDropDownLabel_8.Position = [26 104 62 22];
6814         app.CoastlinecolourDropDownLabel_8.Text = 'Colormap';
6815
6816         % Create hmaxLabel
6817         app.hmaxLabel = uilabel(app.Panel_5);
6818         app.hmaxLabel.WordWrap = 'on';
6819         app.hmaxLabel.FontWeight = 'bold';
6820         app.hmaxLabel.FontAngle = 'italic';
6821         app.hmaxLabel.Tooltip = {'Maximum wave height'};
6822         app.hmaxLabel.Position = [112 122 60 22];
6823         app.hmaxLabel.Text = ' hmax';
6824
6825         % Create MinEditField
6826         app.MinEditField = uieditfield(app.Panel_5, 'numeric');
6827         app.MinEditField.Limits = [0 Inf];
6828         app.MinEditField.ValueDisplayFormat = '%8.1f';
6829         app.MinEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@MinEditFieldValueChanged, true);
6830         app.MinEditField.FontColor = [0.651 0.651 0.651];
6831         app.MinEditField.Enable = 'off';
6832         app.MinEditField.Tooltip = {' '};
6833         app.MinEditField.Position = [112 62 40 16];

```

```

6834
6835     % Create toEditField
6836     app.toEditField = uieditfield(app.Panel_5, 'numeric');
6837     app.toEditField.Limits = [0.1 Inf];
6838     app.toEditField.ValueDisplayFormat = '%.1f';
6839     app.toEditField.ValueChangedFcn = createCallbackFcn(app, @toEditFieldValueChanged, true);
6840     app.toEditField.FontColor = [0.651 0.651 0.651];
6841     app.toEditField.Enable = 'off';
6842     app.toEditField.Tooltip = {''};
6843     app.toEditField.Position = [177 62 40 16];
6844     app.toEditField.Value = 5;
6845
6846     % Create CoastlinecolourDropDownLabel_10
6847     app.CoastlinecolourDropDownLabel_10 = uilabel(app.Panel_5);
6848     app.CoastlinecolourDropDownLabel_10.WordWrap = 'on';
6849     app.CoastlinecolourDropDownLabel_10.FontWeight = 'bold';
6850     app.CoastlinecolourDropDownLabel_10.Tooltip = {'Limit of the average height values to display on the map'};
6851     app.CoastlinecolourDropDownLabel_10.Position = [25 55 67 30];
6852     app.CoastlinecolourDropDownLabel_10.Text = 'Colorbar';
6853
6854     % Create toLabel
6855     app.toLabel = uilabel(app.Panel_5);
6856     app.toLabel.Position = [159 59 16 22];
6857     app.toLabel.Text = 'to';
6858
6859     % Create etaLabel
6860     app.etaLabel = uilabel(app.Panel_5);
6861     app.etaLabel.WordWrap = 'on';
6862     app.etaLabel.FontWeight = 'bold';
6863     app.etaLabel.FontAngle = 'italic';
6864     app.etaLabel.Tooltip = {'Sea surface displacement at time T'};
6865     app.etaLabel.Position = [311 122 57 22];
6866     app.etaLabel.Text = 'eta';
6867
6868     % Create CoastlinecolourDropDown_7
6869     app.CoastlinecolourDropDown_7 = uidropdown(app.Panel_5);
6870     app.CoastlinecolourDropDown_7.Items = {'--- MATLAB default ---', 'autumn', 'bone', 'colorcube', 'cool', 'copper', 'flag', 'gray', 'hot', 'hsv', 'jet', 'parula', 'pink', 'prism', 'spring', 'summer', 'turbo', 'winter', '--- CBREWER 2 ---', '< sequential >', 'blue', 'blue - green', 'blue - purple', 'green - blue', 'greens', 'grays', 'oranges', 'orange - red', 'purple - blue', 'purple - blue - green', 'purple - red', 'purples', 'red - purple', 'reds', 'yellow - green', 'yellow - green - blue', 'yellow - orange - brown', 'yellow - orange - red', '< divergent >', 'brown - teal', 'pink - light green', 'purple - green', 'purple - orange', 'red - blue', 'red - gray', 'red - yellow - blue', 'red - yellow - green', 'spectral', '< qualitative >', 'accent', 'dark 2', 'paired', 'pastel 1', 'pastel 2', 'set 1', 'set 2', 'set 3'};
6871     app.CoastlinecolourDropDown_7.ValueChangedFcn = createCallbackFcn(app, @CoastlinecolourDropDown_7ValueChanged, true);
6872     app.CoastlinecolourDropDown_7.Enable = 'off';
6873     app.CoastlinecolourDropDown_7.Tooltip = {''};
6874     app.CoastlinecolourDropDown_7.Position = [311 106 141 17];
6875     app.CoastlinecolourDropDown_7.Value = 'red - blue';
6876
6877     % Create FlipCheckBox_2
6878     app.FlipCheckBox_2 = uicheckbox(app.Panel_5);

```



```

6879         app.FlipCheckBox_2.ValueChangedFcn = createCallbackFcn(app, ↵
@FlipCheckBox_2ValueChanged, true);
6880         app.FlipCheckBox_2.Enable = 'off';
6881         app.FlipCheckBox_2.Tooltip = {'Reverse the color sequence'};
6882         app.FlipCheckBox_2.Text = 'Flip';
6883         app.FlipCheckBox_2.FontSize = 11;
6884         app.FlipCheckBox_2.Position = [311 85 40 22];
6885         app.FlipCheckBox_2.Value = true;
6886
6887         % Create ColorinterpolationEditField_2
6888         app.ColorinterpolationEditField_2 = uicontrol(app.Panel_5, ↵
'numeric');
6889         app.ColorinterpolationEditField_2.Limits = [0 200];
6890         app.ColorinterpolationEditField_2.ValueDisplayFormat = '%3.0f';
6891         app.ColorinterpolationEditField_2.ValueChangedFcn = ↵
createCallbackFcn(app, @ColorinterpolationEditField_2ValueChanged, true);
6892         app.ColorinterpolationEditField_2.FontColor = [0.651 0.651 0.651];
6893         app.ColorinterpolationEditField_2.Enable = 'off';
6894         app.ColorinterpolationEditField_2.Tooltip = {' '};
6895         app.ColorinterpolationEditField_2.Position = [422 88 30 16];
6896         app.ColorinterpolationEditField_2.Value = 100;
6897
6898         % Create DivisionLabel
6899         app.DivisionLabel = uicontrol(app.Panel_5);
6900         app.DivisionLabel.FontSize = 11;
6901         app.DivisionLabel.Enable = 'off';
6902         app.DivisionLabel.Tooltip = {'Defines the color distribution:'; ↵
'; 'Low value: '; 'Distinct color boundaries'; '; 'Higher value: '; 'Smoother color ↵
transition'};
6903         app.DivisionLabel.Position = [377 85 44 22];
6904         app.DivisionLabel.Text = 'Division';
6905
6906         % Create MaxEditField_2
6907         app.MaxEditField_2 = uicontrol(app.Panel_5, 'numeric');
6908         app.MaxEditField_2.Limits = [0 Inf];
6909         app.MaxEditField_2.ValueDisplayFormat = '%.1f';
6910         app.MaxEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@MaxEditField_2ValueChanged, true);
6911         app.MaxEditField_2.FontColor = [0.651 0.651 0.651];
6912         app.MaxEditField_2.Enable = 'off';
6913         app.MaxEditField_2.Tooltip = {'Suggestion: '; 'Set this value as ↵
the (+) of the min value to place the zero mark in the middle of the colorbar'};
6914         app.MaxEditField_2.Position = [376 62 40 16];
6915         app.MaxEditField_2.Value = 0.5;
6916
6917         % Create MinEditField_2
6918         app.MinEditField_2 = uicontrol(app.Panel_5, 'numeric');
6919         app.MinEditField_2.ValueDisplayFormat = '%8.1f';
6920         app.MinEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@MinEditField_2ValueChanged, true);
6921         app.MinEditField_2.FontColor = [0.651 0.651 0.651];
6922         app.MinEditField_2.Enable = 'off';
6923         app.MinEditField_2.Tooltip = {'Suggestion: '; 'Set this value as ↵
the (-) of the max value to place the zero mark in the middle of the colorbar'};
6924         app.MinEditField_2.Position = [311 62 41 16];
6925         app.MinEditField_2.Value = -0.5;
6926
6927         % Create toLabel 2

```



```

6928     app.toLabel_2 = uilabel(app.Panel_5);
6929     app.toLabel_2.Position = [358 59 16 22];
6930     app.toLabel_2.Text = 'to';
6931
6932     % Create mLabel
6933     app.mLabel = uilabel(app.Panel_5);
6934     app.mLabel.Position = [223 59 25 22];
6935     app.mLabel.Text = 'm';
6936
6937     % Create mLabel_2
6938     app.mLabel_2 = uilabel(app.Panel_5);
6939     app.mLabel_2.Position = [423 59 25 22];
6940     app.mLabel_2.Text = 'm';
6941
6942     % Create CoastlinecolourDropDownLabel_9
6943     app.CoastlinecolourDropDownLabel_9 = uilabel(app.Panel_5);
6944     app.CoastlinecolourDropDownLabel_9.FontWeight = 'bold';
6945     app.CoastlinecolourDropDownLabel_9.Tooltip = {'Required input:','  

'Bathymetry file'};
6946     app.CoastlinecolourDropDownLabel_9.Position = [307 28 69 22];
6947     app.CoastlinecolourDropDownLabel_9.Text = 'Land Color';
6948
6949     % Create HorizontalCheckBox
6950     app.HorizontalCheckBox = uicheckbox(app.Panel_5);
6951     app.HorizontalCheckBox.Text = 'Horizontal';
6952     app.HorizontalCheckBox.Position = [119 17 76 22];
6953
6954     % Create VerticalCheckBox
6955     app.VerticalCheckBox = uicheckbox(app.Panel_5);
6956     app.VerticalCheckBox.Text = 'Vertical';
6957     app.VerticalCheckBox.Position = [199 17 61 22];
6958
6959     % Create FlipBasemapLabel_2
6960     app.FlipBasemapLabel_2 = uilabel(app.Panel_5);
6961     app.FlipBasemapLabel_2.FontWeight = 'bold';
6962     app.FlipBasemapLabel_2.Tooltip = {'Flips the basemap.','';  

'Horizontally (left-right)'; 'Vertically (up-side down)'};
6963     app.FlipBasemapLabel_2.Position = [25 17 87 22];
6964     app.FlipBasemapLabel_2.Text = 'Flip Basemap';
6965
6966     % Create CoastlinecolourDropDown_2
6967     app.CoastlinecolourDropDown_2 = uidropdown(app.Panel_5);
6968     app.CoastlinecolourDropDown_2.Items = {'Black', 'Dark gray',  

'Medium gray', 'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta',  

'White'};
6969     app.CoastlinecolourDropDown_2.ValueChangedFcn = createCallbackFcn(  

app, @CoastlinecolourDropDown_2ValueChanged, true);
6970     app.CoastlinecolourDropDown_2.Enable = 'off';
6971     app.CoastlinecolourDropDown_2.Tooltip = {'';
6972     app.CoastlinecolourDropDown_2.Position = [381 31 106 17];
6973     app.CoastlinecolourDropDown_2.Value = 'Dark gray';
6974
6975     % Create TextSizeLabel_2
6976     app.TextSizeLabel_2 = uilabel(app.Panel_5);
6977     app.TextSizeLabel_2.WordWrap = 'on';
6978     app.TextSizeLabel_2.FontWeight = 'bold';
6979     app.TextSizeLabel_2.Tooltip = {'Font size of the colorbar  

labels'};

```

```

6980     app.TextSizeLabel_2.Position = [307 3 60 28];
6981     app.TextSizeLabel_2.Text = 'Text Size';
6982
6983     % Create ColorbarTextSize
6984     app.ColorbarTextSize = ueditfield(app.Panel_5, 'numeric');
6985     app.ColorbarTextSize.Limits = [1 50];
6986     app.ColorbarTextSize.ValueDisplayFormat = '%2.0f';
6987     app.ColorbarTextSize.ValueChangedFcn = createCallbackFcn(app, ↵
@ColorbarTextSizeValueChanged, true);
6988     app.ColorbarTextSize.FontColor = [0.651 0.651 0.651];
6989     app.ColorbarTextSize.Enable = 'off';
6990     app.ColorbarTextSize.Tooltip = {' '};
6991     app.ColorbarTextSize.Position = [381 9 29 16];
6992     app.ColorbarTextSize.Value = 11;
6993
6994     % Create BasemapLabel
6995     app.BasemapLabel = uilabel(app.WaveHeightMapTab);
6996     app.BasemapLabel.BackgroundColor = [0.9412 0.9412 0.9412];
6997     app.BasemapLabel.FontSize = 15;
6998     app.BasemapLabel.FontWeight = 'bold';
6999     app.BasemapLabel.FontColor = [0.0314 0.3686 0.6];
7000     app.BasemapLabel.Tooltip = {'Modify the visuals of the imported ↵
file/s (eta/hmax)'};
7001     app.BasemapLabel.Position = [38 639 89 22];
7002     app.BasemapLabel.Text = ' Basemap ';
7003
7004     % Create Panel_6
7005     app.Panel_6 = uipanel(app.WaveHeightMapTab);
7006     app.Panel_6.AutoResizeChildren = 'off';
7007     app.Panel_6.Position = [11 39 520 77];
7008
7009     % Create pngCheckBox
7010     app.pngCheckBox = uicheckbox(app.Panel_6);
7011     app.pngCheckBox.Tooltip = {'Resolution is set to 300 DPI'};
7012     app.pngCheckBox.Text = '.png';
7013     app.pngCheckBox.Position = [191 12 46 22];
7014
7015     % Create FileFormatLabel
7016     app.FileFormatLabel = uilabel(app.Panel_6);
7017     app.FileFormatLabel.Position = [69 12 66 22];
7018     app.FileFormatLabel.Text = 'File Format';
7019
7020     % Create jpgCheckBox
7021     app.jpgCheckBox = uicheckbox(app.Panel_6);
7022     app.jpgCheckBox.Tooltip = {'Resolution is set to 300 DPI'};
7023     app.jpgCheckBox.Text = '.jpg';
7024     app.jpgCheckBox.Position = [139 12 42 22];
7025
7026     % Create tifCheckBox_2
7027     app.tifCheckBox_2 = uicheckbox(app.Panel_6);
7028     app.tifCheckBox_2.ValueChangedFcn = createCallbackFcn(app, ↵
@tifCheckBox_2ValueChanged, true);
7029     app.tifCheckBox_2.Enable = 'off';
7030     app.tifCheckBox_2.Tooltip = {'Georeferenced raster file.'; ''; ↵
'Enabled when "Plot separately" is selected.'};
7031     app.tifCheckBox_2.Text = '.tif';
7032     app.tifCheckBox_2.Position = [246 12 35 22];
7033

```

```

7034         % Create Button_2
7035         app.Button_2 = uibutton(app.Panel_6, 'push');
7036         app.Button_2.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_2Pushed, true);
7037         app.Button_2.FontSize = 10;
7038         app.Button_2.FontAngle = 'italic';
7039         app.Button_2.Tooltip = {'Load the preferred directory.'; '';↵
'"OUTPUT_FILES/Figures" folder will be created in the selected directory.'};
7040         app.Button_2.Position = [403 42 19 19];
7041         app.Button_2.Text = '...';
7042
7043         % Create OutputDirectoryLabel
7044         app.OutputDirectoryLabel = uilabel(app.Panel_6);
7045         app.OutputDirectoryLabel.Tooltip = {'Load the preferred↵
directory.'; ''; '"OUTPUT_FILES/Figures" folder will be created in the selected↵
directory.'};
7046         app.OutputDirectoryLabel.Position = [40 40 94 22];
7047         app.OutputDirectoryLabel.Text = 'Output Directory';
7048
7049         % Create OutputDirectoryEditField
7050         app.OutputDirectoryEditField = uieditfield(app.Panel_6, 'text');
7051         app.OutputDirectoryEditField.Editable = 'off';
7052         app.OutputDirectoryEditField.FontColor = [0 0 1];
7053         app.OutputDirectoryEditField.Tooltip = {'Load the preferred↵
directory.'; ''; '"OUTPUT_FILES/Figures" folder will be created in the selected↵
directory.'};
7054         app.OutputDirectoryEditField.Placeholder = 'Default: Desktop';
7055         app.OutputDirectoryEditField.Position = [139 43 257 16];
7056
7057         % Create FramerateEditField_2Label
7058         app.FramerateEditField_2Label = uilabel(app.Panel_6);
7059         app.FramerateEditField_2Label.WordWrap = 'on';
7060         app.FramerateEditField_2Label.Visible = 'off';
7061         app.FramerateEditField_2Label.Tooltip = {'Frames per second.'; '';↵
'Lower value = slower animation.'};
7062         app.FramerateEditField_2Label.Position = [346 10 68 27];
7063         app.FramerateEditField_2Label.Text = 'Frame rate';
7064
7065         % Create FramerateEditField_2
7066         app.FramerateEditField_2 = uieditfield(app.Panel_6, 'numeric');
7067         app.FramerateEditField_2.Limits = [0 100];
7068         app.FramerateEditField_2.ValueDisplayFormat = '%.0f';
7069         app.FramerateEditField_2.FontColor = [0.651 0.651 0.651];
7070         app.FramerateEditField_2.Visible = 'off';
7071         app.FramerateEditField_2.Tooltip = {''};
7072         app.FramerateEditField_2.Position = [412 15 30 17];
7073         app.FramerateEditField_2.Value = 5;
7074
7075         % Create mp4CheckBox
7076         app.mp4CheckBox = uicheckbox(app.Panel_6);
7077         app.mp4CheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@mp4CheckBoxValueChanged, true);
7078         app.mp4CheckBox.Enable = 'off';
7079         app.mp4CheckBox.Tooltip = {'Create an animation.'; ''; 'File is↵
saved as "animation.mp4."'; ''; 'Enabled when "Plot separately" is selected.'};
7080         app.mp4CheckBox.Text = '.mp4';
7081         app.mp4CheckBox.Position = [290 12 49 22];
7082

```

```

7083         % Create SaveMapLabel
7084         app.SaveMapLabel = uilabel(app.WaveHeightMapTab);
7085         app.SaveMapLabel.BackgroundColor = [0.9412 0.9412 0.9412];
7086         app.SaveMapLabel.FontSize = 15;
7087         app.SaveMapLabel.FontWeight = 'bold';
7088         app.SaveMapLabel.FontColor = [0.0314 0.3686 0.6];
7089         app.SaveMapLabel.Position = [38 105 92 22];
7090         app.SaveMapLabel.Text = ' Save Map ';
7091
7092         % Create GENERATEButton
7093         app.GENERATEButton = uibutton(app.WaveHeightMapTab, 'push');
7094         app.GENERATEButton.ButtonPushedFcn = createCallbackFcn(app, @
@GENERATEButtonPushed, true);
7095         app.GENERATEButton.FontSize = 14;
7096         app.GENERATEButton.FontWeight = 'bold';
7097         app.GENERATEButton.Position = [179 6 183 25];
7098         app.GENERATEButton.Text = 'GENERATE';
7099
7100         % Create Panel_19
7101         app.Panel_19 = uipanel(app.WaveHeightMapTab);
7102         app.Panel_19.AutoResizeChildren = 'off';
7103         app.Panel_19.Position = [11 128 520 172];
7104
7105         % Create BoundaryLimitsLabel
7106         app.BoundaryLimitsLabel = uilabel(app.Panel_19);
7107         app.BoundaryLimitsLabel.FontWeight = 'bold';
7108         app.BoundaryLimitsLabel.Tooltip = {'When all textboxes are left
unchanged (all zeroes), the values are automatically set based on the extent of the
uploaded file and the input southwest corner coordinates'};
7109         app.BoundaryLimitsLabel.Position = [25 91 99 22];
7110         app.BoundaryLimitsLabel.Text = 'Boundary Limits';
7111
7112         % Create PlotindegreesCheckBox_2
7113         app.PlotindegreesCheckBox_2 = uicheckbox(app.Panel_19);
7114         app.PlotindegreesCheckBox_2.ValueChangedFcn = createCallbackFcn
(app, @PlotindegreesCheckBox_2ValueChanged, true);
7115         app.PlotindegreesCheckBox_2.Tooltip = {'Change the label format
into geographic coordinates'};
7116         app.PlotindegreesCheckBox_2.Text = 'Plot in degrees';
7117         app.PlotindegreesCheckBox_2.FontSize = 11;
7118         app.PlotindegreesCheckBox_2.Position = [403 56 96 22];
7119
7120         % Create EastEditFieldLabel
7121         app.EastEditFieldLabel = uilabel(app.Panel_19);
7122         app.EastEditFieldLabel.Position = [331 62 29 22];
7123         app.EastEditFieldLabel.Text = 'East';
7124
7125         % Create EastEditField
7126         app.EastEditField = uieditfield(app.Panel_19, 'numeric');
7127         app.EastEditField.ValueDisplayFormat = '%8.4f';
7128         app.EastEditField.ValueChangedFcn = createCallbackFcn(app, @
@EastEditFieldValueChanged, true);
7129         app.EastEditField.FontColor = [0.651 0.651 0.651];
7130         app.EastEditField.Position = [265 65 60 16];
7131
7132         % Create WestEditFieldLabel
7133         app.WestEditFieldLabel = uilabel(app.Panel_19);
7134         app.WestEditFieldLabel.HorizontalAlignment = 'right';

```

```

7135         app.WestEditFieldLabel.Position = [87 62 32 22];
7136         app.WestEditFieldLabel.Text = 'West';
7137
7138         % Create WestEditField
7139         app.WestEditField = uieditfield(app.Panel_19, 'numeric');
7140         app.WestEditField.ValueDisplayFormat = '%8.4f';
7141         app.WestEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@WestEditFieldValueChanged, true);
7142         app.WestEditField.FontColor = [0.651 0.651 0.651];
7143         app.WestEditField.Position = [125 65 60 16];
7144
7145         % Create NorthEditFieldLabel
7146         app.NorthEditFieldLabel = uilabel(app.Panel_19);
7147         app.NorthEditFieldLabel.HorizontalAlignment = 'right';
7148         app.NorthEditFieldLabel.Position = [206 91 35 22];
7149         app.NorthEditFieldLabel.Text = 'North';
7150
7151         % Create NorthEditField
7152         app.NorthEditField = uieditfield(app.Panel_19, 'numeric');
7153         app.NorthEditField.ValueDisplayFormat = '%8.4f';
7154         app.NorthEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@NorthEditFieldValueChanged, true);
7155         app.NorthEditField.FontColor = [0.651 0.651 0.651];
7156         app.NorthEditField.Position = [195 77 60 16];
7157
7158         % Create SouthEditFieldLabel
7159         app.SouthEditFieldLabel = uilabel(app.Panel_19);
7160         app.SouthEditFieldLabel.HorizontalAlignment = 'right';
7161         app.SouthEditFieldLabel.Position = [205 34 37 22];
7162         app.SouthEditFieldLabel.Text = 'South';
7163
7164         % Create SouthEditField
7165         app.SouthEditField = uieditfield(app.Panel_19, 'numeric');
7166         app.SouthEditField.ValueDisplayFormat = '%8.4f';
7167         app.SouthEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@SouthEditFieldValueChanged, true);
7168         app.SouthEditField.FontColor = [0.651 0.651 0.651];
7169         app.SouthEditField.Position = [195 53 60 16];
7170
7171         % Create Height_2
7172         app.Height_2 = uieditfield(app.Panel_19, 'numeric');
7173         app.Height_2.Limits = [1 Inf];
7174         app.Height_2.ValueDisplayFormat = '%2.0f';
7175         app.Height_2.ValueChangedFcn = createCallbackFcn(app, ↵
@Height_2ValueChanged, true);
7176         app.Height_2.FontColor = [0.651 0.651 0.651];
7177         app.Height_2.Enable = 'off';
7178         app.Height_2.Tooltip = {' '};
7179         app.Height_2.Position = [330 12 40 16];
7180         app.Height_2.Value = 11;
7181
7182         % Create ThicknessLabel_8
7183         app.ThicknessLabel_8 = uilabel(app.Panel_19);
7184         app.ThicknessLabel_8.HorizontalAlignment = 'right';
7185         app.ThicknessLabel_8.Position = [283 9 41 22];
7186         app.ThicknessLabel_8.Text = 'Height';
7187
7188         % Create AutoSetCheckBox 2

```

```

7189         app.AutoSetCheckBox_2 = uicheckbox(app.Panel_19);
7190         app.AutoSetCheckBox_2.ValueChangedFcn = createCallbackFcn(app, ↵
@AutoSetCheckBox_2ValueChanged, true);
7191         app.AutoSetCheckBox_2.Text = ' Auto Set';
7192         app.AutoSetCheckBox_2.FontSize = 11;
7193         app.AutoSetCheckBox_2.Position = [102 9 68 22];
7194         app.AutoSetCheckBox_2.Value = true;
7195
7196         % Create Width_2
7197         app.Width_2 = uieditfield(app.Panel_19, 'numeric');
7198         app.Width_2.Limits = [1 Inf];
7199         app.Width_2.ValueDisplayFormat = '%2.0f';
7200         app.Width_2.ValueChangedFcn = createCallbackFcn(app, ↵
@Width_2ValueChanged, true);
7201         app.Width_2.FontColor = [0.651 0.651 0.651];
7202         app.Width_2.Enable = 'off';
7203         app.Width_2.Tooltip = {' '};
7204         app.Width_2.Position = [225 12 41 16];
7205         app.Width_2.Value = 8;
7206
7207         % Create ThicknessLabel_9
7208         app.ThicknessLabel_9 = uilabel(app.Panel_19);
7209         app.ThicknessLabel_9.Tooltip = {'Unit: inches'};
7210         app.ThicknessLabel_9.Position = [188 9 37 22];
7211         app.ThicknessLabel_9.Text = 'Width';
7212
7213         % Create FigureSizeLabel_2
7214         app.FigureSizeLabel_2 = uilabel(app.Panel_19);
7215         app.FigureSizeLabel_2.FontWeight = 'bold';
7216         app.FigureSizeLabel_2.Position = [25 9 69 22];
7217         app.FigureSizeLabel_2.Text = 'Figure Size';
7218
7219         % Create CloseFiguresButton_4
7220         app.CloseFiguresButton_4 = uibutton(app.Panel_19, 'push');
7221         app.CloseFiguresButton_4.ButtonPushedFcn = createCallbackFcn(app, ↵
@CloseFiguresButton_4Pushed2, true);
7222         app.CloseFiguresButton_4.Tooltip = {' '};
7223         app.CloseFiguresButton_4.Position = [403 10 89 20];
7224         app.CloseFiguresButton_4.Text = 'Close Figures';
7225
7226         % Create SettoDefaultButton_2
7227         app.SettoDefaultButton_2 = uibutton(app.Panel_19, 'push');
7228         app.SettoDefaultButton_2.ButtonPushedFcn = createCallbackFcn(app, ↵
@SettoDefaultButton_2Pushed, true);
7229         app.SettoDefaultButton_2.Tooltip = {'Use the boundary limits of ↵
the eta/hmax file'};
7230         app.SettoDefaultButton_2.Position = [403 76 84 20];
7231         app.SettoDefaultButton_2.Text = 'Set to Default';
7232
7233         % Create ButtonGroup_11
7234         app.ButtonGroup_11 = uibuttongroup(app.Panel_19);
7235         app.ButtonGroup_11.AutoResizeChildren = 'off';
7236         app.ButtonGroup_11.SelectionChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_11SelectionChanged, true);
7237         app.ButtonGroup_11.Position = [106 126 311 30];
7238
7239         % Create PlotalldataainonefigureButton
7240         app.PlotalldataainonefigureButton = uiradiobutton(app. ↵

```

```

ButtonGroup_11);
7241         app.PlotalldataainonefigureButton.Tooltip = {'All maps are
displayed as subplots in one window'};
7242         app.PlotalldataainonefigureButton.Text = 'Plot all data in one
figure';
7243         app.PlotalldataainonefigureButton.FontWeight = 'bold';
7244         app.PlotalldataainonefigureButton.Position = [13 4 164 22];
7245         app.PlotalldataainonefigureButton.Value = true;
7246
7247         % Create PlotseparatelyButton_2
7248         app.PlotseparatelyButton_2 = uiradiobutton(app.ButtonGroup_11);
7249         app.PlotseparatelyButton_2.Tooltip = {'One map per window'};
7250         app.PlotseparatelyButton_2.Text = 'Plot separately';
7251         app.PlotseparatelyButton_2.FontWeight = 'bold';
7252         app.PlotseparatelyButton_2.Position = [193 4 108 22];
7253
7254         % Create GeneralLayoutLabel_3
7255         app.GeneralLayoutLabel_3 = uilabel(app.WaveHeightMapTab);
7256         app.GeneralLayoutLabel_3.BackgroundColor = [0.9412 0.9412 0.9412];
7257         app.GeneralLayoutLabel_3.FontSize = 15;
7258         app.GeneralLayoutLabel_3.FontWeight = 'bold';
7259         app.GeneralLayoutLabel_3.FontColor = [0.0314 0.3686 0.6];
7260         app.GeneralLayoutLabel_3.Position = [38 290 131 22];
7261         app.GeneralLayoutLabel_3.Text = ' General Layout ';
7262
7263         % Create GaugeRecordsTab
7264         app.GaugeRecordsTab = uitab(app.TabGroup);
7265         app.GaugeRecordsTab.AutoResizeChildren = 'off';
7266         app.GaugeRecordsTab.Tooltip = {'When ''3 & 4'' is used for Y data,
default values: ''; 'Column 3: Dashed-dotted lines'; 'Column 4: Dotted lines'};
7267         app.GaugeRecordsTab.SizeChangedFcn = createCallbackFcn(app,
@GaugeRecordsTabSizeChanged, true);
7268         app.GaugeRecordsTab.Title = 'Gauge Records';
7269         app.GaugeRecordsTab.Scrollable = 'on';
7270
7271         % Create Panel_10
7272         app.Panel_10 = uipanel(app.GaugeRecordsTab);
7273         app.Panel_10.AutoResizeChildren = 'off';
7274         app.Panel_10.Interruptible = 'off';
7275         app.Panel_10.Position = [11 129 520 562];
7276
7277         % Create PlotPropertiesLabel
7278         app.PlotPropertiesLabel = uilabel(app.Panel_10);
7279         app.PlotPropertiesLabel.FontWeight = 'bold';
7280         app.PlotPropertiesLabel.Position = [50 179 91 22];
7281         app.PlotPropertiesLabel.Text = 'Plot Properties';
7282
7283         % Create ColorDropDownLabel_2
7284         app.ColorDropDownLabel_2 = uilabel(app.Panel_10);
7285         app.ColorDropDownLabel_2.Tooltip = {''; 'When column ''3,4,5'''
is selected for Y axis, the default colors are: ''; '3 - purple'; '4 - green';
'5 - black'};
7286         app.ColorDropDownLabel_2.Position = [50 85 35 22];
7287         app.ColorDropDownLabel_2.Text = 'Color';
7288
7289         % Create GaugeLineColor
7290         app.GaugeLineColor = uidropdown(app.Panel_10);
7291         app.GaugeLineColor.Items = {'Black', 'Dark gray', 'Medium gray',

```



```

'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta'};
7292         app.GaugeLineColor.Visible = 'off';
7293         app.GaugeLineColor.Position = [108 87 89 17];
7294         app.GaugeLineColor.Value = 'Black';
7295
7296         % Create LineStyleLabel
7297         app.LineStyleLabel = uilabel(app.Panel_10);
7298         app.LineStyleLabel.Tooltip = {'Line style of the plotted data.';
7299 '''; 'When column ''3,4,5''' is selected for Y axis, the default styles are:''''; ''';
7300 '3, 4 - dashed lines'; '5 - solid line'};
7301
7302         app.LineStyleLabel.Position = [50 131 28 22];
7303         app.LineStyleLabel.Text = 'Line';
7304
7305         % Create GaugeLineStyle
7306         app.GaugeLineStyle = uidropdown(app.Panel_10);
7307         app.GaugeLineStyle.Items = {'Solid', 'Dashed', 'Dotted', 'Dash-
7308 dotted'};
7309         app.GaugeLineStyle.Tooltip = {''};
7310         app.GaugeLineStyle.Position = [107 134 91 17];
7311         app.GaugeLineStyle.Value = 'Solid';
7312
7313         % Create ButtonGroup_10
7314         app.ButtonGroup_10 = uibuttongroup(app.Panel_10);
7315         app.ButtonGroup_10.AutoResizeChildren = 'off';
7316         app.ButtonGroup_10.SelectionChangedFcn = createCallbackFcn(app,
7317 @ButtonGroup_10SelectionChanged, true);
7318         app.ButtonGroup_10.FontWeight = 'bold';
7319         app.ButtonGroup_10.Position = [104 510 311 30];
7320
7321         % Create PlotalldataainonegraphButton
7322         app.PlotalldataainonegraphButton = uiradiobutton(app.
7323 ButtonGroup_10);
7324         app.PlotalldataainonegraphButton.Tooltip = {'All data are
7325 summarized in one graph'};
7326         app.PlotalldataainonegraphButton.Text = 'Plot all data in one
7327 graph';
7328         app.PlotalldataainonegraphButton.FontWeight = 'bold';
7329         app.PlotalldataainonegraphButton.Position = [13 4 165 22];
7330         app.PlotalldataainonegraphButton.Value = true;
7331
7332         % Create PlotseparatelyButton_4
7333         app.PlotseparatelyButton_4 = uiradiobutton(app.ButtonGroup_10);
7334         app.PlotseparatelyButton_4.Tooltip = {'Data from each ''sta_'''
7335 file is plotted in separate windows'};
7336         app.PlotseparatelyButton_4.Text = 'Plot separately';
7337         app.PlotseparatelyButton_4.FontWeight = 'bold';
7338         app.PlotseparatelyButton_4.Position = [195 4 108 22];
7339
7340         % Create DataLineWidth
7341         app.DataLineWidth = uieditfield(app.Panel_10, 'numeric');
7342         app.DataLineWidth.Limits = [0.1 Inf];
7343         app.DataLineWidth.ValueDisplayFormat = '%8.1f';
7344         app.DataLineWidth.ValueChangedFcn = createCallbackFcn(app,
7345 @DataLineWidthValueChanged, true);
7346         app.DataLineWidth.FontColor = [0.651 0.651 0.651];
7347         app.DataLineWidth.Tooltip = {''};
7348         app.DataLineWidth.Position = [107 111 29 16];
7349         app.DataLineWidth.Value = 1;

```



```

7340
7341 % Create WidthLabel
7342 app.WidthLabel = uilabel(app.Panel_10);
7343 app.WidthLabel.Position = [50 108 37 22];
7344 app.WidthLabel.Text = 'Width';
7345
7346 % Create AxesLabels
7347 app.AxesLabels = uilabel(app.Panel_10);
7348 app.AxesLabels.BackgroundColor = [0.9412 0.9412 0.9412];
7349 app.AxesLabels.FontWeight = 'bold';
7350 app.AxesLabels.Position = [50 297 75 22];
7351 app.AxesLabels.Text = 'Axes Labels';
7352
7353 % Create TextArea2_4
7354 app.TextArea2_4 = uitextarea(app.Panel_10);
7355 app.TextArea2_4.Editable = 'off';
7356 app.TextArea2_4.BackgroundColor = [0.9412 0.9412 0.9412];
7357 app.TextArea2_4.Enable = 'off';
7358 app.TextArea2_4.Position = [173 356 222 109];
7359
7360 % Create xMinLimit
7361 app.xMinLimit = uieditfield(app.Panel_10, 'numeric');
7362 app.xMinLimit.ValueDisplayFormat = '%8.1f';
7363 app.xMinLimit.ValueChangedFcn = createCallbackFcn(app, ↵
@xMinLimitValueChanged, true);
7364 app.xMinLimit.FontColor = [0.651 0.651 0.651];
7365 app.xMinLimit.Enable = 'off';
7366 app.xMinLimit.Tooltip = {'X-axis minimum value'};
7367 app.xMinLimit.Position = [217 356 45 16];
7368
7369 % Create MinimumdepthEditFieldLabel_4
7370 app.MinimumdepthEditFieldLabel_4 = uilabel(app.Panel_10);
7371 app.MinimumdepthEditFieldLabel_4.HorizontalAlignment = 'center';
7372 app.MinimumdepthEditFieldLabel_4.Position = [213 336 51 22];
7373 app.MinimumdepthEditFieldLabel_4.Text = 'X min';
7374
7375 % Create xMaxLimit
7376 app.xMaxLimit = uieditfield(app.Panel_10, 'numeric');
7377 app.xMaxLimit.ValueDisplayFormat = '%8.1f';
7378 app.xMaxLimit.ValueChangedFcn = createCallbackFcn(app, ↵
@xMaxLimitValueChanged, true);
7379 app.xMaxLimit.FontColor = [0.651 0.651 0.651];
7380 app.xMaxLimit.Enable = 'off';
7381 app.xMaxLimit.Tooltip = {'X-axis maximum value'};
7382 app.xMaxLimit.Position = [340 356 55 16];
7383 app.xMaxLimit.Value = 3600;
7384
7385 % Create MaximumdepthEditFieldLabel_5
7386 app.MaximumdepthEditFieldLabel_5 = uilabel(app.Panel_10);
7387 app.MaximumdepthEditFieldLabel_5.HorizontalAlignment = 'center';
7388 app.MaximumdepthEditFieldLabel_5.Position = [343 336 53 22];
7389 app.MaximumdepthEditFieldLabel_5.Text = 'X max';
7390
7391 % Create xLimInterval
7392 app.xLimInterval = uieditfield(app.Panel_10, 'numeric');
7393 app.xLimInterval.Limits = [0 Inf];
7394 app.xLimInterval.ValueDisplayFormat = '%8.1f';
7395 app.xLimInterval.ValueChangedFcn = createCallbackFcn(app, ↵

```

```

@xLimIntervalValueChanged, true);
7396         app.xLimInterval.FontColor = [0.651 0.651 0.651];
7397         app.xLimInterval.Enable = 'off';
7398         app.xLimInterval.Tooltip = {'X-axis tick interval'};
7399         app.xLimInterval.Position = [275 356 51 16];
7400         app.xLimInterval.Value = 1000;
7401
7402         % Create MaximumdepthEditFieldLabel_6
7403         app.MaximumdepthEditFieldLabel_6 = uilabel(app.Panel_10);
7404         app.MaximumdepthEditFieldLabel_6.HorizontalAlignment = 'center';
7405         app.MaximumdepthEditFieldLabel_6.Position = [278 336 45 22];
7406         app.MaximumdepthEditFieldLabel_6.Text = 'Interval';
7407
7408         % Create CheckBox_2
7409         app.CheckBox_2 = uicheckbox(app.Panel_10);
7410         app.CheckBox_2.ValueChangedFcn = createCallbackFcn(app, ↵
@CheckBox_2ValueChanged, true);
7411         app.CheckBox_2.Text = '';
7412         app.CheckBox_2.WordWrap = 'on';
7413         app.CheckBox_2.FontSize = 10;
7414         app.CheckBox_2.Position = [173 353 21 22];
7415         app.CheckBox_2.Value = true;
7416
7417         % Create MaximumdepthEditFieldLabel_4
7418         app.MaximumdepthEditFieldLabel_4 = uilabel(app.Panel_10);
7419         app.MaximumdepthEditFieldLabel_4.HorizontalAlignment = 'right';
7420         app.MaximumdepthEditFieldLabel_4.Position = [110 446 53 22];
7421         app.MaximumdepthEditFieldLabel_4.Text = 'Y max';
7422
7423         % Create yLimInterval
7424         app.yLimInterval = uieditfield(app.Panel_10, 'numeric');
7425         app.yLimInterval.Limits = [0 Inf];
7426         app.yLimInterval.ValueDisplayFormat = '%8.1f';
7427         app.yLimInterval.ValueChangedFcn = createCallbackFcn(app, ↵
@yLimIntervalValueChanged, true);
7428         app.yLimInterval.FontColor = [0.651 0.651 0.651];
7429         app.yLimInterval.Enable = 'off';
7430         app.yLimInterval.Tooltip = {'Y-axis tick interval'};
7431         app.yLimInterval.Position = [173 428 36 16];
7432         app.yLimInterval.Value = 0.5;
7433
7434         % Create MaximumdepthEditFieldLabel_8
7435         app.MaximumdepthEditFieldLabel_8 = uilabel(app.Panel_10);
7436         app.MaximumdepthEditFieldLabel_8.HorizontalAlignment = 'right';
7437         app.MaximumdepthEditFieldLabel_8.Position = [110 425 53 22];
7438         app.MaximumdepthEditFieldLabel_8.Text = 'Interval';
7439
7440         % Create yMinLimit
7441         app.yMinLimit = uieditfield(app.Panel_10, 'numeric');
7442         app.yMinLimit.ValueDisplayFormat = '%8.1f';
7443         app.yMinLimit.ValueChangedFcn = createCallbackFcn(app, ↵
@yMinLimitValueChanged, true);
7444         app.yMinLimit.FontColor = [0.651 0.651 0.651];
7445         app.yMinLimit.Enable = 'off';
7446         app.yMinLimit.Tooltip = {'Y-axis maximum value'};
7447         app.yMinLimit.Position = [173 407 45 16];
7448         app.yMinLimit.Value = -1;
7449

```

```

7450 % Create MinimumdepthEditFieldLabel_3
7451 app.MinimumdepthEditFieldLabel_3 = uilabel(app.Panel_10);
7452 app.MinimumdepthEditFieldLabel_3.HorizontalAlignment = 'right';
7453 app.MinimumdepthEditFieldLabel_3.Position = [112 404 51 22];
7454 app.MinimumdepthEditFieldLabel_3.Text = 'Y min';
7455
7456 % Create yMaxLimit
7457 app.yMaxLimit = uieditfield(app.Panel_10, 'numeric');
7458 app.yMaxLimit.Limits = [0 Inf];
7459 app.yMaxLimit.ValueDisplayFormat = '%8.1f';
7460 app.yMaxLimit.ValueChangedFcn = createCallbackFcn(app, ↵
@yMaxLimitValueChanged, true);
7461 app.yMaxLimit.FontColor = [0.651 0.651 0.651];
7462 app.yMaxLimit.Enable = 'off';
7463 app.yMaxLimit.Tooltip = {'Y-axis maximum value'};
7464 app.yMaxLimit.Position = [173 449 45 16];
7465 app.yMaxLimit.Value = 1;
7466
7467 % Create CheckBox
7468 app.CheckBox = uicheckbox(app.Panel_10);
7469 app.CheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@CheckBoxValueChanged, true);
7470 app.CheckBox.Text = '';
7471 app.CheckBox.WordWrap = 'on';
7472 app.CheckBox.FontSize = 8;
7473 app.CheckBox.FontWeight = 'bold';
7474 app.CheckBox.Position = [173 370 16 31];
7475 app.CheckBox.Value = true;
7476
7477 % Create YAutoSetLabel
7478 app.YAutoSetLabel = uilabel(app.Panel_10);
7479 app.YAutoSetLabel.HorizontalAlignment = 'right';
7480 app.YAutoSetLabel.FontSize = 10;
7481 app.YAutoSetLabel.Tooltip = {'Automatically set the Y axis ↵
limits'};
7482 app.YAutoSetLabel.Position = [110 374 53 22];
7483 app.YAutoSetLabel.Text = 'Y Auto Set';
7484
7485 % Create XAutoSetLabel
7486 app.XAutoSetLabel = uilabel(app.Panel_10);
7487 app.XAutoSetLabel.HorizontalAlignment = 'right';
7488 app.XAutoSetLabel.FontSize = 10;
7489 app.XAutoSetLabel.Tooltip = {'Automatically set the X axis ↵
limits'};
7490 app.XAutoSetLabel.Position = [110 353 53 22];
7491 app.XAutoSetLabel.Text = 'X Auto Set';
7492
7493 % Create LabelEditFieldLabel
7494 app.LabelEditFieldLabel = uilabel(app.Panel_10);
7495 app.LabelEditFieldLabel.Position = [50 252 39 22];
7496 app.LabelEditFieldLabel.Text = 'Y Axis';
7497
7498 % Create YAxisEditField
7499 app.YAxisEditField = uieditfield(app.Panel_10, 'text');
7500 app.YAxisEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@YAxisEditFieldValueChanged, true);
7501 app.YAxisEditField.FontColor = [0.651 0.651 0.651];
7502 app.YAxisEditField.Position = [105 255 108 16];

```

```

7503     app.YAxisEditField.Value = '\eta (m)';
7504
7505     % Create XandYLimitsLabel_2
7506     app.XandYLimitsLabel_2 = uilabel(app.Panel_10);
7507     app.XandYLimitsLabel_2.BackgroundColor = [0.9412 0.9412 0.9412];
7508     app.XandYLimitsLabel_2.FontWeight = 'bold';
7509     app.XandYLimitsLabel_2.Position = [241 467 91 22];
7510     app.XandYLimitsLabel_2.Text = 'X and Y Limits: ';
7511
7512     % Create YAxisLabelEditField_2Label
7513     app.YAxisLabelEditField_2Label = uilabel(app.Panel_10);
7514     app.YAxisLabelEditField_2Label.Position = [50 276 39 22];
7515     app.YAxisLabelEditField_2Label.Text = 'X Axis';
7516
7517     % Create XAxisEditField
7518     app.XAxisEditField = uieditfield(app.Panel_10, 'text');
7519     app.XAxisEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@XAxisEditFieldValueChanged, true);
7520     app.XAxisEditField.ValueChangingFcn = createCallbackFcn(app, ↵
@XAxisEditFieldValueChanging, true);
7521     app.XAxisEditField.FontColor = [0.651 0.651 0.651];
7522     app.XAxisEditField.Position = [105 279 108 16];
7523     app.XAxisEditField.Value = 'Time (sec)';
7524
7525     % Create BackgroundGridLabel
7526     app.BackgroundGridLabel = uilabel(app.Panel_10);
7527     app.BackgroundGridLabel.BackgroundColor = [0.9412 0.9412 0.9412];
7528     app.BackgroundGridLabel.FontWeight = 'bold';
7529     app.BackgroundGridLabel.Position = [288 179 103 22];
7530     app.BackgroundGridLabel.Text = 'Background Grid';
7531
7532     % Create StyleDropDown_4Label
7533     app.StyleDropDown_4Label = uilabel(app.Panel_10);
7534     app.StyleDropDown_4Label.Tooltip = {'Line style of the grid↵
lines'};
7535     app.StyleDropDown_4Label.Position = [289 155 32 22];
7536     app.StyleDropDown_4Label.Text = 'Style';
7537
7538     % Create GridStyle
7539     app.GridStyle = uidropdown(app.Panel_10);
7540     app.GridStyle.Items = {'None', 'X axis only', 'Y axis only', 'Both↵
- major lines', 'Both - with minor lines'};
7541     app.GridStyle.DropDownOpeningFcn = createCallbackFcn(app, ↵
@GridStyleDropDownOpening, true);
7542     app.GridStyle.ValueChangedFcn = createCallbackFcn(app, ↵
@GridStyleValueChanged, true);
7543     app.GridStyle.Tooltip = {' '};
7544     app.GridStyle.Position = [364 157 100 17];
7545     app.GridStyle.Value = 'None';
7546
7547     % Create Lineaty0DropDownLabel
7548     app.Lineaty0DropDownLabel = uilabel(app.Panel_10);
7549     app.Lineaty0DropDownLabel.WordWrap = 'on';
7550     app.Lineaty0DropDownLabel.Tooltip = {'Add a horizontal line at y =↵
0'};
7551     app.Lineaty0DropDownLabel.Position = [289 108 69 22];
7552     app.Lineaty0DropDownLabel.Text = 'Line at y=0';
7553

```

```

7554         % Create Lineaty0DropDown
7555         app.Lineaty0DropDown = uidropdown(app.Panel_10);
7556         app.Lineaty0DropDown.Items = {'None', 'Solid', 'Dashed', 'Dotted', '
'Dash-dotted'};
7557         app.Lineaty0DropDown.ValueChangedFcn = createCallbackFcn(app, '
@Lineaty0DropDownValueChanged', true);
7558         app.Lineaty0DropDown.Tooltip = {' '};
7559         app.Lineaty0DropDown.Position = [364 110 100 17];
7560         app.Lineaty0DropDown.Value = 'Dotted';
7561
7562         % Create LineYThickness
7563         app.LineYThickness = uieditfield(app.Panel_10, 'numeric');
7564         app.LineYThickness.Limits = [0.1 Inf];
7565         app.LineYThickness.ValueDisplayFormat = '%2.1f';
7566         app.LineYThickness.ValueChangedFcn = createCallbackFcn(app, '
@LineYThicknessValueChanged', true);
7567         app.LineYThickness.HorizontalAlignment = 'left';
7568         app.LineYThickness.FontColor = [0.651 0.651 0.651];
7569         app.LineYThickness.Tooltip = {' '};
7570         app.LineYThickness.Position = [364 87 29 16];
7571         app.LineYThickness.Value = 0.5;
7572
7573         % Create ThicknessLabel_3
7574         app.ThicknessLabel_3 = uilabel(app.Panel_10);
7575         app.ThicknessLabel_3.WordWrap = 'on';
7576         app.ThicknessLabel_3.Position = [289 82 71 27];
7577         app.ThicknessLabel_3.Text = 'Width at y=0';
7578
7579         % Create ThicknessLabel_5
7580         app.ThicknessLabel_5 = uilabel(app.Panel_10);
7581         app.ThicknessLabel_5.Position = [50 227 53 22];
7582         app.ThicknessLabel_5.Text = 'Text Size';
7583
7584         % Create GridLabelSize
7585         app.GridLabelSize = uieditfield(app.Panel_10, 'numeric');
7586         app.GridLabelSize.Limits = [1 Inf];
7587         app.GridLabelSize.ValueDisplayFormat = '%2.0f';
7588         app.GridLabelSize.ValueChangedFcn = createCallbackFcn(app, '
@GridLabelSizeValueChanged', true);
7589         app.GridLabelSize.FontColor = [0.651 0.651 0.651];
7590         app.GridLabelSize.Position = [106 230 29 16];
7591         app.GridLabelSize.Value = 12;
7592
7593         % Create GaugeLineColor_multiple
7594         app.GaugeLineColor_multiple = uidropdown(app.Panel_10);
7595         app.GaugeLineColor_multiple.Items = {'autumn', 'bone', '
'colorcube', 'cool', 'copper', 'flag', 'gray', 'hot', 'hsv', 'jet', 'lines', '
'parula', 'pink', 'prism', 'spring', 'summer', 'turbo', 'winter'};
7596         app.GaugeLineColor_multiple.ValueChangedFcn = createCallbackFcn '
(app, @GaugeLineColor_multipleValueChanged, true);
7597         app.GaugeLineColor_multiple.Position = [108 87 89 17];
7598         app.GaugeLineColor_multiple.Value = 'lines';
7599
7600         % Create FlipCheckBox_3
7601         app.FlipCheckBox_3 = uicheckbox(app.Panel_10);
7602         app.FlipCheckBox_3.Tooltip = {'Reverse the color sequence'};
7603         app.FlipCheckBox_3.Text = 'Flip';
7604         app.FlipCheckBox_3.FontSize = 11;

```

```

7605         app.FlipCheckBox_3.Position = [207 85 40 22];
7606
7607         % Create Height
7608         app.Height = uieditfield(app.Panel_10, 'numeric');
7609         app.Height.Limits = [1 Inf];
7610         app.Height.ValueDisplayFormat = '%2.0f';
7611         app.Height.ValueChangedFcn = createCallbackFcn(app, ↵
@HeightValueChanged, true);
7612         app.Height.FontColor = [0.651 0.651 0.651];
7613         app.Height.Enable = 'off';
7614         app.Height.Tooltip = {' '};
7615         app.Height.Position = [283 27 40 16];
7616         app.Height.Value = 8;
7617
7618         % Create ThicknessLabel_6
7619         app.ThicknessLabel_6 = uilabel(app.Panel_10);
7620         app.ThicknessLabel_6.Tooltip = {'Unit: inches'};
7621         app.ThicknessLabel_6.Position = [239 24 41 22];
7622         app.ThicknessLabel_6.Text = 'Height';
7623
7624         % Create FigureSizeLabel
7625         app.FigureSizeLabel = uilabel(app.Panel_10);
7626         app.FigureSizeLabel.FontWeight = 'bold';
7627         app.FigureSizeLabel.Position = [50 43 69 22];
7628         app.FigureSizeLabel.Text = 'Figure Size';
7629
7630         % Create Width
7631         app.Width = uieditfield(app.Panel_10, 'numeric');
7632         app.Width.Limits = [1 Inf];
7633         app.Width.ValueDisplayFormat = '%2.0f';
7634         app.Width.ValueChangedFcn = createCallbackFcn(app, ↵
@WidthValueChanged, true);
7635         app.Width.FontColor = [0.651 0.651 0.651];
7636         app.Width.Enable = 'off';
7637         app.Width.Tooltip = {' '};
7638         app.Width.Position = [182 27 41 16];
7639         app.Width.Value = 11;
7640
7641         % Create ThicknessLabel_7
7642         app.ThicknessLabel_7 = uilabel(app.Panel_10);
7643         app.ThicknessLabel_7.Tooltip = {'Unit: inches'};
7644         app.ThicknessLabel_7.Position = [139 24 37 22];
7645         app.ThicknessLabel_7.Text = 'Width';
7646
7647         % Create AutoSetCheckBox
7648         app.AutoSetCheckBox = uicheckbox(app.Panel_10);
7649         app.AutoSetCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@AutoSetCheckBoxValueChanged, true);
7650         app.AutoSetCheckBox.Text = 'Auto Set';
7651         app.AutoSetCheckBox.Position = [50 24 68 22];
7652         app.AutoSetCheckBox.Value = true;
7653
7654         % Create CloseFiguresButton_2
7655         app.CloseFiguresButton_2 = uibutton(app.Panel_10, 'push');
7656         app.CloseFiguresButton_2.ButtonPushedFcn = createCallbackFcn(app, ↵
@CloseFiguresButton_2Pushed2, true);
7657         app.CloseFiguresButton_2.Tooltip = {' '};
7658         app.CloseFiguresButton_2.Position = [364 24 89 20];

```

```

7659         app.CloseFiguresButton_2.Text = 'Close Figures';
7660
7661         % Create LineYThickness_2
7662         app.LineYThickness_2 = uieditfield(app.Panel_10, 'numeric');
7663         app.LineYThickness_2.Limits = [0.1 Inf];
7664         app.LineYThickness_2.ValueDisplayFormat = '%2.1f';
7665         app.LineYThickness_2.ValueChangedFcn = createCallbackFcn(app, ↵
@LineYThickness_2ValueChanged, true);
7666         app.LineYThickness_2.Editable = 'off';
7667         app.LineYThickness_2.FontColor = [0.651 0.651 0.651];
7668         app.LineYThickness_2.Position = [364 134 29 16];
7669         app.LineYThickness_2.Value = 0.5;
7670
7671         % Create ThicknessLabel_12
7672         app.ThicknessLabel_12 = uilabel(app.Panel_10);
7673         app.ThicknessLabel_12.Position = [289 131 37 22];
7674         app.ThicknessLabel_12.Text = 'Width';
7675
7676         % Create LegendLabel
7677         app.LegendLabel = uilabel(app.Panel_10);
7678         app.LegendLabel.FontWeight = 'bold';
7679         app.LegendLabel.Position = [287 297 48 22];
7680         app.LegendLabel.Text = 'Legend';
7681
7682         % Create legendfirsttextlabel
7683         app.legendfirsttextlabel = uilabel(app.Panel_10);
7684         app.legendfirsttextlabel.Tooltip = {'Text added to data labels, ↵
for example: Station 1, Station 2'};
7685         app.legendfirsttextlabel.Position = [288 276 35 22];
7686
7687         % Create legendfirsttext
7688         app.legendfirsttext = uieditfield(app.Panel_10, 'text');
7689         app.legendfirsttext.ValueChangedFcn = createCallbackFcn(app, ↵
@legendfirsttextValueChanged, true);
7690         app.legendfirsttext.FontColor = [0.651 0.651 0.651];
7691         app.legendfirsttext.Tooltip = {' '};
7692         app.legendfirsttext.Position = [345 279 104 16];
7693         app.legendfirsttext.Value = 'Station';
7694
7695         % Create LocationDropDownLabel
7696         app.LocationDropDownLabel = uilabel(app.Panel_10);
7697         app.LocationDropDownLabel.Position = [288 227 52 22];
7698         app.LocationDropDownLabel.Text = 'Location';
7699
7700         % Create LocationDropDown
7701         app.LocationDropDown = uidropdown(app.Panel_10);
7702         app.LocationDropDown.Items = {'North', 'South', 'East', 'West', ↵
'Northeast', 'Northwest', 'Southeast', 'Southwest', 'Northoutside', 'Southoutside', ↵
'Eastoutside', 'Westoutside', 'Northeastoutside', 'Northwestoutside', ↵
'Southeastoutside', 'Southwestoutside', 'Best', 'Bestoutside', 'None'};
7703         app.LocationDropDown.ValueChangedFcn = createCallbackFcn(app, ↵
@LocationDropDownValueChanged, true);
7704         app.LocationDropDown.Tooltip = {' '};
7705         app.LocationDropDown.Position = [345 230 104 17];
7706         app.LocationDropDown.Value = 'Best';
7707
7708         % Create LegendSize
7709         app.LegendSize = uieditfield(app.Panel_10, 'numeric');

```



```

7710         app.LegendSize.Limits = [1 Inf];
7711         app.LegendSize.ValueDisplayFormat = '%2.0f';
7712         app.LegendSize.ValueChangedFcn = createCallbackFcn(app, ↵
@LegendSizeValueChanged, true);
7713         app.LegendSize.FontColor = [0.651 0.651 0.651];
7714         app.LegendSize.Position = [345 255 29 16];
7715         app.LegendSize.Value = 12;
7716
7717         % Create ThicknessLabel_4
7718         app.ThicknessLabel_4 = uilabel(app.Panel_10);
7719         app.ThicknessLabel_4.Position = [288 252 53 22];
7720         app.ThicknessLabel_4.Text = 'Text Size';
7721
7722         % Create MarkerDropDownLabel
7723         app.MarkerDropDownLabel = uilabel(app.Panel_10);
7724         app.MarkerDropDownLabel.Position = [50 155 43 22];
7725         app.MarkerDropDownLabel.Text = 'Marker';
7726
7727         % Create MarkerDropDown
7728         app.MarkerDropDown = uidropdown(app.Panel_10);
7729         app.MarkerDropDown.Items = {'none', 'o', '+', '*', '.', 'x', '-', ↵
'|', '^', 'v', '>', '<', 'diamond', 'hexagram', 'pentagram', 'square'};
7730         app.MarkerDropDown.Tooltip = {'none'; 'o'; '+'; '*'};
7731         app.MarkerDropDown.Position = [106 157 92 17];
7732         app.MarkerDropDown.Value = 'none';
7733
7734         % Create PLOTButton
7735         app.PLOTButton = uibutton(app.GaugeRecordsTab, 'push');
7736         app.PLOTButton.ButtonPushedFcn = createCallbackFcn(app, ↵
@PLOTButtonPushed, true);
7737         app.PLOTButton.FontSize = 14;
7738         app.PLOTButton.FontWeight = 'bold';
7739         app.PLOTButton.Position = [179 6 183 25];
7740         app.PLOTButton.Text = 'PLOT';
7741
7742         % Create Panel_11
7743         app.Panel_11 = uipanel(app.GaugeRecordsTab);
7744         app.Panel_11.AutoResizeChildren = 'off';
7745         app.Panel_11.SizeChangedFcn = createCallbackFcn(app, ↵
@Panel_11SizeChanged, true);
7746         app.Panel_11.Position = [11 703 520 119];
7747
7748         % Create FilesLabel_2
7749         app.FilesLabel_2 = uilabel(app.Panel_11);
7750         app.FilesLabel_2.FontWeight = 'bold';
7751         app.FilesLabel_2.Tooltip = {'Load the files.'; ''; 'Files must ↵
start with 'sta_''};
7752         app.FilesLabel_2.Position = [44 80 32 22];
7753         app.FilesLabel_2.Text = 'Files';
7754
7755         % Create FileTextArea_2
7756         app.FileTextArea_2 = uitextarea(app.Panel_11);
7757         app.FileTextArea_2.ValueChangedFcn = createCallbackFcn(app, ↵
@FileTextArea_2ValueChanged, true);
7758         app.FileTextArea_2.Editable = 'off';
7759         app.FileTextArea_2.FontColor = [0.651 0.651 0.651];
7760         app.FileTextArea_2.Tooltip = {''};
7761         app.FileTextArea_2.Placeholder = 'sta xxxx';

```



```

7762         app.FileTextArea_2.Position = [44 19 95 60];
7763
7764         % Create Button_16
7765         app.Button_16 = uibutton(app.Panel_11, 'push');
7766         app.Button_16.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_16Pushed, true);
7767         app.Button_16.FontAngle = 'italic';
7768         app.Button_16.FontColor = [0.651 0.651 0.651];
7769         app.Button_16.Tooltip = {'Load the files.'; ''; 'Files must start↵
with 'sta_''};
7770         app.Button_16.Position = [146 60 19 19];
7771         app.Button_16.Text = '...';
7772
7773         % Create XaxisLabel
7774         app.XaxisLabel = uilabel(app.Panel_11);
7775         app.XaxisLabel.Tooltip = {'Select column to use for plotting.';↵
''; 'Unit for each column: '; '1: time (s)'; '2: eta (m)'; '3: u vector (m/s)'; '4:↵
v vector (m/s)'; '5: z vector (m/s)'};
7776         app.XaxisLabel.Position = [188 57 44 22];
7777         app.XaxisLabel.Text = 'X axis: ';
7778
7779         % Create SelectColumnLabel
7780         app.SelectColumnLabel = uilabel(app.Panel_11);
7781         app.SelectColumnLabel.FontWeight = 'bold';
7782         app.SelectColumnLabel.Tooltip = {'Select column to use for↵
plotting.'; ''; 'Unit for each column: '; '1: time (s)'; '2: eta (m)'; '3: u vector↵
(m/s)'; '4: v vector (m/s)'; '5: z vector (m/s)'};
7783         app.SelectColumnLabel.Position = [188 80 89 22];
7784         app.SelectColumnLabel.Text = 'Select Column';
7785
7786         % Create XaxisdataLabel
7787         app.XaxisdataLabel = uilabel(app.Panel_11);
7788         app.XaxisdataLabel.Tooltip = {'Example: '; 'Convert column 1 data↵
from seconds to minutes. '};
7789         app.XaxisdataLabel.Position = [357 56 65 22];
7790         app.XaxisdataLabel.Text = 'X axis data';
7791
7792         % Create ConvertLabel
7793         app.ConvertLabel = uilabel(app.Panel_11);
7794         app.ConvertLabel.FontWeight = 'bold';
7795         app.ConvertLabel.Tooltip = {'Examples: '; 'Convert column 1 data↵
from seconds to minutes. '; ''; 'Convert column 2 data from meters to↵
centimeters.'};
7796         app.ConvertLabel.Position = [333 80 54 22];
7797         app.ConvertLabel.Text = 'Convert: ';
7798
7799         % Create numeratorY
7800         app.numeratorY = uieditfield(app.Panel_11, 'numeric');
7801         app.numeratorY.Limits = [1 Inf];
7802         app.numeratorY.ValueDisplayFormat = '%8.0f';
7803         app.numeratorY.HorizontalAlignment = 'center';
7804         app.numeratorY.FontColor = [0.651 0.651 0.651];
7805         app.numeratorY.Enable = 'off';
7806         app.numeratorY.Position = [445 26 36 16];
7807         app.numeratorY.Value = 100;
7808
7809         % Create denominatorY
7810         app.denominatorY = uieditfield(app.Panel 11, 'numeric');

```

```

7811         app.denominatorY.Limits = [1 Inf];
7812         app.denominatorY.ValueDisplayFormat = '%8.0f';
7813         app.denominatorY.ValueChangedFcn = createCallbackFcn(app, ↵
@denominatorYValueChanged, true);
7814         app.denominatorY.HorizontalAlignment = 'center';
7815         app.denominatorY.FontColor = [0.651 0.651 0.651];
7816         app.denominatorY.Enable = 'off';
7817         app.denominatorY.Position = [445 11 36 16];
7818         app.denominatorY.Value = 1;
7819
7820         % Create Label_9
7821         app.Label_9 = uilabel(app.Panel_11);
7822         app.Label_9.Position = [424 19 13 22];
7823         app.Label_9.Text = 'x';
7824
7825         % Create YaxisEditFieldLabel
7826         app.YaxisEditFieldLabel = uilabel(app.Panel_11);
7827         app.YaxisEditFieldLabel.Tooltip = {'Example: Convert column 2 data ↵
from meters to centimeters.'};
7828         app.YaxisEditFieldLabel.Position = [357 19 65 22];
7829         app.YaxisEditFieldLabel.Text = 'Y axis data';
7830
7831         % Create CheckBox_3
7832         app.CheckBox_3 = uicheckbox(app.Panel_11);
7833         app.CheckBox_3.ValueChangedFcn = createCallbackFcn(app, ↵
@CheckBox_3ValueChanged, true);
7834         app.CheckBox_3.Text = '';
7835         app.CheckBox_3.Position = [333 56 25 22];
7836
7837         % Create CheckBox_4
7838         app.CheckBox_4 = uicheckbox(app.Panel_11);
7839         app.CheckBox_4.ValueChangedFcn = createCallbackFcn(app, ↵
@CheckBox_4ValueChanged, true);
7840         app.CheckBox_4.Tooltip = {'Only the input in the first textbox ↵
under ''Y axis: Use data from column'' will be converted'};
7841         app.CheckBox_4.Text = '';
7842         app.CheckBox_4.Position = [333 19 25 22];
7843
7844         % Create numeratorX
7845         app.numeratorX = uieditfield(app.Panel_11, 'numeric');
7846         app.numeratorX.Limits = [1 Inf];
7847         app.numeratorX.ValueDisplayFormat = '%8.0f';
7848         app.numeratorX.HorizontalAlignment = 'center';
7849         app.numeratorX.FontColor = [0.651 0.651 0.651];
7850         app.numeratorX.Enable = 'off';
7851         app.numeratorX.Position = [445 64 36 16];
7852         app.numeratorX.Value = 1;
7853
7854         % Create denominatorX
7855         app.denominatorX = uieditfield(app.Panel_11, 'numeric');
7856         app.denominatorX.Limits = [1 Inf];
7857         app.denominatorX.ValueDisplayFormat = '%8.0f';
7858         app.denominatorX.HorizontalAlignment = 'center';
7859         app.denominatorX.FontColor = [0.651 0.651 0.651];
7860         app.denominatorX.Enable = 'off';
7861         app.denominatorX.Position = [445 49 36 16];
7862         app.denominatorX.Value = 60;
7863

```

```

7864 % Create Label_5
7865 app.Label_5 = uilabel(app.Panel_11);
7866 app.Label_5.Position = [424 56 13 22];
7867 app.Label_5.Text = 'x';
7868
7869 % Create Label_6
7870 app.Label_6 = uilabel(app.Panel_11);
7871 app.Label_6.FontSize = 22;
7872 app.Label_6.Position = [435 51 10 29];
7873 app.Label_6.Text = '(';
7874
7875 % Create Label_7
7876 app.Label_7 = uilabel(app.Panel_11);
7877 app.Label_7.FontSize = 22;
7878 app.Label_7.Position = [487 52 10 29];
7879 app.Label_7.Text = ')';
7880
7881 % Create Label_10
7882 app.Label_10 = uilabel(app.Panel_11);
7883 app.Label_10.FontSize = 22;
7884 app.Label_10.Position = [435 12 10 29];
7885 app.Label_10.Text = '(';
7886
7887 % Create Label_11
7888 app.Label_11 = uilabel(app.Panel_11);
7889 app.Label_11.FontSize = 22;
7890 app.Label_11.Position = [487 13 10 29];
7891 app.Label_11.Text = ')';
7892
7893 % Create YaxisSetcolumncountLabel
7894 app.YaxisSetcolumncountLabel = uilabel(app.Panel_11);
7895 app.YaxisSetcolumncountLabel.Tooltip = {'Select column to use for
plotting.'; ''; 'Unit for each column: '; '1: time (s)'; '2: eta (m)'; '3: u vector
(m/s)'; '4: v vector (m/s)'; '5: z vector (m/s)'};
7896 app.YaxisSetcolumncountLabel.Position = [188 33 45 22];
7897 app.YaxisSetcolumncountLabel.Text = 'Y axis: ';
7898
7899 % Create YaxisUseDataDropDown
7900 app.YaxisUseDataDropDown = uidropdown(app.Panel_11);
7901 app.YaxisUseDataDropDown.Items = {'1', '2', '3', '4', '5',
'3,4,5'};
7902 app.YaxisUseDataDropDown.DropDownOpeningFcn = createCallbackFcn
(app, @YaxisUseDataDropDownOpening, true);
7903 app.YaxisUseDataDropDown.ValueChangedFcn = createCallbackFcn(app,
@YaxisUseDataDropDownValueChanged, true);
7904 app.YaxisUseDataDropDown.Tooltip = {'';
7905 app.YaxisUseDataDropDown.Position = [231 36 61 16];
7906 app.YaxisUseDataDropDown.Value = '2';
7907
7908 % Create XaxisUseDataDropDown
7909 app.XaxisUseDataDropDown = uidropdown(app.Panel_11);
7910 app.XaxisUseDataDropDown.Items = {'1', '2', '3', '4'};
7911 app.XaxisUseDataDropDown.ValueChangedFcn = createCallbackFcn(app,
@XaxisUseDataDropDownValueChanged, true);
7912 app.XaxisUseDataDropDown.Tooltip = {'';
7913 app.XaxisUseDataDropDown.Position = [231 61 41 16];
7914 app.XaxisUseDataDropDown.Value = '1';
7915

```

```

7916         % Create InputDataLabel_3
7917         app.InputDataLabel_3 = uilabel(app.GaugeRecordsTab);
7918         app.InputDataLabel_3.BackgroundColor = [0.9412 0.9412 0.9412];
7919         app.InputDataLabel_3.FontSize = 15;
7920         app.InputDataLabel_3.FontWeight = 'bold';
7921         app.InputDataLabel_3.FontColor = [0.0314 0.3686 0.6];
7922         app.InputDataLabel_3.Position = [38 813 97 22];
7923         app.InputDataLabel_3.Text = '    Input Data    ';
7924
7925         % Create Panel_13
7926         app.Panel_13 = uipanel(app.GaugeRecordsTab);
7927         app.Panel_13.AutoResizeChildren = 'off';
7928         app.Panel_13.Position = [11 39 520 77];
7929
7930         % Create pngCheckBox_2
7931         app.pngCheckBox_2 = uicheckbox(app.Panel_13);
7932         app.pngCheckBox_2.Tooltip = {'Resolution is set to 300 DPI'};
7933         app.pngCheckBox_2.Text = '.png';
7934         app.pngCheckBox_2.Position = [192 12 46 22];
7935
7936         % Create FileFormatLabel_2
7937         app.FileFormatLabel_2 = uilabel(app.Panel_13);
7938         app.FileFormatLabel_2.Position = [68 12 66 22];
7939         app.FileFormatLabel_2.Text = 'File Format';
7940
7941         % Create jpgCheckBox_2
7942         app.jpgCheckBox_2 = uicheckbox(app.Panel_13);
7943         app.jpgCheckBox_2.Tooltip = {'Resolution is set to 300 DPI'};
7944         app.jpgCheckBox_2.Text = '.jpg';
7945         app.jpgCheckBox_2.Position = [139 12 42 22];
7946
7947         % Create pdfCheckBox_2
7948         app.pdfCheckBox_2 = uicheckbox(app.Panel_13);
7949         app.pdfCheckBox_2.Tooltip = {'Vector-type pdf file'};
7950         app.pdfCheckBox_2.Text = '.pdf';
7951         app.pdfCheckBox_2.Position = [249 12 43 22];
7952
7953         % Create Button_18
7954         app.Button_18 = uibutton(app.Panel_13, 'push');
7955         app.Button_18.ButtonPushedFcn = createCallbackFcn(app, @
@Button_18Pushed, true);
7956         app.Button_18.FontSize = 10;
7957         app.Button_18.FontAngle = 'italic';
7958         app.Button_18.Tooltip = {'Load the preferred directory.'; '';
'"OUTPUT_FILES/Figures" folder will be created in the selected directory.'};
7959         app.Button_18.Position = [403 42 19 19];
7960         app.Button_18.Text = '...';
7961
7962         % Create OutputDirectoryLabel_3
7963         app.OutputDirectoryLabel_3 = uilabel(app.Panel_13);
7964         app.OutputDirectoryLabel_3.Position = [40 40 94 22];
7965         app.OutputDirectoryLabel_3.Text = 'Output Directory';
7966
7967         % Create OutputDirectoryEditField_2
7968         app.OutputDirectoryEditField_2 = uieditfield(app.Panel_13,
'text');
7969         app.OutputDirectoryEditField_2.ValueChangedFcn = createCallbackFcn
(app, @OutputDirectoryEditField_2ValueChanged, true);

```

```

7970         app.OutputDirectoryEditField_2.Editable = 'off';
7971         app.OutputDirectoryEditField_2.FontColor = [0 0 1];
7972         app.OutputDirectoryEditField_2.Tooltip = {'Load the preferred
directory.'; ''; "'OUTPUT_FILES/Figures'" folder will be created in the selected
directory.'};
7973         app.OutputDirectoryEditField_2.Placeholder = 'Default: Desktop';
7974         app.OutputDirectoryEditField_2.Position = [139 43 257 16];
7975
7976         % Create txtCheckBox
7977         app.txtCheckBox = uicheckbox(app.Panel_13);
7978         app.txtCheckBox.ValueChangedFcn = createCallbackFcn(app,
@txtCheckBoxValueChanged, true);
7979         app.txtCheckBox.Enable = 'off';
7980         app.txtCheckBox.Tooltip = {'Save raw data from 'sta_' files as
tab-delimited texts '; '; 'Functionality enabled when "Plot separately" is
selected.'};
7981         app.txtCheckBox.Text = '.txt';
7982         app.txtCheckBox.Position = [356 12 39 22];
7983
7984         % Create epsCheckBox
7985         app.epsCheckBox = uicheckbox(app.Panel_13);
7986         app.epsCheckBox.Tooltip = {'Vector graphic file'};
7987         app.epsCheckBox.Text = '.eps';
7988         app.epsCheckBox.Position = [303 12 45 22];
7989
7990         % Create SavePlotLabel
7991         app.SavePlotLabel = uilabel(app.GaugeRecordsTab);
7992         app.SavePlotLabel.BackgroundColor = [0.9412 0.9412 0.9412];
7993         app.SavePlotLabel.FontSize = 15;
7994         app.SavePlotLabel.FontWeight = 'bold';
7995         app.SavePlotLabel.FontColor = [0.0314 0.3686 0.6];
7996         app.SavePlotLabel.Position = [38 105 92 22];
7997         app.SavePlotLabel.Text = ' Save Plot ';
7998
7999         % Create PlotStyleandLayoutLabel
8000         app.PlotStyleandLayoutLabel = uilabel(app.GaugeRecordsTab);
8001         app.PlotStyleandLayoutLabel.BackgroundColor = [0.9412 0.9412
0.9412];
8002         app.PlotStyleandLayoutLabel.FontSize = 15;
8003         app.PlotStyleandLayoutLabel.FontWeight = 'bold';
8004         app.PlotStyleandLayoutLabel.FontColor = [0.0314 0.3686 0.6];
8005         app.PlotStyleandLayoutLabel.Position = [34 681 174 22];
8006         app.PlotStyleandLayoutLabel.Text = ' Plot Style and Layout ';
8007
8008         % Create VelocityMapTab
8009         app.VelocityMapTab = uitab(app.TabGroup);
8010         app.VelocityMapTab.AutoResizeChildren = 'off';
8011         app.VelocityMapTab.SizeChangedFcn = createCallbackFcn(app,
@VelocityMapTabSizeChanged, true);
8012         app.VelocityMapTab.Title = 'Velocity Map';
8013
8014         % Create GENERATEButton_2
8015         app.GENERATEButton_2 = uibutton(app.VelocityMapTab, 'push');
8016         app.GENERATEButton_2.ButtonPushedFcn = createCallbackFcn(app,
@GENERATEButton_2Pushed, true);
8017         app.GENERATEButton_2.FontSize = 14;
8018         app.GENERATEButton_2.FontWeight = 'bold';
8019         app.GENERATEButton_2.Position = [179 6 183 25];

```

```

8020     app.GENERATEButton_2.Text = 'GENERATE';
8021
8022     % Create Panel_14
8023     app.Panel_14 = uipanel(app.VelocityMapTab);
8024     app.Panel_14.AutoResizeChildren = 'off';
8025     app.Panel_14.Position = [11 550 520 65];
8026
8027     % Create QuiverColorDropDown
8028     app.QuiverColorDropDown = uidropdown(app.Panel_14);
8029     app.QuiverColorDropDown.Items = {'Black', 'Dark gray', 'Medium gray', 'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
8030     app.QuiverColorDropDown.ValueChangedFcn = createCallbackFcn(app, @QuiverColorDropDownValueChanged, true);
8031     app.QuiverColorDropDown.Position = [396 12 87 16];
8032     app.QuiverColorDropDown.Value = 'Black';
8033
8034     % Create XEditField_2Label_3
8035     app.XEditField_2Label_3 = uilabel(app.Panel_14);
8036     app.XEditField_2Label_3.Tooltip = {'Size scaling'};
8037     app.XEditField_2Label_3.Position = [131 9 67 22];
8038     app.XEditField_2Label_3.Text = 'Body Scale';
8039
8040     % Create arrowscale
8041     app.arrowscale = uieditfield(app.Panel_14, 'numeric');
8042     app.arrowscale.Limits = [0 Inf];
8043     app.arrowscale.ValueDisplayFormat = '%8.1f';
8044     app.arrowscale.ValueChangedFcn = createCallbackFcn(app, @arrowscaleValueChanged, true);
8045     app.arrowscale.FontColor = [0.651 0.651 0.651];
8046     app.arrowscale.Tooltip = {'Scaling factor. '; ''; 'Set to 0 to disable auto scaling. '};
8047     app.arrowscale.Position = [228 12 35 16];
8048     app.arrowscale.Value = 10;
8049
8050     % Create ArrowThicknessLabel
8051     app.ArrowThicknessLabel = uilabel(app.Panel_14);
8052     app.ArrowThicknessLabel.HorizontalAlignment = 'right';
8053     app.ArrowThicknessLabel.Tooltip = {'Line thickness'};
8054     app.ArrowThicknessLabel.Position = [281 9 37 22];
8055     app.ArrowThicknessLabel.Text = 'Width';
8056
8057     % Create ArrowThickness
8058     app.ArrowThickness = uieditfield(app.Panel_14, 'numeric');
8059     app.ArrowThickness.Limits = [0 Inf];
8060     app.ArrowThickness.ValueDisplayFormat = '%.1f';
8061     app.ArrowThickness.ValueChangedFcn = createCallbackFcn(app, @ArrowThicknessValueChanged, true);
8062     app.ArrowThickness.FontColor = [0.651 0.651 0.651];
8063     app.ArrowThickness.Tooltip = {' '};
8064     app.ArrowThickness.Position = [335 12 35 16];
8065     app.ArrowThickness.Value = 1;
8066
8067     % Create ColorLabel_2
8068     app.ColorLabel_2 = uilabel(app.Panel_14);
8069     app.ColorLabel_2.Position = [397 30 35 22];
8070     app.ColorLabel_2.Text = 'Color';
8071
8072     % Create ArrowSizeLabel

```

```

8073     app.ArrowSizeLabel = uilabel(app.Panel_14);
8074     app.ArrowSizeLabel.WordWrap = 'on';
8075     app.ArrowSizeLabel.Tooltip = {'Arrowhead size relative to the body'↵
size'};
8076     app.ArrowSizeLabel.Position = [131 27 85 29];
8077     app.ArrowSizeLabel.Text = 'Arrowhead Size';
8078
8079     % Create ArrowHeadSize
8080     app.ArrowHeadSize = uieditfield(app.Panel_14, 'numeric');
8081     app.ArrowHeadSize.Limits = [0 1];
8082     app.ArrowHeadSize.ValueDisplayFormat = '%.1f';
8083     app.ArrowHeadSize.ValueChangedFcn = createCallbackFcn(app, ↵
@ArrowHeadSizeValueChanged, true);
8084     app.ArrowHeadSize.FontColor = [0.651 0.651 0.651];
8085     app.ArrowHeadSize.Tooltip = {''};
8086     app.ArrowHeadSize.Position = [228 33 35 16];
8087     app.ArrowHeadSize.Value = 0.2;
8088
8089     % Create PlotvectorsCheckBox
8090     app.PlotvectorsCheckBox = uicheckbox(app.Panel_14);
8091     app.PlotvectorsCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@PlotvectorsCheckBoxValueChanged, true);
8092     app.PlotvectorsCheckBox.Tooltip = {'Overlay vectors on the map. '↵
'; 'Applicable when U and V vectors are loaded.'};
8093     app.PlotvectorsCheckBox.Text = 'Plot vectors';
8094     app.PlotvectorsCheckBox.FontWeight = 'bold';
8095     app.PlotvectorsCheckBox.Position = [31 19 91 22];
8096
8097     % Create SpacingLabel
8098     app.SpacingLabel = uilabel(app.Panel_14);
8099     app.SpacingLabel.HorizontalAlignment = 'right';
8100     app.SpacingLabel.Tooltip = {'Spacing between arrows to avoid'↵
clustering on the map'};
8101     app.SpacingLabel.Position = [278 30 50 22];
8102     app.SpacingLabel.Text = 'Spacing';
8103
8104     % Create ArrowSpacing
8105     app.ArrowSpacing = uieditfield(app.Panel_14, 'numeric');
8106     app.ArrowSpacing.Limits = [0.001 Inf];
8107     app.ArrowSpacing.ValueDisplayFormat = '%11.2f';
8108     app.ArrowSpacing.ValueChangedFcn = createCallbackFcn(app, ↵
@ArrowSpacingValueChanged, true);
8109     app.ArrowSpacing.FontColor = [0.651 0.651 0.651];
8110     app.ArrowSpacing.Position = [337 33 35 16];
8111     app.ArrowSpacing.Value = 1;
8112
8113     % Create Panel_15
8114     app.Panel_15 = uipanel(app.VelocityMapTab);
8115     app.Panel_15.AutoResizeChildren = 'off';
8116     app.Panel_15.SizeChangedFcn = createCallbackFcn(app, ↵
@Panel_15SizeChanged, true);
8117     app.Panel_15.Position = [11 626 520 196];
8118
8119     % Create FileTextArea_3
8120     app.FileTextArea_3 = uitextarea(app.Panel_15);
8121     app.FileTextArea_3.ValueChangedFcn = createCallbackFcn(app, ↵
@FileTextArea_3ValueChanged, true);
8122     app.FileTextArea_3.Editable = 'off';

```



```

8123     app.FileTextArea_3.Tooltip = {''};
8124     app.FileTextArea_3.Placeholder = 'u_xxxx; umean_xxxx; umax_xxxx';
8125     app.FileTextArea_3.Position = [32 77 84 74];
8126
8127     % Create UvectorsLabel
8128     app.UvectorsLabel = uilabel(app.Panel_15);
8129     app.UvectorsLabel.WordWrap = 'on';
8130     app.UvectorsLabel.FontWeight = 'bold';
8131     app.UvectorsLabel.Tooltip = {'Load files with filenames starting
with 'u''};
8132     app.UvectorsLabel.Position = [33 157 61 22];
8133     app.UvectorsLabel.Text = 'U vectors';
8134
8135     % Create DepthFileEditField_2
8136     app.DepthFileEditField_2 = ueditfield(app.Panel_15, 'text');
8137     app.DepthFileEditField_2.ValueChangedFcn = createCallbackFcn(app,
@DepthFileEditField_2ValueChanged, true);
8138     app.DepthFileEditField_2.Editable = 'off';
8139     app.DepthFileEditField_2.Tooltip = {'Load the bathymetry file used
in the simulation'; ''; 'Format accepted: .txt, .tif'};
8140     app.DepthFileEditField_2.Placeholder = '.txt, .tif., mask_';
8141     app.DepthFileEditField_2.Position = [33 22 96 16];
8142
8143     % Create BathymetryLabel_2
8144     app.BathymetryLabel_2 = uilabel(app.Panel_15);
8145     app.BathymetryLabel_2.FontWeight = 'bold';
8146     app.BathymetryLabel_2.Tooltip = {'Load the file'; ''; 'Select the
bathymetry used in the simulation'; ''; 'Format accepted: .txt, .tif'};
8147     app.BathymetryLabel_2.Position = [33 38 71 22];
8148     app.BathymetryLabel_2.Text = 'Bathymetry';
8149
8150     % Create Button_22
8151     app.Button_22 = uibutton(app.Panel_15, 'push');
8152     app.Button_22.ButtonPushedFcn = createCallbackFcn(app,
@Button_22Pushed, true);
8153     app.Button_22.Tooltip = {'Load the file'; ''; 'Select the
bathymetry used in the simulation'; ''; 'Format accepted: .txt, .tif'};
8154     app.Button_22.Position = [136 20 19 19];
8155     app.Button_22.Text = '...';
8156
8157     % Create SouthwestCornerLabel_2
8158     app.SouthwestCornerLabel_2 = uilabel(app.Panel_15);
8159     app.SouthwestCornerLabel_2.FontWeight = 'bold';
8160     app.SouthwestCornerLabel_2.Tooltip = {'Coordinates of the
southwest corner of the loaded files.'};
8161     app.SouthwestCornerLabel_2.Position = [244 134 110 22];
8162     app.SouthwestCornerLabel_2.Text = 'Southwest Corner';
8163
8164     % Create LongitudeEditField_2Label
8165     app.LongitudeEditField_2Label = uilabel(app.Panel_15);
8166     app.LongitudeEditField_2Label.HorizontalAlignment = 'right';
8167     app.LongitudeEditField_2Label.WordWrap = 'on';
8168     app.LongitudeEditField_2Label.Tooltip = {'West boundary'; '';
'Unit: degrees / meters'};
8169     app.LongitudeEditField_2Label.Position = [246 114 60 22];
8170     app.LongitudeEditField_2Label.Text = 'Longitude';
8171
8172     % Create LongitudeEditField 2

```



```

8173         app.LongitudeEditField_2 = uieditfield(app.Panel_15, 'numeric');
8174         app.LongitudeEditField_2.ValueDisplayFormat = '%8.4f';
8175         app.LongitudeEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@LongitudeEditField_2ValueChanged, true);
8176         app.LongitudeEditField_2.FontColor = [0.651 0.651 0.651];
8177         app.LongitudeEditField_2.Tooltip = {' '};
8178         app.LongitudeEditField_2.Position = [314 117 60 16];
8179
8180         % Create LatitudeEditField_2Label
8181         app.LatitudeEditField_2Label = uilabel(app.Panel_15);
8182         app.LatitudeEditField_2Label.HorizontalAlignment = 'right';
8183         app.LatitudeEditField_2Label.WordWrap = 'on';
8184         app.LatitudeEditField_2Label.Tooltip = {'South boundary'; ''; ↵
'Unit: degrees / meters'};
8185         app.LatitudeEditField_2Label.Position = [251 94 44 17];
8186         app.LatitudeEditField_2Label.Text = 'Latitude';
8187
8188         % Create LatitudeEditField_2
8189         app.LatitudeEditField_2 = uieditfield(app.Panel_15, 'numeric');
8190         app.LatitudeEditField_2.ValueDisplayFormat = '%8.4f';
8191         app.LatitudeEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@LatitudeEditField_2ValueChanged, true);
8192         app.LatitudeEditField_2.FontColor = [0.651 0.651 0.651];
8193         app.LatitudeEditField_2.Tooltip = {' '};
8194         app.LatitudeEditField_2.Position = [314 95 60 16];
8195
8196         % Create GridSizeLabel
8197         app.GridSizeLabel = uilabel(app.Panel_15);
8198         app.GridSizeLabel.FontWeight = 'bold';
8199         app.GridSizeLabel.Tooltip = {'Resolution'; 'Unit: degrees / ↵
meters'};
8200         app.GridSizeLabel.Position = [401 134 57 22];
8201         app.GridSizeLabel.Text = 'Grid Size';
8202
8203         % Create YEditField_2Label_2
8204         app.YEditField_2Label_2 = uilabel(app.Panel_15);
8205         app.YEditField_2Label_2.HorizontalAlignment = 'right';
8206         app.YEditField_2Label_2.Tooltip = {'Y direction'; ''; 'Unit: ↵
degrees / meters'};
8207         app.YEditField_2Label_2.Position = [403 92 13 21];
8208         app.YEditField_2Label_2.Text = 'Y';
8209
8210         % Create gridY_2
8211         app.gridY_2 = uieditfield(app.Panel_15, 'numeric');
8212         app.gridY_2.Limits = [0 Inf];
8213         app.gridY_2.ValueDisplayFormat = '%8.5f';
8214         app.gridY_2.ValueChangedFcn = createCallbackFcn(app, ↵
@gridY_2ValueChanged, true);
8215         app.gridY_2.FontColor = [0.651 0.651 0.651];
8216         app.gridY_2.Tooltip = {' '};
8217         app.gridY_2.Position = [427 94 60 16];
8218
8219         % Create TotalSimuilationTimesecEditFieldLabel_4
8220         app.TotalSimuilationTimesecEditFieldLabel_4 = uilabel(app. ↵
Panel_15);
8221         app.TotalSimuilationTimesecEditFieldLabel_4.FontWeight = 'bold';
8222         app.TotalSimuilationTimesecEditFieldLabel_4.Position = [244 38 98 ↵
22];

```

```

8223         app.TotalSimuilationTimesecEditFieldLabel_4.Text = 'Simulation
Time';
8224
8225         % Create StartTime2
8226         app.StartTime2 = uieditfield(app.Panel_15, 'numeric');
8227         app.StartTime2.Limits = [0 Inf];
8228         app.StartTime2.ValueDisplayFormat = '%8.1f';
8229         app.StartTime2.ValueChangedFcn = createCallbackFcn(app,
@StartTime2ValueChanged, true);
8230         app.StartTime2.FontColor = [0.651 0.651 0.651];
8231         app.StartTime2.Tooltip = {''};
8232         app.StartTime2.Position = [314 22 47 16];
8233
8234         % Create StartLabel_2
8235         app.StartLabel_2 = uilabel(app.Panel_15);
8236         app.StartLabel_2.HorizontalAlignment = 'right';
8237         app.StartLabel_2.Tooltip = {'For imported initial tsunami files
with a non-zero start time, input the value here. Example: for a 30-second
snapshot, input '30'.'; ''; 'Unit: seconds'};
8238         app.StartLabel_2.Position = [275 19 31 22];
8239         app.StartLabel_2.Text = 'Start';
8240
8241         % Create IntervalLabel_2
8242         app.IntervalLabel_2 = uilabel(app.Panel_15);
8243         app.IntervalLabel_2.HorizontalAlignment = 'right';
8244         app.IntervalLabel_2.Tooltip = {'Snapshot interval.'; ''; 'Same as
PLOT_INTV in input.txt'; ''; 'Unit: seconds'};
8245         app.IntervalLabel_2.Position = [372 19 45 22];
8246         app.IntervalLabel_2.Text = 'Interval';
8247
8248         % Create TotalSimuilationTimesecEditField_4
8249         app.TotalSimuilationTimesecEditField_4 = uieditfield(app.Panel_15,
'numeric');
8250         app.TotalSimuilationTimesecEditField_4.Limits = [1 Inf];
8251         app.TotalSimuilationTimesecEditField_4.ValueDisplayFormat = '%8.1
f';
8252         app.TotalSimuilationTimesecEditField_4.ValueChangedFcn =
createCallbackFcn(app, @TotalSimuilationTimesecEditField_4ValueChanged, true);
8253         app.TotalSimuilationTimesecEditField_4.FontColor = [0.651 0.651
0.651];
8254         app.TotalSimuilationTimesecEditField_4.Tooltip = {''};
8255         app.TotalSimuilationTimesecEditField_4.Position = [427 22 60 16];
8256         app.TotalSimuilationTimesecEditField_4.Value = 1;
8257
8258         % Create XEditField_2Label_2
8259         app.XEditField_2Label_2 = uilabel(app.Panel_15);
8260         app.XEditField_2Label_2.HorizontalAlignment = 'right';
8261         app.XEditField_2Label_2.Tooltip = {'X direction'; ''; 'Unit:
degrees / meters'};
8262         app.XEditField_2Label_2.Position = [406 114 13 22];
8263         app.XEditField_2Label_2.Text = 'X ';
8264
8265         % Create gridX_2
8266         app.gridX_2 = uieditfield(app.Panel_15, 'numeric');
8267         app.gridX_2.Limits = [0 Inf];
8268         app.gridX_2.ValueDisplayFormat = '%8.5f';
8269         app.gridX_2.ValueChangedFcn = createCallbackFcn(app,
@gridX_2ValueChanged, true);

```

```

8270     app.gridX_2.FontColor = [0.651 0.651 0.651];
8271     app.gridX_2.Tooltip = {' '};
8272     app.gridX_2.Position = [427 117 60 16];
8273
8274     % Create FileTextArea_4
8275     app.FileTextArea_4 = uitextarea(app.Panel_15);
8276     app.FileTextArea_4.Editable = 'off';
8277     app.FileTextArea_4.Tooltip = {'The V-vector file is automatically↵
searched in the same directory as the corresponding U-vector file.'};
8278     app.FileTextArea_4.Placeholder = '(Auto finds) v_xxxx;↵
vmean_xxxx;';
8279     app.FileTextArea_4.Position = [127 77 84 74];
8280
8281     % Create Button_19
8282     app.Button_19 = uibutton(app.Panel_15, 'push');
8283     app.Button_19.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_19Pushed, true);
8284     app.Button_19.FontAngle = 'italic';
8285     app.Button_19.FontColor = [0.651 0.651 0.651];
8286     app.Button_19.Tooltip = {'Load files with filenames starting with↵
'u' ' '};
8287     app.Button_19.Position = [93 159 19 19];
8288     app.Button_19.Text = '...';
8289
8290     % Create InputDataLabel_4
8291     app.InputDataLabel_4 = uilabel(app.VelocityMapTab);
8292     app.InputDataLabel_4.BackgroundColor = [0.9412 0.9412 0.9412];
8293     app.InputDataLabel_4.FontSize = 15;
8294     app.InputDataLabel_4.FontWeight = 'bold';
8295     app.InputDataLabel_4.FontColor = [0.0314 0.3686 0.6];
8296     app.InputDataLabel_4.Position = [38 813 97 22];
8297     app.InputDataLabel_4.Text = ' Input Data ';
8298
8299     % Create ArrowsLabel
8300     app.ArrowsLabel = uilabel(app.VelocityMapTab);
8301     app.ArrowsLabel.BackgroundColor = [0.9412 0.9412 0.9412];
8302     app.ArrowsLabel.FontSize = 15;
8303     app.ArrowsLabel.FontWeight = 'bold';
8304     app.ArrowsLabel.FontColor = [0.0314 0.3686 0.6];
8305     app.ArrowsLabel.Position = [38 603 73 22];
8306     app.ArrowsLabel.Text = ' Arrows ';
8307
8308     % Create Panel_16
8309     app.Panel_16 = uipanel(app.VelocityMapTab);
8310     app.Panel_16.AutoResizeChildren = 'off';
8311     app.Panel_16.SizeChangedFcn = createCallbackFcn(app, ↵
@Panel_16SizeChanged, true);
8312     app.Panel_16.Position = [11 315 520 222];
8313
8314     % Create TabGroup3
8315     app.TabGroup3 = uitabgroup(app.Panel_16);
8316     app.TabGroup3.AutoResizeChildren = 'off';
8317     app.TabGroup3.Position = [0 -1 520 204];
8318
8319     % Create BasemapTab
8320     app.BasemapTab = uitab(app.TabGroup3);
8321     app.BasemapTab.AutoResizeChildren = 'off';
8322     app.BasemapTab.SizeChangedFcn = createCallbackFcn(app, ↵

```

```

@BasemapTabSizeChanged, true);
8323         app.BasemapTab.Title = 'Basemap';
8324
8325         % Create ButtonGroup_16
8326         app.ButtonGroup_16 = uibuttongroup(app.BasemapTab);
8327         app.ButtonGroup_16.AutoSizeChildren = 'off';
8328         app.ButtonGroup_16.SelectionChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_16SelectionChanged, true);
8329         app.ButtonGroup_16.BorderType = 'none';
8330         app.ButtonGroup_16.FontWeight = 'bold';
8331         app.ButtonGroup_16.Position = [14 54 117 120];
8332
8333         % Create VelocityButton
8334         app.VelocityButton = uiradiobutton(app.ButtonGroup_16);
8335         app.VelocityButton.Tooltip = {'Z component of the velocity'};
8336         app.VelocityButton.Text = 'Z Velocity';
8337         app.VelocityButton.FontWeight = 'bold';
8338         app.VelocityButton.Position = [11 98 78 22];
8339         app.VelocityButton.Value = true;
8340
8341         % Create etaButton
8342         app.etaButton = uiradiobutton(app.ButtonGroup_16);
8343         app.etaButton.Tooltip = {'Sea surface displacement at specific ↵
time snapshot.'; ''; 'The eta files are automatically loaded. '; ''; 'Ensure the ↵
eta files are in the same folder as the U vector files.'};
8344         app.etaButton.Text = 'eta';
8345         app.etaButton.FontWeight = 'bold';
8346         app.etaButton.Position = [10 74 40 22];
8347
8348         % Create hmaxButton
8349         app.hmaxButton = uiradiobutton(app.ButtonGroup_16);
8350         app.hmaxButton.Tooltip = {'Maximum wave height.'; ''; ''; 'The ↵
hmax files are automatically loaded. '; ''; 'Ensure the hmax files are in the same ↵
folder as the U vector files.'};
8351         app.hmaxButton.Text = 'hmax';
8352         app.hmaxButton.FontWeight = 'bold';
8353         app.hmaxButton.Position = [9 50 53 22];
8354
8355         % Create BathymetryButton
8356         app.BathymetryButton = uiradiobutton(app.ButtonGroup_16);
8357         app.BathymetryButton.Text = 'Bathymetry';
8358         app.BathymetryButton.FontWeight = 'bold';
8359         app.BathymetryButton.Position = [8 26 88 22];
8360
8361         % Create VorticityButton
8362         app.VorticityButton = uiradiobutton(app.ButtonGroup_16);
8363         app.VorticityButton.Text = 'Vorticity';
8364         app.VorticityButton.FontWeight = 'bold';
8365         app.VorticityButton.Position = [7 3 69 22];
8366
8367         % Create ThirdTabColorMapPanel
8368         app.ThirdTabColorMapPanel = uipanel(app.BasemapTab);
8369         app.ThirdTabColorMapPanel.AutoSizeChildren = 'off';
8370         app.ThirdTabColorMapPanel.BorderType = 'none';
8371         app.ThirdTabColorMapPanel.Position = [265 32 203 113];
8372
8373         % Create maplabelsize
8374         app.maplabelsize = uieditfield(app.ThirdTabColorMapPanel, ↵

```

```

'numeric');
8375         app.maplabelsize.Limits = [0 50];
8376         app.maplabelsize.ValueDisplayFormat = '%3.0f';
8377         app.maplabelsize.ValueChangedFcn = createCallbackFcn(app, ↵
@maplabelsizeValueChanged, true);
8378         app.maplabelsize.FontColor = [0.651 0.651 0.651];
8379         app.maplabelsize.Tooltip = {''};
8380         app.maplabelsize.Position = [70 11 35 16];
8381         app.maplabelsize.Value = 10;
8382
8383         % Create BackgroundMapColorDropDown
8384         app.BackgroundMapColorDropDown = uidropdown(app.↵
ThirdTabColorMapPanel);
8385         app.BackgroundMapColorDropDown.Items = {'--- MATLAB default ---', ↵
'autumn', 'bone', 'colorcube', 'cool', 'copper', 'flag', 'gray', 'hot', 'hsv', ↵
'jet', 'parula', 'pink', 'prism', 'spring', 'summer', 'turbo', 'winter', '--- ↵
CBREWER 2 ---', '< sequential >', 'blue', 'blue - green', 'blue - purple', 'green - ↵
blue', 'greens', 'grays', 'oranges', 'orange - red', 'purple - blue', 'purple - ↵
blue - green', 'purple - red', 'purples', 'red - purple', 'reds', 'yellow - green', ↵
'yellow - green - blue', 'yellow - orange - brown', 'yellow - orange - red', '< ↵
divergent >', 'brown - teal', 'pink - light green', 'purple - green', 'purple - ↵
orange', 'red - blue', 'red - gray', 'red - yellow - blue', 'red - yellow - green', ↵
'spectral', '< qualitative >', 'accent', 'dark 2', 'paired', 'pastel 1', 'pastel ↵
2', 'set 1', 'set 2', 'set 3'};
8386         app.BackgroundMapColorDropDown.DropDownOpeningFcn = ↵
createCallbackFcn(app, @BackgroundMapColorDropDownOpening, true);
8387         app.BackgroundMapColorDropDown.ValueChangedFcn = createCallbackFcn ↵
(app, @BackgroundMapColorDropDownValueChanged, true);
8388         app.BackgroundMapColorDropDown.Position = [8 92 174 16];
8389         app.BackgroundMapColorDropDown.Value = 'blue - purple';
8390
8391         % Create FlipCheckBox_4
8392         app.FlipCheckBox_4 = uicontrol(app.ThirdTabColorMapPanel);
8393         app.FlipCheckBox_4.Tooltip = {'Reverse the color sequence'};
8394         app.FlipCheckBox_4.Text = 'Flip';
8395         app.FlipCheckBox_4.FontSize = 11;
8396         app.FlipCheckBox_4.Position = [8 67 40 22];
8397
8398         % Create InterpolationDivisionEditFieldLabel_2
8399         app.InterpolationDivisionEditFieldLabel_2 = uilabel(app.↵
ThirdTabColorMapPanel);
8400         app.InterpolationDivisionEditFieldLabel_2.HorizontalAlignment = ↵
'right';
8401         app.InterpolationDivisionEditFieldLabel_2.FontSize = 11;
8402         app.InterpolationDivisionEditFieldLabel_2.Tooltip = {'Defines the ↵
color distribution: '; ' '; 'Low value: '; 'Distinct color boundaries: '; ' '; 'Higher ↵
value: '; 'Smoother color transition'};
8403         app.InterpolationDivisionEditFieldLabel_2.Position = [93 68 48 ↵
22];
8404         app.InterpolationDivisionEditFieldLabel_2.Text = 'Division';
8405
8406         % Create InterpolationDivisionEditField_2
8407         app.InterpolationDivisionEditField_2 = uieditfield(app.↵
ThirdTabColorMapPanel, 'numeric');
8408         app.InterpolationDivisionEditField_2.Limits = [0 Inf];
8409         app.InterpolationDivisionEditField_2.RoundFractionalValues = 'on';
8410         app.InterpolationDivisionEditField_2.ValueDisplayFormat = '%3.0f';
8411         app.InterpolationDivisionEditField_2.ValueChangedFcn = ↵

```

```

createCallbackFcn(app, @InterpolationDivisionEditField_2ValueChanged, true);
8412         app.InterpolationDivisionEditField_2.FontColor = [0.651 0.651 0.651];
8413         app.InterpolationDivisionEditField_2.Tooltip = {' '};
8414         app.InterpolationDivisionEditField_2.Position = [146 72 36 16];
8415         app.InterpolationDivisionEditField_2.Value = 4;
8416
8417         % Create MinBarValue
8418         app.MinBarValue = uieditfield(app.ThirdTabColorMapPanel, 'numeric');
8419         app.MinBarValue.ValueDisplayFormat = '%.2f';
8420         app.MinBarValue.ValueChangedFcn = createCallbackFcn(app, @MinBarValueValueChanged, true);
8421         app.MinBarValue.FontColor = [0.651 0.651 0.651];
8422         app.MinBarValue.Tooltip = {' '};
8423         app.MinBarValue.Position = [50 34 55 16];
8424
8425         % Create toLabel_3
8426         app.toLabel_3 = uilabel(app.ThirdTabColorMapPanel);
8427         app.toLabel_3.Tooltip = {'Define the minimum value to display in the map.'; ' '; 'Unit: meters'};
8428         app.toLabel_3.Position = [111 31 25 22];
8429         app.toLabel_3.Text = 'to';
8430
8431         % Create mLabel_3
8432         app.mLabel_3 = uilabel(app.ThirdTabColorMapPanel);
8433         app.mLabel_3.Tooltip = {'Define the maximum value to display in the map.'; ' '; 'Unit: meters'};
8434         app.mLabel_3.Position = [186 31 25 22];
8435         app.mLabel_3.Text = 'm';
8436
8437         % Create MaxBarValue
8438         app.MaxBarValue = uieditfield(app.ThirdTabColorMapPanel, 'numeric');
8439         app.MaxBarValue.ValueDisplayFormat = '%.2f';
8440         app.MaxBarValue.ValueChangedFcn = createCallbackFcn(app, @MaxBarValueValueChanged, true);
8441         app.MaxBarValue.FontColor = [0.651 0.651 0.651];
8442         app.MaxBarValue.Tooltip = {' '};
8443         app.MaxBarValue.Position = [127 34 55 16];
8444         app.MaxBarValue.Value = 6;
8445
8446         % Create LimitsLabel
8447         app.LimitsLabel = uilabel(app.ThirdTabColorMapPanel);
8448         app.LimitsLabel.Tooltip = {'Define the maximum value to display in the map.'; ' '; 'Unit: meters'};
8449         app.LimitsLabel.Position = [8 31 38 22];
8450         app.LimitsLabel.Text = 'Limits';
8451
8452         % Create TextSizeLabel
8453         app.TextSizeLabel = uilabel(app.ThirdTabColorMapPanel);
8454         app.TextSizeLabel.Tooltip = {'Define the maximum value to display in the map.'; ' '; 'Unit: meters'};
8455         app.TextSizeLabel.Position = [8 8 53 22];
8456         app.TextSizeLabel.Text = 'Text Size';
8457
8458         % Create ColormapLabel
8459         app.ColormapLabel = uilabel(app.BasemapTab);

```

```

8460     app.ColormapLabel.FontWeight = 'bold';
8461     app.ColormapLabel.Position = [195 120 62 22];
8462     app.ColormapLabel.Text = 'Colormap';
8463
8464     % Create ColorbarLabel
8465     app.ColorbarLabel = uilabel(app.BasemapTab);
8466     app.ColorbarLabel.FontWeight = 'bold';
8467     app.ColorbarLabel.Position = [195 62 56 22];
8468     app.ColorbarLabel.Text = 'Colorbar';
8469
8470     % Create FileTextArea_5
8471     app.FileTextArea_5 = uitextarea(app.BasemapTab);
8472     app.FileTextArea_5.ValueChangedFcn = createCallbackFcn(app, ↵
@FileTextArea_5ValueChanged, true);
8473     app.FileTextArea_5.Editable = 'off';
8474     app.FileTextArea_5.FontColor = [0.149 0.149 0.149];
8475     app.FileTextArea_5.BackgroundColor = [0.9412 0.9412 0.9412];
8476     app.FileTextArea_5.Tooltip = {'File preview. '; ''; 'Ensure that ↵
the listed files are in the same folder as the U vector files.'};
8477     app.FileTextArea_5.Placeholder = '(Filename preview)';
8478     app.FileTextArea_5.Position = [21 16 114 35];
8479
8480     % Create HorizontalCheckBox_2
8481     app.HorizontalCheckBox_2 = uicheckbox(app.BasemapTab);
8482     app.HorizontalCheckBox_2.Text = 'Horizontal';
8483     app.HorizontalCheckBox_2.Position = [302 148 76 22];
8484
8485     % Create VerticalCheckBox_2
8486     app.VerticalCheckBox_2 = uicheckbox(app.BasemapTab);
8487     app.VerticalCheckBox_2.Text = 'Vertical';
8488     app.VerticalCheckBox_2.Position = [388 148 61 22];
8489
8490     % Create FlipBasemapLabel
8491     app.FlipBasemapLabel = uilabel(app.BasemapTab);
8492     app.FlipBasemapLabel.FontWeight = 'bold';
8493     app.FlipBasemapLabel.Tooltip = {'Flips the basemap.'; ''; ↵
'Horizontally (left-right)'; 'Vertically (up-side down)'};
8494     app.FlipBasemapLabel.Position = [195 148 83 22];
8495     app.FlipBasemapLabel.Text = 'Flip Basemap';
8496
8497     % Create CoastlinecolourDropDownLabel_12
8498     app.CoastlinecolourDropDownLabel_12 = uilabel(app.BasemapTab);
8499     app.CoastlinecolourDropDownLabel_12.FontWeight = 'bold';
8500     app.CoastlinecolourDropDownLabel_12.Tooltip = {'Required input:'; ↵
'Bathymetry file'};
8501     app.CoastlinecolourDropDownLabel_12.Position = [195 7 69 22];
8502     app.CoastlinecolourDropDownLabel_12.Text = 'Land Color';
8503
8504     % Create LandColor
8505     app.LandColor = uidropdown(app.BasemapTab);
8506     app.LandColor.Items = {'Black', 'Dark gray', 'Medium gray', 'Light ↵
gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
8507     app.LandColor.Enable = 'off';
8508     app.LandColor.Tooltip = {''};
8509     app.LandColor.Position = [273 10 105 16];
8510     app.LandColor.Value = 'Dark gray';
8511
8512     % Create BathymetryContoursTab 2

```



```

8513         app.BathymetryContoursTab_2 = uitab(app.TabGroup3);
8514         app.BathymetryContoursTab_2.AutoResizeChildren = 'off';
8515         app.BathymetryContoursTab_2.Tooltip = {'Enabled when bathymetry
file is uploaded'};
8516         app.BathymetryContoursTab_2.Title = 'Bathymetry Contours';
8517
8518         % Create RangeLabel
8519         app.RangeLabel = uilabel(app.BathymetryContoursTab_2);
8520         app.RangeLabel.FontWeight = 'bold';
8521         app.RangeLabel.Position = [45 109 42 22];
8522         app.RangeLabel.Text = 'Range';
8523
8524         % Create PlotBathymetryContoursCheckBox
8525         app.PlotBathymetryContoursCheckBox = uicheckbox(app.
BathymetryContoursTab_2);
8526         app.PlotBathymetryContoursCheckBox.ValueChangedFcn =
createCallbackFcn(app, @PlotBathymetryContoursCheckBoxValueChanged, true);
8527         app.PlotBathymetryContoursCheckBox.Enable = 'off';
8528         app.PlotBathymetryContoursCheckBox.Text = 'Plot Bathymetry
Contours';
8529         app.PlotBathymetryContoursCheckBox.FontWeight = 'bold';
8530         app.PlotBathymetryContoursCheckBox.Position = [45 141 170 22];
8531
8532         % Create MinimumEditField_2Label
8533         app.MinimumEditField_2Label = uilabel(app.
BathymetryContoursTab_2);
8534         app.MinimumEditField_2Label.Position = [45 86 55 22];
8535         app.MinimumEditField_2Label.Text = 'Minimum';
8536
8537         % Create MinimumEditField_2
8538         app.MinimumEditField_2 = uieditfield(app.BathymetryContoursTab_2,
'numeric');
8539         app.MinimumEditField_2.Limits = [0 Inf];
8540         app.MinimumEditField_2.ValueDisplayFormat = '%8.1f';
8541         app.MinimumEditField_2.ValueChangedFcn = createCallbackFcn(app,
@MinimumEditField_2ValueChanged, true);
8542         app.MinimumEditField_2.FontColor = [0.651 0.651 0.651];
8543         app.MinimumEditField_2.Enable = 'off';
8544         app.MinimumEditField_2.Tooltip = {'0 value = coastline'; ''
'Lower values indicate shallower waters.'};
8545         app.MinimumEditField_2.Position = [102 89 57 16];
8546
8547         % Create MaximumEditField_2Label
8548         app.MaximumEditField_2Label = uilabel(app.
BathymetryContoursTab_2);
8549         app.MaximumEditField_2Label.Position = [45 32 58 22];
8550         app.MaximumEditField_2Label.Text = 'Maximum';
8551
8552         % Create MaximumEditField_2
8553         app.MaximumEditField_2 = uieditfield(app.BathymetryContoursTab_2,
'numeric');
8554         app.MaximumEditField_2.Limits = [0.1 Inf];
8555         app.MaximumEditField_2.ValueDisplayFormat = '%8.1f';
8556         app.MaximumEditField_2.ValueChangedFcn = createCallbackFcn(app,
@MaximumEditField_2ValueChanged, true);
8557         app.MaximumEditField_2.FontColor = [0.651 0.651 0.651];
8558         app.MaximumEditField_2.Enable = 'off';
8559         app.MaximumEditField_2.Tooltip = {'Higher value indicate deeper

```



```

waters'};
8560         app.MaximumEditField_2.Position = [102 35 57 16];
8561         app.MaximumEditField_2.Value = 10000;
8562
8563         % Create IntervalEditField_6Label
8564         app.IntervalEditField_6Label = uilabel(app.
BathymetryContoursTab_2);
8565         app.IntervalEditField_6Label.Position = [45 59 45 22];
8566         app.IntervalEditField_6Label.Text = 'Interval';
8567
8568         % Create IntervalEditField_6
8569         app.IntervalEditField_6 = uieditfield(app.BathymetryContoursTab_2,
'numeric');
8570         app.IntervalEditField_6.Limits = [0 Inf];
8571         app.IntervalEditField_6.ValueDisplayFormat = '%8.1f';
8572         app.IntervalEditField_6.ValueChangedFcn = createCallbackFcn(app,
@IntervalEditField_6ValueChanged, true);
8573         app.IntervalEditField_6.FontColor = [0.651 0.651 0.651];
8574         app.IntervalEditField_6.Enable = 'off';
8575         app.IntervalEditField_6.Tooltip = {'Interval between adjacent
contours'};
8576         app.IntervalEditField_6.Position = [102 62 48 16];
8577         app.IntervalEditField_6.Value = 500;
8578
8579         % Create LineLabel_4
8580         app.LineLabel_4 = uilabel(app.BathymetryContoursTab_2);
8581         app.LineLabel_4.FontWeight = 'bold';
8582         app.LineLabel_4.Position = [196 109 30 22];
8583         app.LineLabel_4.Text = 'Line';
8584
8585         % Create AddLabelCheckBox_2
8586         app.AddLabelCheckBox_2 = uicheckbox(app.BathymetryContoursTab_2);
8587         app.AddLabelCheckBox_2.ValueChangedFcn = createCallbackFcn(app,
@AddLabelCheckBox_2ValueChanged, true);
8588         app.AddLabelCheckBox_2.Enable = 'off';
8589         app.AddLabelCheckBox_2.Text = 'Add Label';
8590         app.AddLabelCheckBox_2.FontWeight = 'bold';
8591         app.AddLabelCheckBox_2.Position = [375 109 80 22];
8592
8593         % Create ColorDropDown_4Label
8594         app.ColorDropDown_4Label = uilabel(app.BathymetryContoursTab_2);
8595         app.ColorDropDown_4Label.Position = [196 32 35 22];
8596         app.ColorDropDown_4Label.Text = 'Color';
8597
8598         % Create ColorDropDown_4
8599         app.ColorDropDown_4 = uidropdown(app.BathymetryContoursTab_2);
8600         app.ColorDropDown_4.Items = {'Black', 'Dark gray', 'Medium gray',
'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
8601         app.ColorDropDown_4.Enable = 'off';
8602         app.ColorDropDown_4.Position = [234 35 104 16];
8603         app.ColorDropDown_4.Value = 'Black';
8604
8605         % Create StyleDropDown_3Label_2
8606         app.StyleDropDown_3Label_2 = uilabel(app.BathymetryContoursTab_2);
8607         app.StyleDropDown_3Label_2.Position = [196 59 32 22];
8608         app.StyleDropDown_3Label_2.Text = 'Style';
8609
8610         % Create StyleDropDown 3

```

```

8611         app.StyleDropDown_3 = uidropdown(app.BathymetryContoursTab_2);
8612         app.StyleDropDown_3.Items = {'Solid', 'Dashed', 'Dotted', 'Dash-
dotted'};
8613         app.StyleDropDown_3.Enable = 'off';
8614         app.StyleDropDown_3.Position = [234 62 104 16];
8615         app.StyleDropDown_3.Value = 'Solid';
8616
8617         % Create WidthEditField_2Label
8618         app.WidthEditField_2Label = uilabel(app.BathymetryContoursTab_2);
8619         app.WidthEditField_2Label.Position = [196 86 37 22];
8620         app.WidthEditField_2Label.Text = 'Width';
8621
8622         % Create WidthEditField_2
8623         app.WidthEditField_2 = uieditfield(app.BathymetryContoursTab_2,
'numeric');
8624         app.WidthEditField_2.Limits = [0 Inf];
8625         app.WidthEditField_2.ValueDisplayFormat = '%3.1f';
8626         app.WidthEditField_2.ValueChangedFcn = createCallbackFcn(app,
@WidthEditField_2ValueChanged, true);
8627         app.WidthEditField_2.FontColor = [0.651 0.651 0.651];
8628         app.WidthEditField_2.Enable = 'off';
8629         app.WidthEditField_2.Tooltip = {'Line thickness'};
8630         app.WidthEditField_2.Position = [234 89 33 16];
8631         app.WidthEditField_2.Value = 0.1;
8632
8633         % Create IntervalEditField_8Label
8634         app.IntervalEditField_8Label = uilabel(app.
BathymetryContoursTab_2);
8635         app.IntervalEditField_8Label.Tooltip = {'Add labels at specified
intervals.'; ''; 'Suggestion: Set at intervals divisible by the contour range
interval'; ''; ''; 'Suggestion: Set at intervals divisible by the depth range
interval'};
8636         app.IntervalEditField_8Label.Position = [375 59 45 22];
8637         app.IntervalEditField_8Label.Text = 'Interval';
8638
8639         % Create LabelSizeCont
8640         app.LabelSizeCont = uieditfield(app.BathymetryContoursTab_2,
'numeric');
8641         app.LabelSizeCont.Limits = [0 30];
8642         app.LabelSizeCont.ValueDisplayFormat = '%8.0f';
8643         app.LabelSizeCont.ValueChangedFcn = createCallbackFcn(app,
@LabelSizeContValueChanged, true);
8644         app.LabelSizeCont.FontColor = [0.651 0.651 0.651];
8645         app.LabelSizeCont.Enable = 'off';
8646         app.LabelSizeCont.Tooltip = {''};
8647         app.LabelSizeCont.Position = [426 89 40 16];
8648         app.LabelSizeCont.Value = 8;
8649
8650         % Create SpacingEditField_4Label_3
8651         app.SpacingEditField_4Label_3 = uilabel(app.
BathymetryContoursTab_2);
8652         app.SpacingEditField_4Label_3.Tooltip = {'Font size'};
8653         app.SpacingEditField_4Label_3.Position = [375 86 28 22];
8654         app.SpacingEditField_4Label_3.Text = 'Size';
8655
8656         % Create SpacingEditField_4Label_2
8657         app.SpacingEditField_4Label_2 = uilabel(app.
BathymetryContoursTab_2);

```

```

8658         app.SpacingEditField_4Label_2.WordWrap = 'on';
8659         app.SpacingEditField_4Label_2.Tooltip = {'Adjust the distance↵
between the labels'; ''; 'Higher values reduce label crowding'};
8660         app.SpacingEditField_4Label_2.Position = [375 29 77 29];
8661         app.SpacingEditField_4Label_2.Text = 'Spacing';
8662
8663         % Create SpacingEditField
8664         app.SpacingEditField = uicontrol(app.BathymetryContoursTab_2, ↵
'numeric');
8665         app.SpacingEditField.Limits = [0 Inf];
8666         app.SpacingEditField.ValueDisplayFormat = '%8.1f';
8667         app.SpacingEditField.ValueChangedFcn = createCallbackFcn(app, ↵
@SpacingEditFieldValueChanged, true);
8668         app.SpacingEditField.FontColor = [0.651 0.651 0.651];
8669         app.SpacingEditField.Enable = 'off';
8670         app.SpacingEditField.Tooltip = {'';
8671         app.SpacingEditField.Position = [426 35 47 16];
8672         app.SpacingEditField.Value = 100;
8673
8674         % Create IntervalEditField_8
8675         app.IntervalEditField_8 = uicontrol(app.BathymetryContoursTab_2, ↵
'numeric');
8676         app.IntervalEditField_8.Limits = [0 Inf];
8677         app.IntervalEditField_8.ValueDisplayFormat = '%8.1f';
8678         app.IntervalEditField_8.ValueChangedFcn = createCallbackFcn(app, ↵
@IntervalEditField_8ValueChanged, true);
8679         app.IntervalEditField_8.FontColor = [0.651 0.651 0.651];
8680         app.IntervalEditField_8.Enable = 'off';
8681         app.IntervalEditField_8.Tooltip = {'';
8682         app.IntervalEditField_8.Position = [426 62 47 16];
8683         app.IntervalEditField_8.Value = 100;
8684
8685         % Create GaugesTab_2
8686         app.GaugesTab_2 = uitab(app.TabGroup3);
8687         app.GaugesTab_2.AutoResizeChildren = 'off';
8688         app.GaugesTab_2.SizeChangedFcn = createCallbackFcn(app, ↵
@GaugesTab_2SizeChanged, true);
8689         app.GaugesTab_2.Title = 'Gauges';
8690
8691         % Create CoastlinecolourDropDownLabel_19
8692         app.CoastlinecolourDropDownLabel_19 = uilabel(app.GaugesTab_2);
8693         app.CoastlinecolourDropDownLabel_19.Position = [45 57 32 22];
8694         app.CoastlinecolourDropDownLabel_19.Text = 'Style';
8695
8696         % Create gaugemarkerVelocityTab
8697         app.gaugemarkerVelocityTab = uidropdown(app.GaugesTab_2);
8698         app.gaugemarkerVelocityTab.Items = {'o', '+', '*', '.', 'x', '-', ↵
'|', '^', 'v', '>', '<', 'diamond', 'hexagram', 'pentagram', 'square'};
8699         app.gaugemarkerVelocityTab.Enable = 'off';
8700         app.gaugemarkerVelocityTab.Tooltip = {'Marker style'};
8701         app.gaugemarkerVelocityTab.Position = [81 60 92 16];
8702         app.gaugemarkerVelocityTab.Value = 'o';
8703
8704         % Create ColorDropDown_7Label
8705         app.ColorDropDown_7Label = uilabel(app.GaugesTab_2);
8706         app.ColorDropDown_7Label.Position = [45 30 35 22];
8707         app.ColorDropDown_7Label.Text = 'Color';
8708

```

```

8709         % Create ColorDropDown_7
8710         app.ColorDropDown_7 = uidropdown(app.GaugesTab_2);
8711         app.ColorDropDown_7.Items = {'Black', 'Dark gray', 'Medium gray', '
'Light gray', 'Red', 'Green', 'Blue', 'Yellow', 'Cyan', 'Magenta', 'White'};
8712         app.ColorDropDown_7.Enable = 'off';
8713         app.ColorDropDown_7.Position = [81 33 103 16];
8714         app.ColorDropDown_7.Value = 'Black';
8715
8716         % Create Button_27
8717         app.Button_27 = uibutton(app.GaugesTab_2, 'push');
8718         app.Button_27.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_27Pushed, true);
8719         app.Button_27.Enable = 'off';
8720         app.Button_27.Tooltip = {'Load the file'; ''; 'File extension must ↵
be .shp or .txt (tab delimited)'};
8721         app.Button_27.Position = [211 116 19 19];
8722         app.Button_27.Text = '...';
8723
8724         % Create GaugeFileLabel_2
8725         app.GaugeFileLabel_2 = uilabel(app.GaugesTab_2);
8726         app.GaugeFileLabel_2.FontWeight = 'bold';
8727         app.GaugeFileLabel_2.Position = [45 114 26 22];
8728         app.GaugeFileLabel_2.Text = 'File';
8729
8730         % Create FileEditField_2
8731         app.FileEditField_2 = uieditfield(app.GaugesTab_2, 'text');
8732         app.FileEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@FileEditField_2ValueChanged, true);
8733         app.FileEditField_2.Editable = 'off';
8734         app.FileEditField_2.Enable = 'off';
8735         app.FileEditField_2.Tooltip = {'File extension must be .shp or . ↵
txt'};
8736         app.FileEditField_2.Placeholder = '.shp or .txt';
8737         app.FileEditField_2.Position = [74 117 130 16];
8738
8739         % Create PlotGaugesCheckBox
8740         app.PlotGaugesCheckBox = uicheckbox(app.GaugesTab_2);
8741         app.PlotGaugesCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@PlotGaugesCheckBoxValueChanged, true);
8742         app.PlotGaugesCheckBox.Text = 'Plot Gauges';
8743         app.PlotGaugesCheckBox.FontWeight = 'bold';
8744         app.PlotGaugesCheckBox.Position = [45 141 92 22];
8745
8746         % Create AddLabelCheckBox
8747         app.AddLabelCheckBox = uicheckbox(app.GaugesTab_2);
8748         app.AddLabelCheckBox.ValueChangedFcn = createCallbackFcn(app, ↵
@AddLabelCheckBoxValueChanged, true);
8749         app.AddLabelCheckBox.Enable = 'off';
8750         app.AddLabelCheckBox.Tooltip = {'The label follows the sequence of ↵
latitude-longitude combinations from the loaded file.'; ''; 'The labels correspond ↵
to the numeric values in the sta_files.'};
8751         app.AddLabelCheckBox.Text = ' Add Label';
8752         app.AddLabelCheckBox.FontWeight = 'bold';
8753         app.AddLabelCheckBox.Position = [254 84 86 22];
8754
8755         % Create FontSizeEditField_2Label
8756         app.FontSizeEditField_2Label = uilabel(app.GaugesTab_2);
8757         app.FontSizeEditField_2Label.Enable = 'off';

```

```

8758     app.FontSizeEditField_2Label.Tooltip = {'Font size'};
8759     app.FontSizeEditField_2Label.Position = [255 57 33 22];
8760     app.FontSizeEditField_2Label.Text = 'Size';
8761
8762     % Create FontSizeEditField_2
8763     app.FontSizeEditField_2 = uieditfield(app.GaugesTab_2, 'numeric');
8764     app.FontSizeEditField_2.Limits = [0 100];
8765     app.FontSizeEditField_2.ValueDisplayFormat = '%8.0f';
8766     app.FontSizeEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@FontSizeEditField_2ValueChanged, true);
8767     app.FontSizeEditField_2.FontColor = [0.651 0.651 0.651];
8768     app.FontSizeEditField_2.Enable = 'off';
8769     app.FontSizeEditField_2.Tooltip = {'Font size'};
8770     app.FontSizeEditField_2.Position = [290 60 30 16];
8771     app.FontSizeEditField_2.Value = 12;
8772
8773     % Create GLspacing_2
8774     app.GLspacing_2 = uieditfield(app.GaugesTab_2, 'numeric');
8775     app.GLspacing_2.Limits = [0 100];
8776     app.GLspacing_2.ValueDisplayFormat = '%8.3f';
8777     app.GLspacing_2.ValueChangedFcn = createCallbackFcn(app, ↵
@GLspacing_2ValueChanged, true);
8778     app.GLspacing_2.FontColor = [0.651 0.651 0.651];
8779     app.GLspacing_2.Enable = 'off';
8780     app.GLspacing_2.Tooltip = {' '};
8781     app.GLspacing_2.Position = [394 60 40 16];
8782     app.GLspacing_2.Value = 0.5;
8783
8784     % Create GLspacingLabel_2
8785     app.GLspacingLabel_2 = uilabel(app.GaugesTab_2);
8786     app.GLspacingLabel_2.Enable = 'off';
8787     app.GLspacingLabel_2.Tooltip = {'Horizontal distance between the ↵
point and the label. ' ; ' ' ; ' A value of 0.5 means that a label for a point at ↵
120°E will be placed at 120.5°E.' ; ' ' ; 'Unit: degrees / meters'};
8788     app.GLspacingLabel_2.Position = [339 57 50 22];
8789     app.GLspacingLabel_2.Text = 'Spacing';
8790
8791     % Create AlignmentLabel_2
8792     app.AlignmentLabel_2 = uilabel(app.GaugesTab_2);
8793     app.AlignmentLabel_2.Enable = 'off';
8794     app.AlignmentLabel_2.Tooltip = {'Horizontal and vertical ↵
alignments relative to the marker location'};
8795     app.AlignmentLabel_2.Position = [255 30 59 22];
8796     app.AlignmentLabel_2.Text = 'Alignment';
8797
8798     % Create CoastlinecolourDropDown_10
8799     app.CoastlinecolourDropDown_10 = uidropdown(app.GaugesTab_2);
8800     app.CoastlinecolourDropDown_10.Items = {'Centre', 'Right', ↵
'Left'};
8801     app.CoastlinecolourDropDown_10.Enable = 'off';
8802     app.CoastlinecolourDropDown_10.Tooltip = {' '};
8803     app.CoastlinecolourDropDown_10.Position = [314 33 70 16];
8804     app.CoastlinecolourDropDown_10.Value = 'Right';
8805
8806     % Create CoastlinecolourDropDown_11
8807     app.CoastlinecolourDropDown_11 = uidropdown(app.GaugesTab_2);
8808     app.CoastlinecolourDropDown_11.Items = {'Centre', 'Top', ↵
'Bottom'};

```

```

8809     app.CoastlinecolourDropDown_11.Enable = 'off';
8810     app.CoastlinecolourDropDown_11.Tooltip = {'';
8811     app.CoastlinecolourDropDown_11.Position = [389 33 70 16];
8812     app.CoastlinecolourDropDown_11.Value = 'Centre';
8813
8814     % Create SizeEditField_2Label
8815     app.SizeEditField_2Label = uilabel(app.GaugesTab_2);
8816     app.SizeEditField_2Label.Position = [45 84 28 22];
8817     app.SizeEditField_2Label.Text = 'Size';
8818
8819     % Create SizeEditField_2
8820     app.SizeEditField_2 = uieditfield(app.GaugesTab_2, 'numeric');
8821     app.SizeEditField_2.Limits = [0 100];
8822     app.SizeEditField_2.ValueDisplayFormat = '%3.0f';
8823     app.SizeEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@SizeEditField_2ValueChanged, true);
8824     app.SizeEditField_2.FontColor = [0.651 0.651 0.651];
8825     app.SizeEditField_2.Enable = 'off';
8826     app.SizeEditField_2.Position = [81 87 30 16];
8827     app.SizeEditField_2.Value = 5;
8828
8829     % Create Panel_17
8830     app.Panel_17 = uipanel(app.VelocityMapTab);
8831     app.Panel_17.AutoResizeChildren = 'off';
8832     app.Panel_17.Position = [11 39 520 77];
8833
8834     % Create pngCheckBox_3
8835     app.pngCheckBox_3 = uicheckbox(app.Panel_17);
8836     app.pngCheckBox_3.Tooltip = {'Resolution is set to 300 DPI'};
8837     app.pngCheckBox_3.Text = '.png';
8838     app.pngCheckBox_3.Position = [186 12 46 22];
8839
8840     % Create FileFormatLabel_3
8841     app.FileFormatLabel_3 = uilabel(app.Panel_17);
8842     app.FileFormatLabel_3.Position = [69 12 66 22];
8843     app.FileFormatLabel_3.Text = 'File Format';
8844
8845     % Create jpegCheckBox_3
8846     app.jpegCheckBox_3 = uicheckbox(app.Panel_17);
8847     app.jpegCheckBox_3.Tooltip = {'Resolution is set to 300 DPI'};
8848     app.jpegCheckBox_3.Text = '.jpeg';
8849     app.jpegCheckBox_3.Position = [138 12 42 22];
8850
8851     % Create Button_26
8852     app.Button_26 = uibutton(app.Panel_17, 'push');
8853     app.Button_26.ButtonPushedFcn = createCallbackFcn(app, ↵
@Button_26Pushed, true);
8854     app.Button_26.FontSize = 10;
8855     app.Button_26.FontAngle = 'italic';
8856     app.Button_26.Tooltip = {'Load the preferred directory.'; '';↵
'"OUTPUT_FILES/Figures" folder will be created in the selected directory.'};
8857     app.Button_26.Position = [403 42 19 19];
8858     app.Button_26.Text = '...';
8859
8860     % Create FramerateEditFieldLabel
8861     app.FramerateEditFieldLabel = uilabel(app.Panel_17);
8862     app.FramerateEditFieldLabel.WordWrap = 'on';
8863     app.FramerateEditFieldLabel.Visible = 'off';

```



```

8864         app.FramerateEditFieldLabel.Tooltip = {'Frames per second.'; '';
'Lower value = slower animation.'};
8865         app.FramerateEditFieldLabel.Position = [383 7 77 33];
8866         app.FramerateEditFieldLabel.Text = 'Frame rate';
8867
8868         % Create FramerateEditField
8869         app.FramerateEditField = ueditfield(app.Panel_17, 'numeric');
8870         app.FramerateEditField.Limits = [0 100];
8871         app.FramerateEditField.ValueDisplayFormat = '%.0f';
8872         app.FramerateEditField.ValueChangedFcn = createCallbackFcn(app,
@FramerateEditFieldValueChanged, true);
8873         app.FramerateEditField.FontColor = [0.651 0.651 0.651];
8874         app.FramerateEditField.Visible = 'off';
8875         app.FramerateEditField.Tooltip = {''};
8876         app.FramerateEditField.Position = [448 15 30 17];
8877         app.FramerateEditField.Value = 5;
8878
8879         % Create mp4CheckBox_2
8880         app.mp4CheckBox_2 = uicheckbox(app.Panel_17);
8881         app.mp4CheckBox_2.ValueChangedFcn = createCallbackFcn(app,
@mp4CheckBox_2ValueChanged, true);
8882         app.mp4CheckBox_2.Enable = 'off';
8883         app.mp4CheckBox_2.Tooltip = {'Create an animation.'; ''; 'File is
saved as "animation.mp4."'; ''; 'Enabled when "Plot separately" is selected.'};
8884         app.mp4CheckBox_2.Text = '.mp4';
8885         app.mp4CheckBox_2.Position = [324 12 49 22];
8886
8887         % Create OutputDirectoryLabel_2
8888         app.OutputDirectoryLabel_2 = uilabel(app.Panel_17);
8889         app.OutputDirectoryLabel_2.Tooltip = {'Load the preferred
directory.'; ''; ''OUTPUT_FILES/Figures'' folder will be created in the selected
directory.'};
8890         app.OutputDirectoryLabel_2.Position = [40 40 94 22];
8891         app.OutputDirectoryLabel_2.Text = 'Output Directory';
8892
8893         % Create OutputDirectoryEditField_3
8894         app.OutputDirectoryEditField_3 = ueditfield(app.Panel_17,
'text');
8895         app.OutputDirectoryEditField_3.ValueChangedFcn = createCallbackFcn
(app, @OutputDirectoryEditField_3ValueChanged, true);
8896         app.OutputDirectoryEditField_3.Editable = 'off';
8897         app.OutputDirectoryEditField_3.FontColor = [0 0 1];
8898         app.OutputDirectoryEditField_3.Tooltip = {''};
8899         app.OutputDirectoryEditField_3.Placeholder = 'Default: Desktop';
8900         app.OutputDirectoryEditField_3.Position = [139 43 257 16];
8901
8902         % Create txtCheckBox_2
8903         app.txtCheckBox_2 = uicheckbox(app.Panel_17);
8904         app.txtCheckBox_2.Enable = 'off';
8905         app.txtCheckBox_2.Tooltip = {'Save raw data of U and V vector
files as tab-delimited texts.'; ''; 'It is enabled when "Plot separately" is
selected.'};
8906         app.txtCheckBox_2.Text = '.txt';
8907         app.txtCheckBox_2.Position = [238 12 39 22];
8908
8909         % Create tifCheckBox
8910         app.tifCheckBox = uicheckbox(app.Panel_17);
8911         app.tifCheckBox.Enable = 'off';

```

```

8912         app.tifCheckBox.Tooltip = { 'Georeferenced raster file.'; '' ;
'Enabled when "Plot separately" is selected.' };
8913         app.tifCheckBox.Text = '.tif';
8914         app.tifCheckBox.Position = [283 12 35 22];
8915
8916         % Create SaveMapLabel_2
8917         app.SaveMapLabel_2 = uilabel(app.VelocityMapTab);
8918         app.SaveMapLabel_2.BackgroundColor = [0.9412 0.9412 0.9412];
8919         app.SaveMapLabel_2.FontSize = 15;
8920         app.SaveMapLabel_2.FontWeight = 'bold';
8921         app.SaveMapLabel_2.FontColor = [0.0314 0.3686 0.6];
8922         app.SaveMapLabel_2.Position = [38 105 92 22];
8923         app.SaveMapLabel_2.Text = ' Save Map ';
8924
8925         % Create BasemapandOverlaysLabel
8926         app.BasemapandOverlaysLabel = uilabel(app.VelocityMapTab);
8927         app.BasemapandOverlaysLabel.BackgroundColor = [0.9412 0.9412 0.9412];
8928         app.BasemapandOverlaysLabel.FontSize = 15;
8929         app.BasemapandOverlaysLabel.FontWeight = 'bold';
8930         app.BasemapandOverlaysLabel.FontColor = [0.0314 0.3686 0.6];
8931         app.BasemapandOverlaysLabel.Position = [38 527 187 22];
8932         app.BasemapandOverlaysLabel.Text = ' Basemap and Overlays ';
8933
8934         % Create Panel_18
8935         app.Panel_18 = uipanel(app.VelocityMapTab);
8936         app.Panel_18.AutoResizeChildren = 'off';
8937         app.Panel_18.Position = [11 128 520 172];
8938
8939         % Create BoundaryLimitsLabel_2
8940         app.BoundaryLimitsLabel_2 = uilabel(app.Panel_18);
8941         app.BoundaryLimitsLabel_2.FontWeight = 'bold';
8942         app.BoundaryLimitsLabel_2.Tooltip = { 'When all textboxes are left
unchanged (all zeroes), the values are automatically set based on the extent of the
uploaded file and the input southwest corner coordinates' };
8943         app.BoundaryLimitsLabel_2.Position = [25 91 99 22];
8944         app.BoundaryLimitsLabel_2.Text = 'Boundary Limits';
8945
8946         % Create PlotindegreesCheckBox
8947         app.PlotindegreesCheckBox = uicheckbox(app.Panel_18);
8948         app.PlotindegreesCheckBox.ValueChangedFcn = createCallbackFcn(app,
@PlotindegreesCheckBoxValueChanged, true);
8949         app.PlotindegreesCheckBox.Tooltip = { 'Change the label format into
geographic coordinates' };
8950         app.PlotindegreesCheckBox.Text = 'Plot in degrees';
8951         app.PlotindegreesCheckBox.FontSize = 11;
8952         app.PlotindegreesCheckBox.Position = [403 56 96 22];
8953
8954         % Create EastEditField_2Label
8955         app.EastEditField_2Label = uilabel(app.Panel_18);
8956         app.EastEditField_2Label.Position = [331 62 29 22];
8957         app.EastEditField_2Label.Text = 'East';
8958
8959         % Create EastEditField_2
8960         app.EastEditField_2 = uieditfield(app.Panel_18, 'numeric');
8961         app.EastEditField_2.ValueDisplayFormat = '%8.4f';
8962         app.EastEditField_2.ValueChangedFcn = createCallbackFcn(app,
@EastEditField_2ValueChanged, true);

```



```

8963     app.EastEditField_2.FontColor = [0.651 0.651 0.651];
8964     app.EastEditField_2.Position = [265 65 60 16];
8965
8966     % Create WestEditField_2Label
8967     app.WestEditField_2Label = uilabel(app.Panel_18);
8968     app.WestEditField_2Label.HorizontalAlignment = 'right';
8969     app.WestEditField_2Label.Position = [87 62 32 22];
8970     app.WestEditField_2Label.Text = 'West';
8971
8972     % Create WestEditField_2
8973     app.WestEditField_2 = uieditfield(app.Panel_18, 'numeric');
8974     app.WestEditField_2.ValueDisplayFormat = '%8.4f';
8975     app.WestEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@WestEditField_2ValueChanged, true);
8976     app.WestEditField_2.FontColor = [0.651 0.651 0.651];
8977     app.WestEditField_2.Position = [125 65 60 16];
8978
8979     % Create NorthEditField_2Label
8980     app.NorthEditField_2Label = uilabel(app.Panel_18);
8981     app.NorthEditField_2Label.HorizontalAlignment = 'right';
8982     app.NorthEditField_2Label.Position = [206 91 35 22];
8983     app.NorthEditField_2Label.Text = 'North';
8984
8985     % Create NorthEditField_2
8986     app.NorthEditField_2 = uieditfield(app.Panel_18, 'numeric');
8987     app.NorthEditField_2.ValueDisplayFormat = '%8.4f';
8988     app.NorthEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@NorthEditField_2ValueChanged, true);
8989     app.NorthEditField_2.FontColor = [0.651 0.651 0.651];
8990     app.NorthEditField_2.Position = [195 77 60 16];
8991
8992     % Create SouthEditField_2Label
8993     app.SouthEditField_2Label = uilabel(app.Panel_18);
8994     app.SouthEditField_2Label.HorizontalAlignment = 'right';
8995     app.SouthEditField_2Label.Position = [205 34 37 22];
8996     app.SouthEditField_2Label.Text = 'South';
8997
8998     % Create SouthEditField_2
8999     app.SouthEditField_2 = uieditfield(app.Panel_18, 'numeric');
9000     app.SouthEditField_2.ValueDisplayFormat = '%8.4f';
9001     app.SouthEditField_2.ValueChangedFcn = createCallbackFcn(app, ↵
@SouthEditField_2ValueChanged, true);
9002     app.SouthEditField_2.FontColor = [0.651 0.651 0.651];
9003     app.SouthEditField_2.Position = [195 53 60 16];
9004
9005     % Create Height_3
9006     app.Height_3 = uieditfield(app.Panel_18, 'numeric');
9007     app.Height_3.Limits = [1 Inf];
9008     app.Height_3.ValueDisplayFormat = '%2.0f';
9009     app.Height_3.ValueChangedFcn = createCallbackFcn(app, ↵
@Height_3ValueChanged, true);
9010     app.Height_3.FontColor = [0.651 0.651 0.651];
9011     app.Height_3.Enable = 'off';
9012     app.Height_3.Tooltip = {' '};
9013     app.Height_3.Position = [330 12 40 16];
9014     app.Height_3.Value = 8;
9015
9016     % Create ThicknessLabel 10

```

```

9017     app.ThicknessLabel_10 = uilabel(app.Panel_18);
9018     app.ThicknessLabel_10.Tooltip = {'Unit: inches'};
9019     app.ThicknessLabel_10.Position = [283 9 41 22];
9020     app.ThicknessLabel_10.Text = 'Height';
9021
9022     % Create AutoSetCheckBox_3
9023     app.AutoSetCheckBox_3 = uicheckbox(app.Panel_18);
9024     app.AutoSetCheckBox_3.ValueChangedFcn = createCallbackFcn(app, ↵
@AutoSetCheckBox_3ValueChanged, true);
9025     app.AutoSetCheckBox_3.Text = ' Auto Set';
9026     app.AutoSetCheckBox_3.FontSize = 11;
9027     app.AutoSetCheckBox_3.Position = [102 9 68 22];
9028     app.AutoSetCheckBox_3.Value = true;
9029
9030     % Create Width_3
9031     app.Width_3 = uieditfield(app.Panel_18, 'numeric');
9032     app.Width_3.Limits = [1 Inf];
9033     app.Width_3.ValueDisplayFormat = '%2.0f';
9034     app.Width_3.ValueChangedFcn = createCallbackFcn(app, ↵
@Width_3ValueChanged, true);
9035     app.Width_3.FontColor = [0.651 0.651 0.651];
9036     app.Width_3.Enable = 'off';
9037     app.Width_3.Tooltip = {' '};
9038     app.Width_3.Position = [225 12 41 16];
9039     app.Width_3.Value = 11;
9040
9041     % Create ThicknessLabel_11
9042     app.ThicknessLabel_11 = uilabel(app.Panel_18);
9043     app.ThicknessLabel_11.Tooltip = {'Unit: inches'};
9044     app.ThicknessLabel_11.Position = [188 9 37 22];
9045     app.ThicknessLabel_11.Text = 'Width';
9046
9047     % Create FigureSizeLabel_3
9048     app.FigureSizeLabel_3 = uilabel(app.Panel_18);
9049     app.FigureSizeLabel_3.FontWeight = 'bold';
9050     app.FigureSizeLabel_3.Position = [25 9 69 22];
9051     app.FigureSizeLabel_3.Text = 'Figure Size';
9052
9053     % Create CloseFiguresButton_3
9054     app.CloseFiguresButton_3 = uibutton(app.Panel_18, 'push');
9055     app.CloseFiguresButton_3.ButtonPushedFcn = createCallbackFcn(app, ↵
@CloseFiguresButton_3Pushed, true);
9056     app.CloseFiguresButton_3.Tooltip = {' '};
9057     app.CloseFiguresButton_3.Position = [403 10 89 20];
9058     app.CloseFiguresButton_3.Text = 'Close Figures';
9059
9060     % Create SettoDefaultButton
9061     app.SettoDefaultButton = uibutton(app.Panel_18, 'push');
9062     app.SettoDefaultButton.ButtonPushedFcn = createCallbackFcn(app, ↵
@SettoDefaultButtonPushed, true);
9063     app.SettoDefaultButton.Tooltip = {'Use the boundary limits of the ↵
U-vector file'};
9064     app.SettoDefaultButton.Position = [403 76 84 20];
9065     app.SettoDefaultButton.Text = 'Set to Default';
9066
9067     % Create ButtonGroup_12
9068     app.ButtonGroup_12 = uibuttongroup(app.Panel_18);
9069     app.ButtonGroup_12.AutoSizeChildren = 'off';

```

```

9070         app.ButtonGroup_12.SelectionChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_12SelectionChanged, true);
9071         app.ButtonGroup_12.SizeChangedFcn = createCallbackFcn(app, ↵
@ButtonGroup_12SizeChanged, true);
9072         app.ButtonGroup_12.Position = [106 126 311 30];
9073
9074         % Create PlotalldataainonefigureButton_2
9075         app.PlotalldataainonefigureButton_2 = uiradiobutton(app. ↵
ButtonGroup_12);
9076         app.PlotalldataainonefigureButton_2.Tooltip = {'All maps are ↵
displayed as subplots in one window'};
9077         app.PlotalldataainonefigureButton_2.Text = 'Plot all data in one ↵
figure';
9078         app.PlotalldataainonefigureButton_2.FontWeight = 'bold';
9079         app.PlotalldataainonefigureButton_2.Position = [13 4 164 22];
9080         app.PlotalldataainonefigureButton_2.Value = true;
9081
9082         % Create PlotseparatelyButton_3
9083         app.PlotseparatelyButton_3 = uiradiobutton(app.ButtonGroup_12);
9084         app.PlotseparatelyButton_3.Tooltip = {'One map per window'};
9085         app.PlotseparatelyButton_3.Text = 'Plot separately';
9086         app.PlotseparatelyButton_3.FontWeight = 'bold';
9087         app.PlotseparatelyButton_3.Position = [193 4 108 22];
9088
9089         % Create GeneralLayoutLabel_2
9090         app.GeneralLayoutLabel_2 = uilabel(app.VelocityMapTab);
9091         app.GeneralLayoutLabel_2.BackgroundColor = [0.9412 0.9412 0.9412];
9092         app.GeneralLayoutLabel_2.FontSize = 15;
9093         app.GeneralLayoutLabel_2.FontWeight = 'bold';
9094         app.GeneralLayoutLabel_2.FontColor = [0.0314 0.3686 0.6];
9095         app.GeneralLayoutLabel_2.Position = [38 290 131 22];
9096         app.GeneralLayoutLabel_2.Text = '  General Layout  ';
9097
9098         % Show the figure after all components are created
9099         app.UIFigure.Visible = 'on';
9100     end
9101 end
9102
9103 % App creation and deletion
9104 methods (Access = public)
9105
9106     % Construct app
9107     function app = FUNMAP_Output
9108
9109         runningApp = getRunningApp(app);
9110
9111         % Check for running singleton app
9112         if isempty(runningApp)
9113
9114             % Create UIFigure and components
9115             createComponents(app)
9116
9117             % Register the app with App Designer
9118             registerApp(app, app.UIFigure)
9119
9120             % Execute the startup function
9121             runStartupFcn(app, @startupFcn)
9122         else

```

```
9123
9124         % Focus the running singleton app
9125         figure(runningApp.UIFigure)
9126
9127         app = runningApp;
9128     end
9129
9130     if nargin == 0
9131         clear app
9132     end
9133 end
9134
9135 % Code that executes before app deletion
9136 function delete(app)
9137
9138     % Delete UIFigure when app is deleted
9139     delete(app.UIFigure)
9140 end
9141 end
9142 end
```